RF & Microwave Engineering

An Interactive Textbook

with MRI Physics & Coil Engineering

10 Chapters ▪ 89 Sections ▪ 124 figures

with companion interactive calculators

RF Toolbox

rftoolbox.ca

Compiled from the RF Toolbox reference library.

Preface

This book collects the RF Toolbox reference library — sixty-eight illustrated articles on radio-frequency engineering, microwave design, and magnetic-resonance imaging — into a single, sequenced volume, and extends it with further chapters on active-device, receiver, communication-system, antenna, and test engineering. Each reference topic began as a standalone web article at rftoolbox.ca; here they are re-ordered into a foundations-to-advanced progression, organised into ten chapters of themed sections, and typeset with their original diagrams.

The material spans classical electromagnetism, transmission-line theory and impedance matching, guided-wave structures, passive components and filters, noise and amplifier design, signal generation and frequency conversion, antennas, complete RF systems, electromagnetic compatibility, and a full treatment of MRI physics and RF coil engineering. Every chapter is paired online with an interactive calculator that lets you put its equations to work with your own numbers.

How to Use This Book

The ten chapters are arranged so that each builds on the last, but their sections are also written to stand alone for reference. If you are new to RF engineering, read Chapters 1 and 2 in order — waves, decibels, transmission lines, VSWR, the Smith chart, and matching are the vocabulary everything else uses. Practitioners can jump directly to the section they need.

Each section opens with *Learning Objectives* and closes with a *Worked Example*, *Key Takeaways*, and *Exercises*; worked answers to every exercise are collected in *Solutions to Exercises* at the back, and each section points to standard texts in the *References*. Displayed equations are boxed set-off results; symbols follow the conventions in the Notation section below. Wherever this book shows a formula, the online edition offers a live calculator for it: for example, the VSWR / return-loss converter, the Smith chart, the cascaded noise-figure and IP3 tools, the antenna calculators, and the MRI coil and SAR estimators. Visit rftoolbox.ca to use them.

Notation

Z_0	Characteristic impedance (ohms), typically 50 or 75 Ω.
Γ	Voltage reflection coefficient, $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$.
VSWR	Voltage standing-wave ratio, $(1 + \lvert\Gamma\rvert)/(1 - \lvert\Gamma\rvert)$.
λ	Wavelength; $\beta = 2\pi/\lambda$ is the phase constant.
$f,\ \omega$	Frequency (Hz) and angular frequency $\omega = 2\pi f$.
S_{ij}	Scattering parameters (S-parameters).
NF, F	Noise figure (dB) and noise factor (linear).
$B_0,\ B_1$	MRI static field and RF transmit field.
γ	Gyromagnetic ratio (MRI); also used for propagation constant in line theory.

Decibel quantities: dBm is power referenced to 1 milliwatt; dBc is relative to a carrier; dBi is antenna gain relative to an isotropic radiator.

Contents

Chapter 1

Electromagnetic Foundations

1.1 Maxwell's Equations

LEARNING OBJECTIVES

- State Maxwell's four equations and name the physical law each expresses.
- Explain how the displacement-current term makes self-sustaining electromagnetic waves possible.
- Derive the speed of light and the impedance of free space from the constants ε_0 and μ_0.

Maxwell's equations are the four partial differential equations that completely describe classical electromagnetism. They were formulated by James Clerk Maxwell in 1865 and unified electricity, magnetism, and optics into a single theoretical framework.

1.1.1 The Four Equations

$\nabla \cdot \mathbf{E} = \dfrac{\rho}{\varepsilon_0}$ — *Gauss's Law: electric field diverges from charges.*
$\nabla \cdot \mathbf{B} = 0$ — *Gauss's Law for Magnetism: no magnetic monopoles.*
$\nabla \times \mathbf{E} = -\dfrac{\partial \mathbf{B}}{\partial t}$ — *Faraday's Law: a changing magnetic field induces an electric field.*
$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \dfrac{\partial \mathbf{E}}{\partial t}$ — *Ampère–Maxwell Law: currents and changing electric fields create magnetic fields.*

1.1.2 Displacement Current

Maxwell's key insight was adding the displacement current term $\varepsilon_0 \partial \mathbf{E}/\partial t$ to Ampère's law. This predicts that a time-varying electric field produces a magnetic field even in the absence of physical current — allowing electromagnetic waves to propagate through a vacuum.

1.1.3 Electromagnetic Wave Speed

Taking the curl of Faraday's law and substituting Ampère–Maxwell gives a wave equation. The wave speed is:

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}} \approx 2.998 \times 10^8 \text{ m/s}$$

Maxwell recognised this as the speed of light, demonstrating that light is an electromagnetic wave.

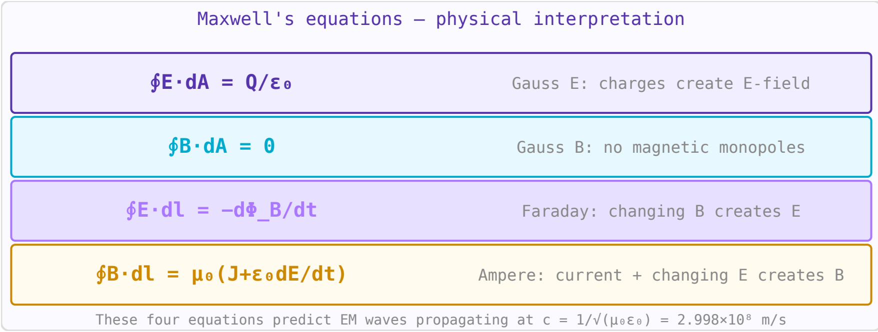


Figure 1.1: Maxwell's equations in integral form. Together they are the complete foundation of classical electromagnetism.

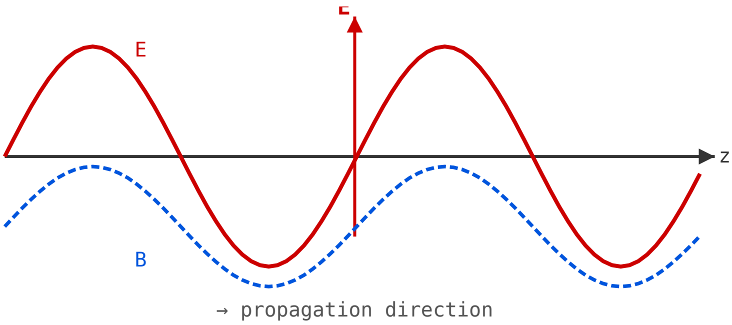


Figure 1.2: Transverse EM wave: E (red) and B (blue dashed) oscillate perpendicular to each other and to the propagation direction z.

1.1.4 In Materials

In a dielectric medium with relative permittivity ε_r and relative permeability μ_r, the wave speed becomes $v = c/\sqrt{\varepsilon_r \mu_r}$. This slowing is why wavelengths in PCB substrates are shorter than in free space.

WORKED EXAMPLE: THE SPEED OF LIGHT FROM FIRST PRINCIPLES

The Ampère–Maxwell and Faraday laws combine into a wave equation whose propagation speed is set entirely by the two constants of free space,

$$c = \frac{1}{\sqrt{\mu_0 \varepsilon_0}}.$$

With $\mu_0 = 4\pi \times 10^{-7}$ H/m and $\varepsilon_0 = 8.854 \times 10^{-12}$ F/m,

$$c = \frac{1}{\sqrt{(1.2566 \times 10^{-6})(8.854 \times 10^{-12})}} = \frac{1}{\sqrt{1.1127 \times 10^{-17}}} = 2.998 \times 10^{8} \text{ m/s}.$$

The same two constants fix the wave impedance of free space, $\eta_0 = \sqrt{\mu_0/\varepsilon_0} = 376.7\ \Omega \approx 120\pi\ \Omega$. That electromagnetism alone predicts the speed of light was Maxwell's decisive insight.

KEY TAKEAWAYS

- The four laws: Gauss (electric field diverges from charge), Gauss for magnetism (no monopoles), Faraday (a changing $\mathbf{B}$ induces $\mathbf{E}$), and Ampère–Maxwell (current *and* a changing $\mathbf{E}$ create $\mathbf{B}$).
- The displacement-current term $\varepsilon_0\,\partial\mathbf{E}/\partial t$ closes the loop between the two curl equations and yields travelling-wave solutions.
- $c = 1/\sqrt{\mu_0\varepsilon_0}$; in a medium the phase velocity is $v = c/\sqrt{\varepsilon_r\mu_r}$.

Exercises

1. Compute the intrinsic impedance of free space, $\eta_0 = \sqrt{\mu_0/\varepsilon_0}$.
2. A plane wave travels in a lossless dielectric with $\varepsilon_r = 4$, $\mu_r = 1$. Find its phase velocity and its wavelength at 1 GHz.
3. Which of Maxwell's equations underlies the operation of a transformer, and why?

Further reading: Jackson [3] (ch. 6–7); Pozar [1] (ch. 1); Ramo *et al.* [4].

1.2 Electromagnetic Waves

LEARNING OBJECTIVES

- Relate the electric and magnetic fields of a plane wave through the intrinsic impedance.
- Compute wavelength, phase velocity, and time-average power density (the Poynting flux).
- Distinguish linear, circular, and elliptical polarization.

An electromagnetic wave is a self-propagating oscillation of electric and magnetic fields, perpendicular to each other and to the direction of travel. No medium is required — EM waves propagate in vacuum at the speed of light.

1.2.1 Wave Properties

A plane wave travelling in the z-direction has:

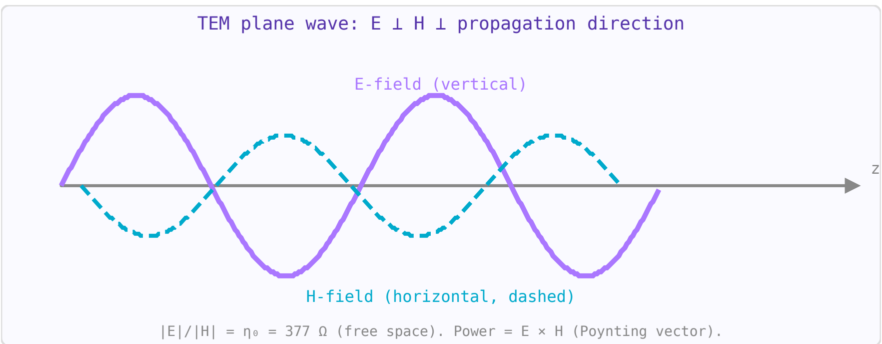


Figure 1.3: TEM plane wave: E and H oscillate perpendicular to each other and to the propagation direction z.

$$\mathbf{E}(z,t) = E_0 \cos(\omega t - kz)\,\hat{x} \quad \mathbf{B}(z,t) = \frac{E_0}{c}\cos(\omega t - kz)\,\hat{y}$$

The wavenumber $k = 2\pi/\lambda = \omega/c$, angular frequency $\omega = 2\pi f$, wavelength $\lambda = c/f$.

1.2.2 The EM Spectrum

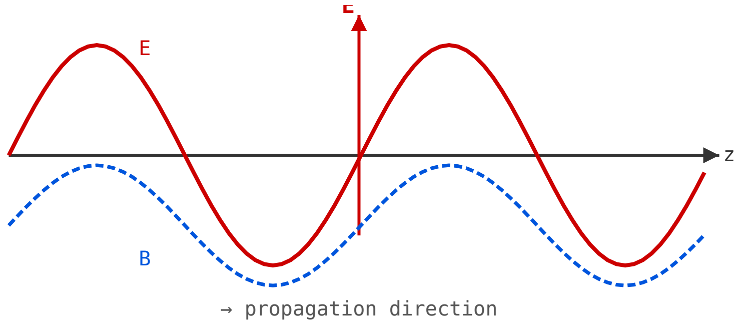


Figure 1.4: Linearly polarised plane wave propagating in the z-direction.

- ELF / VLF < 30 kHz — submarine communications, power lines
- MF / HF 300 kHz–30 MHz — AM radio, shortwave
- VHF / UHF 30 MHz–3 GHz — FM radio, TV, WiFi, cellular
- Microwave 3–300 GHz — radar, satellite, 5G, microwave ovens, MRI
- Infrared / Visible / UV above 300 GHz
- X-ray / Gamma very high frequencies — ionising radiation

1.2.3 Polarisation

The polarisation of an EM wave describes the orientation of the electric field vector. Linear polarisation: E oscillates in one plane. Circular polarisation: E rotates, tracing a helix. In MRI, circular (quadrature) polarisation of the B1 field is used to improve SNR by 40%.

1.2.4 Poynting Vector

Power flow per unit area is given by the Poynting vector: $\mathbf{S} = \mathbf{E} \times \mathbf{H}$ (W/m^2). Time-averaged power: $\langle S \rangle = E_0^2/(2\eta)$ where $\eta = \sqrt{\mu/\varepsilon}$ is the wave impedance (377 Ω in free space).

WORKED EXAMPLE: FIELDS AND POWER IN A PLANE WAVE

A uniform plane wave in free space has a peak electric field $E_0 = 10$ V/m. The magnetic field follows from the intrinsic impedance,

$$H_0 = \frac{E_0}{\eta_0} = \frac{10}{376.7} = 2.65 \times 10^{-2} \text{ A/m}.$$

The time-average power density carried by the wave is

$$S_{\text{avg}} = \frac{E_0^2}{2\eta_0} = \frac{10^2}{2(376.7)} = 0.133 \text{ W/m}^2.$$

$\mathbf{E}$, $\mathbf{H}$, and the direction of propagation are mutually perpendicular, and the power flows along $\mathbf{S} = \mathbf{E} \times \mathbf{H}$.

KEY TAKEAWAYS

- In a TEM wave $\mathbf{E} \perp \mathbf{H} \perp$ direction of travel, and $|E|/|H| = \eta$ (the intrinsic impedance).
- Free space: $\eta_0 = 120\pi \approx 377\ \Omega$; in a medium $\eta = \eta_0\sqrt{\mu_r/\varepsilon_r}$.
- Time-average power density $S_{\text{avg}} = E_0^2/(2\eta)$; polarization is the tip-path traced by $\mathbf{E}$ over one cycle.

Exercises

1. A wave in free space has a power density of 1 mW/cm^2 (= 10 W/m^2). Find its peak electric field.
2. What is the free-space wavelength at 2.45 GHz?
3. Two equal-amplitude, orthogonal linear components combine to give circular polarization. What phase difference between them is required?

Further reading: Balanis [14] (ch. 4); Pozar [1] (ch. 1); Ramo *et al.* [4].

1.3 Dielectrics and Permittivity

LEARNING OBJECTIVES

- Define relative permittivity ε_r and the loss tangent $\tan\delta$.
- Compute how a dielectric slows waves and shortens wavelength.
- Choose a substrate for low loss at high frequency.

A dielectric is an insulating material that, when placed in an electric field, stores energy by polarising its molecules. The key RF properties of a dielectric are its relative permittivity (dielectric constant) and loss tangent.

1.3.1 Relative Permittivity (ε_r)

The dielectric constant ε_r relates the permittivity of the material to that of free space: $\varepsilon = \varepsilon_r\varepsilon_0$. It determines how much wavelengths are shortened in the material: $\lambda = \lambda_0/\sqrt{\varepsilon_r}$.

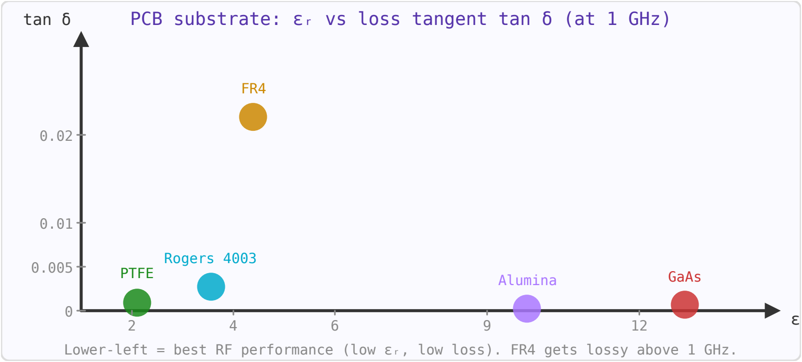


Figure 1.5: PCB substrate map. FR4 is cheapest but lossy; Rogers and PTFE offer much lower $\tan\delta$ for microwave applications.

1.3.2 Loss Tangent ($\tan\delta$)

A real dielectric also dissipates energy. The loss tangent represents the ratio of the imaginary to real part of the complex permittivity:

$$\tan\delta = \frac{\varepsilon_r''}{\varepsilon_r'}$$

Lower $\tan\delta$ means lower dielectric loss. Attenuation in a transmission line increases with loss tangent and frequency.

1.3.3 Common PCB Substrate Materials

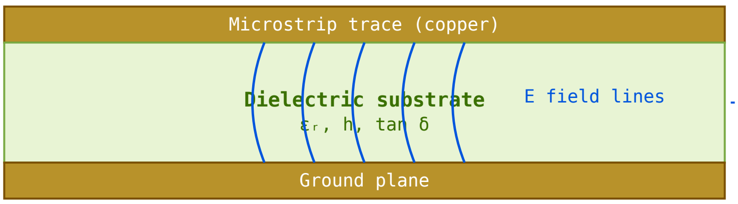


Figure 1.6: Microstrip cross-section. Field lines extend partly into air above the trace, giving εeff < εr.

- FR4 — $\varepsilon_r \approx 4.4$, $\tan\delta \approx 0.02$. Standard PCB material, adequate to ~5 GHz.
- Rogers 4003C — $\varepsilon_r = 3.55$, $\tan\delta = 0.0027$. Low loss, stable with temperature, suitable to 77 GHz.
- PTFE (Teflon) — $\varepsilon_r \approx 2.1$, \(\tan\delta < 0.001\). Excellent high-frequency performance, used in coaxial cable dielectrics.
- Alumina (Al_2O_3) — $\varepsilon_r \approx 9.8$, $\tan\delta \approx 0.001$. Used in MMIC substrates.

1.3.4 Frequency Dependence

Both ε_r and $\tan\delta$ vary with frequency. For PCB design above 1 GHz, always use substrate data measured at the operating frequency rather than the low-frequency value printed on datasheets.

WORKED EXAMPLE: WAVELENGTH AND VELOCITY IN A SUBSTRATE

A signal propagates through bulk PTFE, $\varepsilon_r = 2.1$ (a low-loss microwave dielectric). The wave slows by $\sqrt{\varepsilon_r}$, so the *velocity factor* is

$$\mathrm{VF} = \frac{v}{c} = \frac{1}{\sqrt{\varepsilon_r}} = \frac{1}{\sqrt{2.1}} = 0.69,$$

i.e. the wave travels at 69% of the free-space speed. At 5 GHz the wavelength inside the material is

$$\lambda = \frac{c}{f\sqrt{\varepsilon_r}} = \frac{3 \times 10^8}{(5 \times 10^9)(1.449)} = 41.4 \text{ mm},$$

versus 60 mm in free space. (On a real microstrip the relevant quantity is the *effective* permittivity, treated in Part III.)

KEY TAKEAWAYS

- A dielectric slows waves: $v = c/\sqrt{\varepsilon_r}$, so wavelength shrinks by $\sqrt{\varepsilon_r}$.
- The loss tangent $\tan\delta = \varepsilon''/\varepsilon'$ measures dielectric loss; smaller is better, and it matters more as frequency rises.
- Both ε_r and $\tan\delta$ drift with frequency (dispersion), so substrates are specified at a stated test frequency.

Exercises

1. A coaxial cable uses solid PTFE ($\varepsilon_r = 2.1$). What is its velocity factor?
2. Find the wavelength of a 5 GHz signal inside bulk Rogers RO4350 ($\varepsilon_r = 3.66$).
3. At 10 GHz, which loses less energy per wavelength: FR-4 ($\tan\delta \approx 0.02$) or PTFE ($\tan\delta \approx 0.001$)?

Further reading: Pozar [1] (ch. 1); Ludwig and Bogdanov [6] (ch. 2).

1.4 Magnetic Materials

LEARNING OBJECTIVES

- Define relative permeability μ_r and its effect on inductance.
- Explain why ferrites — not laminated iron — are used at RF.
- Compute the inductance of a wound magnetic core.

Magnetic materials are used in RF and power electronics to make inductors, transformers, ferrite beads, and circulators. Understanding their properties is essential for choosing the right core material.

1.4.1 Relative Permeability (μ_r)

Permeability μ_r relates the magnetic flux density to the applied field: $B = \mu_r\mu_0 H$. High μ_r increases inductance but also limits useful frequency range (permeability drops at high frequencies).

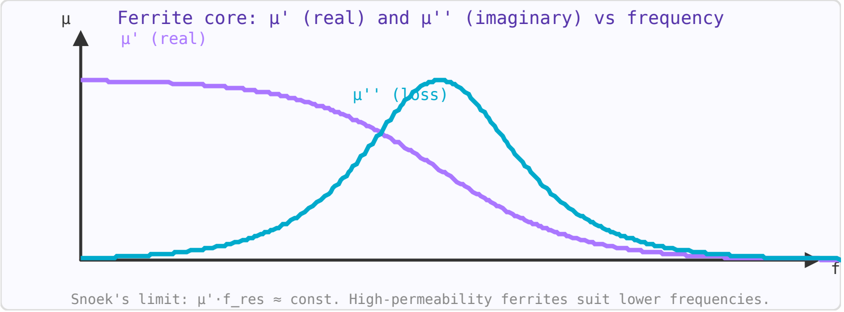


Figure 1.7: Ferrite permeability: μ' falls above resonance; μ" (loss) peaks at resonance. Use below resonance for low-loss performance.

1.4.2 Ferrites

Ferrites are ceramic magnetic materials with high resistivity (minimising eddy current losses at RF). They are used in:

- **Ferrite beads** — suppress high-frequency noise on power lines and signal traces by presenting high impedance above a target frequency.
- **Ferrite cores** — increase inductance of RF chokes and RF transformers, particularly below 50 MHz.
- **Circulators and isolators** — exploit the non-reciprocal properties of magnetised ferrite (Faraday rotation) to route signals directionally.

1.4.3 Core Saturation

At high flux densities, magnetic cores saturate — μ_r collapses, inducing harmonics and power loss. Always verify that peak flux density is below the saturation flux density B_{sat} of the chosen material under worst-case conditions.

1.4.4 Core Loss

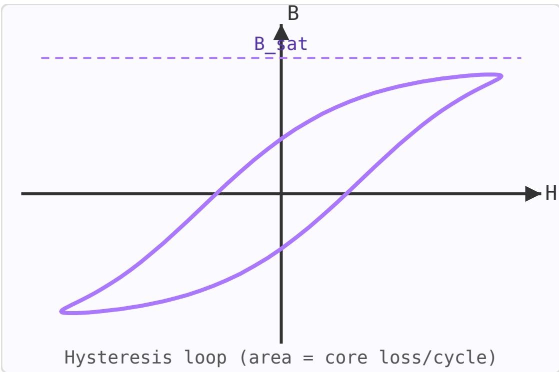


Figure 1.8: Hysteresis loop. Area enclosed = energy dissipated as heat per cycle.

Core loss (hysteresis + eddy current losses) is characterised by the complex permeability $\mu = \mu' - j\mu''$. The loss factor μ''/μ' (analogous to $\tan\delta$) increases with frequency. Ferrite materials are categorised by their intended frequency range (e.g. MnZn ferrites for <5 MHz, NiZn for 1–100 MHz).

WORKED EXAMPLE: INDUCTANCE OF A FERRITE TOROID

A toroidal ferrite core has relative permeability $\mu_r = 125$, cross-sectional area $A = 0.10\ \text{cm}^2 = 1.0 \times 10^{-5}\ \text{m}^2$, and mean magnetic path length $l = 3.0\ \text{cm} = 0.030\ \text{m}$. Wound with $N = 20$ turns, its inductance is

$$L = \frac{\mu_0\,\mu_r\,N^2 A}{l} = \frac{(4\pi \times 10^{-7})(125)(20^2)(1.0 \times 10^{-5})}{0.030}.$$

Evaluating: $\mu_0\mu_r = 1.571 \times 10^{-4}$; times $N^2A = 4.0 \times 10^{-3}$ gives 6.28×10^{-7}; divided by l gives

$$L = 2.09 \times 10^{-5}\ \text{H} = 20.9\ \mu\text{H}.$$

Without the core ($\mu_r = 1$) the same winding would give only 0.17 μH — the ferrite multiplies inductance by μ_r.

KEY TAKEAWAYS

- Permeability multiplies inductance ($L \propto \mu_r$) and, because $L \propto N^2$, inductance scales with the *square* of the turns.
- Ferrites combine high μ_r with high resistivity, which suppresses the eddy-current losses that make laminated iron unusable at RF.
- Cores saturate: beyond the rated flux μ collapses. Ferrite beads deliberately add frequency-dependent *loss* to damp common-mode RF.

Exercises

1. You double the number of turns on a fixed core. By what factor does the inductance change?
2. Why is laminated silicon-steel unsuitable for an inductor at 100 MHz?
3. A ferrite bead is rated to present about 100 Ω at 100 MHz. At that frequency is the impedance mainly reactive or resistive, and why?

Further reading: Hayward [7] (ch. 2); Ott [18] (ferrites for EMC).

1.5 Skin Effect

LEARNING OBJECTIVES

- Compute the skin depth of a conductor at a given frequency.
- Explain how the skin effect raises a conductor's AC resistance.
- Estimate the fraction of current flowing within one skin depth.

The skin effect is the tendency of alternating electric current to flow primarily near the surface of a conductor. At higher frequencies, the current density decreases exponentially with depth from the surface, concentrating in a thin layer called the skin depth.

1.5.1 Skin Depth Formula

$$\delta = \sqrt{\frac{2}{\omega\mu\sigma}} = \sqrt{\frac{1}{\pi f \mu\sigma}}$$

where $\omega = 2\pi f$ is angular frequency, μ is permeability, and σ is conductivity. Current density falls to $1/e \approx 37\%$ of its surface value at depth δ.

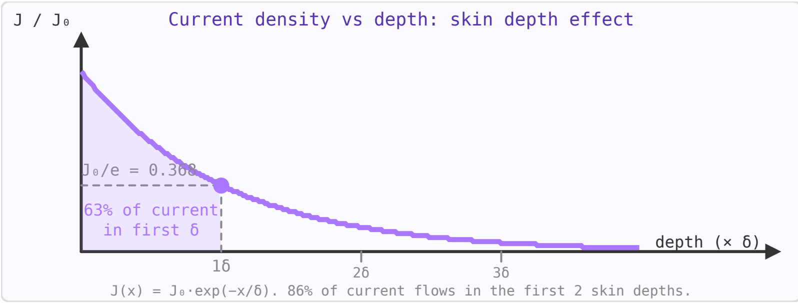


Figure 1.9: Current density decays exponentially with depth. At depth = δ, J has fallen to 1/e $\approx$ 37% of its surface value.

1.5.2 Values for Common Conductors at 100 MHz

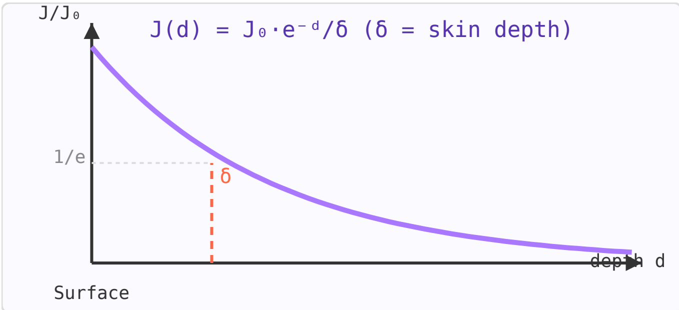


Figure 1.10: Current density J vs depth d. At one skin depth δ, J falls to $1/e \approx 37\%$ of its surface value.

- Copper ($\sigma = 5.8 \times 10^7$ S/m): $\delta \approx 6.6$ μm
- Aluminium ($\sigma = 3.8 \times 10^7$ S/m): $\delta \approx 8.2$ μm
- Silver ($\sigma = 6.3 \times 10^7$ S/m): $\delta \approx 6.3$ μm

1.5.3 Impact on RF Resistance

Because current only flows in a thin annulus near the surface, the effective cross-sectional area of the conductor is reduced, increasing resistance compared to DC. The AC resistance of a round wire of radius $r \gg \delta$ is approximately:

$$R_{AC} \approx R_{DC} \cdot \frac{r}{2\delta}$$

For a 1 mm diameter copper wire at 100 MHz, $R_{AC}/R_{DC} \approx 38$. This is why multi-strand (Litz) wire is used in high-Q inductors at lower RF frequencies — many thin strands, each smaller than the skin depth, provide more effective conductor area.

1.5.4 Design Implications

- Conductor thickness in PCBs should be at least 3–5 skin depths to minimise resistive loss.

- Silver plating adds only marginal benefit over copper above ~100 MHz (skin depth ~6 μm) as long as the copper is thicker than a few skin depths.
- In MRI coils at 128 MHz (3 T), copper tubing (thick wall) is preferred over solid wire for high-Q resonators.

WORKED EXAMPLE: SKIN DEPTH IN COPPER AT 1 GHZ

The skin depth is

$$\delta = \frac{1}{\sqrt{\pi f \mu \sigma}},$$

with copper conductivity $\sigma = 5.8 \times 10^7$ S/m and $\mu = \mu_0 = 4\pi \times 10^{-7}$ H/m. At $f = 1$ GHz,

$$\pi f \mu \sigma = (\pi)(10^9)(1.2566 \times 10^{-6})(5.8 \times 10^7) = 2.29 \times 10^{11},$$

$$\delta = \frac{1}{\sqrt{2.29 \times 10^{11}}} = \frac{1}{4.78 \times 10^5} = 2.09 \times 10^{-6} \text{ m} \approx 2.1\ \mu\text{m}.$$

Almost all of the RF current is packed into the outer couple of microns of the conductor — which is why plating quality and surface roughness dominate loss at microwave frequencies.

KEY TAKEAWAYS

- $\delta = 1/\sqrt{\pi f \mu \sigma}$; skin depth falls as $1/\sqrt{f}$.
- About 63% $(1 - e^{-1})$ of the current flows within one skin depth of the surface.
- AC resistance therefore rises as $\sqrt{f}$; this favours large-surface or silver-plated conductors and makes surface finish critical.

Exercises

1. Copper's skin depth is 2.1 μm at 1 GHz. What is it at 100 MHz?
2. If the operating frequency increases by a factor of 100, how does the skin depth change?
3. Gold ($\sigma \approx 4.1 \times 10^7$ S/m) is plated over copper. Estimate gold's skin depth at 1 GHz relative to copper's 2.1 μm.

Further reading: Pozar [1] (ch. 1); Ramo *et al.* [4].

Chapter 2

Transmission Lines and Impedance

2.1 Decibels, dBm and RF Power Units

LEARNING OBJECTIVES

- Convert between watts, dBm, and dBW.
- Add gains and losses in dB along a signal chain.
- Distinguish dBm, dBc, dBi, and dBW.

The decibel (dB) is the most important unit in RF engineering. It turns multiplication into addition, making cascade analysis practical. Once you're fluent in dB arithmetic, working with gain, loss, and power levels becomes fast and intuitive.

2.1.1 Why Logarithmic Scales?

A receiver handles signals ranging from 10^{-15} W (noise floor) to 10^{-3} W (nearby transmitter) — a range of 10^{12}. Writing and plotting these on a linear scale is impractical. Logarithms compress that 10^{12} ratio to a 120 dB range that fits on a graph. More importantly, a cascade of gain and loss stages multiplies powers together — in dB you just add.

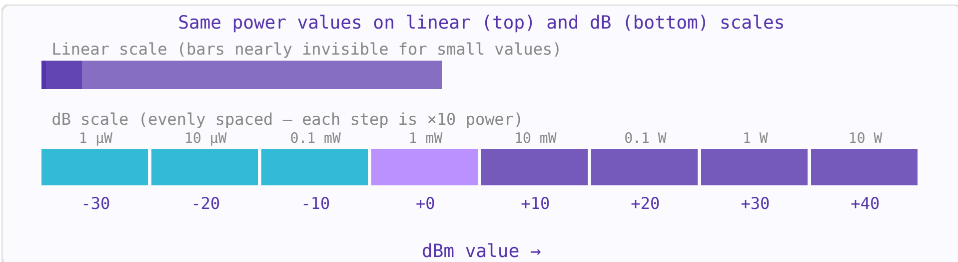


Figure 2.1: The dB scale compresses a million-fold power range into readable 10 dB steps.

2.1.2 The Decibel (dB)

The decibel expresses a *ratio* between two power values:

$$\text{ratio (dB)} = 10\log_{10}\left(\frac{P_2}{P_1}\right)$$

For voltage or current ratios (into the same impedance), the factor is 20 instead of 10 because power is proportional to voltage squared:

$$\text{ratio (dB)} = 20\log_{10}\left(\frac{V_2}{V_1}\right)$$

Key values to memorise:

Power ratio	Voltage ratio	dB	Memory aid
10×	3.16×	+10 dB	One order of magnitude
2×	1.41×	+3 dB	Double the power
1.26×	1.12×	+1 dB	Smallest audible difference
1×	1×	0 dB	No change
0.5×	0.71×	−3 dB	Half the power
0.1×	0.316×	−10 dB	One tenth
0.001×	0.0316×	−30 dB	One thousandth

2.1.3 Absolute Power: dBm and dBW

dB alone is a ratio — it needs a reference to become an absolute level. **dBm** uses 1 milliwatt as the reference:

$$P_{\text{dBm}} = 10\log_{10}\left(\frac{P_{\text{watts}}}{1\text{ mW}}\right)$$

dBW uses 1 watt. The conversion: P(dBm) = P(dBW) + 30.

Power	dBm	dBW	Context
1 kW	+60 dBm	+30 dBW	AM broadcast transmitter
10 W	+40 dBm	+10 dBW	Handheld radio TX
1 W	+30 dBm	0 dBW	Bluetooth TX max
100 mW	+20 dBm	−10 dBW	Wi-Fi typical
1 mW	0 dBm	−30 dBW	Reference level
1 μW	−30 dBm	−60 dBW	Strong receiver input
1 nW	−60 dBm	−90 dBW	Typical receiver input
−100 dBm	−100 dBm	−130 dBW	Near noise floor
10^{-15} W = 1 fW	−120 dBm	−150 dBW	Thermal noise floor, 1 MHz BW, room temperature

2.1.4 dB Arithmetic in Cascades

This is where dB pays off. A system with a 20 dB LNA, a 7 dB cable loss, a 10 dB mixer loss, and 30 dB of IF gain, driven by a −90 dBm signal:

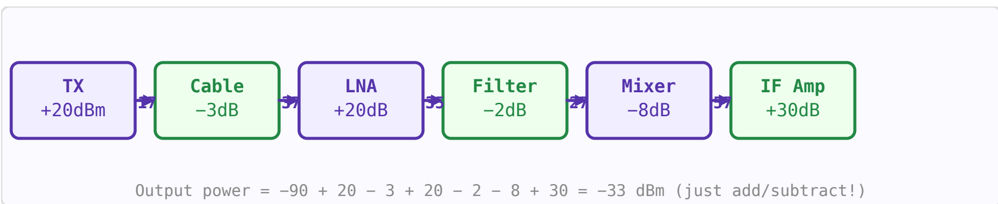


Figure 2.2: In dB arithmetic, cascade gain/loss is simple addition. No need to multiply or divide linear values.

$$P_{out} = -90 + 20 - 7 - 10 + 30 = -57\text{ dBm}$$

Compare doing this in linear: $10^{-12} \times 100 \times 0.2 \times 0.1 \times 1000 = 2{\times}10^{-12}$ W. Much harder to get right mentally.

2.1.5 Other dB Variants

Unit	Reference	Use
dBm	1 mW	Absolute RF power, transmitters and receivers
dBW	1 W	Satellite EIRP specifications
dBc	Carrier power	Relative: spurious, harmonics, phase noise
dBi	Isotropic antenna	Antenna gain (absolute)
dBd	Half-wave dipole	Antenna gain (relative to dipole). 0 dBd = 2.15 dBi
dBμV	1 μV	EMC and receiver sensitivity specifications in Europe
dBμV/m	1 μV/m	Field strength in EMC measurements
dBFS	Full scale	Digital audio and ADC levels (always $\leq$ 0 dBFS)

2.1.6 Noise Floor and kTB

Thermal noise power available from a resistor at temperature T in bandwidth B:

$$P_{noise} = kTB \approx -174 \text{ dBm/Hz at } 290 \text{ K}$$

In a bandwidth B (Hz): $P_{noise} = -174 + 10\log_{10}(B)$ dBm. In 1 MHz bandwidth: $-174 + 60 = -114$ dBm. In 20 MHz (Wi-Fi): $-174 + 73 = -101$ dBm. Add the receiver noise figure to get the actual noise floor.

2.1.7 Converting Back to Linear

$$P_{\text{watts}} = 10^{P_{dBm}/10} \times 10^{-3} \text{ W}$$

Shortcut: 0 dBm = 1 mW, every +10 dB multiplies by 10, every +3 dB doubles.

WORKED EXAMPLE: A GAIN CHAIN IN DBM

An antenna delivers -40 dBm to a front end with a $+15$ dB LNA, a -6 dB filter, and $+30$ dB of IF gain. Because gains multiply as powers but *add* as decibels, the output level is

$$P_{out} = -40 + 15 - 6 + 30 = -1 \text{ dBm},$$

i.e. $P = 10^{-0.1} = 0.79$ mW. Working in dB turns a product of linear power ratios into a running sum.

KEY TAKEAWAYS

- $P_{\text{dBm}} = 10\log_{10}(P/1\,\text{mW})$; 0 dBm = 1 mW, +30 dBm = 1 W.
- Cascaded gains and losses simply add in dB.
- dBc is relative to a carrier, dBi is antenna gain vs. isotropic, dBW = dBm -30.

Exercises

1. Convert +43 dBm to watts.
2. A signal is -70 dBm and a spur is -90 dBm. What is the spur level in dBc?

3. Express +30 dBm in dBW.

Further reading: Pozar [1] (ch. 10); Hayward [7].

2.2 Complex Impedance and Reactance

LEARNING OBJECTIVES

- Compute the reactance of an inductor or capacitor at a frequency.
- Combine resistance and reactance into a complex impedance and find its magnitude and phase.
- Relate quality factor to reactance and loss.

At DC, resistance is the only opposition to current flow. At RF, inductors and capacitors also oppose current — but in a frequency-dependent and phase-shifting way. The complete description requires complex numbers: impedance $Z = R + jX$, where the imaginary part X is the reactance.

2.2.1 Phasors and Sinusoidal Steady State

In a circuit driven by a sinusoidal source at frequency f, all voltages and currents are sinusoids at the same frequency. We represent them as phasors — complex numbers whose magnitude gives amplitude and whose angle gives phase. Multiplying a phasor by j rotates it 90° (a quarter cycle), which is exactly what an inductor does: it makes current lag voltage by 90°.

2.2.2 Reactance

Reactance X is the imaginary part of impedance. It stores and releases energy without dissipating it:

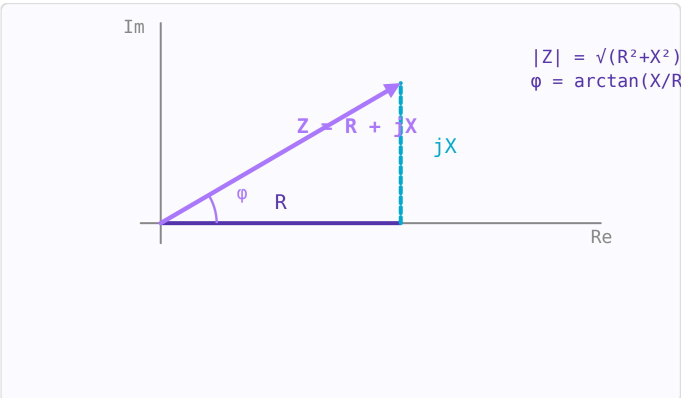


Figure 2.3: Impedance $Z = R + jX$ as a phasor in the complex plane. R is resistance, X is reactance.

Component	Impedance Z	Reactance X	Phase
Resistor R	R	0	V and I in phase
Inductor L	$j\omega L$	$+\omega L = +2\pi fL$	I lags V by 90°
Capacitor C	$1/(j\omega C)$	$-1/(\omega C) = -1/(2\pi fC)$	I leads V by 90°

At resonance ($\omega L = 1/\omega C$), the inductive and capacitive reactances cancel and the impedance is purely resistive.

2.2.3 Admittance Y = 1/Z

For parallel circuit analysis, admittance Y = G + jB is more convenient than impedance. G is the conductance (1/R) and B is the susceptance (−1/X for an inductor, +1/X for a capacitor — note the sign flip). On a Smith Chart, the admittance chart is the impedance chart rotated 180°.

$$Y = \frac{1}{Z} = \frac{1}{R + jX} = \frac{R}{R^2 + X^2} - j\frac{X}{R^2 + X^2} = G + jB$$

2.2.4 Series vs Parallel Equivalents

A real inductor has both inductance L and series resistance R_s. It can be represented equivalently as a parallel combination of L_p and R_p, where the two representations are equivalent at one frequency. This conversion matters when designing matching networks:

$$Q = \frac{X_s}{R_s} = \frac{R_p}{X_p} \qquad R_p = R_s(1 + Q^2) \qquad X_p = X_s\frac{Q^2 + 1}{Q^2}$$

2.2.5 Quality Factor Q

The Q factor of a reactive element describes how "ideal" it is — the ratio of energy stored to energy lost per cycle:

$$Q = \frac{|X|}{R} = \frac{\omega_0 L}{R_s} = \frac{1}{\omega_0 C R_s}$$

High Q means low loss. A typical SMD inductor at 100 MHz has Q = 30–80. A quartz crystal resonator has Q = 10,000–100,000. Q also sets bandwidth: $BW = f_0/Q$.

2.2.6 Frequency Dependence

At higher frequencies, parasitics dominate. A real capacitor has series inductance (ESL) that causes it to become inductive above its self-resonant frequency (SRF). A real inductor has inter-winding capacitance that causes it to self-resonate. Always check the SRF of a component against your operating frequency:

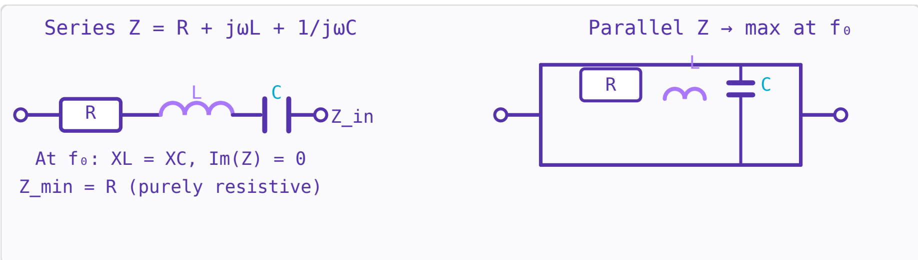


Figure 2.4: Series RLC: minimum impedance at resonance. Parallel RLC: maximum impedance at resonance.

Component value	Typical SRF	Notes
10 μF electrolytic	500 kHz	Useless for RF bypassing
100 nF ceramic (0402)	50 MHz	Good for audio, limited at RF
10 nF ceramic (0402)	200 MHz	OK for HF/VHF bypassing
100 pF ceramic (0402)	2 GHz	Good RF bypass to ~1 GHz
10 pF ceramic (0402)	8 GHz	Good for microwave matching

1 pF ceramic (0402)	20+ GHz	Usable at mmWave

2.2.7 Impedance vs Frequency: What to Expect

A 50 Ω transmission line port expects 50 + j0 Ω. Impedance that deviates from this causes reflections. The Smith Chart provides an intuitive visual representation of complex impedance across frequency — a single-frequency impedance is a point, and an impedance varying with frequency traces a curve. Understanding complex impedance is the prerequisite for reading and using the Smith Chart.

WORKED EXAMPLE: IMPEDANCE OF A SERIES RL BRANCH

A 50 Ω resistor is in series with a 10 nH inductor at 1 GHz. The inductive reactance is

$$X_L = 2\pi f L = 2\pi(10^9)(10 \times 10^{-9}) = 62.8\ \Omega,$$

so $Z = 50{+}j62.8\ \Omega$, with $|Z| = \sqrt{50^2 + 62.8^2} = 80.3\ \Omega$ and $\angle Z = \arctan(62.8/50) = 51.5°$.

KEY TAKEAWAYS

- $X_L = \omega L$ rises with frequency; $X_C = 1/(\omega C)$ falls with frequency.
- $Z = R + jX$: magnitude $\sqrt{R^2 + X^2}$, phase $\arctan(X/R)$.
- A reactive element's quality factor is $Q = |X|/R$.

Exercises

1. Find the reactance of a 2 pF capacitor at 2.4 GHz.
2. At what frequency does a 10 nH inductor present 50 Ω of reactance?
3. A branch has $Z = 30 - j40\ \Omega$. Give its magnitude and phase.

Further reading: Pozar [1] (ch. 1); Ludwig and Bogdanov [6] (ch. 1).

2.3 Transmission Line Theory

LEARNING OBJECTIVES

- Relate characteristic impedance to a line's per-unit-length inductance and capacitance.
- Compute the reflection coefficient of a loaded line.
- Use a quarter-wave transformer as an impedance inverter.

A transmission line is a structure that guides electromagnetic waves from one point to another. At RF frequencies, ordinary wire connections behave as transmission lines — understanding this is essential for signal integrity and impedance matching.

2.3.1 Telegrapher's Equations

A transmission line is modelled as a ladder network of distributed resistance R, inductance L, capacitance C, and conductance G per unit length. The resulting wave equations are:

$$\frac{\partial V}{\partial z} = -(R + j\omega L)I \qquad \frac{\partial I}{\partial z} = -(G + j\omega C)V$$

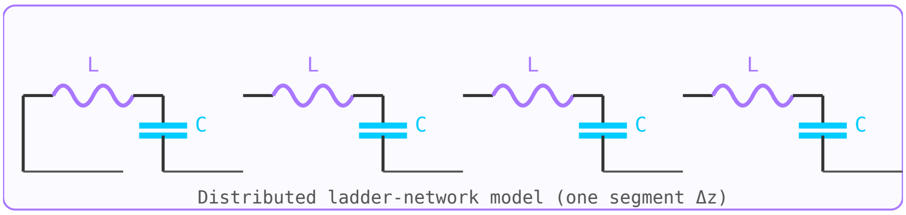


Figure 2.5: Distributed-element model: series L, shunt C per unit length Δz.

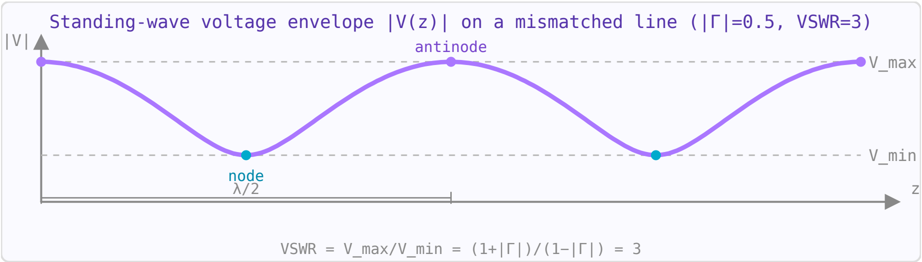


Figure 2.6: Standing wave on a mismatched line: V_max/V_min = VSWR = 3 for |Γ|=0.5.

2.3.2 Characteristic Impedance

$$Z_0 = \sqrt{\frac{R + j\omega L}{G + j\omega C}} \xrightarrow{\text{lossless}} \sqrt{\frac{L}{C}}$$

Coax with 50 Ω characteristic impedance: $Z_0 = (60/\sqrt{\varepsilon_r}) \ln(D/d)$.

2.3.3 Reflection Coefficient

When a wave hits a load impedance $Z_L \neq Z_0$, some power is reflected. The voltage reflection coefficient is:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

Return loss $= -20 \log_{10} |\Gamma|$ dB. VSWR $= (1 + |\Gamma|)/(1 - |\Gamma|)$.

2.3.4 Standing Waves

The superposition of forward and reflected waves creates a standing wave pattern on the line. The voltage varies with position; the ratio of maximum to minimum voltage is the VSWR. A matched load ($\Gamma = 0$) has VSWR $= 1$ and no standing waves.

2.3.5 Input Impedance

A transmission line of length ℓ terminated in Z_L presents input impedance:

$$Z_{in} = Z_0 \frac{Z_L + jZ_0 \tan(\beta\ell)}{Z_0 + jZ_L \tan(\beta\ell)}$$

Special cases: $\ell = \lambda/4 \rightarrow Z_{in} = Z_0^2/Z_L$ (quarter-wave transformer); $\ell = \lambda/2 \rightarrow Z_{in} = Z_L$ (half-wave section is transparent).

WORKED EXAMPLE: QUARTER-WAVE TRANSFORMER

To match a 100 Ω load to a 50 Ω line, insert a quarter-wavelength section of impedance

$$Z_1 = \sqrt{Z_0 Z_L} = \sqrt{50 \times 100} = 70.7\ \Omega.$$

At the design frequency the $\lambda/4$ line transforms the load to $Z_1^2/Z_L = 70.7^2/100 = 50\ \Omega$ — a perfect match. The match degrades away from that frequency.

KEY TAKEAWAYS

- $Z_0 = \sqrt{L/C}$ from the per-unit-length L and C.
- $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$; a matched load ($Z_L = Z_0$) gives $\Gamma = 0$.
- A $\lambda/4$ line inverts impedance: $Z_{in} = Z_0^2/Z_L$.

Exercises

1. Find the reflection coefficient of a 75 Ω load on a 50 Ω line.
2. What Z_0 quarter-wave section matches 25 Ω to 50 Ω?
3. A 70.7 Ω quarter-wave line is terminated in 50 Ω. Find its input impedance.

Further reading: Pozar [1] (ch. 2); Collin [2] (ch. 3).

2.4 Transmission Line Loss and Attenuation

LEARNING OBJECTIVES

- Distinguish conductor loss from dielectric loss.
- Convert an attenuation rate (dB/m) into total loss over a length.
- Estimate propagation delay from the velocity factor.

Every real transmission line attenuates signals as they travel. Two independent mechanisms cause this loss: **conductor loss** (resistive heating in the metal) and **dielectric loss** (power absorbed in the substrate or insulator). Total attenuation $\alpha = \alpha c + \alpha d$ in Np/m; multiply by 8.686 to convert to dB/m.

2.4.1 Conductor Loss & Skin Effect

At RF frequencies current crowds into a thin surface layer of thickness $\delta s = \sqrt{}(1/\pi f \mu_0 \sigma)$ — the skin depth. AC resistance rises as $\sqrt{}$f. The surface resistance (resistance per square) is:

$$R_s = \sqrt{\frac{\pi f \mu_0}{\sigma}} \quad [\Omega/\square]$$

For a **coaxial line** with inner conductor radius a and outer radius b:

$$\alpha_c = \frac{R_s}{4\pi Z_0}\left(\frac{1}{a} + \frac{1}{b}\right) \quad [\text{Np/m}]$$

For **microstrip** the conductor loss depends on the effective strip width w and involves a correction for current crowding at the edges; simplified for w/h > 1:

$$\alpha_c \approx \frac{R_s}{Z_0\, w} \quad [\mathrm{Np/m}]$$

2.4.2 Dielectric Loss

The loss tangent $\tan\delta$ of the substrate causes power absorption proportional to frequency. For **microstrip** (Pozar, Microwave Engineering eq. 3.30):

$$\alpha_d = \frac{k_0\, \varepsilon_r\, (\varepsilon_{\mathrm{eff}} - 1)\, \tan\delta}{2\sqrt{\varepsilon_{\mathrm{eff}}}\, (\varepsilon_r - 1)} \quad [\mathrm{Np/m}]$$

where $k_0 = 2\pi f/c$ and εeff is the effective permittivity. For **coaxial** line:

$$\alpha_d = \frac{\pi f \sqrt{\varepsilon_r}}{c}\, \tan\delta \quad [\mathrm{Np/m}]$$

Dielectric loss is proportional to frequency and often dominates conductor loss at GHz frequencies in lossy substrates such as FR4 ($\tan\delta \approx 0.02$).

2.4.3 Total Loss & Unit Conversion

$$\alpha_{\mathrm{total}} = \alpha_c + \alpha_d \quad [\mathrm{Np/m}] \qquad \alpha_{\mathrm{dB}} = 8.686\, \alpha_{\mathrm{total}} \quad [\mathrm{dB/m}]$$

Insertion loss over a physical length L is simply α_dB × L dB. Both frequency and length increase loss linearly (for dB/m at fixed f, or for fixed length scaling with f).

2.4.4 Propagation Delay

Signals travel at the phase velocity vph = c/$\sqrt{\varepsilon}$eff, giving a time delay:

$$t_d = \frac{L\sqrt{\varepsilon_{\mathrm{eff}}}}{c} \quad [\mathrm{s}]$$

On FR4 microstrip (εeff ≈ 3.5), vph ≈ 0.53c — roughly 6 ps/mm of delay. Delay is independent of frequency for a non-dispersive TEM line.

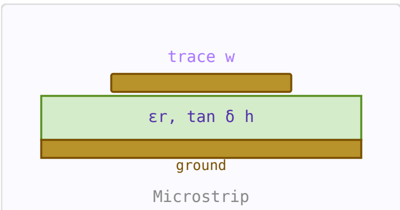


Figure 2.7: Microstrip cross-section showing substrate height h, trace width w.

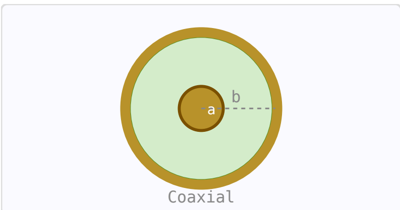


Figure 2.8: Coax cross-section. Conductor loss from inner (a) and outer (b) conductors.

2.4.5 Practical Values

- FR4 microstrip 50 Ω, 1.6 mm substrate: ≈ 0.5 dB/cm at 10 GHz
- Rogers RO4003C microstrip 50 Ω: ≈ 0.09 dB/cm at 10 GHz ($\tan\delta = 0.0027$)
- RG-58 coax: ≈ 0.22 dB/m at 100 MHz; ≈ 1.1 dB/m at 2.4 GHz
- LMR-400 coax: ≈ 0.068 dB/m at 1 GHz; ≈ 0.12 dB/m at 2.4 GHz

WORKED EXAMPLE: CABLE LOSS BUDGET

A 10 m coax run is rated 0.2 dB/m at 1 GHz, so the total attenuation is $0.2 \times 10 = 2$ dB: a +10 dBm signal arrives at +8 dBm (6.3 mW of the original 10 mW, about 63%). With velocity factor 0.66, the one-way delay over the run is

$$t = \frac{\ell}{\mathrm{VF} \cdot c} = \frac{10}{0.66 \times 3 \times 10^8} = 50.5 \text{ ns}.$$

KEY TAKEAWAYS

- Total loss (dB) = per-metre loss × length; dB losses add.
- Conductor loss grows as $\sqrt{f}$ (skin effect); dielectric loss grows roughly linearly with f.
- Delay $t = \ell/(\mathrm{VF} \cdot c)$; a lower velocity factor means more delay.

Exercises

1. A cable loses 0.15 dB/m. What is the total loss over 20 m?
2. What fraction of the power survives 3 dB of loss?
3. What is the delay through 5 m of coax with velocity factor 0.66?

Further reading: Pozar [1] (ch. 2); Ludwig and Bogdanov [6].

2.5 VSWR, Return Loss and Reflection Coefficient

LEARNING OBJECTIVES

- Convert among $|\Gamma|$, VSWR, and return loss.
- Relate reflected power to $|\Gamma|$.
- Choose a return-loss target for an application.

VSWR, return loss, reflection coefficient, and mismatch loss are four equivalent ways to describe the same physical phenomenon: how well two impedances are matched. They are all simple algebraic transformations of each other. Understanding all four is essential for RF work because different instruments and standards use different conventions.

2.5.1 Reflection Coefficient Γ

When a travelling wave encounters a discontinuity — a change in impedance — some energy is reflected. The complex voltage reflection coefficient at the load is:

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

where Z_0 is the reference (usually 50 Ω) and Z_L is the load impedance. $|\Gamma|$ ranges from 0 (perfect match) to 1 (total reflection). The angle of Γ gives the phase of the reflection.

2.5.2 Return Loss (RL)

Return loss is the magnitude of the reflection expressed in dB, always as a positive number for a passive load:

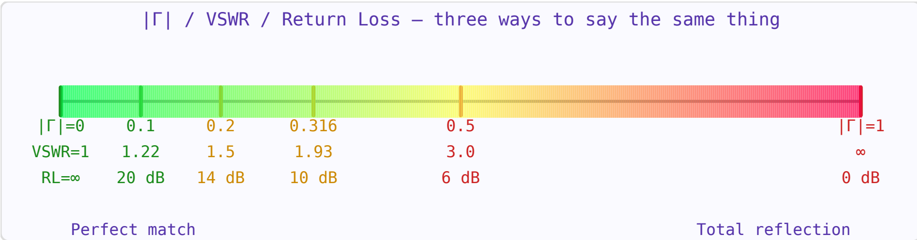


Figure 2.9: The match quality spectrum from perfect (green) to total reflection (red).

$$RL = -20 \log_{10} |\Gamma| \text{ dB}$$

Higher return loss means a better match. Typical design targets:

Return Loss	$\|\Gamma\|$	VSWR	Mismatch Loss	Context
∞ dB	0	1.00:1	0 dB	Perfect match (theoretical)
20 dB	0.100	1.22:1	0.04 dB	Good — typical antenna target
14 dB	0.200	1.50:1	0.18 dB	Acceptable for many systems
10 dB	0.316	1.93:1	0.46 dB	Minimum acceptable for most RF
6 dB	0.500	3.00:1	1.25 dB	Poor — significant power loss
0 dB	1.000	∞	∞ dB	Total reflection (open or short)

2.5.3 VSWR — Voltage Standing Wave Ratio

When forward and reflected waves coexist on a transmission line, they superimpose to create a standing wave pattern. The ratio of the maximum to minimum voltage amplitude along the line is the VSWR:

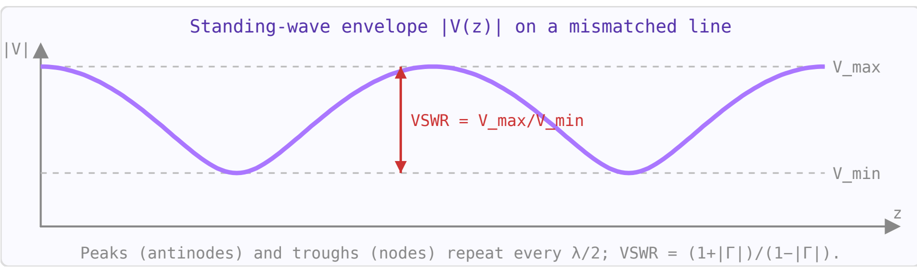


Figure 2.10: Voltage amplitude along a mismatched line. VSWR = ratio of maximum to minimum voltage.

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|}$$

A VSWR of 1 means no standing waves (perfect match). VSWR is measured with a slotted line or a VNA. It is always ≥ 1.

2.5.4 Mismatch Loss

Even without any dissipative loss, a mismatch causes a reduction in delivered power because some fraction of the incident power is reflected and not absorbed by the load:

$$ML = -10 \log_{10}(1 - |\Gamma|^2) \text{ dB}$$

This is also called the *transmission coefficient* $T = 1 - |\Gamma|^2$. At RL = 10 dB, mismatch loss is only 0.46 dB — often acceptable. But at RL = 6 dB ($|\Gamma| = 0.5$), 25% of the power is reflected and mismatch loss reaches 1.25 dB.

2.5.5 The S_{11} Connection

On a VNA, S_{11} is the complex reflection coefficient Γ measured at port 1. Its magnitude in dB equals the negative of the return loss: $|S_{11}|$_dB = −RL. A Smith Chart plots Γ directly in the complex plane.

2.5.6 Physical Interpretation

The standing wave on a mismatched line creates voltage maxima and minima that can exceed the incident amplitude. At VSWR = 2 the peak voltage is twice the matched case — this matters for power handling, dielectric breakdown, and cable ratings. High-power transmitters specify maximum allowed VSWR to protect their output stages.

WORKED EXAMPLE: A 75 Ω LOAD ON A 50 Ω LINE

For $Z_L = 75\ \Omega$ on $Z_0 = 50\ \Omega$, $\Gamma = (75 - 50)/(75 + 50) = 0.20$, hence

$$\text{VSWR} = \frac{1 + |\Gamma|}{1 - |\Gamma|} = \frac{1.2}{0.8} = 1.5, \qquad \text{RL} = -20 \log_{10}(0.2) = 14.0 \text{ dB}.$$

The reflected power is $|\Gamma|^2 = 4\%$ and the mismatch loss is $-10 \log_{10}(1 - 0.04) = 0.18$ dB.

KEY TAKEAWAYS

- $|\Gamma|$, VSWR, RL, and mismatch loss are one physical fact in four algebras.
- Reflected-power fraction $= |\Gamma|^2$; VSWR $= (1 + |\Gamma|)/(1 - |\Gamma|)$.
- RL ≥ 10 dB (VSWR ≤ 1.9) is a common practical floor.

Exercises

1. For VSWR = 3.0, find $|\Gamma|$, return loss, and reflected power.
2. A return loss of 20 dB corresponds to what $|\Gamma|$ and VSWR?
3. What VSWR and return loss does a perfectly matched load have?

Further reading: Pozar [1] (ch. 2); Ludwig and Bogdanov [6] (ch. 2).

2.6 The Smith Chart

LEARNING OBJECTIVES

- Read a normalized impedance from a point on the chart.
- Convert impedance to admittance with a 180° rotation.
- Use constant-$|\Gamma|$ circles and the wavelength scale for matching.

The Smith Chart is a polar plot of the reflection coefficient Γ with constant-resistance circles and constant-reactance arcs overlaid. Every point on the chart represents a unique normalised impedance $z = Z/Z_0$. Once you understand its structure, the chart makes impedance matching design immediate and visual.

2.6.1 Chart Structure

The chart is a circle of radius 1 (the $|\Gamma| = 1$ boundary, representing total reflection). The centre is the perfect match point ($z = 1$, $\Gamma = 0$). The right-most point is an open circuit ($z = \infty$). The left-most point is a short circuit ($z = 0$).

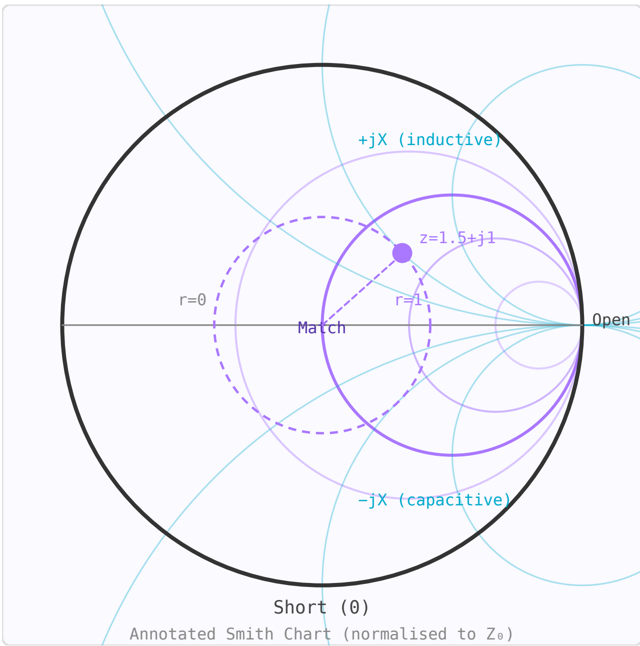


Figure 2.11: Smith Chart: constant-resistance circles (purple) and constant-reactance arcs (teal). Upper half = inductive; lower half = capacitive.

- **Constant-resistance circles** (drawn in purple on the RF Toolbox chart): Circles tangent to the right side. The $r = 1$ circle passes through the centre and represents all impedances with real part equal to Z_0.
- **Constant-reactance arcs** (drawn in teal): Arcs emanating from the right-most point. Upper half = inductive (positive X), lower half = capacitive (negative X). The real axis ($X = 0$) represents purely resistive impedances.

2.6.2 Reading an Impedance

To plot $Z = 75 + j50\ \Omega$ in a $50\ \Omega$ system, first normalise: $z = (75 + j50)/50 = 1.5 + j1.0$. Find the intersection of the $r = 1.5$ circle and the $x = +1.0$ arc. That point is in the upper half of the chart (inductive).

2.6.3 Moving Along a Transmission Line

Moving along a *lossless* transmission line toward the generator rotates the impedance point *clockwise* around the centre of the chart. The radius stays constant ($|\Gamma|$ is constant because the line has no loss). One full revolution = $\lambda/2$ of travel. Moving $\lambda/4$ toward the generator rotates 180°.

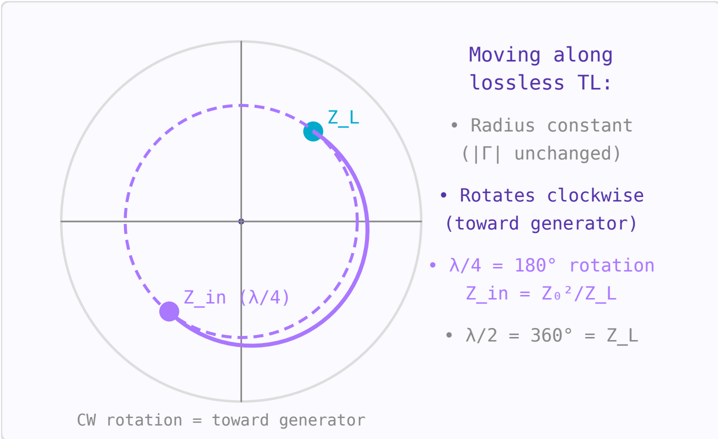


Figure 2.12: Transmission line moves impedance clockwise on a constant-$|\Gamma|$ circle. A $\lambda/4$ section rotates 180°.

$$\Gamma_{in} = \Gamma_L \cdot e^{-j2\beta\ell}$$

This is the basis of the quarter-wave transformer: rotate 180°, and impedance Z_L becomes $Z_0{}^2$/Z_L at the input.

2.6.4 Reading $|\Gamma|$, Return Loss, and VSWR

The radius of the point from the chart centre equals $|\Gamma|$. The dashed constant-$|\Gamma|$ circle that passes through the operating point is the "VSWR circle" — the VSWR equals $(1 + |\Gamma|)/(1 - |\Gamma|)$. Return loss = $-20 \log_{10}|\Gamma|$ dB. A point right at the centre has $|\Gamma| = 0$, VSWR = 1, RL = ∞ dB (perfect match).

2.6.5 Admittance: The Rotated Chart

The admittance Smith Chart is identical to the impedance chart but rotated 180° around the centre. On a combined Z/Y chart, the admittance of any impedance point is found by rotating that point 180° through the centre. This is useful for adding shunt (parallel) elements — on the admittance chart, adding a shunt susceptance moves the point along a constant-conductance circle.

2.6.6 Matching Network Design — Step by Step

To match Z_L = 25 − j30 Ω to 50 Ω using an L-network:

1. Normalise: z_L = (25 − j30)/50 = 0.5 − j0.6. Plot this point.
2. Adding a series inductor moves the point clockwise along the constant r = 0.5 circle (toward more positive X). Add series L until you reach the r = 1 circle.

3. The new point is on r = 1 with some positive X. Now add a shunt capacitor: switch to the admittance chart (rotate 180°), move along constant g = 1 circle to the centre by adding susceptance. Read off the capacitor value.

4. Alternatively, use the RF Toolbox L-Network calculator to get exact values, then verify on the Smith Chart.

2.6.7 Stub Matching

A shunt stub matching network uses a transmission line stub to add the right susceptance to cancel the load's reactive part, bringing the point to the real axis, then a series line section to rotate it to the centre. On the Smith Chart this is done graphically: move along the constant-$|\Gamma|$ circle until you hit the g = 1 circle (for shunt matching), then add stub susceptance to reach the centre.

2.6.8 Practical Tips

- An impedance that sweeps a clockwise circle on the Smith Chart as frequency increases is dominated by transmission line behaviour.
- A small anticlockwise hook in the sweep indicates a series resonance.
- The upper half of the chart = inductive = store energy in magnetic field. Lower half = capacitive = store energy in electric field.
- The outer rim ($|\Gamma| = 1$) represents lossless reactive terminations: shorts, opens, and purely reactive loads.

WORKED EXAMPLE: NORMALIZED IMPEDANCE AND REFLECTION

A load $Z_L = 75 + j50\ \Omega$ in a $50\ \Omega$ system normalizes to $z = Z_L/Z_0 = 1.5 + j1.0$. Its reflection coefficient is

$$\Gamma = \frac{z-1}{z+1} = \frac{0.5 + j1.0}{2.5 + j1.0} = 0.38\angle 39^\circ,$$

so $|\Gamma| = 0.38$ (VSWR ≈ 2.2). On the chart the point lies in the upper, inductive half, on the $r = 1.5$ resistance circle.

KEY TAKEAWAYS

- The chart plots Γ; constant-r circles and constant-x arcs overlay it.
- A $\lambda/4$ line (a 180° trip in Γ) turns z into its admittance $y = 1/z$.
- Motion toward the generator is clockwise; one full revolution is $\lambda/2$.

Exercises

1. Normalize $Z_L = 25 - j50\ \Omega$ in a $50\ \Omega$ system.
2. Where on the chart are a short circuit and an open circuit?
3. A load lies on the $r = 1$ circle. What does that imply for matching with a series reactance?

Further reading: Pozar [1] (ch. 2); Steer [5].

2.7 S-Parameters

LEARNING OBJECTIVES

- Interpret S_{11} and S_{21} as reflection and transmission.
- Relate S_{11} to return loss and S_{21} to insertion loss or gain.
- Recognize reciprocity and the role of the reference impedance.

S-parameters (scattering parameters) describe how RF signals are scattered by a network. Unlike Z or Y parameters, S-parameters are measured with all ports terminated in a reference impedance Z_0 (typically 50 Ω), making them practical to measure with a vector network analyser (VNA) at microwave frequencies.

2.7.1 Definition

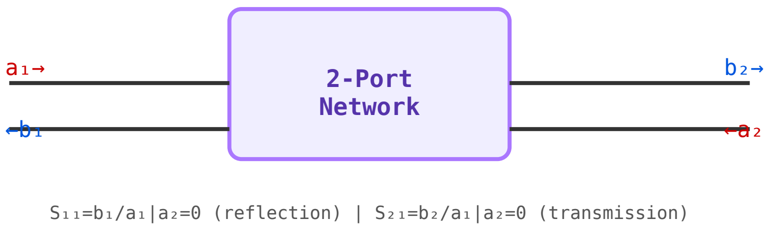


Figure 2.13: 2-port S-parameter definitions. All ports terminated in Z_0 during measurement.

For a 2-port network, S-parameters relate incident (a) and reflected/transmitted (b) wave amplitudes:

$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

- S_{11} — input reflection coefficient (return loss at port 1)
- S_{21} — forward transmission (insertion loss or gain)
- S_{12} — reverse isolation
- S_{22} — output reflection coefficient

2.7.2 Smith Chart

The Smith Chart maps the complex reflection coefficient Γ onto a plot of normalised impedance. Constant-resistance circles are in the horizontal centre; constant-reactance arcs run vertically. Moving clockwise along a constant-$|\Gamma|$ circle corresponds to adding transmission line length toward the generator.

2.7.3 Touchstone Files

Measured S-parameters are stored in Touchstone format (.s1p, .s2p, etc.). Each line contains a frequency point followed by pairs of values encoding each Sij in MA (magnitude-angle), RI (real-imaginary), or DB (dB-angle) format.

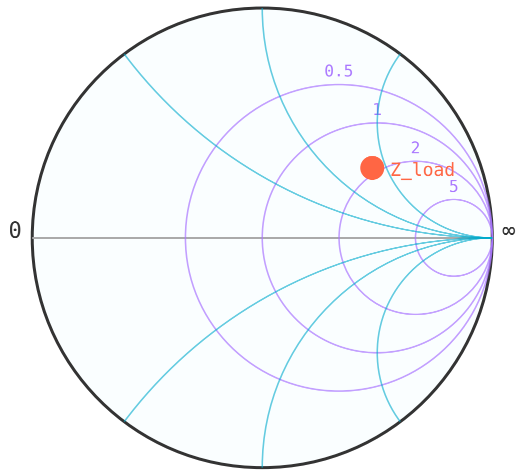


Figure 2.14: Smith Chart: constant-resistance circles (purple), constant-reactance arcs (blue). Example load shown as orange dot.

WORKED EXAMPLE: READING A TWO-PORT

An amplifier in a 50 Ω system shows $S_{11} = 0.1$ and $S_{21} = 10$. The input return loss is $-20\log_{10}|S_{11}| = 20$ dB (well matched) and the forward gain is $20\log_{10}|S_{21}| = 20$ dB. Because $|S_{21}| > 1$ the device is active; a passive reciprocal network would have $S_{21} = S_{12}$ with $|S_{21}| \leq 1$.

KEY TAKEAWAYS

- $S_{11} = b_1/a_1$ (reflection); $S_{21} = b_2/a_1$ (transmission), other ports matched.
- $\mathrm{RL} = -20\log|S_{11}|$; insertion loss/gain $= 20\log|S_{21}|$.
- Reciprocal networks satisfy $S_{ij} = S_{ji}$; S-parameters are defined against a reference impedance.

Exercises

1. $S_{11} = 0.316$. What is the return loss?
2. A filter has $S_{21} = 0.5$. What is its insertion loss in dB?
3. For a lossless two-port, what is $|S_{11}|^2 + |S_{21}|^2$?

Further reading: Pozar [1] (ch. 4); Gonzalez [10] (ch. 1).

2.8 Impedance Matching

LEARNING OBJECTIVES

- Explain why matching maximizes power transfer and minimizes reflections.
- Design a two-element L-network for a resistive mismatch.
- Choose among L-network, stub, and quarter-wave methods.

Maximum power transfer from source to load occurs when the load impedance is the complex conjugate of the source impedance. Impedance matching is the process of inserting a network between source and load to achieve this condition at the desired frequency.

2.8.1 Why Matching Matters

- Maximises power transfer to the load
- Minimises reflections and standing waves
- Prevents damage from reflected power in high-power systems
- Optimises noise figure in low-noise amplifiers (noise match $\neq$ power match)

2.8.2 L-Network

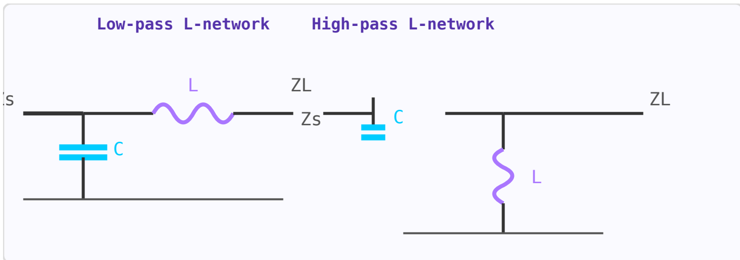


Figure 2.15: L-network topologies. Left: low-pass (shunt C + series L). Right: high-pass (series C + shunt L).

Two reactive elements (one series, one shunt) form the simplest matching network. The Q factor is fixed by the impedance ratio: $Q = \sqrt{R_{high}/R_{low} - 1}$. Only one frequency can be matched; the fractional bandwidth $\approx 1/Q$.

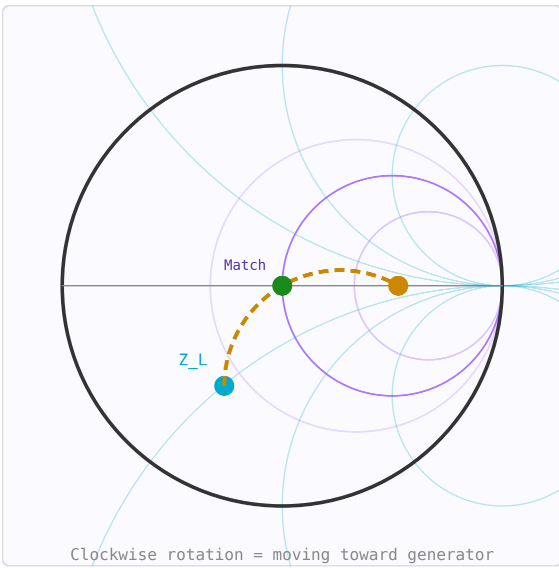


Figure 2.16: Smith Chart matching path: L_load (blue) $\rightarrow$ $\lambda/4$ rotation (orange) $\rightarrow$ matched (green).

2.8.3 Pi and T Networks

Three-element networks allow Q to be chosen independently of the impedance ratio. A Pi network has two shunt elements; a T network has two series elements. Higher Q gives better harmonic suppression but narrower bandwidth.

2.8.4 Quarter-Wave Transformer

A $\lambda/4$ transmission line section with $Z_0 = \sqrt{Z_1 Z_2}$ matches two real impedances at the design frequency. Bandwidth is typically 10–20% for VSWR < 2.

2.8.5 Stub Matching

A shorted or open transmission line stub in shunt with the main line presents a pure susceptance. Single-stub matching can match any load, but requires moving the stub to the correct point on the line (found using the Smith Chart).

WORKED EXAMPLE: L-NETWORK, 100 Ω TO 50 Ω

To match $R_L = 100\ \Omega$ down to $R_0 = 50\ \Omega$, the network quality factor is $Q = \sqrt{R_L/R_0 - 1} = \sqrt{2-1} = 1$. The series reactance is $X_s = QR_0 = 50\ \Omega$ and the shunt reactance $X_p = R_L/Q = 100\ \Omega$ (opposite signs). At 1 GHz a series 50 Ω inductor is $L = X_s/\omega = 7.96$ nH and a shunt 100 Ω capacitor is $C = 1/(\omega X_p) = 1.59$ pF.

KEY TAKEAWAYS

- Matching delivers maximum power (conjugate match) and drives $\Gamma \to 0$.
- L-network $Q = \sqrt{R_{\text{high}}/R_{\text{low}} - 1}$ sets the two reactances.
- L-networks are narrowband; stubs and $\lambda/4$ transformers are distributed alternatives.

Exercises

1. Find the L-network Q to match 200 Ω to 50 Ω.
2. Which is inherently broader band: a single L-network or a multi-section transformer?
3. Conjugate-matching a source $Z_s = 50 + j30\ \Omega$ needs what load impedance?

Further reading: Pozar [1] (ch. 5); Ludwig and Bogdanov [6] (ch. 8).

Chapter 3

Guided-Wave Structures and Interconnect

3.1 Microwave Transmission Lines

LEARNING OBJECTIVES

- Compare microstrip, stripline, CPW, and coax.
- Explain the effective permittivity of microstrip.
- Relate line geometry to characteristic impedance.

At microwave frequencies, transmission lines must be treated as distributed elements. Several planar and non-planar structures are used depending on frequency, power, loss, and fabrication constraints.

3.1.1 Microstrip

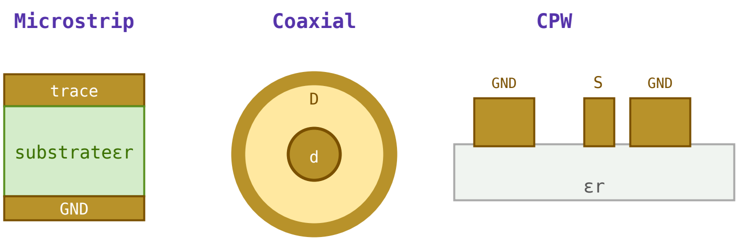


Figure 3.1: Cross-sections: microstrip (left), coaxial cable (centre), coplanar waveguide CPW (right).

The most common planar line in PCB design. A conductor trace on top of a dielectric substrate with a ground plane below. Fields are partly in the substrate and partly in air, giving an effective dielectric constant \(\varepsilon_{eff} < \varepsilon_r\). Easy to fabricate but has higher radiation loss at millimetre-wave frequencies.

3.1.2 Stripline

Conductor buried between two ground planes in a homogeneous dielectric. All fields are in the dielectric so $\varepsilon_{eff} = \varepsilon_r$ exactly. Lower radiation loss than microstrip but harder to access for shunt connections.

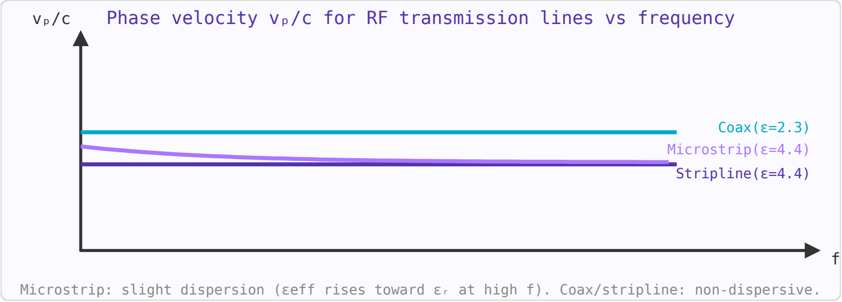


Figure 3.2: Phase velocity for common transmission lines. Microstrip is slightly dispersive; coax and stripline are not.

3.1.3 Coplanar Waveguide (CPW)

Signal conductor flanked by ground conductors on the same surface. Easy shunt connections (no via needed), suitable for MMIC (monolithic microwave integrated circuit) fabrication. Widely used above 10 GHz.

3.1.4 Coaxial Line

TEM mode, no cutoff frequency (down to DC). The dominant transmission medium for test equipment, cables, and connectors. Standard impedances: 50 Ω (minimum attenuation/maximum power tradeoff), 75 Ω (minimum attenuation in air-filled coax).

3.1.5 Waveguide

Hollow metallic tube supporting TE and TM modes. Has a cutoff frequency below which propagation does not occur. Very low loss at microwave/millimetre-wave frequencies. Used in high-power radar and satellite systems.

WORKED EXAMPLE: EFFECTIVE PERMITTIVITY OF MICROSTRIP

Microstrip fields are partly in air and partly in the substrate, so the effective permittivity lies between 1 and ε_r. A common estimate is

$$\varepsilon_{\text{eff}} \approx \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2} \frac{1}{\sqrt{1 + 12h/W}}.$$

On FR-4 ($\varepsilon_r = 4.4$) with $W/h = 2$: $\varepsilon_{\text{eff}} = 2.7 + 1.7/\sqrt{7} = 3.34$. The guided wavelength is then $\lambda_g = \lambda_0/\sqrt{\varepsilon_{\text{eff}}}$.

KEY TAKEAWAYS

- Microstrip's mixed air/dielectric fields give an *effective* permittivity between 1 and ε_r.
- Stripline is fully enclosed (TEM, non-dispersive); CPW has coplanar grounds; coax is shielded and broadband.
- Wider lines (larger W/h) give lower Z_0.

Exercises

1. A stripline sits in a uniform $\varepsilon_r = 4$ dielectric. What is its effective permittivity?

2. Does widening a microstrip raise or lower Z_0?

3. Find the guided wavelength at 2 GHz for $\varepsilon_{\text{eff}} = 3.34$.

Further reading: Pozar [1] (ch. 3); Steer [5].

3.2 RF PCB Design

LEARNING OBJECTIVES

- Design controlled-impedance traces (microstrip and stripline).
- Account for via and bond-wire parasitics.
- Apply RF grounding and return-path rules.

At RF frequencies, PCB traces behave as transmission lines. Their characteristic impedance must be controlled to prevent reflections, signal integrity problems, and radiation. The four standard controlled-impedance structures are microstrip, stripline, coplanar waveguide (CPW), and differential pairs. Each has different shielding characteristics, loss properties, and PCB layer requirements.

3.2.1 Comparison of PCB Transmission Line Types

Type	Layers needed	Shielding	Via needed for termination?	Best for
Microstrip	2 (trace + ground)	Partial (one side open)	No	General RF up to ~10 GHz, antennas
Stripline	3+ (trace between 2 grounds)	Full TEM, no radiation	Yes (for shunt elements)	High isolation, sensitive circuits, digital backplanes
CPW (coplanar waveguide)	1 (trace + ground coplanar)	Partial to full (GCPW)	No — shunt connections on same layer	MMIC, mmWave, no-via shunt connections
Differential pair	2	Partial (common-mode rejects noise)	No	USB, HDMI, LVDS, balanced RF

3.2.2 Coplanar Waveguide (CPW)

CPW places the signal trace between two coplanar ground planes. No ground plane is needed below the substrate, making it ideal for flip-chip and MMIC die attachment where ground connections must be made on the surface. The characteristic impedance depends on the trace width w and gap s to the ground planes:

$$Z_0 = \frac{30\pi}{\sqrt{\varepsilon_{eff}}}\frac{K(k')}{K(k)}, \quad k = \frac{w}{w + 2s}$$

where K(k) is the complete elliptic integral of the first kind. Grounded CPW (GCPW) adds a bottom ground plane, suppressing the parasitic microstrip mode and is preferred for multilayer PCBs.

3.2.3 Differential Pair Impedance

Edge-coupled microstrip differential pairs carry balanced signals. The key parameter is the differential impedance $Z_{diff} = 2Z_{odd}$, where Z_{odd} is the odd-mode impedance. Standard targets: 100 Ω differential for USB/HDMI/LVDS, 90 Ω for PCIe.

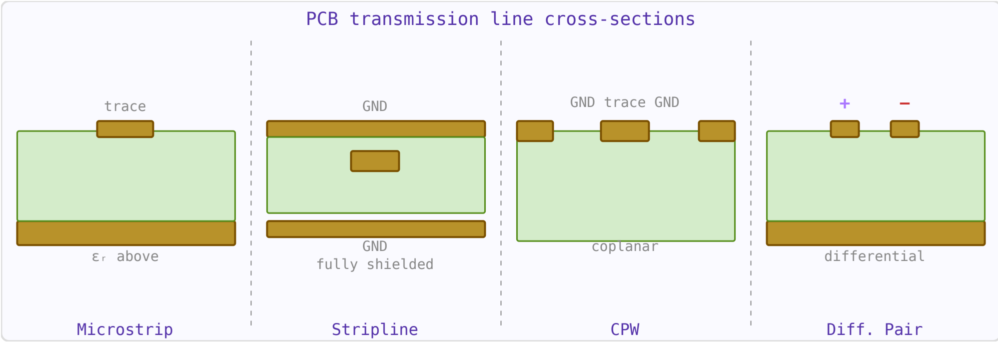


Figure 3.3: Cross-sections of the four standard PCB RF transmission line types.

$$Z_{diff} = 2Z_{odd} \approx 2Z_0 \left(1 - 0.347e^{-2.655s/h}\right)$$

where s is the edge-to-edge spacing and h is the substrate height. The even-mode impedance $Z_{even} \approx 2Z_0/(1 + 0.347e^{-2.655s/h})$ determines common-mode behaviour. Tight spacing reduces Z_diff; looser spacing increases it toward 2×Z_0_single.

3.2.4 Stripline

Stripline is a flat conductor centred symmetrically between two ground planes. It is a TEM mode structure with no dispersion and no radiation — unlike microstrip whose fields extend into air above the trace. The impedance:

$$Z_0 = \frac{60}{\sqrt{\varepsilon_r}} \ln\left(\frac{4b}{0.67\pi w_{eff}(0.8 + t/w)}\right)$$

where b is the ground-plane separation and w_{eff} accounts for trace thickness. Because the medium is uniform, $v_p = c/\sqrt{\varepsilon_r}$ with no effective permittivity correction.

3.2.5 Via Parasitics

A PCB via connecting signal layers has inductance and capacitance:

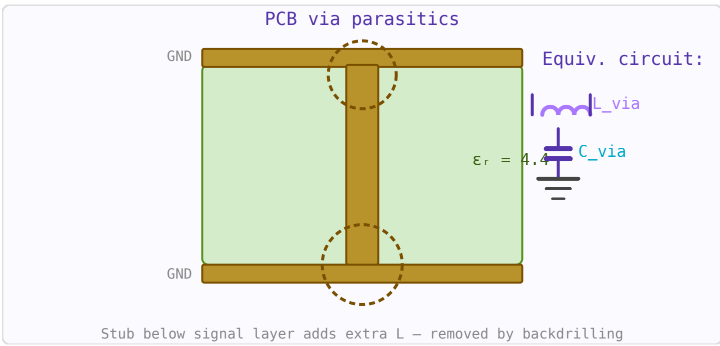


Figure 3.4: Via cross-section showing inner conductor, antipad, and the parasitic inductance + capacitance equivalent circuit.

$$L_{via} \approx 0.2h \ln\left(\frac{4h}{d}\right) \text{ nH}, \quad C_{via} \approx \frac{1.41\varepsilon_r h}{\ln(D/d)} \text{ pF}$$

where h is the via height (mm), d the drill diameter (mm), and D the antipad diameter (mm). At 1 GHz, a standard FR4 via (0.3 mm drill, 1.6 mm deep) has about 0.5 nH inductance and 0.15 pF capacitance — enough to cause significant reflections at microwave frequencies. Via stubs (unused portions below the signal layer) are particularly problematic and are removed by backdrilling.

3.2.6 Bond Wire Inductance

Gold or aluminium bond wires connecting a die to a package or PCB have parasitic inductance (Grover formula):

$$L_{wire} = 0.2\ell\left[\ln\left(\frac{2\ell}{d}\right) - 0.75\right] \text{ nH}$$

where ℓ and d are in mm. A typical 1.5 mm gold bond wire (d = 25 μm) has about 1.2 nH inductance, giving 7.5 Ω reactance at 1 GHz — often requiring resonating capacitors in MMIC matching networks.

3.2.7 Common Material Properties

Substrate	ε_r	tan δ	Typical use
FR4	4.2–4.6	0.018–0.025	General PCB, < 3 GHz
Rogers 4003C	3.55	0.0027	RF PCB, antennas, up to 30 GHz
Rogers 4350B	3.66	0.0037	Low-cost RF alternative
PTFE (Teflon)	2.1	0.001	High-frequency, low-loss
Alumina (Al_2O_3)	9.8	0.0002	MMICs, hybrid circuits
GaAs	12.9	0.0006	MMIC substrates

WORKED EXAMPLE: VIA INDUCTANCE

A plated through-via ($h = 1.6$ mm, radius $r = 0.15$ mm) has an approximate inductance

$$L \approx \frac{\mu_0 h}{2\pi}\left[\ln\frac{2h}{r} + 0.5\right] = 3.20 \times 10^{-10}\,[\ln(21.3) + 0.5] = 1.14 \text{ nH}.$$

At 5 GHz that via presents $X_L = 2\pi f L = 36\ \Omega$ — far from negligible, so RF vias are kept short and often paralleled.

KEY TAKEAWAYS

- Trace impedance is set by width, dielectric height, and ε_r; keep it controlled.
- Vias (∼1 nH) and bond wires (∼1 nH/mm) add series inductance that bites at GHz.
- Give the return current a continuous path directly under the signal trace.

Exercises

1. A 2 mm bond wire is ∼ 2 nH. What is its reactance at 5 GHz?
2. How does placing two identical vias in parallel change the inductance?
3. Why route an RF trace over a solid ground plane rather than a slotted one?

Further reading: Pozar [1] (ch. 3); Ott [18].

3.3 Coaxial Cable

LEARNING OBJECTIVES

- Relate coax dimensions and dielectric to Z_0 and velocity factor.
- Read attenuation and power ratings from cable tables.
- Choose a cable for a frequency, loss, and power requirement.

Coaxial cable is the primary transmission line for RF signals above a few MHz. Choosing the wrong cable — even for a short run — can add significant insertion loss, degrade shielding, or limit power handling. This page is a quick reference for the most common types.

3.3.1 Cable Comparison Table

Type	**OD (mm)**	$\mathbf{Z_0}$	**Vel. factor**	**Atten. @ 1 GHz**	**Max power**	**Best use**
RG-174	2.8	50 Ω	0.66	1.5 dB/m	25 W	Pigtails, U.FL connections, very flexible low-power
RG-316	2.6	50 Ω	0.69	1.3 dB/m	30 W	PTFE insulation; flexible, low loss vs RG-174
RG-58 (C/U)	5.0	50 Ω	0.66	0.55 dB/m	200 W	General RF cable, lab use, instrumentation up to ~500 MHz
RG-59	6.1	75 Ω	0.66	0.6 dB/m	200 W	Video, CATV, 75 Ω systems
RG-213 (U)	10.3	50 Ω	0.66	0.28 dB/m	1 kW	Base station feedlines, amateur radio HF/VHF
LMR-195	4.95	50 Ω	0.80	0.43 dB/m	250 W	Low-loss flexible for short runs, jumpers
LMR-400	10.3	50 Ω	0.85	0.135 dB/m	1.1 kW	Low-loss runs to 1 GHz; antenna feedlines
LMR-600	15.9	50 Ω	0.87	0.087 dB/m	2.5 kW	Long runs, base station, cellular infrastructure
Semi-rigid UT-085	2.2	50 Ω	0.695	0.84 dB/m	100 W	PCB-to-PCB, MMIC, excellent shielding, non-flexible

Semi-rigid UT-141	3.6	50 Ω	0.695	0.44 dB/m	300 W	Microwave benching, test fixtures

3.3.2 Attenuation Scaling with Frequency

Coaxial cable attenuation increases approximately as $\sqrt{f}$ (due to skin effect in the conductors) plus a smaller f-dependent term from dielectric loss:

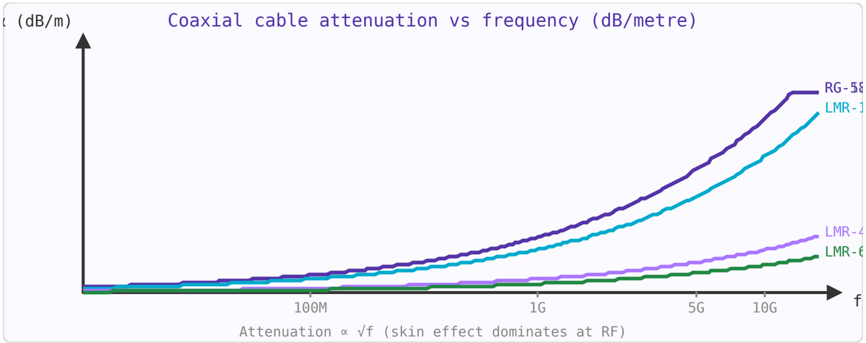


Figure 3.5: Coaxial cable attenuation increases as $\sqrt{f}$. LMR-400 has 5× less loss than RG-174 at the same frequency.

$$\alpha \approx \alpha_1 \sqrt{f} + \alpha_2 f \quad [\text{dB/m}]$$

At 100 MHz, LMR-400 has ~0.044 dB/m. At 2.4 GHz it's ~0.22 dB/m. A 10-metre run at 2.4 GHz loses 2.2 dB — worth using LMR-600 or a tower-mounted LNA for long runs.

3.3.3 Velocity Factor and Electrical Length

The velocity factor (VF) is the ratio of signal propagation speed in the cable to the speed of light. VF depends on the dielectric: solid PE ≈ 0.66, foam PE ≈ 0.78–0.88, PTFE ≈ 0.69. Electrical length = physical length / VF × 360° per wavelength. A 1-metre RG-58 cable at 300 MHz is one electrical wavelength long (0.66 × 1 m / 1 m = 0.66λ — 237°).

3.3.4 Power Handling Limits

Two limits apply: **thermal** (conductor heating → insulation damage, typically rated at 40°C ambient) and **voltage breakdown** (at peak voltage, the dielectric arcs over). For most cables the thermal limit is lower. Power derating at high temperatures: reduce rated power by ~1% per °C above 40°C. At altitude (lower air pressure), the breakdown voltage decreases — critical for airborne equipment.

3.3.5 Practical Selection Guide

- **0–30 MHz, any run length**: RG-213 or LMR-400. Loss is not critical.
- **VHF/UHF, < 3 m**: RG-58 or LMR-195. Adequate loss, easy to handle.
- **VHF/UHF, > 10 m**: LMR-400 or LMR-600 to keep feedline loss below 1–2 dB.
- **Microwave, > 2 GHz, critical loss budget**: LMR-400 or semi-rigid.
- **Pigtails and internal connections**: RG-316 or UT-085 semi-rigid.

- **High power (> 500 W)**: LMR-600 or larger hardline; ensure good connector torque.

WORKED EXAMPLE: CHARACTERISTIC IMPEDANCE OF COAX

For coax, $Z_0 = (138/\sqrt{\varepsilon_r})\log_{10}(D/d)$. With PTFE ($\varepsilon_r = 2.1$), inner conductor $d = 0.9$ mm and shield inner diameter $D = 2.95$ mm,

$$Z_0 = \frac{138}{\sqrt{2.1}}\log_{10}\frac{2.95}{0.9} = 95.2 \times 0.516 = 49\ \Omega,$$

a 50 Ω cable. Its velocity factor is $1/\sqrt{2.1} = 0.69$.

KEY TAKEAWAYS

- $Z_0 = (138/\sqrt{\varepsilon_r})\log_{10}(D/d)$; the diameter ratio sets impedance.
- Velocity factor $= 1/\sqrt{\varepsilon_r}$; foamed dielectrics raise it and cut loss.
- Attenuation rises with frequency; larger cable (e.g. LMR-400) is lower-loss but stiffer.

Exercises

1. Velocity factor of solid-PE coax ($\varepsilon_r = 2.25$)?
2. Does doubling D/d (same dielectric) raise or lower Z_0?
3. RG-58 loses ~ 0.5 dB/m at 1 GHz. What is the loss over 6 m?

Further reading: Pozar [1] (ch. 2); Ludwig and Bogdanov [6].

3.4 RF Connectors

LEARNING OBJECTIVES

- Match a connector to a frequency range and cable.
- Explain why connector geometry sets the upper frequency.
- Apply torque and mating-cycle good practice.

Connector choice often determines system performance at microwave frequencies. Every connector junction contributes insertion loss, a small impedance discontinuity (which worsens with frequency), and a potential point of failure. Choosing the right connector requires balancing frequency range, power handling, mechanical robustness, and cost.

3.4.1 Connector Comparison Table

Connector	Max frequency	Impedance	Power (CW)	Torque spec	Notes
BNC	4 GHz	50 Ω / 75 Ω	80 W @ 1 GHz	Bayonet lock	Quick-connect; RF lab, oscilloscopes, video (75 Ω)
TNC	12 GHz	50 Ω / 75 Ω	100 W	1.13 N·m	BNC with threaded coupling; more rugged

N-type	18 GHz	50 Ω / 75 Ω	300 W @ 1 GHz	1.36 N·m	Workhorse for VHF/UHF/microwave; weatherproof versions
SMA	18 GHz (26 GHz semi-precision)	50 Ω	100 W @ 1 GHz	0.9 N·m	Most common RF connector; keep to 12 GHz in production
SMB	4 GHz	50 Ω	25 W	Snap-on	Compact snap-on; consumer electronics
MMCX	6 GHz	50 Ω	50 W	Snap-on	Miniature; Wi-Fi modules, GPS receivers
U.FL / IPEX	6 GHz	50 Ω	10 W	Snap-on	Very small; < 30 insertion cycles; PCB to antenna pigtail
2.92 mm (K)	40 GHz	50 Ω	10 W	0.9 N·m	SMA-compatible mating; 26–40 GHz range
2.4 mm	50 GHz	50 Ω	8 W	0.9 N·m	Fragile; Ka-band test and mmWave
1.85 mm (V)	67 GHz	50 Ω	5 W	0.9 N·m	Very fragile; V-band measurements
1.0 mm	110 GHz	50 Ω	2 W	0.45 N·m	W-band; lab use only

3.4.2 Insertion Loss Rules of Thumb

Every connector pair (male + female) contributes roughly:

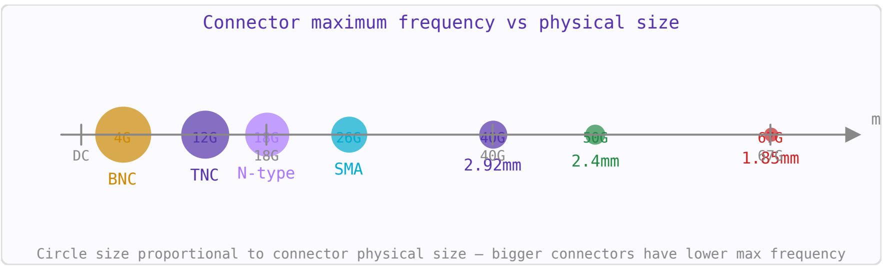


Figure 3.6: RF connector maximum frequency vs physical size. Smaller connectors enable higher frequencies but are more fragile.

- SMA pair: 0.1 dB at 10 GHz, 0.2 dB at 18 GHz
- N-type pair: 0.05 dB at 10 GHz
- 2.92 mm pair: 0.15 dB at 30 GHz, 0.3 dB at 40 GHz
- Each adapter (e.g. SMA-to-N): doubles the connector insertion loss

In a 10-connector measurement path at 18 GHz, connector losses alone can exceed 2 dB — easily dominating the DUT's own loss.

3.4.3 Torque and Connector Longevity

Over-torquing damages centre pin contacts. Under-torquing causes intermittent connections and unpredictable impedance discontinuities. Always use a torque wrench for precision measurements. SMA connectors are rated for ~500 mating cycles; precision SMA (0.9 N·m spec) for ~2000 cycles. 2.4 mm and 1.85 mm connectors are easily damaged by cross-threading — always align carefully before tightening.

3.4.4 PCB Launch Connectors

The transition from a PCB trace to a connector is a discontinuity. Edge-launch SMA connectors must be carefully matched — the launch geometry, board thickness, and connector pin dimension must all be designed together. At 20+ GHz, a poorly designed SMA launch can add 0.5–1 dB of loss and create a visible ripple in S_{11} measurements. Some vendors provide field-solver-optimised footprints for their connectors at specific frequencies.

3.4.5 75 Ω Systems

Cable television, broadcast video, and some telecommunications infrastructure use 75 Ω impedance (derived from the impedance of a centre-loaded dipole in free space). BNC-75, TNC-75, N-75, and F-connectors are used. Never mate 50 Ω and 75 Ω connectors — the impedance mismatch causes a permanent reflection, and physically incompatible types (N-50 vs N-75) can be damaged if forced.

WORKED EXAMPLE: CONNECTOR FREQUENCY CEILING

A coaxial connector supports a single TEM mode only up to the frequency where higher-order modes appear, set by its air-gap dimensions. Standard SMA is usable to about 18 GHz; the smaller 2.92 mm ("K") connector reaches $\sim$ 40 GHz, and 1.85 mm ("V") $\sim$ 65 GHz. Smaller air dimensions push the moding frequency higher — at the cost of power handling and ruggedness.

KEY TAKEAWAYS

- Precision connectors trade size for bandwidth: SMA $\sim$18 GHz, 2.92 mm $\sim$40 GHz, 1.85 mm $\sim$65 GHz.
- Impedance (50 or 75 Ω) and dielectric interface must match the cable.
- Torque to spec and limit mating cycles; a damaged interface spoils return loss.

Exercises

1. Which connector suits a 26 GHz link: SMA or 2.92 mm?
2. Are SMA and 2.92 mm mechanically mateable?
3. Why does over-torquing or excessive mating degrade RF performance?

Further reading: Pozar [1]; Steer [5].

3.5 Waveguides

LEARNING OBJECTIVES

- Compute the dominant TE_{10} cutoff frequency.
- Explain single-mode operation between f_c and $2f_c$.
- Relate guide dimensions to the operating band.

A waveguide is a hollow metallic tube that confines and guides electromagnetic waves. Unlike coaxial line (which supports TEM mode down to DC), waveguides have a cutoff frequency below which propagation does not occur. Above cutoff, waveguides offer very low loss, making them the preferred transmission medium at millimetre-wave frequencies and for high-power radar systems.

3.5.1 Modes

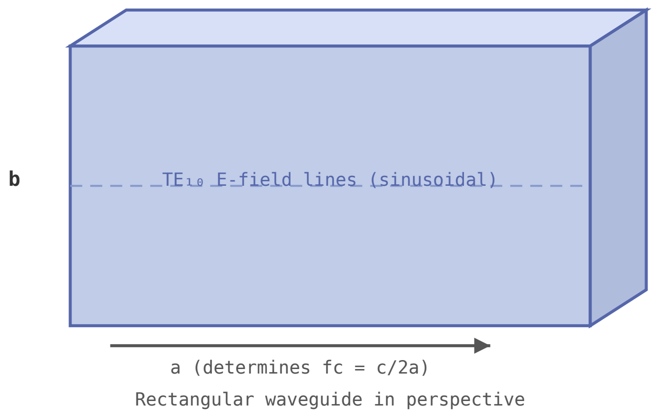


Figure 3.7: Rectangular waveguide. Cutoff frequency fc = c/(2a) for the dominant TE_{10} mode.

Rectangular waveguides support TE (transverse electric) and TM (transverse magnetic) modes. The dominant mode TE_{10} has the lowest cutoff frequency and is used in virtually all single-mode waveguide applications. The cutoff frequency of the TE_{10} mode in a rectangular waveguide of width a is:

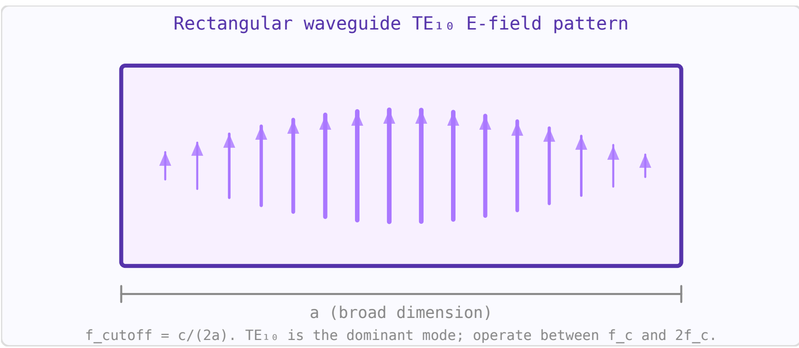


Figure 3.8: TE_{10} mode: E-field is vertical, sinusoidal in x, zero at walls. f_c = c/(2a) — only this mode propagates near cutoff.

$$f_c = \frac{c}{2a}$$

3.5.2 Standard Waveguide Bands

- WR-90 X-band: 8.2–12.4 GHz
- WR-42 K-band: 18–26.5 GHz
- WR-28 Ka-band: 26.5–40 GHz
- WR-10 W-band: 75–110 GHz

3.5.3 Attenuation

Waveguide losses are due to finite conductivity of the walls. Attenuation is minimum at approximately 1.8× the cutoff frequency and rises toward both the cutoff and at very high frequencies. Rectangular waveguide attenuation is typically 0.01–0.5 dB/m, much lower than coax at the same frequency.

3.5.4 Applications

Waveguides are used in high-power radar systems, satellite ground station feeds, millimetre-wave communication links, and as the connecting structure between antenna feeds and LNAs in satellite receivers (where minimum loss before the LNA is critical for noise figure).

WORKED EXAMPLE: WR-90 CUTOFF

The dominant TE_{10} cutoff depends only on the broad wall a: $f_c = c/(2a)$. WR-90 has $a = 22.86$ mm, so

$$f_c = \frac{3 \times 10^8}{2(22.86 \times 10^{-3})} = 6.56 \text{ GHz}.$$

Single-mode operation runs from f_c to about $2f_c = 13.1$ GHz; the standard X-band rating 8.2–12.4 GHz sits inside that window.

KEY TAKEAWAYS

- TE_{10} cutoff $f_c = c/(2a)$; below it the guide will not propagate.
- Useful single-mode band is roughly f_c to $2f_c$.
- Waveguide is low-loss and high-power but bulky and band-limited.

Exercises

1. For $a = 15$ mm, find the TE_{10} cutoff.
2. Can a 5 GHz signal propagate in WR-90 ($f_c = 6.56$ GHz)?
3. Roughly where is the top of WR-90's single-mode band?

Further reading: Pozar [1] (ch. 3); Collin [2] (ch. 3).

3.6 VNA Measurements and Calibration

LEARNING OBJECTIVES

- Explain SOL calibration and the reference plane.
- Interpret a measured S_{11} trace.
- Use port extension and time-domain gating.

A vector network analyser (VNA) measures the complex S-parameters of a device under test (DUT) — both magnitude and phase — across a swept frequency range. Understanding what it measures, and how calibration moves the reference plane to your device, is essential for making accurate RF measurements.

3.6.1 What a VNA Measures

A 2-port VNA applies a stimulus at port 1 and measures the signals at both ports simultaneously. It computes:

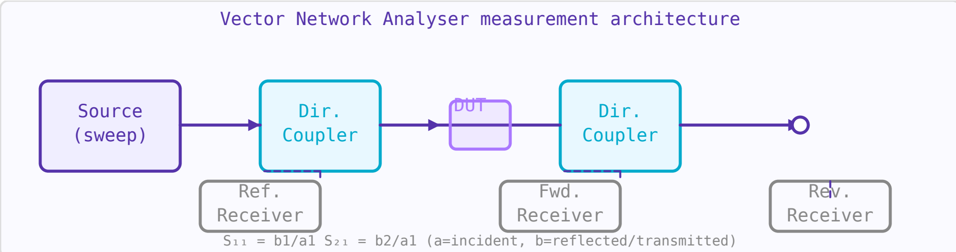


Figure 3.9: VNA architecture: swept source, directional couplers separate incident/reflected/-transmitted waves, receivers measure amplitude and phase.

Parameter	Physical meaning	Typical display
S_{11}	Input reflection coefficient. How much is reflected from port 1.	Smith Chart, log mag (RL)
S_{21}	Forward transmission. Gain or insertion loss from port 1 to port 2.	Log magnitude (dB)
S_{12}	Reverse transmission. Isolation.	Log magnitude (dB)
S_{22}	Output reflection coefficient.	Smith Chart, log mag

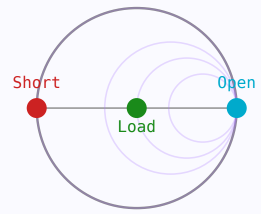
3.6.2 Calibration: Moving the Reference Plane

Without calibration, a VNA measures its own cables, connectors, and switch imperfections. Calibration removes these systematic errors by measuring known standards and computing correction factors. The most common calibration is **SOL** (Short-Open-Load) or **SOLT** (Short-Open-Load-Through):

- **Short**: $\Gamma = -1$ (all reflected, 180° phase). Defines the reference plane position.
- **Open**: $\Gamma = +1$ (all reflected, 0° phase, adjusted for fringing capacitance).
- **Load**: $\Gamma = 0$ (50 Ω termination, no reflection). Sets the amplitude reference.
- **Through**: Connects port 1 to port 2 directly. Characterises transmission path.

Always calibrate at the ends of the test cables, not at the VNA ports — calibration moves the reference plane to wherever you connect the standards.

SOL calibration standards and where they place Γ on the Smith Chart



Standard	|Γ|	Phase	Purpose
Short	1	180°	Reference plane position
Open	1	0°	Phase reference (with fringing C corr.)
50Ω Load	0	–	Amplitude reference, sets return loss scale
Through	–	0°	2-port: transmission reference

Figure 3.10: SOL calibration standards place known Γ values on the Smith Chart rim and centre, allowing systematic error correction.

3.6.3 Port Extension

If you cannot calibrate at the device terminals (e.g. a connector is on a PCB and you calibrate at the cable end), port extension electrically "moves" the reference plane by adding a known electrical delay. It rotates the phase of S_{11} and S_{22} to compensate for the cable length. It is an approximation — it assumes the line is lossless and dispersionless — but is adequate for many applications.

3.6.4 Time-Domain Gating

The VNA can apply an inverse Fourier transform to its frequency-domain data to show the time-domain response (like a TDR). This reveals reflections from connectors and discontinuities separated in time (i.e. distance). Gating in the time domain — windowing to select only a specific time region — then transforming back to frequency allows you to isolate the response of a specific section of the DUT and remove connector effects. This is particularly useful for antenna measurements and filter characterisation.

3.6.5 Interpreting S_{11} on a Smith Chart

A single frequency point on the Smith Chart shows the impedance at that frequency. As frequency sweeps:

- A resistor traces a point on the real axis.
- A capacitor traces clockwise along the $X = 0$ boundary toward the short-circuit point.
- A series LC traces a loop: inductive (upper half), resonant (real axis), capacitive (lower half).
- A transmission line section traces a clockwise circle centred on the origin.
- An antenna shows a small loop near the edge of the chart, touching or nearly touching the real axis at resonance.

3.6.6 Practical Measurement Tips

- Torque connectors to spec (SMA: 0.9 N·m / 8 in·lb). Under/over-torquing causes repeatable but incorrect results.
- Let the VNA warm up for 15–30 minutes before calibrating for best stability.
- Calibrate with the cable attached in its final position — bending it after calibration introduces errors.

- Use at least 201 frequency points across your band; use 1601+ for time-domain measurements to maximise time resolution.
- When measuring low-loss filters (< 0.5 dB), use a full 4-port SOLR or TRL calibration for best accuracy.
- Check calibration by measuring a known open or short — it should show a perfect circle on the Smith Chart rim.

WORKED EXAMPLE: RETURN LOSS TO VSWR ON A VNA

A VNA reads $S_{11} = -14$ dB at band center. That magnitude is $|\Gamma| = 10^{-14/20} = 0.20$, so

$$\mathrm{VSWR} = \frac{1 + 0.20}{1 - 0.20} = 1.5.$$

A short-open-load calibration first moves the reference plane to the probe tips, so the reading describes the device rather than the cables.

KEY TAKEAWAYS

- SOL (short-open-load) calibration sets the reference plane and removes systematic cable error.
- S_{11} in dB is return loss; convert to $|\Gamma|$ and VSWR as needed.
- Port extension shifts the reference plane; time-domain gating isolates a reflection in space.

Exercises

1. $S_{11} = -20$ dB. What is the VSWR?
2. Which three standards define a basic one-port calibration?
3. Why calibrate at the probe tips rather than at the instrument port?

Further reading: Pozar [1] (ch. 4); Steer [5].

Chapter 4

Passive Components, Resonators and Filters

4.1 E-Series Standard Component Values

LEARNING OBJECTIVES

- Explain the logarithmic spacing of E-series values.
- Pick the nearest standard value and its tolerance.
- Choose an E-series for a required precision.

Resistors, capacitors, and inductors are manufactured in standardised value sets called **E-series**, defined by IEC 60063. Rather than covering every decimal value, the series places values at logarithmically equal intervals so that any calculated value falls within a specified tolerance of the nearest standard value.

4.1.1 Logarithmic Spacing

An E-series with N values per decade places mantissas at:

$$v_k = 10^{k/N} \quad k = 0, 1, \ldots, N-1$$

The ratio between consecutive values is always 10^(1/N). This means worst-case error is at most half that ratio, or approximately 1/(2N) × 100% of a decade — which corresponds directly to the component tolerance.

4.1.2 Series Summary

Series	Values/decade	Tolerance	Typical use
E6	6	±20%	General decoupling capacitors
E12	12	±10%	General-purpose resistors
E24	24	±5%	Most common resistor series
E48	48	±2%	Precision resistors
E96	96	±1%	Precision resistors, filter components
E192	192	±0.5%	High-precision networks

4.1.3 E6 and E12 Values (first decade)

E6 mantissas: 1.0, 1.5, 2.2, 3.3, 4.7, 6.8

E12 mantissas (includes all E6): 1.0, 1.2, 1.5, 1.8, 2.2, 2.7, 3.3, 3.9, 4.7, 5.6, 6.8, 8.2

E24 adds further interleaving: 1.0, 1.1, 1.2, 1.3, 1.5, 1.6, 1.8, 2.0, 2.2, 2.4, 2.7, 3.0, 3.3, 3.6, 3.9, 4.3, 4.7, 5.1, 5.6, 6.2, 6.8, 7.5, 8.2, 9.1

4.1.4 Reading E-Series Values

Multiply any mantissa by a decade: a resistor marked **4.7 kΩ** uses mantissa 4.7 in the kΩ decade. A capacitor of **22 nF** uses mantissa 2.2 in the 10 nF decade. The E-series is the same for resistors, capacitors, and inductors — only the unit decade changes.

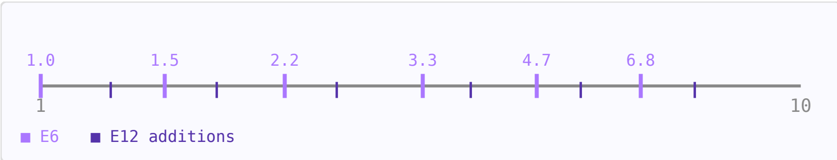


Figure 4.1: Log-scale number line for one decade showing E6 (purple) and E12 (dark) value positions.

4.1.5 Choosing a Series

- **E12**: adequate for bias resistors, pull-ups, and non-critical dividers
- **E24**: standard choice for RF filter component selection
- **E96 / E192**: use when the component value directly sets a frequency, gain, or impedance with <1% accuracy requirement
- Higher-N series cost more and have longer lead times — use only where the tolerance budget demands it

4.1.6 Finding the Nearest Value

To find the nearest E-series value to a calculated target R_target: (1) compute the decade exponent exp = floor($\log_{10}$(R_target)), (2) extract the mantissa m = R_target / 10^exp, (3) scan all N mantissas and pick the closest. The E-Series Finder calculator automates this for any R, C, or L target.

WORKED EXAMPLE: NEAREST E12 VALUE

The E12 series has 12 values per decade, spaced by $10^{1/12} = 1.21$ (21% steps, matching ±10% parts). Suppose a design needs $R = 2.6$ kΩ. The E12 neighbours are 2.2 and 2.7 kΩ; 2.7 kΩ is closest (3.8% away, inside the 10% band), so it is the value to fit.

KEY TAKEAWAYS

- En has n values per decade, spaced by $10^{1/n}$; larger n means tighter steps.
- E6/E12/E24 pair with 20/10/5% tolerances; E96 is the 1% series.
- Round a computed value to the nearest series member within tolerance.

Exercises

1. What is the step ratio of the E24 series?
2. What is the nearest E12 value to 4.5 kΩ?
3. Which series would you use for 1% resistors?

Further reading: Ludwig and Bogdanov [6]; IEC 60063 (preferred numbers).

4.2 Coupled Resonators

LEARNING OBJECTIVES

- Explain mode splitting when two resonators couple.
- Relate the coupling coefficient to bandwidth.
- Recognize critical coupling.

When two resonant LC circuits are brought near each other they exchange energy through mutual inductance or capacitive coupling. This splits the single resonant frequency into two modes and produces a characteristic double-peaked bandpass response. Understanding coupled resonators is essential for designing IF transformers, bandpass filters, and resonator-based sensors.

4.2.1 Coupling Mechanisms

- **Magnetic (inductive)**: mutual inductance M between the two coils. $k = M/L$ for identical resonators ($M < L$, so $0 < k < 1$). Dominant in IF transformers and wound coil pairs.
- **Electric (capacitive)**: a gap capacitor Cm between the two tanks. Used in high-Q cavity and ceramic resonator filters.
- **Mixed**: both mechanisms simultaneously; common in printed resonators.

4.2.2 Mode Splitting

For two identical resonators (same L, C, resonant frequency $f_0 = 1/(2\pi\sqrt{LC})$), coupling splits the resonant frequency into an even mode (lower, f_-) and an odd mode (upper, f_+):

$$f_\pm = \frac{f_0}{\sqrt{1 \mp k}}$$

The mode splitting $\Delta f = f_+ - f_-$ grows with coupling coefficient k. For small k: $\Delta f \approx k{\cdot}f_0$.

4.2.3 Passband Bandwidth

For high-Q resonators the −3 dB passband bandwidth of the coupled pair equals:

$$\mathrm{BW} \approx k \cdot f_0 \qquad Q_{\text{loaded}} = \frac{f_0}{\mathrm{BW}} = \frac{1}{k}$$

A larger coupling coefficient k gives a wider, flatter passband; a smaller k gives a narrower passband with higher loaded Q.

4.2.4 Critical Coupling

The transfer response shape depends on the ratio of coupling to resonator loss. Define the **critical coupling coefficient**:

$$k_{\text{crit}} = \frac{1}{Q} \qquad Q = \frac{\omega_0 L}{R} \text{ (unloaded Q)}$$

- **Under-coupled** (k < k_crit): single peaked response, insertion loss at f_0
- **Critically coupled** (k = k_crit): maximally flat single peak, maximum power transfer at f_0
- **Over-coupled** (k > k_crit): double-peaked response, dip at f_0

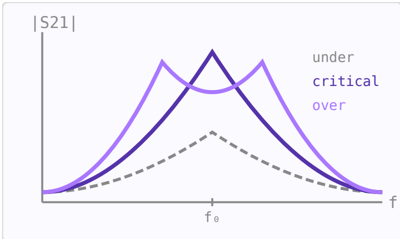


Figure 4.2: Transfer response shapes for three coupling regimes.

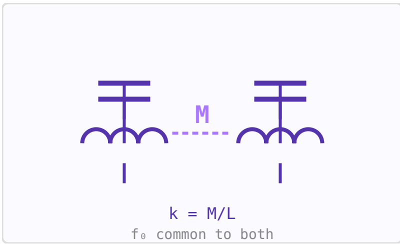


Figure 4.3: Two magnetically coupled LC tanks with mutual inductance M.

4.2.5 Bandpass Filter Design

A two-resonator bandpass filter is designed by choosing k to achieve the desired bandwidth and response shape. For a maximally flat (Butterworth) 2-pole bandpass filter, critical coupling gives the flattest passband. Chebyshev response requires slight over-coupling. Multi-resonator filters chain additional coupled pairs, with each coupling coefficient k_{i,i+1} set from the prototype g-values.

4.2.6 IF Transformers

The coupled resonator principle is the basis of the IF transformer found in superheterodyne receivers. The primary and secondary coils are wound on a shared ferrite core and are slightly over-coupled to achieve a flat-topped passband around the IF frequency (commonly 455 kHz or 10.7 MHz). Adjusting the core position changes the coupling coefficient and therefore the bandwidth.

WORKED EXAMPLE: BANDWIDTH FROM COUPLING

Two identical LC resonators at $f_0 = 100$ MHz couple with coefficient $k = 0.03$. The coupling splits the response into two peaks whose spacing sets the passband:

$$\text{BW} \approx k f_0 = 0.03 \times 100 \text{ MHz} = 3 \text{ MHz}.$$

Weaker coupling narrows the band (peaks merge); stronger coupling widens it into a double hump.

KEY TAKEAWAYS

- Coupling two resonators splits the single peak into two modes.
- Fractional bandwidth $\approx k$, so BW $\approx k f_0$.
- Critical coupling gives the flattest single passband; over-coupling adds ripple.

Exercises

1. For $k = 0.05$ and $f_0 = 1$ GHz, estimate the bandwidth.
2. To halve the bandwidth, how should k change?
3. Which coupling state gives a maximally flat two-resonator response?

Further reading: Pozar [1] (ch. 6); Collin [2].

4.3 RF Attenuators

LEARNING OBJECTIVES

- Design Pi and T resistive attenuators for a target attenuation.
- Preserve the Z_0 match while attenuating.
- Account for power dissipation in the pad.

An attenuator is a passive network that reduces signal power by a precise, frequency-independent amount. Unlike a simple resistive divider, a well-designed attenuator maintains its characteristic impedance at all ports across a wide frequency range. Attenuators are used to set signal levels, improve source or load impedance, isolate stages from reflections, and calibrate measurement systems.

4.3.1 Key Parameters

Parameter	Symbol	Typical values
Attenuation	A (dB)	1–40 dB in 1 dB steps for switched pads; fixed values for inline pads
Characteristic impedance	Z_0	50 Ω (RF/microwave), 75 Ω (cable TV/video)
Power rating	P_max (W)	0.1–50 W for most SMD/connectorised types
Frequency range	DC–f_max	DC to 18 GHz for SMA types; 40+ GHz for 2.92 mm types

VSWR	Typically $<$ 1.2:1 across rated bandwidth

4.3.2 Pi (Π) Attenuator

The Pi pad uses two equal shunt resistors flanking a series resistor. It is the most common topology for RF PCBs. For attenuation A dB with system impedance Z_0:

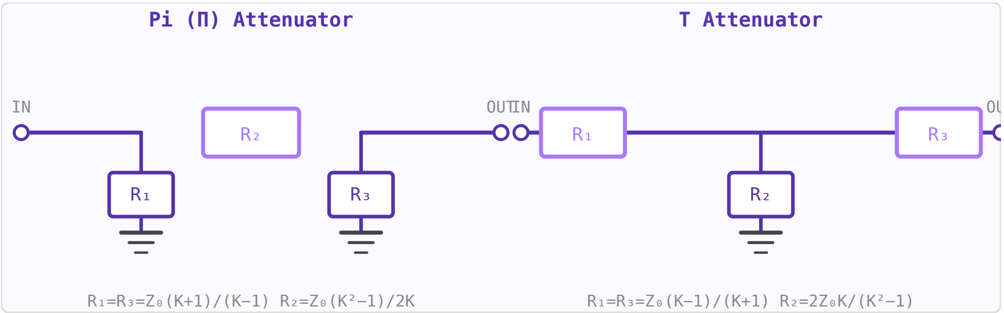


Figure 4.4: Pi attenuator (shunt-series-shunt) and T attenuator (series-shunt-series). K = 10^(A/20).

$$K = 10^{A/20}, \quad R_{shunt} = Z_0 \frac{K+1}{K-1}, \quad R_{series} = Z_0 \frac{K^2-1}{2K}$$

4.3.3 T Attenuator

The T pad uses two equal series resistors with a shunt resistor to ground at the midpoint. Better suited for cable assemblies where both conductors need to be attenuated symmetrically:

$$R_{series} = Z_0 \frac{K-1}{K+1}, \quad R_{shunt} = \frac{2Z_0K}{K^2-1}$$

4.3.4 L-pad (Impedance-Matching Attenuator)

An L-pad matches two different source and load impedances while also providing attenuation. Unlike the Pi and T, the L-pad is inherently asymmetric. The attenuation is fixed once the impedances are set; it cannot be chosen independently. One series resistor and one shunt resistor are needed:

$$R_{series} = \sqrt{Z_S Z_L}\,\frac{\sqrt{Z_S/Z_L}-1}{1}, \quad R_{shunt} = \frac{Z_S Z_L}{R_{series}+Z_S}$$

(exact formula depends on which port is the high-Z side).

4.3.5 Bridged-T Attenuator

The Bridged-T uses the system impedance Z_0 as one of its elements, making it efficient for large attenuation values with fewer components. With a bridging resistor $R_1 = Z_0(K-1)$ and a shunt resistor $R_2 = Z_0/(K-1)$ plus two Z_0 arms.

4.3.6 Temperature and Power Dissipation

Each resistor in the attenuator dissipates a fraction of the incident power. For high-power attenuators, the power in each element must be calculated from the impedance equations. The series element in a Pi pad dissipates $(1-1/K^2)$ of the input power.

Component values for Pi attenuator in 50 Ω system

A (dB)	K	R_shunt (Ω)	R_series (Ω)
1 dB	1.122	869.5 Ω	5.8 Ω
2 dB	1.259	436.2 Ω	11.6 Ω
3 dB	1.413	292.4 Ω	17.6 Ω
6 dB	1.995	150.5 Ω	37.4 Ω
10 dB	3.162	96.2 Ω	71.2 Ω
20 dB	10.000	61.1 Ω	247.5 Ω
30 dB	31.623	53.3 Ω	789.8 Ω

Figure 4.5: Standard Pi attenuator values. R_shunt = $Z_0(K+1)/(K-1)$, R_series = $Z_0(K^2-1)/2K$.

4.3.7 Practical Notes

- Standard RF attenuator values: 1, 2, 3, 6, 10, 20, 30 dB.
- Switched attenuators in LNAs and signal generators use PIN diode or MEMS switching to change attenuation state.
- Surface-mount 0402/0603 resistors are used up to ~6 GHz; above that, thin-film chip attenuators are preferred.
- Place bypass capacitors on the shunt resistors to maintain flat response into high frequencies.

WORKED EXAMPLE: 3 DB PI ATTENUATOR IN 50 Ω

For attenuation A (voltage ratio $K = 10^{A/20}$) in a Z_0 system, the Pi-pad resistors are

$$R_{\text{series}} = Z_0\frac{K^2-1}{2K}, \qquad R_{\text{shunt}} = Z_0\frac{K+1}{K-1}.$$

For $A = 3$ dB, $K = 1.413$: $R_{\text{series}} = 17.6\ \Omega$ and $R_{\text{shunt}} = 292\ \Omega$. The pad attenuates 3 dB while presenting 50 Ω at both ports.

KEY TAKEAWAYS

- Resistive Pi/T pads attenuate while keeping both ports matched to Z_0.
- Values follow from $K = 10^{A/20}$; symmetric pads are bidirectional.
- The pad dissipates the lost power — size the resistors for it.

Exercises

1. What voltage ratio K corresponds to a 6 dB pad?
2. What fraction of input power reaches the load through a 10 dB pad?
3. A 20 dB pad passes 1 W in. Roughly how much must it dissipate?

Further reading: Pozar [1]; Hayward [7].

4.4 Power Dividers and Combiners

LEARNING OBJECTIVES

- Design an equal-split Wilkinson divider.
- Explain isolation and the role of the resistor.
- Compare the Wilkinson to hybrid couplers.

Power dividers split one signal into two or more outputs. Power combiners do the reverse: sum multiple signals into one output. The Wilkinson divider is the classic design because it achieves simultaneous matching, isolation, and lossless operation using only transmission line sections and a single resistor.

4.4.1 Wilkinson Power Divider

The equal-split Wilkinson divider uses two $\lambda/4$ transmission line arms of impedance $Z_0\sqrt{2}$ ($\approx$ 70.7 Ω in a 50 Ω system) and an isolation resistor $R = 2Z_0$ between the two output ports:

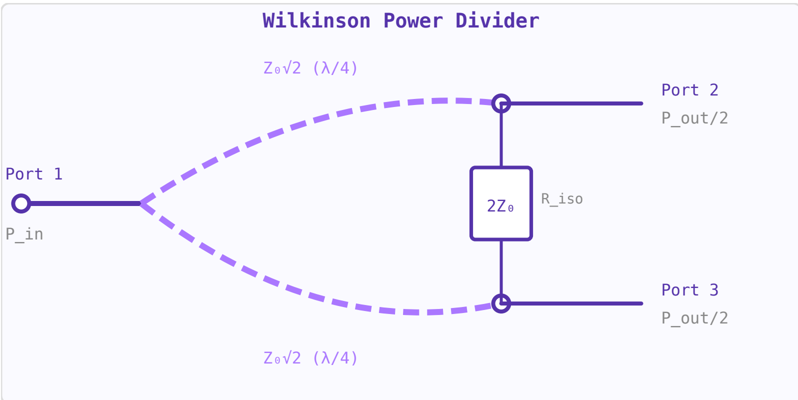


Figure 4.6: Equal-split Wilkinson divider: two $\lambda/4$ arms at $Z_0\sqrt{2} \approx 70.7$ Ω, isolation resistor $2Z_0 = 100$ Ω (for 50 Ω system).

$$Z_{arm} = Z_0\sqrt{2}, \quad R_{iso} = 2Z_0$$

Key properties: all three ports are simultaneously matched; the output ports are isolated from each other (signals applied to port 2 are absorbed by R and do not appear at port 3); the divider is lossless when port 2 and port 3 signals are in phase (the isolation resistor carries no current).

4.4.2 Unequal-Split Wilkinson

For a power split ratio $K^2 : 1$ (port 2 gets K^2 times more power than port 3):

$$Z_1 = Z_0\sqrt{\frac{1+K^2}{K^3}}, \quad Z_2 = Z_0\sqrt{K(1+K^2)}, \quad R_{iso} = Z_0\left(K + \frac{1}{K}\right)$$

For K=1 (equal split): $Z_1 = Z_2 = Z_0\sqrt{2}$, $R = 2Z_0$. For K=2 (6 dB more to port 2): $Z_1 \approx 35.4$ Ω, $Z_2 \approx 112$ Ω, $R = 125$ Ω (in 50 Ω system).

4.4.3 N-Way Dividers

An N-way Wilkinson divider can be built by cascading 2-way sections or using a star junction with N arms. The N-way lumped design uses N $\lambda/4$ arms of impedance $Z_0\sqrt{N}$ and isolation resistors NZ_0 between adjacent output ports.

4.4.4 Hybrid Couplers

A 90° hybrid (branch-line coupler) is a four-port device where the two outputs are 90° apart in phase. It is used in balanced amplifiers and I/Q mixers. A 180° hybrid (rat-race or magic T) gives outputs that are either in-phase or anti-phase, used for sum-difference networks and balanced mixers.

Hybrid coupler types and phase splits

Type	Phase split	BW	Use
Wilkinson	0°	N/A	in-phase split
90° Hybrid	0°/90°	~20%	I/Q mixers, balanced amps
180° Rat-race	0°/180°	~20%	Balanced mixers, Σ/Δ
Lange Coupler	0°/90°	octave	Wideband balanced amps

Figure 4.7: Comparison of common RF power divider and hybrid coupler types.

Device	Phase split	Bandwidth	Use
Wilkinson divider	0° (in-phase)	Moderate (~20%)	Power splitting, combining
90° hybrid coupler	0°/90°	~20%	Balanced amps, I/Q mixers
180° rat-race	0°/180°	~20%	Balanced mixers, sum/difference
Lange coupler	0°/90° (3 dB)	Wide (octave)	Wideband balanced amps

4.4.5 Coherent Power Combining

When N amplifiers feed a combining network, the combined output power is:

$$P_{out} = \eta N P_{each}$$

where η is the combining efficiency (0.93–0.98 for corporate networks). The gain over a single amplifier is $10\log_{10}(N) + 10\log_{10}(\eta)$ dB. If k amplifiers fail, coherent combining means power drops as $(N-k)^2/N^2$ of the full output — graceful degradation rather than total failure.

4.4.6 Physical Implementation

- **PCB microstrip**: The most common implementation. $\lambda/4$ at 2.45 GHz on FR4 ($\varepsilon_r \approx 4.4$) is about 18–20 mm long.
- **Lumped element**: Below ~1 GHz, LC equivalents replace transmission line sections, saving board space.
- **SMD Wilkinson**: Available as pre-packaged components from manufacturers at fixed frequencies.

WORKED EXAMPLE: WILKINSON DIVIDER VALUES

An equal-split two-way Wilkinson uses two quarter-wave arms of impedance $Z_{\text{arm}} = Z_0\sqrt{2} = 50\sqrt{2} = 70.7\ \Omega$ and an isolation resistor $R = 2Z_0 = 100\ \Omega$ between the outputs. It splits input power equally (-3 dB per port), matches all ports, and isolates the outputs; for in-phase equal signals the resistor carries no current, so the divider is lossless in normal use.

KEY TAKEAWAYS

- Equal Wilkinson: $\lambda/4$ arms of $Z_0\sqrt{2}$, isolation resistor $2Z_0$.
- Each output gets half the power (-3 dB); outputs are isolated and matched.
- Hybrids ($90°/180°$) also split power but impose a fixed phase relationship.

Exercises

1. What arm impedance does a 75 Ω Wilkinson use?
2. What is the ideal split loss to each output?
3. What isolation-resistor value does a 50 Ω Wilkinson use?

Further reading: Pozar [1] (ch. 7).

4.5 BALUNs and Hybrids

LEARNING OBJECTIVES

- Explain balanced-to-unbalanced conversion.
- Relate turns ratio to impedance ratio.
- Recognize 90° and 180° hybrids.

A BALUN (BALanced-UNbalanced) converts between single-ended (coaxial) and balanced (differential) connections. Hybrids are 4-port networks that split power with a defined phase relationship. Both are critical building blocks in RF systems.

4.5.1 Why BALUNs are Needed

A dipole antenna is a balanced structure (symmetric about its feed point). Connecting it directly to coaxial cable (unbalanced) allows current to flow on the cable outer conductor, distorting the radiation pattern and causing interference. A BALUN prevents this by presenting high common-mode impedance.

4.5.2 LC BALUN

Uses inductors and capacitors to provide the balanced-to-unbalanced conversion with impedance transformation. The lattice topology uses four elements in a bridge circuit, providing ±45° phase shift on the two balanced outputs and excellent amplitude balance over a moderate bandwidth.

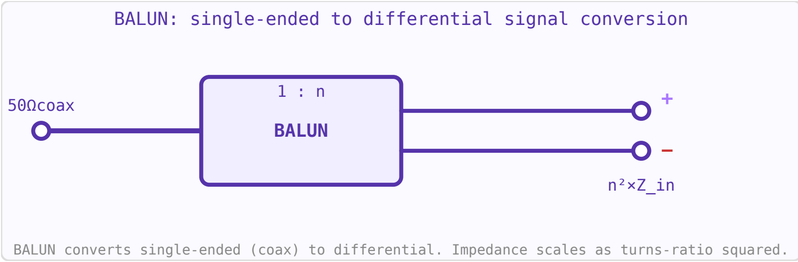


Figure 4.8: BALUN circuit symbol. Impedance ratio n^2:1 allows simultaneous conversion and impedance transformation.

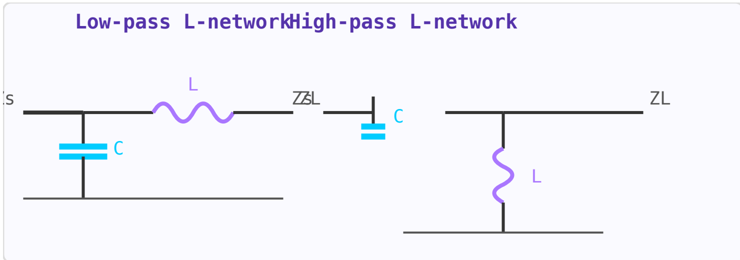


Figure 4.9: L-network as a BALUN building block. Two reactive elements provide impedance transformation and unbalanced-to-balanced conversion.

4.5.3 Marchand BALUN

Two coupled quarter-wave resonators provide wideband balun action (often 2-3 octaves). The open-circuited stubs resonate at f_0 and present the appropriate impedance transformation. Widely used in wideband power amplifiers, mixers, and antenna feeds.

4.5.4 90° Hybrid (Branch-Line)

A 4-port network where power splits equally between two output ports with a 90° phase difference. Used in quadrature modulators, I/Q signal processing, and phased array feed networks. Implemented as a square of four $\lambda/4$ transmission line sections.

4.5.5 180° Hybrid (Rat-Race / Magic-T)

Splits power equally with 0° or 180° phase shift. Used in balanced amplifiers, mixers, and Butler matrices for phased array beamforming.

WORKED EXAMPLE: A 4:1 BALUN IMPEDANCE TRANSFORM

A balun often also transforms impedance. Matching a 200 Ω balanced antenna (e.g. a folded dipole) to a 50 Ω unbalanced feed needs a ratio $Z_{\text{bal}}/Z_{\text{unbal}} = 200/50 = 4$, i.e. a turns ratio of $\sqrt{4} = 2{:}1$. Beyond impedance, the balun forces equal-and-opposite currents on the balanced side, suppressing common-mode current on the feedline.

KEY TAKEAWAYS

- A balun couples a balanced load to an unbalanced line and suppresses common-mode current.
- Impedance ratio = (turns ratio)^2: a 2:1 turns balun is 4:1 in impedance.
- 90° and 180° hybrids give fixed phase splits for mixers and balanced amplifiers.

Exercises

1. What turns ratio gives a 9:1 impedance balun?
2. What impedance-ratio balun matches a 300 Ω folded dipole to 75 Ω coax?
3. What does a 180° hybrid provide that a Wilkinson does not?

Further reading: Pozar [1] (ch. 7); Balanis [14].

4.6 Microwave Passive Components

LEARNING OBJECTIVES

- Identify couplers, circulators, and isolators by function.
- Read coupling and directivity of a directional coupler.
- Use a circulator or isolator to protect a source.

Beyond transmission lines and filters, a range of passive components are essential in microwave systems: couplers, dividers, circulators, and attenuators.

4.6.1 Directional Couplers

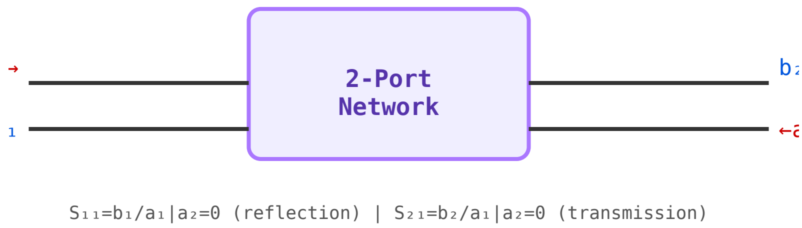


Figure 4.10: A directional coupler is a 4-port device — extending the 2-port S-matrix concept to include forward and reverse coupled ports.

A directional coupler samples a fraction of the forward or reverse traveling wave on a transmission line without significantly disturbing it. Key parameters: coupling factor (C, dB), directivity (D, dB), and insertion loss. Used for power monitoring, signal sampling, and reflectometer circuits in VNAs.

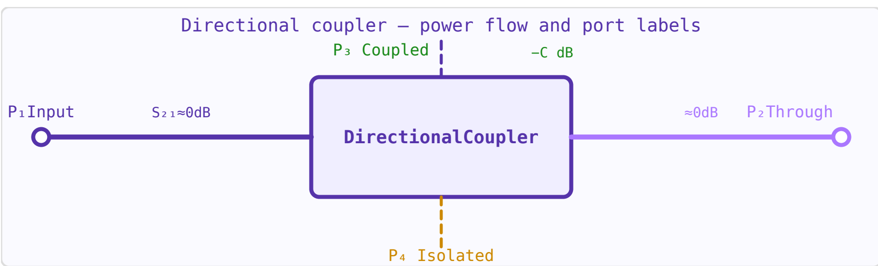


Figure 4.11: Directional coupler: power flows port 1→2 (through, ≈0 dB loss) and 1→3 (coupled, −C dB). Port 4 is isolated.

4.6.2 Power Dividers / Combiners

A Wilkinson power divider splits input power equally into two output ports (−3 dB each) while maintaining isolation between the two outputs. The isolation resistor absorbs any power imbalance between ports. Used in antenna arrays, balanced amplifiers, and mixer feeds.

4.6.3 Circulator

A three-port non-reciprocal device: power entering port 1 exits port 2; power entering port 2 exits port 3; power entering port 3 exits port 1. Uses magnetised ferrite. Critical applications: isolating a PA from antenna reflections, separating transmit and receive paths (T/R switch replacement), and protecting LNAs from high-power pulses in radar.

4.6.4 Attenuators

Fixed attenuators (pads) reduce signal power by a known amount while maintaining Z_0 impedance. Pi and T resistive networks are common topologies. Used for impedance bridging, reducing reflections, and setting test signal levels. Accuracy and flatness with frequency are key specifications.

4.6.5 PIN Diode Switches

PIN diodes operate as variable resistors at RF: forward biased → low resistance (ON); reverse biased → high resistance (OFF). Used for fast antenna switching, T/R switching in radar, and variable attenuators. Switching speed: nanoseconds to microseconds depending on carrier lifetime.

4.6.6 Self-Resonant Frequency (SRF)

No real component is purely one thing. Every capacitor has a parasitic series inductance (its ESL, from the body and mounting), and every inductor has a parasitic parallel capacitance — so each behaves as a series or parallel LC that resonates at its *self-resonant frequency*, $f_{SRF} = 1/(2\pi\sqrt{LC})$. Below its SRF a capacitor is capacitive; at the SRF its reactances cancel and it looks purely resistive (just the ESR, the impedance minimum); above the SRF it turns *inductive* and stops decoupling. The practical rule is to operate components well below their SRF, and to parallel capacitors of different values so their SRFs stagger and keep the impedance low across a wide band. Smaller packages push the SRF higher, which is why RF bypass capacitors use tiny 0201/0402 parts.

WORKED EXAMPLE: DIRECTIONAL COUPLER TAP

A directional coupler samples a fixed fraction of the forward wave. A "20 dB coupler" delivers $P_{\text{coupled}} = P_{in} - 20$ dB, i.e. 1% of the input power to the coupled port. Its *directivity* — the ratio of forward to reverse coupling — should exceed 20 dB so the coupled port samples mainly the forward wave.

KEY TAKEAWAYS

- A directional coupler samples the forward (or reverse) wave; coupling in dB sets the tap fraction.
- Directivity measures how well forward and reverse are separated; higher is better.
- A circulator routes power one way port-to-port; terminate one port to make an isolator.

Exercises

1. What fraction of input power does a 10 dB coupler tap?
2. What component protects a transmitter from reflected power?
3. What is the coupled-port level for +30 dBm into a 20 dB coupler?

Further reading: Pozar [1] (ch. 7); Collin [2].

4.7 RF Filters

LEARNING OBJECTIVES

- Distinguish Butterworth and Chebyshev responses.
- Read filter order from the roll-off rate.
- Realize an LC-ladder low-pass.

RF filters are two-port networks that pass signals in certain frequency bands while attenuating others. They are fundamental in every RF system for channel selection, harmonic suppression, and image rejection.

4.7.1 Filter Types

- **Low-pass (LPF)** — passes below cutoff fc; used for harmonic suppression, anti-aliasing
- **High-pass (HPF)** — passes above fc; used to block DC or low-frequency interference
- **Band-pass (BPF)** — passes a band centred on f_0; used for channel selection
- **Band-stop / Notch** — rejects a specific band; used to suppress interferers

4.7.2 Butterworth Response

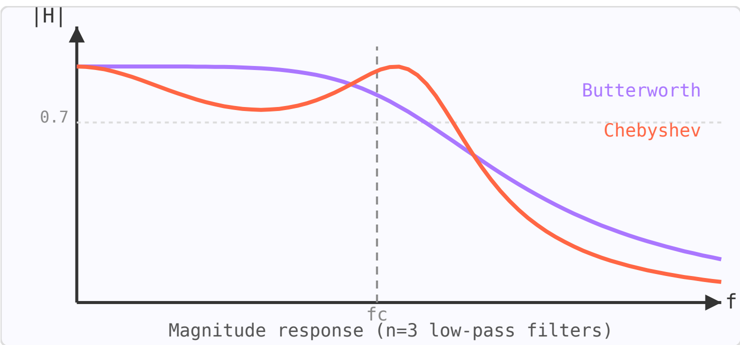


Figure 4.12: n=3 low-pass filter responses: Butterworth (flat passband, purple) vs Chebyshev (steeper roll-off, orange).

Maximally flat magnitude in the passband — no ripple. The magnitude response is:

$$|H(j\omega)| = \frac{1}{\sqrt{1 + (\omega/\omega_c)^{2n}}}$$

Roll-off: 20n dB/decade beyond ω_c. Gentle transition band.

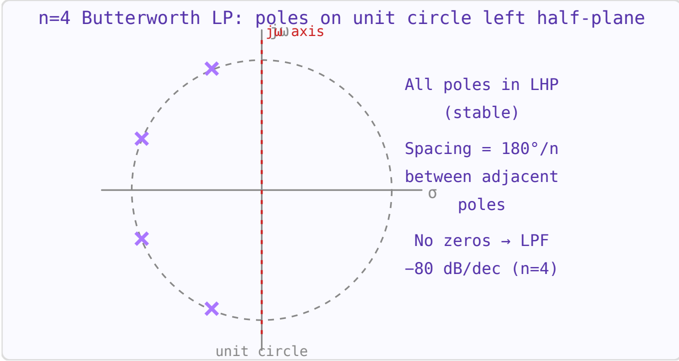


Figure 4.13: Butterworth LP pole-zero plot. n=4 poles equally spaced on the left half of the unit circle in the s-plane.

4.7.3 Chebyshev Response

Allows equiripple in the passband in exchange for a steeper transition band. For the same order, Chebyshev gives more attenuation in the stopband than Butterworth. The ripple (dB) is a design parameter.

4.7.4 LC Ladder Synthesis

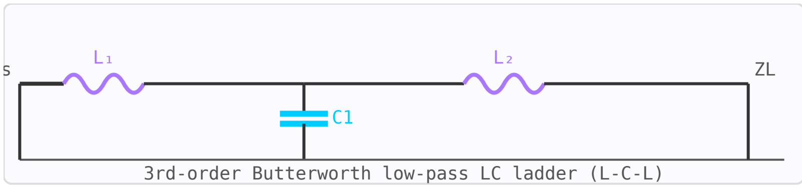


Figure 4.14: 3rd-order LC ladder low-pass filter between source and load impedances.

Prototype low-pass filter element values g_k are tabulated for Butterworth and Chebyshev responses. These normalised values are frequency- and impedance-scaled to the target cutoff frequency and port impedance using:

$$L_k = \frac{g_k R_0}{\omega_c} \quad C_k = \frac{g_k}{R_0 \omega_c}$$

WORKED EXAMPLE: BUTTERWORTH ROLL-OFF AND ORDER

A Butterworth low-pass rolls off at $20n$ dB/decade for order n. To get 60 dB of rejection one decade above cutoff, $n = 60/20 = 3$. A 3rd-order LC ladder (L–C–L or C–L–C) realizes it. A Chebyshev design reaches the same rejection at lower order by allowing passband ripple, at the cost of group-delay flatness.

KEY TAKEAWAYS

- Butterworth is maximally flat; Chebyshev trades passband ripple for steeper roll-off.
- Roll-off $\approx 20n$ dB/decade ($6n$ dB/octave).
- Ladder networks (series L, shunt C) implement LC filters; order = number of reactive elements.

Exercises

1. What is the roll-off of a 5th-order Butterworth, in dB/decade?
2. How many reactive elements are in a 3rd-order LC ladder?
3. Which reaches 40 dB of rejection at lower order: Butterworth or Chebyshev?

Further reading: Pozar [1] (ch. 8); Ludwig and Bogdanov [6].

4.8 Group Delay and Phase Linearity

LEARNING OBJECTIVES

- Define group delay as the negative derivative of phase.
- Explain why group-delay flatness matters for wideband and pulsed signals.
- Compare filter families by delay flatness.

A filter's magnitude response tells you which frequencies get through; its *phase* response tells you *when* they get through. Group delay is the practical measure of that timing — the delay a filter imposes on a signal's envelope or modulation. A filter with flat magnitude but wildly varying group delay will still smear a pulse or corrupt a data eye, which is why group delay is a first-class specification for anything carrying wideband or pulsed signals.

4.8.1 Definition

Group delay is the negative slope of the phase response with respect to angular frequency:

$$\tau_g(\omega) = -\frac{d\phi(\omega)}{d\omega}$$

If the phase is perfectly linear ($\phi = -\omega t_0$), the group delay is a constant t_0 at every frequency — every spectral component is held up by the same time, so the waveform passes through undistorted, just delayed. Where the phase curves, different frequencies leave at different times and the waveform spreads. This frequency-dependent delay is called dispersion.

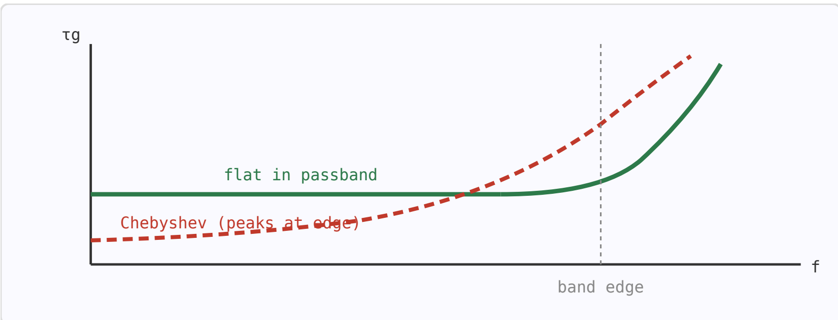


Figure 4.15: Group delay vs frequency: a Bessel/flat-delay response (green) stays constant across the passband, while a sharp Chebyshev (red) peaks strongly near the band edge.

4.8.2 Why It Matters

- **Pulses & radar**: unequal delay across the pulse spectrum broadens and distorts the pulse, degrading range resolution.

- **Digital data**: group-delay ripple causes inter-symbol interference (ISI) and closes the eye diagram, raising BER.
- **Video & analog**: differential delay shifts colour/luma or produces ringing and overshoot on edges.

A useful rule of thumb: keep the peak-to-peak group-delay variation across the signal band to a small fraction of the symbol period (or pulse width).

4.8.3 The Filter-Family Trade-off

Selectivity and flat delay pull in opposite directions. The sharper a filter cuts off, the more its poles cluster near the band edge and the more its group delay peaks there:

Response	Magnitude	Group delay	Step response
Bessel / Thomson	Gentle rolloff	Maximally flat (best)	No overshoot
Butterworth	Maximally flat	Moderate rise at edge	Slight overshoot
Chebyshev	Steep, equiripple	Strong peak at edge	Ringing
Elliptic (Cauer)	Steepest	Worst delay ripple	Heavy ringing

Bessel filters are designed for maximally flat delay and are the default when waveform fidelity matters more than a sharp skirt. When a sharp filter is unavoidable, a *group-delay equaliser* (an all-pass network) can be cascaded to flatten the composite delay at the cost of added absolute delay.

4.8.4 Delay at Band Center

For an all-pole lowpass prototype, the band-center (DC) group delay follows from the pole locations $p_k = \sigma_k + j\omega_k$:

$$\tau_g(0) = \frac{1}{\omega_c} \sum_k \frac{-\sigma_k}{\sigma_k^2 + \omega_k^2}$$

Higher order and more ripple both increase this delay. A 3rd-order Butterworth, for example, has a normalized DC delay of 2 (i.e. $2/\omega_c$); a Chebyshev of the same order delays more. Group delay scales inversely with cutoff — a wider filter delays less.

WORKED EXAMPLE: WHY LINEAR PHASE PRESERVES A PULSE

Group delay is $\tau_g = -\,d\phi/d\omega$. A network with linear phase $\phi = -\omega\tau$ has constant $\tau_g = \tau$, so every frequency component is delayed by the same amount and the waveform passes through undistorted. Near a filter's band edge the phase bends sharply and τ_g peaks, delaying some components more than others and smearing the signal in time. Bessel filters maximise delay flatness (at the cost of roll-off); Chebyshev and elliptic filters give the steepest roll-off but the worst delay ripple.

KEY TAKEAWAYS

- $\tau_g = -\,d\phi/d\omega$; constant group delay means distortionless (linear-phase) transmission.
- Group delay peaks near band edges and near high-Q resonances.
- Bessel $\rightarrow$ flat delay; Butterworth $\rightarrow$ moderate; Chebyshev/elliptic $\rightarrow$ steep roll-off but large delay ripple.

Exercises

1. What is the group delay of a network whose phase is $\phi = -\omega\tau$?
2. Which filter family gives the flattest group delay?
3. Why does group-delay variation degrade a digital data signal?

Further reading: Pozar [1] (ch. 8); Hong [33].

4.9 Stepped-Impedance Low-Pass Filters

LEARNING OBJECTIVES

- Explain the stepped-impedance (hi-lo) low-pass realization.
- Map high-Z sections to series L and low-Z to shunt C.
- Understand where the approximation breaks down.

A stepped-impedance low-pass filter (SI-LPF) replaces the lumped inductors and capacitors of a conventional LC ladder with sections of high-impedance and low-impedance transmission line. This approach is practical on PCB microstrip where lumped components at GHz frequencies behave poorly, and no vias or discrete parts are needed.

4.9.1 Operating Principle

A short section of **high-impedance** line (narrow trace, small physical width) behaves as a series inductor because its shunt capacitance is small relative to its series reactance. A short section of **low-impedance** line (wide trace) behaves as a shunt capacitor. Alternating these sections realises the LC ladder prototype.

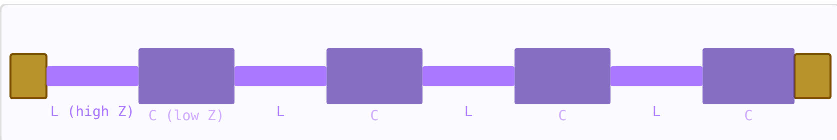


Figure 4.16: Top view of a stepped-impedance LPF: narrow (high-Z, purple) and wide (low-Z, dark) alternating sections.

4.9.2 Element Mapping

The prototype g-values are mapped to electrical lengths using the system impedance Z_0, high-line impedance Zh, and low-line impedance Zl:

$$\beta\ell_{\text{high}} = \frac{g_k\, Z_0}{Z_h} \qquad \text{(series inductor element)}$$

$$\beta\ell_{\text{low}} = \frac{g_k\, Z_l}{Z_0} \qquad \text{(shunt capacitor element)}$$

Here $\beta\ell$ is the electrical length in radians, evaluated at the cutoff frequency fc. The physical length is:

$$\ell = \frac{\beta\ell}{2\pi} \cdot \frac{c}{f_c\sqrt{\varepsilon_{\text{eff}}}}$$

4.9.3 Prototype g-Values

Use standard Butterworth or Chebyshev low-pass prototype tables. For a 3rd-order Butterworth: $g_1 = 1.0$, $g_2 = 2.0$, $g_3 = 1.0$. For Chebyshev 0.5 dB ripple, 5th-order: $g_1 = 1.7058$, $g_2 = 1.2296$, $g_3 = 2.5408$, $g_4 = 1.2296$, $g_5 = 1.7058$. Higher order gives steeper roll-off but longer board area.

4.9.4 Design Rules

- Choose Zh/Z_0 as high as fabrication allows (typically Zh = 80–120 Ω for 50 Ω system)
- Choose Zl as low as practical (typically Zl = 15–30 Ω); limited by board width constraints
- Higher Zh/Zl ratio → shorter sections → better approximation to lumped elements
- All sections must have electrical lengths $\beta\ell < 45°$ at the cutoff frequency for the lumped-element approximation to be valid
- Spurious passband appears at 2fc (or 4fc for symmetric designs); stepped-Z is not suitable above 2–3× cutoff

4.9.5 Microstrip Trace Width

The required trace width for a target impedance Z on a substrate of thickness h and permittivity εr is found by the inverse Hammerstad-Jensen formulas (implemented in the Microstrip Calculator). As a rough guide: for 50 Ω on FR4 (εr ≈ 4.4, h = 1.6 mm), w ≈ 3 mm; for 100 Ω, w ≈ 0.5 mm; for 25 Ω, w ≈ 10 mm.

WORKED EXAMPLE: HI-Z / LO-Z SECTIONS

A stepped-impedance low-pass alternates short high-impedance lines (series inductors) and low-impedance lines (shunt capacitors). A short line of length ℓ contributes

$$L \approx \frac{Z_h\,\ell}{v_p}, \qquad C \approx \frac{\ell}{Z_l\,v_p},$$

so a higher Z_h raises the equivalent series L and a lower Z_l raises the shunt C. The pattern approximates an LC ladder in microstrip with no lumped parts.

KEY TAKEAWAYS

- High-Z (narrow) lines emulate series inductors; low-Z (wide) lines emulate shunt capacitors.
- It builds an LC low-pass from microstrip alone — easy to fabricate.
- Accuracy falls near and above cutoff where the lumped approximation breaks down.

Exercises

1. A narrow, high-Z microstrip section behaves as which element?
2. A wide, low-Z section behaves as which element?
3. To increase the emulated series inductance, raise or lower the section's Z_0?

Further reading: Pozar [1] (ch. 8).

4.10 Advanced Filters

LEARNING OBJECTIVES

- Match distributed, cavity/DR, and acoustic filters to requirements.
- Relate resonator Q to insertion loss and selectivity.
- Explain duplexers/diplexers/multiplexers.

Beyond lumped LC and stepped-impedance filters lies a family of high-performance structures chosen for their Q, size, or steepness.

4.11 Distributed Filters

At microwave frequencies filters are built from coupled transmission lines: coupled-line, combline, interdigital, and hairpin topologies realize bandpass responses in microstrip. They avoid lumped parts and are printed directly on the board.

4.12 High-Q Resonator Filters

Where low loss and high power matter, cavity and dielectric-resonator filters provide unloaded Q in the thousands — versus a couple hundred for microstrip — giving very low insertion loss and sharp skirts for base-station duplexers.

4.13 Acoustic Filters

Surface-acoustic-wave (SAW) and bulk-acoustic-wave (BAW/FBAR) filters convert the signal to an acoustic wave whose wavelength is 10^5 times shorter, packing a very steep, tiny filter into a chip — the technology inside every phone's RF front end. Crystal and ceramic filters serve narrowband IF and lower-frequency needs.

4.14 Specifications and Multiplexers

Filters are judged on insertion loss, selectivity (shape factor), and power handling, all tied to resonator Q. Combining filters into duplexers, diplexers, and multiplexers lets one antenna serve transmit and receive or many bands at once.

WORKED EXAMPLE: LOADED Q FOR A CHANNEL FILTER

A bandpass filter at $f_0 = 2$ GHz must pass a 20 MHz channel. Its loaded quality factor is

$$Q_L = \frac{f_0}{\mathrm{BW}} = \frac{2 \times 10^9}{20 \times 10^6} = 100.$$

For low insertion loss the resonators' *unloaded* Q must far exceed this ($Q_u \gg Q_L$). Microstrip ($Q_u \sim 200$) would be lossy here; a cavity or BAW resonator (Q_u in the thousands) is the right choice.

KEY TAKEAWAYS

- Distributed (coupled-line/combline/interdigital) filters print in microstrip; no lumped parts.
- Cavity/DR give Q_u in the thousands (low loss, sharp); SAW/BAW pack steep filters into a chip.
- Insertion loss falls as Q_u/Q_L rises; duplexers/diplexers share one antenna across TX/RX or bands.

Exercises

1. Loaded Q for a 1% fractional-bandwidth filter?
2. Which filter technology sits in a smartphone's RF front end for steep, tiny filters?
3. Why does a base-station duplexer use cavity resonators rather than microstrip?

Further reading: Hong [33]; Pozar [1] (ch. 8).

4.15 Switches, Phase Shifters and Ferrite Devices

LEARNING OBJECTIVES

- Compare PIN, FET, and MEMS RF switches.
- Compute the resolution of a digital phase shifter.
- Explain nonreciprocal ferrite circulators and isolators.

Control components route, attenuate, phase-shift, and protect RF signals — the switches and dials of a radio front end.

4.16 Switches

RF switches (PIN-diode, FET, or MEMS) select paths, band-switch, and share an antenna between transmit and receive. They are judged on insertion loss, isolation, linearity (IP3), and switching speed; MEMS give the best loss and linearity but switch slowly.

4.17 Phase Shifters

Phase shifters set the progressive phase that steers a phased array. Switched-line, loaded-line, reflection-type, and vector-modulator designs provide either discrete bits or continuous control. A digital phase shifter of n bits has a least significant step of $360°/2^n$.

4.18 Ferrite Devices

Magnetized ferrites are *nonreciprocal*: a circulator routes power port-to-port in one rotational direction, and terminating one port makes an isolator that passes forward power while absorbing reflections — protecting a transmitter from a bad load. YIG spheres make electronically tunable filters and oscillators.

4.19 Attenuators and Limiters

Variable attenuators (PIN or FET) set gain in AGC loops; limiters (PIN-diode) protect sensitive receivers by clipping large input spikes.

WORKED EXAMPLE: DIGITAL PHASE-SHIFTER RESOLUTION

A 6-bit digital phase shifter divides the full circle into

$$\mathrm{LSB} = \frac{360^\circ}{2^6} = 5.625^\circ,$$

so it can command any phase from 0° to 354.375° in 5.625° steps. Finer beam pointing in a phased array needs more bits; array calibration corrects the residual phase errors between elements.

KEY TAKEAWAYS

- Switches (PIN/FET/MEMS): trade insertion loss, isolation, IP3, and speed.
- An n-bit phase shifter has $\mathrm{LSB} = 360^\circ/2^n$.
- Ferrites are nonreciprocal: circulators route one way; terminate a port for an isolator that protects the source.

Exercises

1. Least-significant step of a 4-bit phase shifter?
2. What component protects a transmitter from power reflected by a bad antenna?
3. Which switch technology offers the lowest insertion loss but the slowest switching?

Further reading: Pozar [1] (ch. 7, 10); Steer [5].

Chapter 5

Active Devices and Amplifier Design

5.1 RF Transistors

LEARNING OBJECTIVES

- Compare the major RF transistor technologies (Si BJT/SiGe, CMOS, GaAs, GaN).
- Define f_T and f_{max} and relate them to g_m and capacitance.
- Explain the purpose of the bias network and its trade-offs.

Every active RF function — amplification, oscillation, mixing — rests on a three-terminal transistor that turns a small control signal into a larger controlled current. Choosing the right device technology is the first design decision.

5.2 Device Families

Silicon BJT and SiGe HBT devices offer low cost, low $1/f$ noise, and good linearity into the tens of GHz (SiGe). *CMOS* enables single-chip integration of RF and digital, at the price of higher noise and lower breakdown. *GaAs MESFET/pHEMT* give very low noise figures and reach millimetre waves; *GaN HEMT* combines high breakdown voltage with high power density, dominating modern power amplifiers.

5.3 Figures of Merit

The transition frequency f_T is where the short-circuit current gain falls to unity,

$$f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})},$$

where $g_m = \partial I_D/\partial V_{GS}$ is the transconductance. The maximum oscillation frequency f_{max}, where the unilateral power gain reaches unity, is the more relevant ceiling for amplifiers and typically exceeds f_T. A useful device runs well below both.

5.4 Biasing

The transistor must be held at a quiescent operating point (bias) that sets its g_m, linearity, and noise. Bias is delivered through RF chokes and bypass capacitors that feed DC while blocking RF, and the network must itself be stable and temperature-tolerant. Class-A bias (continuous conduction) gives the best linearity and noise; lower conduction angles trade linearity for efficiency (Part V).

5.5 Small-Signal Model and Technology Choice

Near the operating point the device is described by a linear equivalent circuit (a transconductance g_m, input and feedback capacitances, output conductance), or equivalently by its measured S-parameters at that bias. Match the technology to the job: SiGe or GaAs pHEMT for a low-noise front end, GaN for a power stage, and CMOS when integration with baseband matters most.

WORKED EXAMPLE: TRANSITION FREQUENCY

A device has transconductance $g_m = 50$ mS and total input capacitance $C_{gs}+C_{gd} = 0.2$ pF. Its transition frequency is

$$f_T = \frac{g_m}{2\pi(C_{gs} + C_{gd})} = \frac{0.05}{2\pi(0.2 \times 10^{-12})} = 39.8 \text{ GHz}.$$

Useful gain is available well below f_T; f_{max} sets the true upper limit.

KEY TAKEAWAYS

- $f_T = g_m/[2\pi(C_{gs} + C_{gd})]$; f_{max} (unity power gain) is the practical ceiling.
- GaN for power, GaAs pHEMT / SiGe for low noise, CMOS for integration.
- Bias sets g_m, linearity, and noise; deliver it through chokes/bypass without spoiling stability.

Exercises

1. A device has $g_m = 80$ mS and $C_{in} = 0.1$ pF. Find f_T.
2. Which technology would you pick for a high-power PA?
3. Is f_T or f_{max} the better indicator of usable amplifier bandwidth?

Further reading: Razavi [8] (ch. 2, 5); Gonzalez [10] (ch. 2).

5.6 RF Diodes and Varactors

LEARNING OBJECTIVES

- Match Schottky, varactor, and PIN diodes to their RF roles.
- Compute a varactor's capacitance and tuning ratio versus bias.
- Explain why a PIN diode acts as an RF resistor, not a rectifier.

Diodes are the workhorse two-terminal nonlinear elements of RF: they detect, mix, switch, tune, and generate harmonics.

5.7 Schottky Diodes

A Schottky (metal–semiconductor) diode has no minority-carrier storage, so it switches extremely fast and rectifies well into the millimetre-wave range. Its low, sharp turn-on makes it the standard device for power detectors and for diode (passive) mixers.

5.8 Varactors

A varactor is a reverse-biased junction used as a voltage-variable capacitor,

$$C(V) = \frac{C_0}{(1 + V/\phi)^n},$$

with built-in potential ϕ and grading coefficient n ($n = 1/2$ abrupt, $n \to 1$ hyperabrupt). Varactors tune VCOs and filters; the useful metric is the capacitance (tuning) ratio over the control-voltage range.

5.9 PIN Diodes

A PIN diode has an intrinsic layer that, when forward biased, stores charge and behaves at RF as a *current-controlled resistor* — low resistance when on, high when off — but it does not rectify the RF because the carriers cannot follow it. This makes the PIN diode the standard element for RF switches, variable attenuators, and phase shifters.

5.10 Other Devices

Step-recovery diodes snap off sharply to generate rich harmonic combs (comb generators). Tunnel and Gunn diodes exhibit negative differential resistance and can build simple oscillators. Each exploits a specific nonlinearity of the junction.

WORKED EXAMPLE: VARACTOR TUNING RATIO

An abrupt-junction varactor ($n = 1/2$) has $C_0 = 10$ pF and $\phi = 0.7$ V. At a reverse bias of 4 V,

$$C(4) = \frac{10}{(1 + 4/0.7)^{1/2}} = \frac{10}{\sqrt{6.71}} = 3.86 \text{ pF}.$$

The tuning ratio from 0 to 4 V is $C(0)/C(4) = 10/3.86 = 2.6$, which sets the frequency tuning range of a VCO built with it.

KEY TAKEAWAYS

- Schottky: fast, low-drop — detectors and passive mixers.
- Varactor $C(V) = C_0/(1 + V/\phi)^n$ — VCO/filter tuning; ratio sets range.
- PIN: charge-controlled RF resistor — switches, attenuators, phase shifters.

Exercises

1. An abrupt varactor has $C_0 = 8$ pF, $\phi = 0.7$ V. Find C at 3 V.
2. Which diode would you use to build a broadband RF switch?
3. Why does a PIN diode not demodulate the RF passing through it?

Further reading: Maas [12]; Pozar [1] (ch. 10).

5.11 RF Amplifiers

LEARNING OBJECTIVES

- Distinguish LNA, gain, and power amplifiers.
- Read gain, P1dB, and IP3 from a datasheet.
- Understand the gain-noise-linearity trade.

RF amplifiers increase signal power. Key specifications include gain, noise figure, output power, linearity, and stability. Different applications demand very different tradeoffs: a receiver LNA optimises noise figure; a transmit PA optimises efficiency and output power.

5.11.1 Low-Noise Amplifier (LNA)

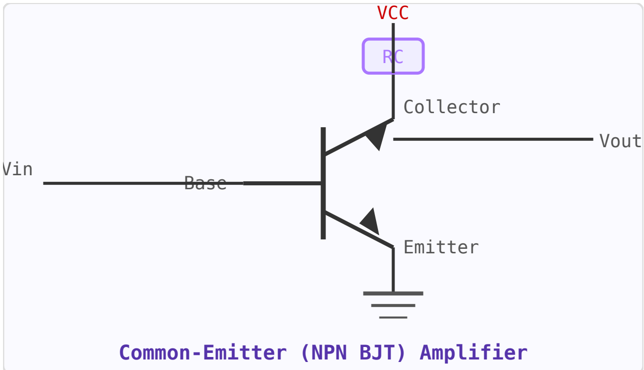


Figure 5.1: Common-emitter NPN amplifier. RC = collector load; voltage divider sets quiescent base current.

The first active stage in a receiver chain. Its noise figure dominates the overall system NF (Friis formula), so minimising LNA NF is critical. The input matching network is designed for noise match rather than power match. Typical specifications: NF = 0.3–1.5 dB, gain = 15–25 dB.

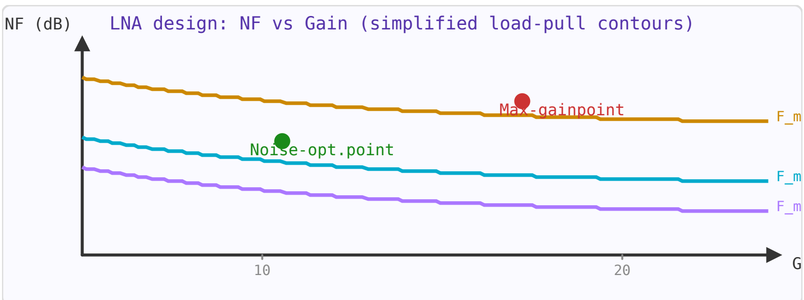


Figure 5.2: LNA tradeoff: minimum NF and maximum gain are different operating points. For first-stage LNA, prioritise NF.

5.11.2 Power Amplifier (PA)

The final stage of a transmitter. PA efficiency (PAE — power added efficiency) is critical for battery life in mobile devices. Class A amplifiers have low distortion but poor efficiency (~25–30%). Class B/AB amplifiers improve efficiency (~50–60%). Class E/F switching amplifiers can achieve theoretical efficiencies above 90%.

5.11.3 Gain and Stability

Transistor amplifiers are potentially unstable at certain frequencies due to internal feedback. The Rollett stability factor $K > 1$ (and \(|\Delta| < 1\)) is the condition for unconditional stability. Amplifiers are often stabilised using resistive loading or feedback networks.

5.11.4 Linearity: P1dB and IP3

The 1 dB compression point (P1dB) is the input power at which the gain has dropped by 1 dB from its small-signal value. The third-order intercept point (IP3) characterises intermodulation distortion — the spurious mixing products generated when two closely spaced signals both enter the amplifier. As a rule of thumb, $IIP3 \approx P1dB + 10$ dBm.

WORKED EXAMPLE: 1 DB COMPRESSION

An amplifier has 20 dB small-signal gain and an output 1 dB compression point OP1dB = +18 dBm. Driven at $P_{in} = -5$ dBm the linear output would be +15 dBm, below OP1dB, so it is still roughly linear. Push toward $P_{in} = -1$ dBm (linear output +19 dBm) and it compresses, clipping and generating distortion.

KEY TAKEAWAYS

- LNAs optimize noise figure; PAs optimize output power and efficiency; both need stability.
- P1dB marks the onset of gain compression (the large-signal limit).
- Higher linearity (IP3, P1dB) usually costs DC power and/or gain.

Exercises

1. Gain = 12 dB, $P_{in} = -30$ dBm. What is the small-signal output?
2. Above P1dB, does the gain rise or fall?
3. Which amplifier goes first in a receive chain, and why?

Further reading: Gonzalez [10]; Razavi [8] (ch. 5).

5.12 Amplifier Stability

LEARNING OBJECTIVES

- Compute the Rollett K-factor from S-parameters.
- Apply the K–$|\Delta|$ unconditional-stability test.
- Recognize stabilization methods.

An RF amplifier can oscillate if the reflection coefficient seen looking into either port exceeds unity magnitude for some source or load impedance. Stability analysis identifies whether this can happen using the device's S-parameters, without having to check every possible termination.

5.12.1 S-Parameter Determinant (Δ)

The first quantity to compute is the two-port determinant:

$$\Delta = S_{11}S_{22} - S_{12}S_{21}$$

This combines all four S-parameters into a single complex number whose magnitude appears in both the K-factor and μ-factor criteria.

5.12.2 Rollett Stability Factor K

The Rollett K-factor was the first widely adopted stability criterion (1962):

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2\,|S_{12}|\,|S_{21}|}$$

The device is **unconditionally stable** (stable for any passive source and load) when both:

$$K > 1 \quad \text{AND} \quad |\Delta| < 1$$

Both conditions must hold simultaneously. K $>$ 1 alone is necessary but not sufficient. When K $<$ 1 the device is only conditionally stable — it may oscillate with certain terminations.

5.12.3 μ-Factor (Edwards-Sinsky)

The μ-factor (1992) provides a single-parameter unconditional stability test that is both necessary and sufficient:

$$\mu = \frac{1 - |S_{11}|^2}{|S_{22} - \Delta\, S_{11}^*| + |S_{12}\, S_{21}|}$$

The device is unconditionally stable if and only if $\mu > 1$. A larger μ indicates a greater stability margin. The output-port version μ' uses S22 in the numerator and S11 in the denominator symmetrically.

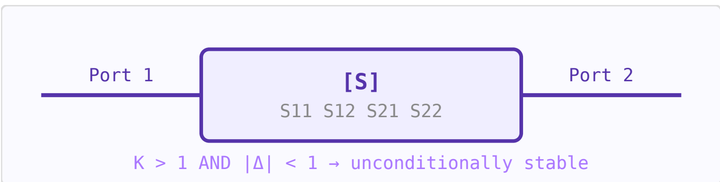


Figure 5.3: Two-port device described by S-parameters. Stability is assessed from S11, S12, S21, S22 alone.

5.12.4 Maximum Available Gain (MAG)

When the device is unconditionally stable (K $>$ 1, $|\Delta| < 1$), there exists an optimal simultaneous conjugate match at both ports. The resulting maximum available gain is:

$$G_{\text{MAG}} = \left|\frac{S_{21}}{S_{12}}\right| \left(K - \sqrt{K^2 - 1}\right)$$

5.12.5 Maximum Stable Gain (MSG)

When K $<$ 1 (conditionally stable), MAG is undefined. The maximum stable gain is the theoretical upper bound achievable by adding loss to bring K = 1:

$$G_{\text{MSG}} = \left|\frac{S_{21}}{S_{12}}\right|$$

MSG is always higher than the achievable gain in practice — it is an optimistic upper bound used to compare devices before choosing stabilisation resistors.

5.12.6 Practical Notes

- S-parameters are frequency-dependent — check stability across the entire frequency range, not just the operating band
- Low-noise transistors often have K < 1 at low frequencies; add series or shunt resistors to stabilise out-of-band
- Stability circles on the Smith Chart show the boundary of terminations that cause |Γin| = 1 or |Γout| = 1
- A device with $\mu = 1.5$ at a given frequency has 50% more stability margin than one with $\mu = 1.01$

WORKED EXAMPLE: ROLLETT STABILITY FACTOR

Unconditional stability requires $K > 1$ and $|\Delta| < 1$, where $\Delta = S_{11}S_{22} - S_{12}S_{21}$ and

$$K = \frac{1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2}{2\,|S_{12}S_{21}|}.$$

A device with $K = 0.8$ is *potentially unstable* — some source/load impedances will oscillate. A small series or shunt resistor (usually at the output) raises K above 1, trading a little gain and noise for stability.

KEY TAKEAWAYS

- Unconditional stability: $K > 1$ *and* $|\Delta| < 1$.
- $K < 1$ means some terminations cause oscillation (potentially unstable).
- Resistive loading or feedback stabilizes, trading gain/NF for stability.

Exercises

1. A device has $K = 1.4$, $|\Delta| = 0.3$. Is it unconditionally stable?
2. What does $K = 0.6$ indicate?
3. Name one way to raise K.

Further reading: Gonzalez [10] (ch. 3); Pozar [1] (ch. 12).

5.13 Small-Signal Amplifier Design

LEARNING OBJECTIVES

- Distinguish transducer, available, and operating power gain.
- Compute the maximum unilateral transducer gain.
- Explain why stability must be checked before matching.

Once a transistor is chosen and biased, small-signal amplifier design is the art of choosing the source and load terminations that deliver the wanted gain, stability, and match.

5.14 Power Gains

Three gains matter, all computable from the S-parameters and the terminations Γ_s, Γ_L: the *transducer gain* G_T (delivered/available from source), the *available gain* G_A, and the *operating gain* G_P. For a unilateral device ($S_{12} \approx 0$) the maximum transducer gain factors cleanly:

$$G_{TU,\max} = \frac{|S_{21}|^2}{(1-|S_{11}|^2)(1-|S_{22}|^2)},$$

achieved by conjugately matching input and output.

5.15 Stability First

A design is only meaningful if it is stable (Chapter on Amplifier Stability): check $K > 1$ and $|\Delta| < 1$. If the device is only conditionally stable, restrict Γ_s, Γ_L to the stable regions or add resistive loading.

5.16 Bilateral Design

When S_{12} is not negligible, input and output interact. The simultaneous conjugate match Γ_{MS}, Γ_{ML} solves a coupled pair of equations and yields the maximum available gain $G_{MA} = |S_{21}/S_{12}|(K - \sqrt{K^2 - 1})$ — defined only when the device is unconditionally stable.

5.17 Design for a Specified Gain

Rarely do we want maximum gain; more often a specific, flat gain across a band. Constant-gain circles on the Smith chart show the loci of Γ_s (or Γ_L) that give a chosen gain, letting the designer trade gain for bandwidth, match, or noise.

WORKED EXAMPLE: MAXIMUM UNILATERAL GAIN

A device has $|S_{21}| = 4$, $|S_{11}| = 0.5$, $|S_{22}| = 0.4$ (and $S_{12} \approx 0$). Its maximum unilateral transducer gain is

$$G_{TU,\max} = \frac{|S_{21}|^2}{(1-|S_{11}|^2)(1-|S_{22}|^2)} = \frac{16}{(1-0.25)(1-0.16)} = \frac{16}{0.63} = 25.4,$$

i.e. 14.0 dB, obtained by conjugately matching both ports.

KEY TAKEAWAYS

- $G_{TU,\max} = |S_{21}|^2/[(1-|S_{11}|^2)(1-|S_{22}|^2)]$ with input/output conjugate match.
- Check stability ($K > 1$, $|\Delta| < 1$) before choosing terminations.
- Constant-gain circles trade gain for bandwidth, match, or noise.

Exercises

1. $|S_{21}| = 3$, $|S_{11}| = 0.4$, $|S_{22}| = 0.3$. Find $G_{TU,\max}$ in dB.
2. Why can maximum available gain be undefined for a real device?
3. What does a constant-gain circle represent?

Further reading: Gonzalez [10] (ch. 3); Pozar [1] (ch. 12).

5.18 Low-Noise Amplifier Design

LEARNING OBJECTIVES

- State the four noise parameters F_{min}, Γ_{opt}, R_n.
- Explain why the noise match differs from the gain/input match.
- Describe inductive source degeneration.

The low-noise amplifier sets the noise figure of the whole receiver (Friis), so its design optimizes noise before gain.

5.19 Noise Parameters

A two-port's noise is fully described by four numbers: the minimum noise figure F_{min}, the optimum source reflection Γ_{opt} that achieves it, and the equivalent noise resistance R_n that says how quickly noise degrades away from Γ_{opt}:

$$F = F_{min} + \frac{4R_n}{Z_0} \frac{|\Gamma_s - \Gamma_{opt}|^2}{(1 - |\Gamma_s|^2)\,|1 + \Gamma_{opt}|^2}.$$

5.20 The Noise–Gain Trade

In general the source impedance for minimum noise (Γ_{opt}) is *not* the one for maximum gain or best input match. Noise circles and gain circles on the Smith chart make the compromise visible; a good LNA source match lands near Γ_{opt} and accepts slightly less than maximum gain.

5.21 Inductive Source Degeneration

The elegant trick used in CMOS/SiGe LNAs is a small source inductor L_s. It generates a real input resistance $\approx \omega_T L_s$ *without adding thermal noise* of its own, so the input can be matched to $50\,\Omega$ at the same time as the noise match — the celebrated simultaneous noise-and-input match.

5.22 Output and Bias

With the source chosen for low noise, the output is conjugately matched for gain, and a clean bias network keeps the stage stable. The result is a stage with a noise figure within a few tenths of a dB of F_{min} and enough gain to suppress the noise of everything after it.

WORKED EXAMPLE: NOISE FIGURE OFF THE OPTIMUM

An LNA device has $F_{min} = 0.5$ dB ($= 1.122$), $R_n = 20\ \Omega$ in a $50\ \Omega$ system. If the source is placed so that $|\Gamma_s - \Gamma_{opt}|^2/[(1 - |\Gamma_s|^2)|1 + \Gamma_{opt}|^2] = 0.1$, the noise factor is

$$F = 1.122 + \frac{4(20)}{50}(0.1) = 1.122 + 0.16 = 1.282,$$

i.e. NF $= 1.08$ dB — only 0.58 dB above the minimum, a typical compromise for a good input match.

KEY TAKEAWAYS

- Noise: $F = F_{min} + (4R_n/Z_0)\,|\Gamma_s - \Gamma_{opt}|^2/[(1-|\Gamma_s|^2)|1+\Gamma_{opt}|^2]$.
- The low-noise source match (Γ_{opt}) generally differs from the gain/input match.
- Inductive source degeneration creates a real input resistance ($\approx \omega_T L_s$) noiselessly.

Exercises

1. If $\Gamma_s = \Gamma_{opt}$, what is the noise figure?
2. Why is L_s degeneration preferred over a physical resistor for input matching?
3. Which stage in a receiver most determines system NF?

Further reading: Razavi [8] (ch. 5); Gonzalez [10] (ch. 4).

5.23 Noise in RF Systems

LEARNING OBJECTIVES

- Compute thermal-noise power in a bandwidth.
- Relate noise figure to SNR degradation.
- Apply Friis to a cascade.

Noise ultimately limits the sensitivity of any RF receiver. Understanding noise sources, how they combine in a cascaded system, and how to minimise their contribution is fundamental to receiver design.

5.23.1 Thermal (Johnson-Nyquist) Noise

Any resistor at temperature T generates noise power due to thermal agitation of electrons. The available noise power in bandwidth B is:

$$P_n = kTB$$

At room temperature (290 K): $kT = -174$ dBm/Hz. This is the noise floor of any passive system at room temperature.

5.23.2 Noise Figure

Noise Figure (NF) is the degradation of SNR through a component, referenced to a 290 K source:

$NF = SNR_{in} - SNR_{out}\ (dB)$

A perfect noiseless amplifier has NF = 0 dB. A 50 Ω resistive attenuator of X dB has NF = X dB.

5.23.3 Friis Formula for Cascaded Systems

$$F_{total} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots$$

The first stage dominates. This is why an LNA with low NF and high gain at the front of a receive chain is critical. Lossy elements (cables, switches) before the LNA directly add to system NF.

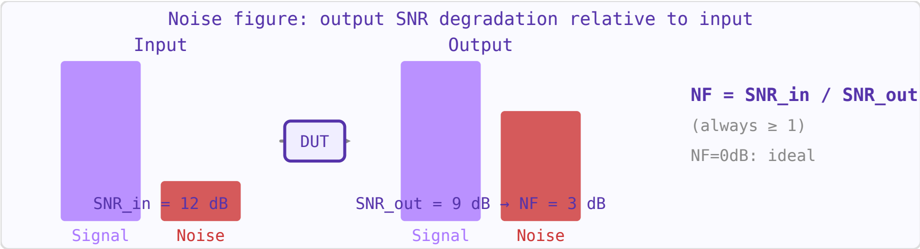


Figure 5.4: Noise figure quantifies how much a device degrades SNR. A 3 dB NF device adds as much noise as a matched resistor at 290 K.

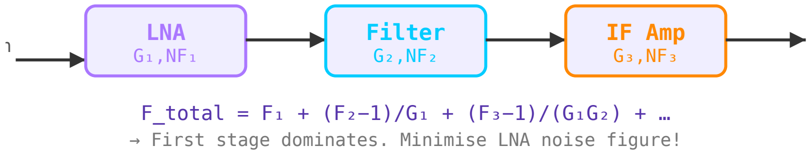


Figure 5.5: Cascaded receiver chain. LNA noise figure dominates because later stages are divided by preceding gains.

5.23.4 System Noise Temperature

Equivalent noise temperature $T_e = (F - 1) \times 290$ K. Useful for low-noise systems such as radio telescopes and satellite receivers where temperatures of a few kelvin are achieved.

WORKED EXAMPLE: THERMAL NOISE FLOOR AND NF

The available thermal-noise power in bandwidth B is $P_n = kTB$. At $T = 290$ K, $kT = -174$ dBm/Hz. In $B = 1$ MHz (60 dB-Hz),

$$P_n = -174 + 60 = -114 \text{ dBm}.$$

An amplifier with NF $= 3$ dB raises the effective floor to -111 dBm and degrades SNR by 3 dB.

KEY TAKEAWAYS

- Thermal noise is -174 dBm/Hz at 290 K; add $10 \log_{10} B$ for the bandwidth.
- NF is the SNR degradation a stage adds; $F = \mathrm{SNR}_{in}/\mathrm{SNR}_{out}$.
- Friis: $F_{\text{tot}} = F_1 + (F_2 - 1)/G_1 + \ldots$ — the first stage dominates.

Exercises

1. What is the noise floor in a 10 MHz bandwidth at 290 K?
2. A chain has $\mathrm{NF}_1 = 2$ dB, $G_1 = 20$ dB, $\mathrm{NF}_2 = 10$ dB. What dominates the cascade NF?
3. An LNA (15 dB gain, 1 dB NF) precedes a mixer (10 dB NF). Which sets the system NF?

Further reading: Pozar [1] (ch. 10); Razavi [8] (ch. 2).

5.24 Power Amplifier Design

LEARNING OBJECTIVES

- Compare PA classes A, B, AB, and switching classes.
- Relate conduction angle to efficiency.
- Explain back-off and linearity.

Power amplifiers (PAs) are among the most challenging RF circuits to design because efficiency and linearity pull in opposite directions. A perfectly linear PA wastes most of its supply power as heat. A highly efficient PA distorts the signal. Modern PA design — especially for cellular and wireless standards — is the art of managing this tradeoff.

5.24.1 PA Classes

PA classes are defined by the transistor's conduction angle — the fraction of the RF cycle during which the device conducts current:

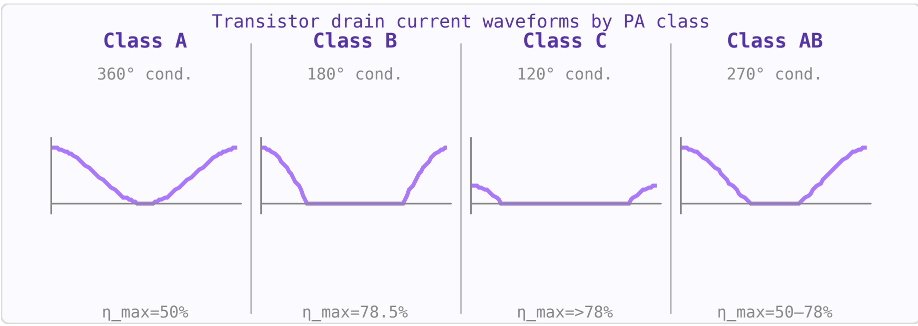


Figure 5.6: Drain current waveforms by amplifier class. Less conduction → higher efficiency but more distortion.

Class	Conduction angle	Max efficiency	Linearity	Use
A	360° (always on)	50%	Excellent	Small-signal, wideband, instrumentation
AB	180°–360°	50–78%	Good	Most practical linear RF PAs (LTE, Wi-Fi)
B	180°	78.5%	Moderate	Push-pull audio and some RF
C	$< 180°$	$> 78.5\%$	Poor	Narrowband constant-envelope (FM, CW radar)
D	Switch	100% (ideal)	Needs filtering	Audio switching amplifiers, some RF
E	Switch + resonant	100% (ideal)	Needs filtering	High efficiency narrowband RF
F	Resonant harmonics	100% (ideal)	Narrowband	High efficiency RF (mobile base stations)

5.24.2 Drain Efficiency vs Power-Added Efficiency (PAE)

$$\eta_{drain} = \frac{P_{out}}{P_{DC}} \qquad \text{PAE} = \frac{P_{out} - P_{in}}{P_{DC}}$$

PAE is the more meaningful figure for RF PAs because it accounts for the drive power required. For a PA with 15 dB gain, the difference between the two is small. For a PA with only 6 dB gain (common at mmWave), PAE is significantly lower than drain efficiency.

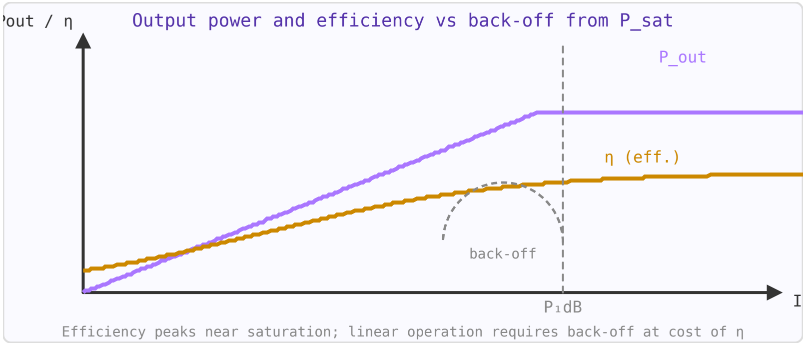


Figure 5.7: PA output power (blue) and efficiency (orange) vs input drive. P_1dB marks onset of compression.

5.24.3 1-dB Compression and Saturation

As input power increases, the output eventually saturates. The $\mathbf{P_1dB}$ (1-dB compression point) is where the gain has dropped by 1 dB from its small-signal value. Above P_1dB the device is nonlinear and generates harmonics and intermodulation products. For linear operation, PAs are backed off from P_1dB by 6–10 dB, accepting lower efficiency.

5.24.4 Linearity Requirements by Modulation

Modulation	PAPR	Typical back-off	Notes
CW / FM	0 dB	0 dB	Constant envelope — no linearity needed
BPSK/QPSK	3–4 dB	3–6 dB	Low PAPR, tolerates moderate nonlinearity
16-QAM	6–7 dB	6–8 dB	Moderate requirements
64-QAM	8–9 dB	8–10 dB	LTE typical; high back-off needed
OFDM (Wi-Fi/5G)	10–12 dB	10–12 dB	High PAPR; DPD or Doherty needed for efficiency

PAPR = Peak-to-Average Power Ratio.

5.24.5 Doherty Architecture

The Doherty PA is the dominant architecture for modern base station and handset PAs because it maintains high efficiency at back-off. It uses two amplifiers — a "main" (carrier) PA that operates in class AB/B, and a "peaking" PA that only turns on near peak power. At back-off, only the main PA operates (high efficiency). At peak power, both combine, restoring efficiency. A Doherty PA can achieve > 50% efficiency at 6 dB back-off, compared to 12–15% for a class A PA.

5.24.6 Digital Predistortion (DPD)

DPD applies an inverse of the PA's nonlinearity digitally, before the signal reaches the PA, so that the output is linear. It requires a feedback path to measure and adapt to the PA's actual behaviour. DPD allows a PA to operate closer to compression (higher efficiency) while meeting EVM and spectral mask requirements. It is standard in modern cellular base stations.

5.24.7 Output Matching

PA transistors are typically low-impedance devices that need an output matching network to present the optimal load impedance for maximum power and efficiency. The load line resistance $R_{opt} = V_{dd}^2/(2P_{out})$ sets the ideal load for a class A stage. For a 28 V, 10 W PA: $R_{opt} = 784/20 = 39.2\ \Omega$ — conveniently near 50 Ω, which is why GaN on 28 V supply is so popular for base station PAs.

WORKED EXAMPLE: CLASS EFFICIENCY LIMITS

Ideal drain efficiency depends on conduction angle. Class A conducts the whole cycle (360°) with a theoretical maximum of 50%; Class B conducts a half cycle (180°) with $\pi/4 = 78.5\%$; Class AB sits between. Switching classes (D, E, F) can approach 100% ideally but require hard switching. Backing a linear PA off 6 dB from saturation trades much of that efficiency for low distortion.

KEY TAKEAWAYS

- Efficiency rises as conduction angle falls: A (50%) $<$ AB $<$ B (78.5%) $<$ switching.
- Linearity and efficiency trade off; back-off buys linearity at lower efficiency.
- Doherty and digital predistortion recover efficiency/linearity for modulated signals.

Exercises

1. What is the ideal maximum efficiency of Class A?
2. What is the ideal maximum efficiency of Class B?
3. Backing a PA off from saturation improves what, at what cost?

Further reading: Cripps [11]; Pozar [1] (ch. 12).

5.25 Thermal Management and Junction Temperature

LEARNING OBJECTIVES

- Build the thermal-resistance chain from junction to ambient.
- Compute junction temperature from dissipated power.
- Relate junction temperature to reliability.

An RF power amplifier turns much of its DC power into heat, and that heat has to escape from a tiny transistor die to the outside world. The temperature the die actually reaches — the *junction temperature* T_j — governs both performance (gain, efficiency and output power all fade as the device heats) and reliability (failure rate rises steeply with temperature). Thermal design is therefore inseparable from PA design.

5.25.1 The Thermal-Resistance Chain

Heat flows from junction to ambient through a series of thermal resistances, exactly like current through series electrical resistors, with power as the "current" and temperature rise as the "voltage":

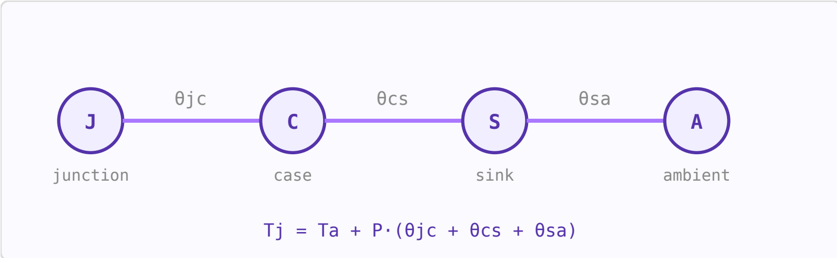


Figure 5.8: The junction-to-ambient thermal chain: die → case → heat sink → air, each stage a thermal resistance in °C/W.

$$\theta_{ja} = \theta_{jc} + \theta_{cs} + \theta_{sa}, \qquad T_j = T_a + P_{diss}\,\theta_{ja}$$

Here θ_{jc} (junction-to-case) is fixed by the device and package, θ_{cs} (case-to-sink) is the interface — thermal grease, a pad, or solder — and θ_{sa} (sink-to-ambient) is the heat sink and airflow, the part the designer most controls. The dissipated power P_{diss} is the DC input minus the RF output: a 40 % efficient 10 W PA dissipates 15 W.

5.25.2 Maximum Power and Margin

Every device has a maximum rated $T_{j,max}$ (often 150–200 °C for silicon, higher for GaN). Turning the equation around gives the most power the thermal path can carry, and the margin at a given operating point:

$$P_{max} = \frac{T_{j,max} - T_a}{\theta_{ja}}, \qquad \text{margin} = T_{j,max} - T_j$$

Good practice is to derate — design for a junction well below the rating (e.g. $T_j \leq 125$ °C) to buy reliability and headroom for a hot ambient. Note the strong lever of ambient temperature: the same amplifier that is fine on the bench at 25 °C can exceed its rating in a sealed enclosure at 70 °C.

5.25.3 Temperature and Reliability

Semiconductor failure mechanisms (electromigration, diffusion) accelerate roughly with an Arrhenius law — a common rule of thumb is that failure rate doubles for every ~10 °C rise in junction temperature. Keeping T_j down is therefore the single biggest lever on the mean-time-between-failures of a power stage. High temperature also shifts bias points and lowers gain and P1dB, so thermal and electrical performance must be co-designed.

5.25.4 Practical Techniques

- **Minimise interface resistance**: thin, high-conductivity thermal interface material, flat mating surfaces, adequate mounting pressure.
- **Spread the heat**: thermal vias under the die pad, copper coins or heat spreaders, direct die-attach to the flange.

- **Size the sink**: choose θ_{sa} (finned extrusion + airflow) for the required P_{max}; forced air or liquid cooling for high power.
- **Watch duty cycle**: pulsed operation uses *transient* thermal impedance, which is lower than the steady-state value for short pulses — a pulsed PA can run higher peak power than its CW rating.

WORKED EXAMPLE: JUNCTION TEMPERATURE OF A 5 W DEVICE

Junction temperature follows $T_j = T_a + P\,\theta_{ja}$, where the junction-to-ambient resistance is the series sum $\theta_{ja} = \theta_{jc}+\theta_{cs}+\theta_{sa}$ of the junction-to-case, case-to-sink and sink-to-ambient resistances. A device dissipating 5 W with $\theta_{jc} = 2$, $\theta_{cs} = 0.5$ and $\theta_{sa} = 3$ °C/W in 25 °C air reaches

$$T_j = 25 + 5\,(2 + 0.5 + 3) = 52.5\ ^\circ\text{C}.$$

Thermal resistance is the direct analogue of electrical resistance — power is the “current” and temperature the “voltage.” Because lifetime falls roughly exponentially with T_j, lowering θ_{sa} with a bigger heatsink buys reliability directly.

KEY TAKEAWAYS

- $T_j = T_a + P\,\theta_{ja}$ with $\theta_{ja} = \theta_{jc} + \theta_{cs} + \theta_{sa}$ (series thermal resistances, °C/W).
- Thermal resistance is the electrical analogue of resistance: power is the “current,” temperature the “voltage.”
- Reliability falls roughly exponentially with T_j (Arrhenius); derating keeps T_j well below the rated maximum.

Exercises

1. A device dissipates 8 W with a total θ_{ja} of 6 °C/W in 30 °C ambient. Find T_j.
2. Which term of θ_{ja} do you reduce by bolting on a bigger heatsink?
3. Why derate a device below its maximum rated junction temperature?

Further reading: Cripps [11]; Gonzalez [10].

5.26 Intermodulation Distortion and IP3

LEARNING OBJECTIVES

- Locate third-order intermodulation products.
- Relate IIP3 to IMD and headroom.
- Cascade IP3.

When two or more signals pass through a nonlinear device (an amplifier, mixer, or any component with a nonlinear transfer function), new frequency components are created. These intermodulation (IM) products can fall within the passband of the system and cannot be removed by filtering. Third-order intermodulation (IM3) is the dominant concern in most RF receivers and transmitters.

5.26.1 Why Nonlinearity Matters

A perfectly linear device produces outputs only at its input frequencies. A nonlinear device with input $x(t) = a_1 \cos(\omega_1 t) + a_2 \cos(\omega_2 t)$ produces output terms at frequencies $n\omega_1 \pm m\omega_2$ for all integers n, m. The third-order products at $2\omega_1 - \omega_2$ and $2\omega_2 - \omega_1$ are close to the original signals and fall in-band.

5.26.2 Third-Order Intercept Point (IP3)

In a two-tone test, both the desired output and the IM3 products are plotted as a function of input power. On a log-log scale, the desired output has slope 1 (1 dB output per 1 dB input); the IM3 products have slope 3 (3 dB output per 1 dB input). Extrapolating these two lines, they intersect at the **IP3 point**.

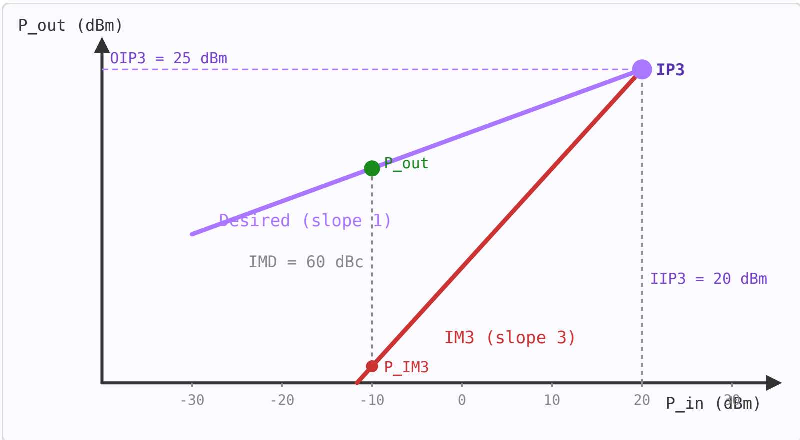


Figure 5.9: Output power vs input power. Signal: slope 1. IM3 products: slope 3. They extrapolate to the IP3 intercept point.

$$P_{IM3,out} = 3P_{in} - 2 \cdot IIP3 + G \qquad OIP3 = IIP3 + G$$

where all powers are in dBm and G is the gain in dB. The OIP3 is typically 10–15 dB above the 1 dB compression point.

5.26.3 Input and Output IP3

Parameter	Symbol	Typical values	Notes
Input IP3	IIP3	−10 to +40 dBm	Referred to device input
Output IP3	OIP3	0 to +50 dBm	OIP3 = IIP3 + Gain
1 dB compression	P_1dB	≈ IIP3 − 9.6 dB	Empirical approximation
Low-noise amplifier	IIP3	0 to +15 dBm	Trades with noise figure
Power amplifier	OIP3	P_1dB + 10 dB	Backed off from saturation
Passive mixer	IIP3	+15 to +25 dBm	Higher than active

5.26.4 IMD Ratio & Dynamic Range

The intermodulation distortion ratio (IMD) is the power difference between the desired signal and the IM3 product at the output:

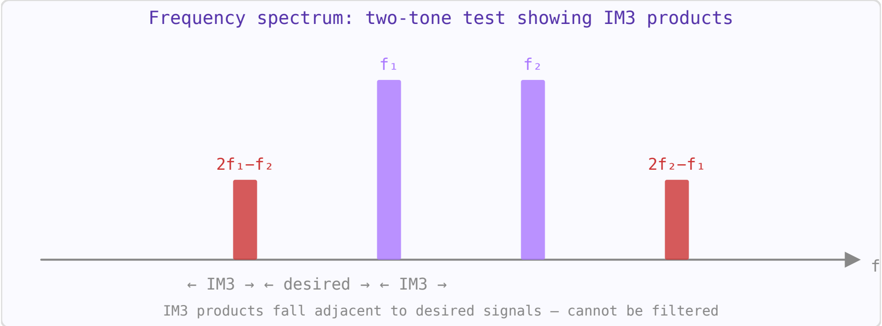


Figure 5.10: Two-tone test: input tones f_1 and f_2 (purple) generate IM3 products at $2f_1-f_2$ and $2f_2-f_1$ (red) close to the passband.

$$\text{IMD} = 2(IIP3 - P_{in})\ \text{dBc}$$

The spurious-free dynamic range (SFDR) is the input power range over which the system can operate without IM3 products exceeding the noise floor:

$$\text{SFDR} = \tfrac{2}{3}(IIP3 - P_{noise})$$

5.26.5 Cascaded IP3

For a chain of stages, the cascaded IIP3 is dominated by later stages (because the signal is amplified). The reciprocal rule gives:

$$\frac{1}{IIP3_{total}} = \frac{1}{IIP3_1} + \frac{G_1}{IIP3_2} + \frac{G_1 G_2}{IIP3_3} + \cdots$$

where all quantities are in linear (not dB) units. This is why the last high-gain stage in a receiver chain typically limits linearity.

5.26.6 Practical Implications

- **LNA IIP3**: typically +5 to +15 dBm. Must handle the full received signal range without generating IM3 products that mask weak desired signals.
- **PA linearity**: power amplifiers operate backed off from their P_1dB to maintain acceptable IM3. ACPR (adjacent channel power ratio) is the PA equivalent of IMD.
- **Mixer IIP3**: passive mixers (diode or FET ring) have higher IP3 than active mixers but higher conversion loss.

WORKED EXAMPLE: IMD FROM IIP3

Two equal tones at input power P_{in} per tone drive a device with input third-order intercept IIP3. The third-order products fall below each tone by

$$\text{IMD} = 2(\text{IIP3} - P_{in}).$$

For IIP3 = +20 dBm and $P_{in} = -10$ dBm per tone, IMD = $2(20 - (-10)) = 60$ dBc. Every 1 dB rise in tone power worsens IMD by 3 dB (the products grow with slope 3).

KEY TAKEAWAYS

- IM3 products at $2f_1 - f_2$ and $2f_2 - f_1$ land in-band and cannot be filtered out.
- IMD (dBc) $= 2(\text{IIP3} - P_{in})$; IM3 grows 3 dB per dB of input.
- In a cascade, late high-gain stages usually dominate the overall IP3.

Exercises

1. IIP3 $= +10$ dBm, $P_{in} = -20$ dBm/tone. Find the IMD.
2. Raise each tone by 3 dB. How much do the IM3 products rise?
3. Which stage tends to dominate cascade IP3: early low-gain or late high-gain?

Further reading: Razavi [8] (ch. 2); Pozar [1] (ch. 10).

Chapter 6

Signal Generation and Frequency Conversion

6.1 Oscillators

LEARNING OBJECTIVES

- State the Barkhausen conditions for oscillation.
- Relate loaded Q to frequency stability and phase noise.
- Recognize common feedback and negative-resistance topologies.

An oscillator generates a periodic signal without an external input. In RF systems, oscillators provide the local oscillator (LO) signal for frequency conversion and the carrier signal for transmitters. Phase noise (short-term frequency instability) is the key quality metric.

6.1.1 Oscillation Conditions

The Barkhausen criterion: a feedback amplifier oscillates when loop gain $|A\beta| = 1$ and loop phase shift is $0°$ (or $360°$). Common topologies: Colpitts (capacitive voltage divider), Hartley (inductive divider), Clapp, Pierce (crystal oscillators).

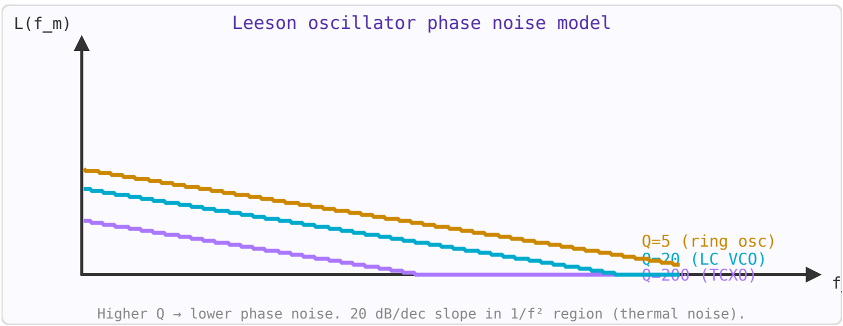


Figure 6.1: Leeson's model: phase noise L(f_m) for oscillators with different Q factors. Higher Q = lower noise at all offsets.

6.1.2 Phase Noise

Phase noise $\mathcal{L}(f_m)$ (dBc/Hz) describes the spectral purity of an oscillator — the noise power in a 1 Hz bandwidth at offset f_m from the carrier, relative to the carrier power. The Leeson model:

$$\mathcal{L}(f_m) = 10 \log \left[\frac{2FkT}{P_s} \left(1 + \frac{f_0^2}{4Q_L^2 f_m^2} \right) \right]$$

Phase noise decreases with increasing loaded Q (Q_L) and oscillator power (P_s).

6.1.3 Voltage-Controlled Oscillator (VCO)

A VCO is an oscillator whose frequency is tuned by a control voltage, via a varactor diode (voltage-variable capacitor). Tuning sensitivity K_v (MHz/V) characterises the VCO. Higher K_v increases frequency range but also increases phase noise sensitivity to supply noise.

6.1.4 Phase-Locked Loop (PLL)

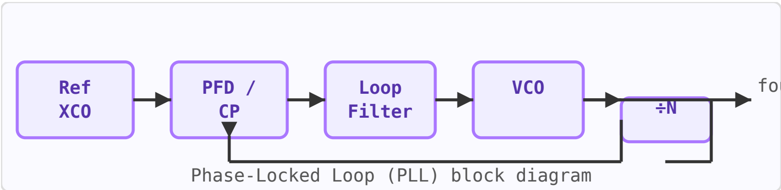


Figure 6.2: PLL block diagram: VCO locked to reference crystal via phase-frequency detector and loop filter.

A PLL phase-locks a VCO to a stable reference oscillator (usually a crystal), allowing the VCO's tunable frequency range with the reference's spectral purity. Components: phase-frequency detector (PFD), charge pump, loop filter, VCO, frequency divider (÷N). Within the loop bandwidth, VCO phase noise is suppressed to near the reference noise floor.

WORKED EXAMPLE: BARKHAUSEN CRITERION

A feedback oscillator sustains oscillation when the loop gain magnitude is at least unity and the loop phase is a multiple of 360°:

$$|A\beta| \geq 1, \qquad \angle A\beta = 0^\circ \pmod{360^\circ}.$$

At start-up the small-signal loop gain must exceed 1 so noise builds up; amplitude then grows until nonlinearity trims the effective gain to exactly 1 in steady state. The frequency is set where the loop phase is zero — usually a high-Q resonator.

KEY TAKEAWAYS

- Oscillation needs loop gain ≥ 1 and $0^\circ \pmod{360^\circ}$ loop phase (Barkhausen).
- Start-up needs gain > 1; steady state settles to unity via nonlinearity.
- A higher-Q tank gives better frequency stability and lower phase noise.

Exercises

1. What loop phase shift sustains oscillation?
2. Why must the start-up loop gain exceed 1?
3. Does a higher-Q tank help or hurt phase noise?

Further reading: Razavi [8] (ch. 8); Lee [9] (ch. 14).

6.2 Quartz Crystal Resonators

LEARNING OBJECTIVES

- Describe the Butterworth–Van Dyke equivalent circuit of a quartz crystal.
- Distinguish series and parallel resonance.
- Explain load capacitance and frequency pullability.

A quartz crystal is a piezoelectric plate whose mechanical resonance is extraordinarily stable and high-Q — the reason it sets the frequency of nearly every clock, radio, and microprocessor. Electrically it behaves as a very sharp resonant circuit, and to the surrounding oscillator it looks like the lumped model below. Understanding that model is the key to choosing a crystal, hitting a target frequency, and trimming it with a load capacitor.

6.2.1 The Butterworth–Van Dyke Model

The crystal is modelled by a *motional* branch — the electrical image of the mechanical resonance — in parallel with the static *shunt* capacitance C_0 of the electrodes and holder:

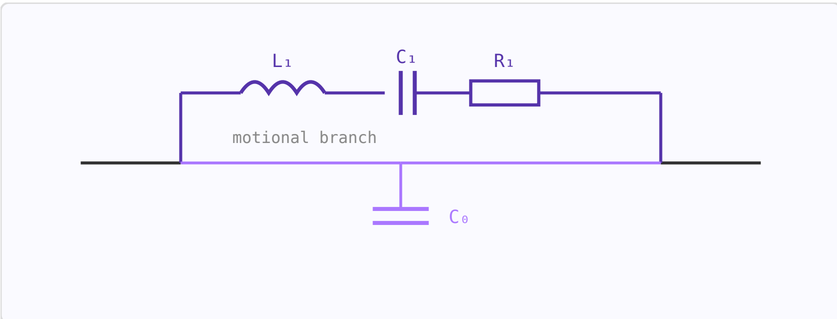


Figure 6.3: Butterworth–Van Dyke model: motional L_1–C_1–R_1 (the mechanical resonance) shunted by the static holder capacitance C_0.

The motional inductance L_1 is huge (henries) and the motional capacitance C_1 tiny (femtofarads) — impossible with real coils and capacitors, which is exactly why the mechanical resonance is so much sharper than an LC tank.

6.2.2 Series and Parallel Resonance

The motional branch alone resonates at the *series* frequency f_s. Adding C_0 creates a second, slightly higher *parallel* (anti-resonant) frequency f_p; the crystal is inductive only in the narrow window between them, where oscillators operate:

$$f_s = \frac{1}{2\pi\sqrt{L_1 C_1}}, \qquad f_p = f_s\sqrt{1 + \frac{C_1}{C_0}}$$

The fractional split $f_p - f_s$ is only a few hundred ppm because $C_1 \ll C_0$. The capacitance ratio $r = C_0/C_1$ (typically 200–1000) sets how far the crystal can be pulled.

6.2.3 Quality Factor

The motional resistance R_1 (the mechanical loss) sets a Q that dwarfs any LC circuit:

$$Q = \frac{2\pi f_s L_1}{R_1} = \frac{1}{2\pi f_s C_1 R_1}$$

Fundamental-mode crystals reach Q of 10^4–10^6; precision SC-cut resonators exceed 10^6. This enormous Q is what gives quartz oscillators their part-per-million (or better) stability.

6.2.4 Pulling with a Load Capacitor

A "parallel-resonant" crystal is specified with a nominal load capacitance C_L; the oscillator's external capacitance shifts the operating frequency by:

$$\frac{\Delta f}{f_s} \approx \frac{C_1}{2\,(C_0 + C_L)}$$

Increasing C_L lowers the frequency toward f_s; decreasing it raises the frequency toward f_p. This is how a trimmer or varactor tunes a crystal oscillator (a VCXO), and why using the wrong load capacitance leaves a crystal off-frequency.

6.2.5 Typical Parameters

Type	Frequency	C_0	R_1 (ESR)	Q
Watch tuning fork	32.768 kHz	1–2 pF	30–50 kΩ	10^4–10^5
AT-cut fundamental	1–30 MHz	3–7 pF	10–100 Ω	10^4–10^5
AT-cut overtone	30–200 MHz	3–6 pF	20–80 Ω	10^4–10^5
SC-cut (OCXO)	5–10 MHz	2–5 pF	50–150 Ω	$>10^6$

Beyond about 30–40 MHz, fundamental plates become too thin, so crystals run on the 3rd, 5th, or 7th overtone. Frequency also drifts slightly with temperature (set by the crystal cut) and ages over time — the reasons TCXOs and OCXOs add temperature compensation or an oven.

WORKED EXAMPLE: THE TWO RESONANCES OF A CRYSTAL

The Butterworth–Van Dyke model puts a motional arm L_1, C_1, R_1 in series, shunted by the electrode capacitance C_0. The series resonance $f_s = 1/(2\pi\sqrt{L_1 C_1})$ is where the motional arm is purely resistive; the parallel resonance f_p sits slightly above, where the motional arm resonates against C_0. Because the ratio C_1/C_0 is tiny (often $\sim 1/300$), f_p is only a few hundred parts-per-million above f_s. The mechanical (acoustic) resonance has extraordinarily low loss, giving quartz a Q of 10^4 to 10^6 — orders of magnitude beyond any LC tank — and hence its excellent frequency stability.

KEY TAKEAWAYS

- Butterworth–Van Dyke model: motional L_1, C_1, R_1 in series with a shunt C_0.
- Series resonance $f_s = 1/(2\pi\sqrt{L_1 C_1})$; parallel resonance $f_p > f_s$ set by C_0.
- Q is enormous (10^4–10^6); a specified load capacitance pulls the operating frequency between f_s and f_p.

Exercises

1. Which is higher in frequency, series or parallel resonance?
2. Why does a crystal have far higher Q than an LC tank of the same frequency?
3. What does increasing the load capacitance do to a parallel-mode crystal's frequency?

Further reading: Razavi [8]; Rohde [13].

6.3 Oscillator Design

LEARNING OBJECTIVES

- Relate the feedback and negative-resistance views of oscillation.
- Compute the resonant frequency of a Colpitts tank.
- Explain how resonator Q governs phase noise and stability.

An oscillator turns DC into a periodic RF signal. Design means guaranteeing start-up, setting the frequency, and minimizing phase noise.

6.4 Two Views of the Same Thing

The *feedback* view (Barkhausen): loop gain ≥ 1 with $0°$ (mod $360°$) phase. The equivalent *negative-resistance* view: the active device presents a negative resistance $-R_d$ that cancels the resonator loss R_{res}. Oscillation starts when $|R_d| > R_{res}$ and settles when, through nonlinear compression, $|R_d| = R_{res}$ at the resonant frequency.

6.5 Topologies

The Colpitts, Hartley, and Clapp oscillators differ only in how the tank taps feedback (a capacitive divider, an inductive divider, or a series-tuned variant). For the best stability, a *crystal* replaces the LC tank: its enormous Q (10^4–10^6) pins the frequency and slashes phase noise. Dielectric-resonator (DRO) and YIG oscillators serve the microwave range.

6.6 Design Flow

Choose a resonator with the highest practical loaded Q; bias the device for sufficient small-signal negative resistance (start-up margin $\sim 3\times$); set the tank for the target frequency; and buffer the output so the load does not pull the frequency. Leeson's model (Chapter on Phase Noise) ties the achievable phase noise to Q and the device's flicker noise — high Q and a low-$1/f$ device win.

WORKED EXAMPLE: COLPITTS FREQUENCY

A Colpitts oscillator uses a tank inductor $L = 10$ nH with a capacitive divider $C_1 = C_2 =$ 2 pF. The feedback capacitors appear in series, $C_s = C_1C_2/(C_1 + C_2) = 1$ pF, so

$$f_0 = \frac{1}{2\pi\sqrt{L\,C_s}} = \frac{1}{2\pi\sqrt{(10 \times 10^{-9})(1 \times 10^{-12})}} = 1.59 \text{ GHz}.$$

The divider ratio sets the feedback needed to sustain oscillation.

KEY TAKEAWAYS

- Start-up: $|R_d| > R_{res}$; steady state: $|R_d| = R_{res}$ (nonlinear limiting).
- Colpitts: $f_0 = 1/(2\pi\sqrt{LC_s})$ with $C_s = C_1C_2/(C_1 + C_2)$.
- Crystals give huge Q — best stability and lowest phase noise.

Exercises

1. A Colpitts tank has $L = 5$ nH, $C_1 = C_2 = 4$ pF. Find f_0.
2. Why buffer an oscillator's output?
3. Why does a crystal oscillator have far lower phase noise than an LC one?

Further reading: Lee [9] (ch. 14–15); Razavi [8] (ch. 8).

6.7 Phase Noise and Jitter

LEARNING OBJECTIVES

- Define $\mathcal{L}(f)$ as single-sideband phase noise.
- Identify the Leeson slope regions.
- Integrate phase noise to RMS jitter.

Phase noise and jitter are two equivalent descriptions of timing uncertainty in an oscillator or clock signal. Phase noise describes instability in the frequency domain — the spectral purity of an oscillator. Jitter describes it in the time domain — the variation of edge transitions from their ideal positions. They are related by an integral transformation.

6.7.1 Phase Noise Definition

The single-sideband (SSB) phase noise L(f) is defined as the ratio of power in a 1 Hz bandwidth at offset f from the carrier to the total carrier power:

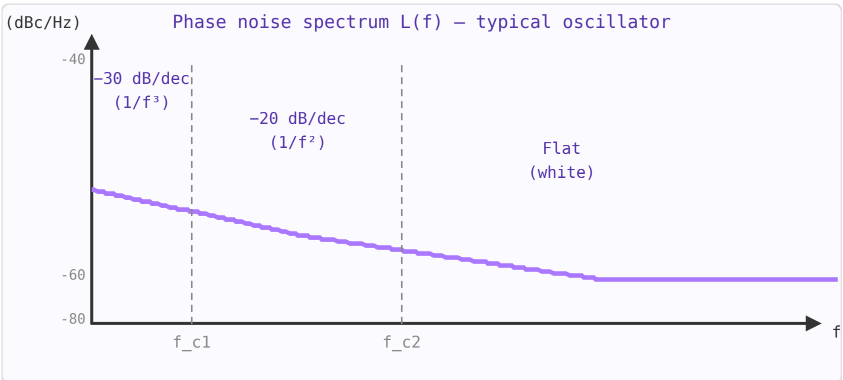


Figure 6.4: Phase noise L(f) showing three regions: $1/f^3$ (flicker FM), $1/f^2$ (white FM / Leeson), and flat (white phase noise floor).

$$\mathcal{L}(f) = \frac{P_{noise,SSB}(f_0 + f,\ 1\text{ Hz BW})}{P_{carrier}} \quad [\text{dBc/Hz}]$$

Phase noise is plotted on a log-log scale. Typical regions:

Offset region	Slope	Dominant mechanism
Very close-in (< 10 Hz)	$1/f^3$ (−30 dB/decade)	Flicker FM noise
Close-in (10 Hz – 10 kHz)	$1/f^2$ (−20 dB/decade)	White FM, Leeson's model
Far-from-carrier	Flat (0 dB/decade)	White phase noise (thermal)

6.7.2 Leeson's Model

For a feedback oscillator with resonator Q factor and noise figure F:

$$\mathcal{L}(f_m) = 10 \log_{10} \left[\frac{2FkT}{P_s} \left(\frac{f_0}{2Q_L f_m} \right)^2 \right]$$

where f_m is the offset frequency, f_0 is the carrier frequency, Q_L is the loaded Q, P_s is the signal power, F is the amplifier noise figure (linear), k = 1.38×10^{-23} J/K (Boltzmann), and T is the temperature in Kelvin. Higher Q and higher signal power both reduce phase noise.

6.7.3 Phase Noise to Jitter Conversion

RMS jitter integrates the phase noise over a bandwidth from f_1 to f_2:

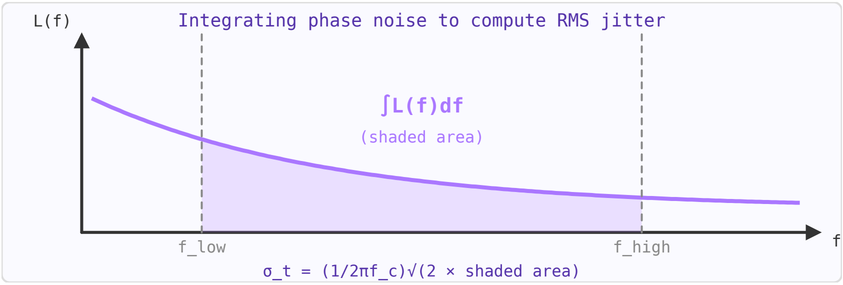


Figure 6.5: RMS jitter equals the square root of twice the integrated phase noise (linear) over the specified bandwidth.

$$\sigma_t = \frac{1}{2\pi f_0} \sqrt{2 \int_{f_1}^{f_2} \mathcal{L}(f)\, df} \quad \text{[s RMS]}$$

where L(f) is in linear units (not dBc/Hz). The factor 2 accounts for double-sideband power. The integration limits define what jitter you care about — typically the Nyquist band for a data converter application, or the loop bandwidth for a PLL.

6.7.4 Typical Values

Oscillator type	Phase noise @ 1 kHz	Phase noise @ 1 MHz	RMS jitter
MEMS oscillator	−120 dBc/Hz	−160 dBc/Hz	200–500 fs
TCXO (100 MHz)	−130 dBc/Hz	−165 dBc/Hz	50–200 fs
OCXO (100 MHz)	−150 dBc/Hz	−170 dBc/Hz	10–50 fs
LC VCO (2.4 GHz)	−85 dBc/Hz	−135 dBc/Hz	1–5 ps
Ring oscillator (1 GHz)	−50 dBc/Hz	−100 dBc/Hz	10–50 ps

6.7.5 Jitter Types

- **Period jitter**: Variation of consecutive clock periods. Relevant for digital setup/hold timing.
- **Cycle-to-cycle jitter**: Difference between adjacent periods. Related to close-in phase noise.
- **Long-term jitter (accumulated)**: Deviation over many cycles. Related to far-from-carrier noise.
- **RMS jitter**: Standard deviation σ. Peak-to-peak jitter $\approx 6\sigma$ to 7σ at 10^{-12} BER.

6.7.6 Practical Impact

For an ADC sampling at frequency f_s, phase noise causes an SNR limit called the *aperture jitter limit*:

$$\mathrm{SNR}_{jitter} = -20 \log_{10}(2\pi f_{in}\, \sigma_t)$$

At f_in = 1 GHz with σ_t = 100 fs, the jitter-limited SNR is only 24 dB — equivalent to a 4-bit ADC. This is why high-frequency ADC clock sources must have sub-100 fs jitter.

WORKED EXAMPLE: PHASE NOISE TO RMS JITTER

RMS timing jitter follows from integrated phase noise:

$$\sigma_t = \frac{1}{2\pi f_c}\sqrt{2\int \mathcal{L}(f)\, df}.$$

Take $\mathcal{L} = -100$ dBc/Hz (10^{-10}/Hz) flat over a 1 MHz band around a 1 GHz carrier. The integrated phase variance is $2 \times 10^{-10} \times 10^6 = 2 \times 10^{-4}$ rad^2, so $\sigma_\phi = 0.0141$ rad and $\sigma_t = \sigma_\phi/(2\pi f_c) = 2.25$ ps.

KEY TAKEAWAYS

- $\mathcal{L}(f)$ is SSB phase noise (dBc/Hz), decreasing with offset.
- Leeson: close-in $1/f^3$, then $1/f^2$, then a flat floor.
- RMS jitter $\sigma_t = \sigma_\phi/(2\pi f_c)$ from the integrated phase noise.

Exercises

1. Does phase noise rise or fall with offset frequency?
2. In the Leeson $1/f^2$ region, how many dB does $\mathcal{L}(f)$ drop per decade?
3. For a fixed integrated phase error, does a higher carrier give more or less time jitter?

Further reading: Lee [9] (ch. 17); Razavi [8] (ch. 8).

6.8 Phase-Locked Loops and Frequency Synthesis

LEARNING OBJECTIVES

- Identify the PLL building blocks.
- Relate the divide ratio to output frequency.
- Understand the loop-bandwidth trade-off.

A phase-locked loop (PLL) is a feedback system that synchronises an oscillator to a reference signal in both frequency and phase. PLLs are ubiquitous in RF systems: they generate the local oscillator frequency in every radio receiver and transmitter, provide clock recovery in serial data links, and clean up noisy frequency sources.

6.8.1 PLL Building Blocks

Block	Function	Key parameter
Phase detector (PD) / PFD	Compares phase of reference and divided VCO output. Generates error signal.	Gain K_φ (A/rad or V/rad)
Charge pump (CP)	Converts PFD pulses to current. Charges loop filter.	Current I_CP (μA–mA)
Loop filter (LF)	Integrates CP output. Sets loop bandwidth and stability.	Bandwidth f_n, phase margin
VCO	Voltage-controlled oscillator. Output frequency $\propto$ control voltage.	Gain K_VCO (MHz/V)
Divider ÷N	Divides VCO frequency by N to match reference. N sets output frequency.	Division ratio N

6.8.2 How It Works

The PFD compares the phase of the reference (f_ref) to the divided VCO output (f_VCO/N). When they match, the loop is locked and f_VCO = N × f_ref. If the VCO drifts high, the PFD generates a correction signal that reduces the control voltage, pulling the frequency back. The loop bandwidth determines how fast this correction happens.

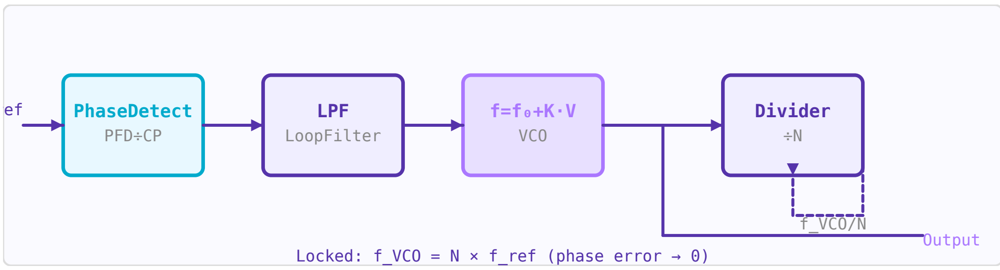


Figure 6.6: PLL block diagram: PFD compares f_ref to f_VCO/N; loop filter drives VCO to lock frequency and phase.

6.8.3 Open-Loop Transfer Function

$$G(s) = \frac{K_\phi I_{CP}}{2\pi} \cdot F(s) \cdot \frac{K_{VCO}}{s} \cdot \frac{1}{N}$$

where F(s) is the loop filter transfer function. The loop bandwidth f_n is approximately where |G(jω)| = 1. Phase margin should be 45–60° for stable, well-damped response.

6.8.4 Lock Time and Bandwidth Tradeoff

The loop bandwidth controls a fundamental tradeoff:

- **Wide bandwidth**: Fast lock time, good VCO noise suppression, poor reference spur rejection, more reference noise passes through.
- **Narrow bandwidth**: Slow lock time, poor VCO noise suppression at close-in offsets, excellent reference spur rejection.

Typical loop bandwidths: 1–100 kHz for frequency synthesisers, 1–10 MHz for fast-hopping systems.

6.8.5 Phase Noise in a PLL

Inside the loop bandwidth, the PLL output tracks the reference and the output phase noise follows the reference noise amplified by $20\cdot\log_{10}(N)$. Outside the loop bandwidth, the output phase noise equals the free-running VCO phase noise. The total output phase noise is the worst of the two at each offset:

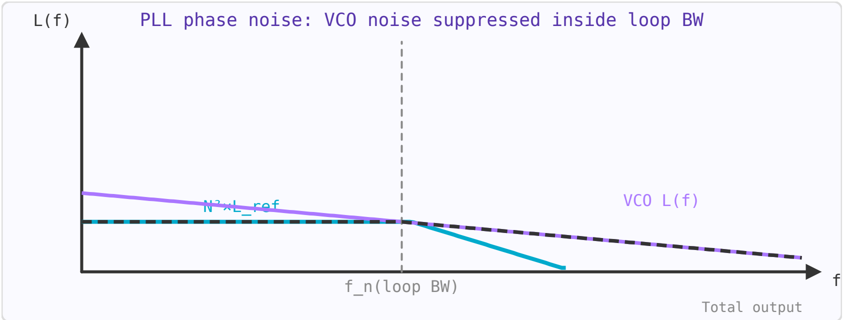


Figure 6.7: PLL output phase noise: inside loop BW = reference noise × N^2. Outside = free-running VCO noise. Total = worst of the two.

$$\mathcal{L}_{out}(f_m) \approx \begin{cases} \mathcal{L}_{ref}(f_m) + 20\log_{10} N & f_m \ll f_n \\ \mathcal{L}_{VCO}(f_m) & f_m \gg f_n \end{cases}$$

6.8.6 Integer-N vs Fractional-N

Architecture	Channel spacing	Phase noise	Reference spurs
Integer-N	= f_ref	20·log(N) above reference noise	Low
Fractional-N (Σ-Δ)	< f_ref (any)	Higher N → worse noise; Σ-Δ adds quantisation noise	Fractional spurs

Integer-N synthesis requires f_ref = channel spacing. For a 200 kHz GSM channel with a 200 kHz reference, the division ratio N can be 4000–9000 for 800–1800 MHz output — the $20\cdot\log_{10}(N)$ penalty is 72–79 dB above the reference noise. Fractional-N uses a Σ-Δ modulator to rapidly switch between integer values of N, achieving fractional average division ratios and allowing wider f_ref for better phase noise.

6.8.7 Reference Spurs

Reference spurs are discrete spectral lines in the VCO output at offsets of ±f_ref (and harmonics) from the carrier. They arise from imperfect charge pump matching — leakage current causes small periodic control voltage ripples at the reference rate. Typical specification: < -70 dBc. Reference spurs cause interference to adjacent channels.

6.8.8 Practical PLL Design Steps

1. Choose f_ref = GCD of all required output frequencies (integer-N) or use fractional-N for fine frequency resolution.

2. Select VCO with adequate tuning range, low K_VCO (for low sensitivity to supply noise), and target phase noise.

3. Choose loop bandwidth: 1/10 of f_ref as a starting point to reject reference spurs.

4. Design a 3rd or 4th order loop filter (2 poles, 1–2 zeros) for 50° phase margin.

5. Simulate phase noise, lock time, and spur levels before fabricating.

WORKED EXAMPLE: INTEGER-N FREQUENCY SYNTHESIS

An integer-N PLL locks its divided output to the reference: $f_{out} = N f_{ref}$. To synthesize 2.44 GHz from a 1 MHz reference, $N = 2440$, and the channel step equals $f_{ref} = 1$ MHz. Inside the loop bandwidth the output tracks the clean reference (multiplied by N); outside it, the VCO's own phase noise dominates. Reference spurs appear at multiples of f_{ref}.

KEY TAKEAWAYS

- PLL: phase detector, charge pump, loop filter, VCO, and $\div N$ feedback.
- Integer-N: $f_{out} = N f_{ref}$; channel spacing equals f_{ref}.
- Loop bandwidth trades spurs/settling against VCO-noise suppression; fractional-N decouples step size from f_{ref}.

Exercises

1. Output of an integer-N PLL with $N = 100$, $f_{ref} = 10$ MHz?

2. For 1 MHz channel spacing in integer-N, what is f_{ref}?

3. Widening the loop bandwidth suppresses which noise more: reference or VCO?

Further reading: Rohde [13]; Razavi [8] (ch. 9).

6.9 PLL and Synthesizer Design

LEARNING OBJECTIVES

- Describe the charge-pump PLL and the role of loop bandwidth and phase margin.
- Explain integer-N vs fractional-N vs DDS synthesis.
- Compute DDS output frequency and resolution.

A frequency synthesizer produces a programmable, low-noise carrier locked to a stable reference. The PLL chapter introduced the loop; here we design it.

6.10 The Charge-Pump Loop

A modern synthesizer is a type-II charge-pump PLL: a phase-frequency detector drives a charge pump into a loop filter that tunes a VCO, whose output is divided by N and compared back to the reference. The open-loop gain sets the loop bandwidth ω_c and the phase margin (typically 45–60° for good settling without peaking).

6.11 Loop-Filter Trade-offs

The loop filter (second- or third-order) sets the bandwidth, which trades two things: a *wide* loop settles fast and suppresses VCO phase noise inside the band, but passes more reference spur; a *narrow* loop is cleaner but slow. The filter is designed around the chosen ω_c and phase margin.

6.12 Integer-N, Fractional-N, and DDS

Integer-N forces the channel step to equal f_{ref}. Fractional-N divides by a non-integer average using a delta-sigma modulator, decoupling step size from f_{ref} and shaping the resulting quantization noise to high offsets. Direct digital synthesis (DDS) takes a different route entirely: a phase accumulator addresses a sine table feeding a DAC, giving very fine, fast-hopping frequency control,

$$f_{out} = \frac{M}{2^n} f_{clk},$$

for an n-bit accumulator and tuning word M.

WORKED EXAMPLE: DDS RESOLUTION

A direct digital synthesizer has an $n = 32$-bit phase accumulator clocked at $f_{clk} = 100$ MHz. Its frequency resolution is

$$\Delta f = \frac{f_{clk}}{2^{32}} = \frac{10^8}{4.29 \times 10^9} = 0.023 \text{ Hz},$$

and a tuning word $M = 2^{30}$ produces $f_{out} = (2^{30}/2^{32})\, f_{clk} = 0.25 \times 100 = 25$ MHz. DDS trades this fine, fast control against DAC spurs and a Nyquist limit of $f_{clk}/2$.

KEY TAKEAWAYS

- Charge-pump type-II PLL; design around loop bandwidth ω_c and 45–60° phase margin.
- Wide loop: fast, low VCO noise, more spur; narrow loop: clean but slow.
- Fractional-N decouples step from f_{ref}; DDS: $f_{out} = (M/2^n) f_{clk}$, resolution $f_{clk}/2^n$.

Exercises

1. A 24-bit DDS clocked at 50 MHz: what is its frequency resolution?
2. Which synthesizer type lets the channel step be far finer than the reference?
3. Name one penalty of widening the PLL loop bandwidth.

Further reading: Rohde [13]; Razavi [8] (ch. 9).

6.13 Mixers and Frequency Conversion

LEARNING OBJECTIVES

- Explain frequency conversion by multiplication.
- Locate sum, difference, and image frequencies.
- Distinguish conversion loss and mixer types.

A mixer multiplies two signals in the time domain, producing sum and difference frequency components. This allows frequency conversion — moving signals from one frequency band to another. Mixers are at the heart of every superheterodyne radio, radar, and wireless communication system.

6.13.1 Basic Operation

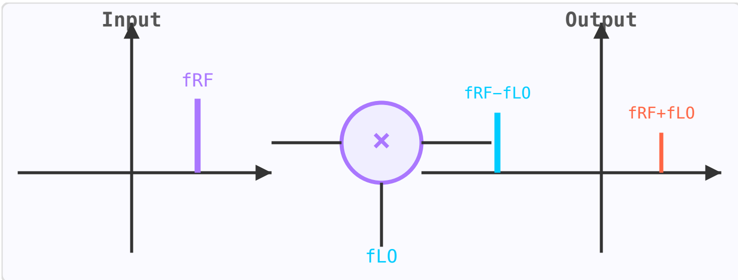


Figure 6.8: Mixer produces sum (fRF+fLO, orange) and difference (fRF−fLO, blue) frequency components.

An ideal mixer with RF input at f_{RF} and local oscillator (LO) at f_{LO} produces output at $f_{IF} = f_{RF} \pm f_{LO}$. The desired sideband is selected by the IF filter; the unwanted image frequency falls at $f_{LO} - f_{IF}$ (for a low-side LO) and must be rejected.

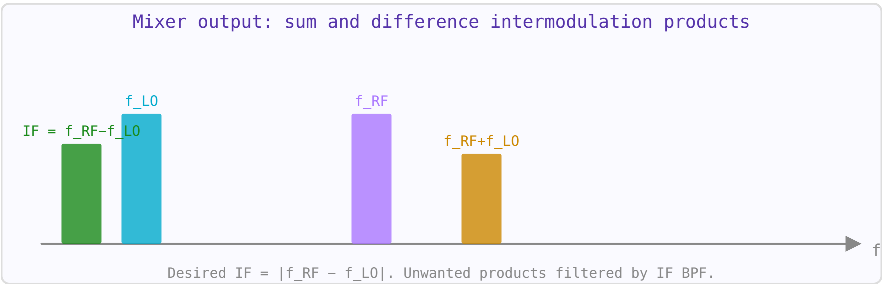


Figure 6.9: Mixing products: desired IF (green), input tones (blue/purple), and upper sideband (orange).

6.13.2 Conversion Loss and Noise Figure

Passive mixers (using diodes or FETs) have conversion loss — the IF output is weaker than the RF input. Typical conversion loss: 5–8 dB for a diode ring mixer. The NF of a passive mixer equals approximately its conversion loss. Active mixers provide conversion gain (0–15 dB) at the cost of higher DC power and potentially more distortion.

6.13.3 Image Rejection Mixer (IQ Mixer)

An IQ (in-phase/quadrature) mixer uses two mixers with LO signals 90° apart, producing in-phase (I) and quadrature (Q) baseband outputs. When the outputs are combined correctly, the image frequency is cancelled. Image rejection of 30–40 dB is achievable in practice; better rejection requires careful amplitude and phase balance.

6.13.4 Spurs and Isolation

Real mixers generate mixing products at $mf_{LO} \pm nf_{RF}$ (spurious responses or "spurs"). Mixer datasheets specify LO-to-RF, LO-to-IF, and RF-to-IF isolation to characterise feedthrough.

Double-balanced mixers provide better isolation and spur rejection than single-ended designs.

WORKED EXAMPLE: DOWNCONVERSION AND THE IMAGE

A mixer multiplies RF and LO, producing outputs at $f_{LO} \pm f_{RF}$. For $f_{RF} = 2000$ MHz and $f_{LO} = 1800$ MHz, the IF is $|f_{RF} - f_{LO}| = 200$ MHz. The *image* at $f_{LO} - f_{IF} = 1600$ MHz also lands at that IF and must be rejected by filtering ahead of the mixer.

KEY TAKEAWAYS

- A mixer produces $f_{LO} \pm f_{RF}$; the wanted IF is the sum or difference.
- The image ($f_{LO} \mp f_{IF}$) maps onto the same IF and must be filtered or rejected.
- Passive diode mixers have conversion loss; active mixers can have gain but add noise.

Exercises

1. $f_{RF} = 900$ MHz, $f_{LO} = 800$ MHz. What is the IF?
2. For that low-side-LO case, where is the image?
3. What blocks the image before it reaches the mixer?

Further reading: Maas [12]; Razavi [8] (ch. 6).

6.14 Modulation Schemes

LEARNING OBJECTIVES

- Distinguish amplitude, frequency, and phase modulation.
- Relate constellation size to bits per symbol.
- Understand the bandwidth-vs-robustness trade.

Modulation encodes information onto a carrier wave by varying its amplitude, frequency, or phase. The choice of modulation scheme involves tradeoffs between spectral efficiency (bits/Hz), power efficiency (Eb/N0 required), and robustness to noise and interference.

6.14.1 Amplitude Modulation (AM)

The carrier amplitude varies with the message signal. Simple to demodulate but spectrally inefficient and susceptible to amplitude noise. Double-sideband suppressed carrier (DSB-SC) removes the carrier for power efficiency; single-sideband (SSB) halves the bandwidth.

6.14.2 Frequency Modulation (FM)

The carrier frequency varies with the message. Provides noise immunity above the FM threshold but requires wider bandwidth than AM. Used in broadcast radio (±75 kHz deviation) and VHF voice communications.

6.14.3 Phase-Shift Keying (PSK)

Digital data is encoded as discrete phase shifts. BPSK (2 phases, 1 bit/symbol) is robust; QPSK (4 phases, 2 bits/symbol) doubles spectral efficiency. Higher orders (8-PSK, 16-PSK) pack more bits per symbol but require higher SNR.

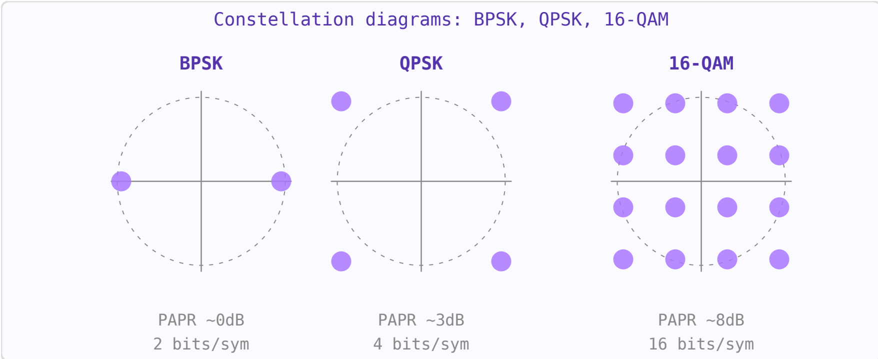


Figure 6.10: Constellation diagrams. More dots per unit area = higher spectral efficiency but requires better SNR.

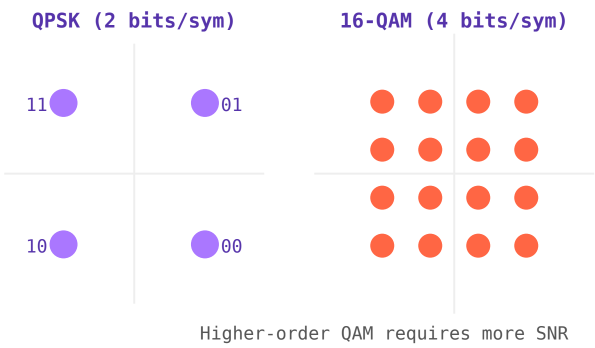


Figure 6.11: QPSK (4 symbols, 2 bits/sym) and 16-QAM (16 symbols, 4 bits/sym) constellation diagrams.

6.14.4 Quadrature Amplitude Modulation (QAM)

Combines amplitude and phase modulation. 16-QAM carries 4 bits/symbol; 64-QAM carries 6 bits/symbol; 256-QAM carries 8 bits/symbol. Used extensively in LTE, 5G, WiFi (802.11ax uses 1024-QAM), and cable TV. Higher QAM orders require much higher SNR and are sensitive to phase noise and nonlinearity.

6.14.5 OFDM

Orthogonal Frequency Division Multiplexing splits the channel into many narrow subcarriers, each modulated with QAM. The cyclic prefix guards against multipath inter-symbol interference. OFDM is the basis of WiFi (802.11a/g/n/ac/ax), LTE, 5G NR, and digital broadcast (DVB-T, DAB).

WORKED EXAMPLE: BITS PER SYMBOL IN QAM

A scheme with M constellation points carries $\log_2 M$ bits per symbol. 16-QAM ($M = 16$) carries $\log_2 16 = 4$ bits/symbol; 64-QAM carries 6. At symbol rate R_s the bit rate is $R_b = R_s \log_2 M$: a 1 Msym/s 16-QAM link carries 4 Mbit/s in about the same bandwidth as 1 Mbit/s BPSK — higher order buys spectral efficiency at the cost of SNR margin.

KEY TAKEAWAYS

- AM varies amplitude; FM/PM vary frequency/phase; digital schemes map bits to constellation points.
- Bits per symbol $= \log_2 M$; higher-order QAM is more spectrally efficient but needs more SNR.
- OFDM spreads data across many orthogonal subcarriers for multipath robustness.

Exercises

1. How many bits per symbol does QPSK carry?
2. What is the bit rate of 64-QAM at 2 Msym/s?
3. Which needs higher SNR for the same BER: QPSK or 256-QAM?

Further reading: Razavi [8] (ch. 3); Steer [5].

Chapter 7

Antennas and Propagation

7.1 Antenna Fundamentals

LEARNING OBJECTIVES

- Define gain, directivity, and efficiency.
- Relate aperture area to gain.
- Read a radiation pattern (HPBW, sidelobes).

An antenna is a transducer that converts between guided electromagnetic waves (in a transmission line) and free-space radiation. Every antenna has a reciprocal receive and transmit behaviour described by the same parameters.

7.1.1 Key Parameters

- **Gain (G)** — ratio of radiation intensity in the peak direction to that of an isotropic radiator, measured in dBi.
- **Directivity (D)** — same as gain but ignoring losses. $G = \eta \cdot D$ where η is radiation efficiency.
- **Beamwidth (HPBW)** — angular width of the main beam between half-power (-3 dB) points.
- **Bandwidth** — frequency range over which VSWR < 2 (or another specification).
- **Polarisation** — orientation of the E-field: linear, circular, or elliptical.
- **Radiation resistance** — equivalent resistance that would dissipate the same power as is radiated.

7.1.2 Half-Wave Dipole

The reference antenna. Total length $\approx \lambda/2$ (slightly less due to end effects). Gain = 2.15 dBi. Radiation resistance $\approx 73\ \Omega$. Omnidirectional in the plane perpendicular to its axis.

7.1.3 Patch (Microstrip) Antenna

A rectangular conductor on a PCB substrate, resonant when the patch length $L \approx \lambda/(2\sqrt{\varepsilon \text{reff}})$. Gain $\approx$ 5–8 dBi, bandwidth 2–5% (can be increased with thicker substrates). Widely used in GPS, WiFi, and cellular handsets due to its flat, PCB-compatible form factor.

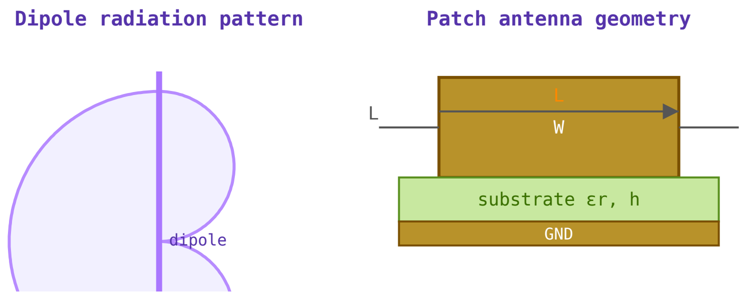


Figure 7.1: Left: dipole E-plane figure-8 radiation pattern (gain ≈ 2.15 dBi). Right: rectangular patch antenna geometry.

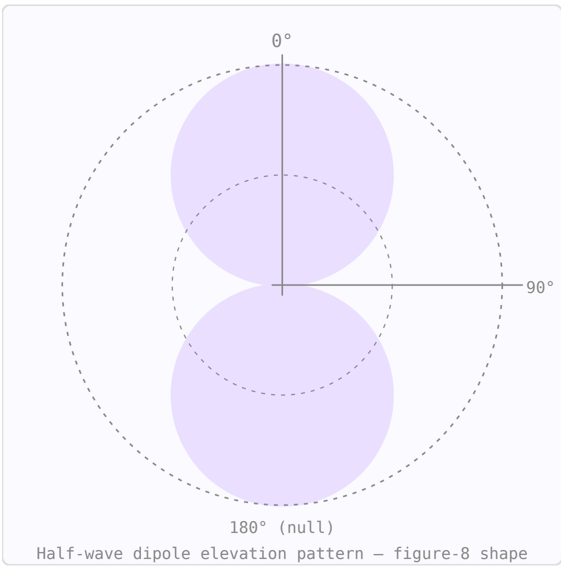


Figure 7.2: Dipole radiation pattern: maximum at broadside (90°), nulls at 0° and 180°.

7.1.4 Array Factor

When N identical elements are arranged in a line with spacing d and progressive phase shift δ, the combined pattern is the element pattern multiplied by the array factor AF(θ). The main beam can be steered electronically by changing δ without physically moving the array — the principle of phased array radar and 5G beamforming.

WORKED EXAMPLE: GAIN OF AN APERTURE ANTENNA

An aperture of physical area A and efficiency η_a has gain $G = \eta_a \, 4\pi A/\lambda^2$. A 1 m^2 dish at 10 GHz ($\lambda = 3$ cm) with $\eta_a = 0.6$ gives

$$G = 0.6 \cdot \frac{4\pi(1)}{(0.03)^2} = 8.38 \times 10^3 = 39.2 \text{ dBi}.$$

Gain rises as $(D/\lambda)^2$, so larger apertures and higher frequencies focus the beam more tightly.

KEY TAKEAWAYS

- Directivity is peak focusing; gain $= \eta_a \times$ directivity (efficiency included).
- Aperture gain $G = \eta_a 4\pi A/\lambda^2$; larger A/λ^2 means higher gain, narrower beam.
- Isotropic (dBi) and half-wave dipole (dBd) references differ by 2.15 dB.

Exercises

1. Express a gain of 100 in dBi.
2. Doubling frequency at fixed dish area changes gain by how many dB?
3. Convert 12 dBd to dBi.

Further reading: Balanis [14] (ch. 2); Kraus [15].

7.2 Antenna Polarisation

LEARNING OBJECTIVES

- Distinguish linear, circular, and elliptical polarization.
- Compute polarization-mismatch loss.
- Read axial ratio.

Polarisation describes the orientation of the electric field vector of the radiated electromagnetic wave. Mismatched polarisation between transmit and receive antennas causes signal loss, regardless of alignment. In MRI, the transmit/receive coil polarisation determines imaging efficiency.

7.2.1 Linear Polarisation

A half-wave dipole radiates a wave whose E-field oscillates in a fixed plane. A vertical dipole is vertically polarised. A horizontal dipole is horizontally polarised. Two dipoles at right angles with the same signal but 0° phase difference are co-polarised along their common axis; with 90° phase difference they produce circular polarisation.

7.2.2 Circular Polarisation (CP)

In circular polarisation, the E-field vector rotates at the carrier frequency. The rotation can be right-hand (RHCP — E rotates clockwise looking in the direction of propagation) or left-hand (LHCP). CP antennas are used for satellite links (independent of receiver orientation), GPS (RHCP), and MRI body coils (CP mode gives $\sqrt{2}$ more B_1 efficiency than linear).

Figure 7.3: RHCP: E-field rotates clockwise (CW) when viewed looking in direction of propagation. LHCP: counter-clockwise.

$$\text{Axial ratio (AR)} = \frac{|E_{major}|}{|E_{minor}|}$$

Perfect circular polarisation: AR = 1 (0 dB). Linear polarisation: AR = ∞. Typical CP antenna spec: AR $<$ 3 dB. A RHCP antenna rejects LHCP signals by the cross-polarisation isolation — ideally infinite, practically 20–30 dB.

7.2.3 Polarisation Mismatch Loss

When the polarisation of the incident wave does not match the antenna, received power is reduced. The polarisation mismatch loss (or polarisation efficiency η_{p}):

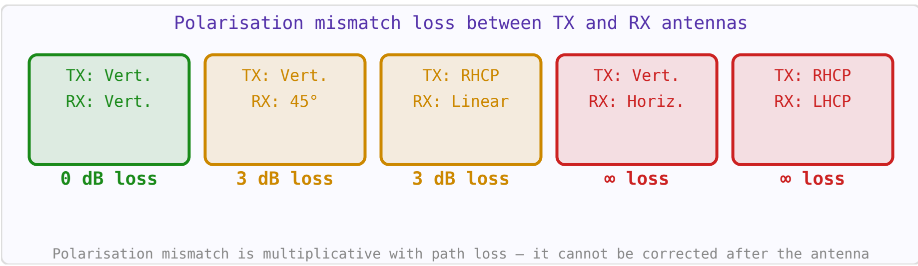


Figure 7.4: Polarisation mismatch loss for common antenna orientation combinations.

$$\eta_p = |\hat{\rho}_{\text{ant}} \cdot \hat{\rho}_{\text{wave}}|^2$$

For linear/circular mismatch: 3 dB loss (worst case). For RHCP/LHCP: theoretically infinite loss. Vertical/horizontal linear: infinite loss at 90° cross-polarisation.

TX polarisation	RX polarisation	Mismatch loss
Vertical linear	Vertical linear	0 dB
Vertical linear	Horizontal linear	∞ dB (no coupling)
RHCP	RHCP	0 dB
RHCP	LHCP	∞ dB (ideal)

RHCP	Vertical linear	3 dB
45° slant linear	Vertical linear	3 dB

7.2.4 Reading Radiation Pattern Plots

A radiation pattern shows antenna gain or field intensity as a function of angle, typically in a polar plot. Key features:

- **Main lobe**: Direction of maximum radiation. Peak value is the antenna gain.
- **−3 dB beamwidth (HPBW)**: Angular width where gain is within 3 dB of the peak. Narrower = more directive.
- **Sidelobe level (SLL)**: Gain of the strongest sidelobe relative to main lobe. Specified in dBc.
- **Front-to-back ratio (F/B)**: Gain of main lobe vs gain directly behind. Important for interference rejection.
- **Null**: Direction of minimum radiation. Can be steered for interference cancellation.

7.2.5 Gain, Directivity and Efficiency

$$G = \eta_r \cdot D \qquad D = \frac{4\pi}{\Omega_A}$$

where η_r is radiation efficiency (0–1) and Ω_A is the beam solid angle in steradians. Gain differs from directivity by the radiation efficiency — a highly directive antenna with 50% efficiency has 3 dB less gain than its directivity. An isotropic radiator (theoretical, impossible to build) has D = 1 = 0 dBi.

7.2.6 Polarisation in MRI

MRI uses circularly polarised B_1 fields. A linearly polarised coil couples to only one sense of rotation — half the available RF power is wasted driving the "wrong" CP mode. A quadrature (CP) coil uses two orthogonal loops fed 90° apart in phase, coupling efficiently to both the transmit field and the NMR signal. This gives $\sqrt{2}$ improvement in SNR and requires half the transmit power for the same flip angle.

WORKED EXAMPLE: POLARIZATION-MISMATCH LOSS

Two linearly polarized antennas misaligned by angle θ couple power as $\cos^2\theta$. For $\theta = 45°$, loss $= -10\log_{10}(\cos^2 45°) = -10\log_{10}(0.5) = 3$ dB. Cross-polarized (90°) antennas couple $\cos^2 90° = 0$ (infinite loss in theory), while linear-to-ideal-circular always loses 3 dB regardless of rotation.

KEY TAKEAWAYS

- Linear-to-linear mismatch loss $= -10\log_{10}(\cos^2\theta)$; 90° gives (ideally) no coupling.
- Linear-to-circular is a fixed 3 dB loss.
- Axial ratio (dB) measures circularity; 0 dB is perfectly circular.

Exercises

1. Mismatch loss for two linear antennas 30° apart?
2. Loss between a linear and an ideal circular antenna?
3. Axial ratio of a perfectly circular wave?

Further reading: Balanis [14] (ch. 2); Kraus [15].

7.3 Yagi-Uda Antennas

LEARNING OBJECTIVES

- Identify reflector, driven element, and directors.
- Relate element count to gain.
- Understand feed impedance and matching.

The Yagi-Uda antenna (commonly called a Yagi) is a directional antenna composed of a fed dipole, a reflector behind it, and one or more directors in front. Only the driven element is connected to the feed line — the others are parasitic and couple electromagnetically. It is the dominant design for VHF/UHF terrestrial and amateur radio links and satellite ground stations.

7.3.1 Element Roles

- **Reflector**: one element, positioned ~0.25λ behind the driven element. Slightly longer than $\lambda/2$ ($\approx 0.505\lambda$). Reflects energy forward, increasing F/B ratio.
- **Driven element**: the only fed element, ≈ 0.47–0.48λ long. Input impedance in the array is ~25 Ω (straight dipole) or ~200 Ω (folded dipole).
- **Directors**: shorter than $\lambda/2$ (≈ 0.43–0.46λ), spaced 0.25–0.42λ forward of the driven element. Each director adds roughly 1–2 dBd of gain. Gain saturates above ~9–10 elements.

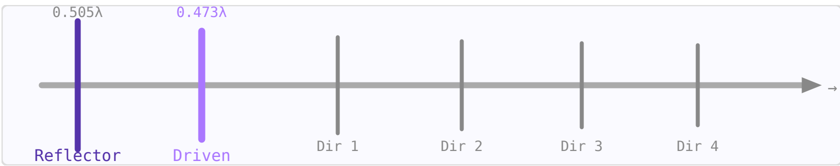


Figure 7.5: 5-element Yagi: reflector (dark), driven element (purple, fed), and three directors (grey).

7.3.2 Gain vs Element Count (NBS/Viezbicke Tables)

The NBS Technical Note 688 (Viezbicke, 1976) tabulates optimised element lengths and spacings for maximum gain, derived from moment-method (NEC) calculations for element diameter $d/\lambda = 0.0085$.

Elements	Gain (dBd)	Gain (dBi)	F/B (approx)

2	3.8	5.9	5 dB
3	5.9	8.1	10 dB
5	8.2	10.4	14 dB
7	10.0	12.2	18 dB
9	11.3	13.5	22 dB

dBi = dBd + 2.15. Adding directors past ~9–10 elements yields diminishing returns (<0.5 dB per element) for significant boom length cost.

7.3.3 Feed Impedance and Matching

The driven-element feed impedance is reduced from the free-space dipole value (~73 Ω) by the mutual coupling of adjacent elements:

- **Straight dipole**: Zin ≈ 25–35 Ω in the array — requires a matching network (gamma match, beta match) to 50 Ω coax
- **Folded dipole**: naturally 4× higher impedance → Zin ≈ 200–300 Ω → use a 4:1 balun or $\lambda/4$ transformer to 50 Ω

7.3.4 Bandwidth and Scaling

A well-optimised Yagi has an operating bandwidth of approximately 2–5% of the centre frequency — narrow enough that element length must be scaled accurately. To move a design to a new frequency, scale all dimensions by the ratio of the original wavelength to the new wavelength (f_old/f_new).

WORKED EXAMPLE: YAGI ELEMENT LENGTHS

In a Yagi-Uda array the reflector is slightly longer than the driven element and the directors slightly shorter, so lengths order as reflector > driven > directors. Typical near-resonance values: reflector $\approx 0.505\lambda$, driven $\approx 0.473\lambda$, directors $\approx 0.44\lambda$. Adding directors raises gain with diminishing returns; a 5-element Yagi gives roughly 10 dBi, rising only slowly beyond about ten elements.

KEY TAKEAWAYS

- Reflector (longest) > driven element > directors (shortest).
- Gain grows with element count but with diminishing returns past ∼10 elements.
- The driven-element impedance is low; a matching network (gamma/hairpin) transforms it to 50 Ω.

Exercises

1. Which element is the longest?
2. Are the directors longer or shorter than the driven element?
3. Is a 5-element Yagi's gain closer to 3 dBi or 10 dBi?

Further reading: Balanis [14] (ch. 10); Viezbicke [16].

7.4 Helical Antennas

LEARNING OBJECTIVES

- Recognize axial-mode operation.
- Apply the Kraus gain formula.
- Understand circular polarization from a helix.

The helical antenna is a wire wound in a helix and fed at one end against a ground plane. Depending on the circumference-to-wavelength ratio it can operate in two distinct modes. The **axial mode** (end-fire) is the most useful for gain antennas — it produces a circularly polarised beam along the helix axis and is widely used in satellite ground stations, CubeSats, and amateur radio.

7.4.1 Normal Mode vs Axial Mode

- **Normal mode**: circumference C $\ll \lambda$; radiation broadside (perpendicular to axis); linearly polarised; low gain. Used as a compact antenna when space is limited.
- **Axial mode**: circumference C $\approx \lambda$; radiation along the axis; circularly polarised; high gain. This is the practical gain antenna.

7.4.2 Axial-Mode Condition

The helix operates in axial mode when:

$$0.75 < \frac{C}{\lambda} < 1.33 \quad \text{where } C = \pi D$$

The optimal pitch angle (angle of the helix wire relative to the horizontal plane) is approximately $\alpha \approx 14°$, which corresponds to a turn spacing S $\approx \lambda/4$ when C $\approx \lambda$.

7.4.3 Gain and Beamwidth (Kraus Formulas)

$$G \approx 15\, N \left(\frac{C}{\lambda}\right)^2 \frac{S}{\lambda} \quad \text{(linear, not dB)}$$

$$\text{HPBW} \approx \frac{52°}{\frac{C}{\lambda}\sqrt{N\frac{S}{\lambda}}}$$

where N is the number of turns. Gain increases with N — a 10-turn helix at C = λ, S = $\lambda/4$ gives G $\approx$ 15×10×1×0.25 = 37.5 ($\approx$ 15.7 dBi).

7.4.4 Input Impedance

The input impedance of the axial-mode helix is nearly resistive and approximated by:

$$Z_{\text{in}} \approx 140\, \frac{C}{\lambda} \quad [\Omega]$$

At the optimum C/λ = 1, Zin $\approx$ 140 Ω. A quarter-wave stripline transformer between the helix feed and the 50 Ω coax connector matches the impedance: the transformer impedance is Z_t = √(140 × 50) $\approx$ 84 Ω.

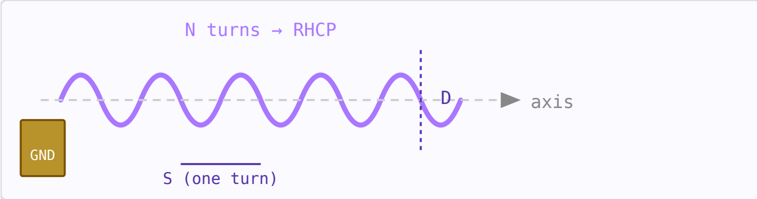


Figure 7.6: Axial-mode helix: N turns of diameter D, turn spacing S, fed against a ground plane.

7.4.5 Circular Polarisation

The axial-mode helix produces circular polarisation naturally. The sense (RHCP or LHCP) is determined by the winding direction:

- **Right-hand winding** (thumb along axis, fingers curl in direction of travel) → RHCP
- **Left-hand winding** → LHCP

For satellite reception (LEO downlinks, GPS), RHCP is standard. The axial ratio of a well-designed axial-mode helix is typically better than 1 dB (very close to perfect circular polarisation) over the 3 dB beamwidth.

7.4.6 Design Procedure

1. Choose target frequency f → compute λ = c/f
2. Set D = λ/π (so C = λ) and S = $\lambda/4$ for optimal pitch angle
3. Choose N for desired gain: G ≈ 15N×1×0.25 = 3.75N → e.g. N = 10 gives G ≈ 37.5 (15.7 dBi)
4. Compute Zin ≈ 140 Ω; design $\lambda/4$ transformer to 50 Ω (Z_t ≈ 84 Ω stripline)
5. Ground plane diameter should be at least 0.75λ to 1λ

WORKED EXAMPLE: AXIAL-MODE HELIX GAIN

An axial-mode helix radiates a circularly polarized beam along its axis when the circumference $C \approx \lambda$. Kraus's gain estimate is

$$G \approx 15\,N\,\frac{C^2}{\lambda^2}\,\frac{S}{\lambda}.$$

For $N = 10$ turns, $C = \lambda$, and spacing $S = 0.25\lambda$: $G \approx 15 \cdot 10 \cdot 1 \cdot 0.25 = 37.5 = 15.7$ dBi. More turns give more gain and a narrower axial beam.

KEY TAKEAWAYS

- Axial mode needs $C \approx \lambda$ and pitch ∼12–14°; it is naturally circularly polarized.
- Kraus: $G \approx 15N(C/\lambda)^2(S/\lambda)$; more turns N raise gain.
- Input impedance is near-resistive ($\sim 140\,C/\lambda$ Ω), matched with a taper.

Exercises

1. What circumference (in λ) puts a helix in axial mode?
2. Doubling the turns changes the Kraus gain by how many dB?
3. What polarization does an axial-mode helix radiate?

Further reading: Kraus [15] (ch. 7); Balanis [14].

7.5 Aperture Antennas: Reflectors and Horns

LEARNING OBJECTIVES

- Compute the gain of an aperture antenna from its area and efficiency.
- Relate parabolic-dish diameter to gain and half-power beamwidth.
- Explain horn feeds and the notion of optimum aperture efficiency.

An aperture antenna radiates through a physical opening much larger than a wavelength — a dish's projected disc or a horn's flared mouth. Because the whole aperture radiates in phase, the gain grows with the aperture area measured in square wavelengths, and the beam narrows in proportion to the aperture width. Dishes and horns dominate microwave point-to-point links, satellite ground stations, radar, and radio astronomy, and horns serve as the standard-gain references of antenna measurement.

7.5.1 Gain from Effective Aperture

Any antenna's gain is tied to its *effective aperture* A_e — the area from which it collects (or radiates) power — through the universal relation:

$$G = \frac{4\pi A_e}{\lambda^2}, \qquad A_e = \eta_{ap}\, A_{phys}$$

where A_{phys} is the physical aperture area and η_{ap} is the aperture efficiency (0–1) that lumps illumination taper, spillover, blockage, and surface error. Rearranging gives the effective aperture that a stated gain corresponds to, $A_e = G\lambda^2/4\pi$ — the same quantity used in the Friis link equation.

7.5.2 Parabolic Reflectors

A parabola reflects rays from a feed at its focus into a collimated plane wave. For a circular dish of diameter D:

$$G = \eta_{ap}\left(\frac{\pi D}{\lambda}\right)^2, \qquad \theta_{HPBW} \approx \frac{70\lambda}{D}\ (^\circ), \qquad \theta_{FNBW} \approx \frac{2.44\lambda}{D}\ \text{rad}$$

Gain rises 6 dB per doubling of diameter (or of frequency), while the half-power beamwidth halves. Typical dish aperture efficiency is 0.55–0.70; the illumination is a compromise — a strongly tapered feed reduces sidelobes and spillover but under-uses the rim, lowering efficiency. The focal-length-to-diameter ratio f/D (typically 0.3–0.5) sets how much of the sphere the feed must illuminate.

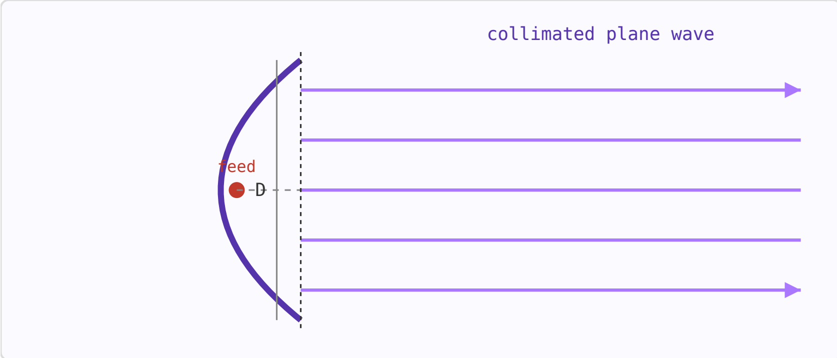


Figure 7.7: A parabolic reflector converts the spherical wave from a focal feed into a plane wave across the aperture, forming a narrow pencil beam.

7.5.3 Horn Antennas

A horn flares a waveguide out to a larger mouth, tapering the impedance to free space and forming a clean, low-sidelobe beam. Its gain follows the same aperture law with an efficiency near 0.51 for an *optimum* horn (the flare length that keeps the aperture phase error to a quarter wavelength):

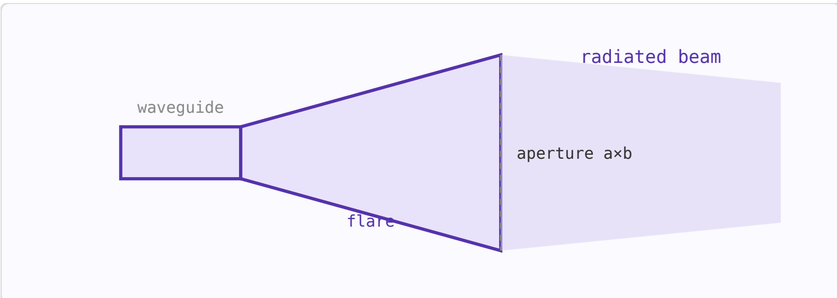


Figure 7.8: A pyramidal horn tapers the guide to a rectangular aperture a×b; a conical horn flares a circular guide to a diameter D.

$$G \approx \eta_{ap} \frac{4\pi A}{\lambda^2}, \quad \eta_{ap} \approx 0.51; \qquad A = a\,b \text{ (pyramidal)}, \quad A = \frac{\pi D^2}{4} \text{ (conical)}$$

Pyramidal horns have unequal E- and H-plane beamwidths, roughly $\theta_E \approx 56\lambda/b$ and $\theta_H \approx 67\lambda/a$ degrees. Corrugated and dielectric-loaded horns raise efficiency and symmetry and are common as reflector feeds. A calibrated "standard-gain horn" is the reference against which other antennas are measured.

7.5.4 EIRP and the Link

Once the gain is known, the effective isotropic radiated power combines it with the transmit power, and the effective aperture feeds the receive side of the link:

$$\text{EIRP} = P_t + G \text{ [dBm/dBi]}; \qquad A_e = \frac{G\lambda^2}{4\pi}; \qquad R_{ff} = \frac{2D^2}{\lambda} \text{ (far-field distance)}$$

The far-field (Fraunhofer) distance $2D^2/\lambda$ marks where the beam is fully formed — for large apertures this can be hundreds of metres, which matters when range-testing or siting antennas.

7.5.5 Typical Numbers

Antenna	Size	Freq	Gain	HPBW
Parabolic dish (η=0.6)	1 m	10 GHz	≈38 dBi	≈2.1°
Parabolic dish (η=0.6)	3 m	6 GHz	≈43 dBi	≈1.2°
Standard-gain horn	15×11 cm	10 GHz	≈20 dBi	≈15°
Conical horn feed	5 cm dia	12 GHz	≈13 dBi	≈35°

WORKED EXAMPLE: GAIN AND BEAMWIDTH OF A 1 M DISH

A 1 m parabolic dish operates at 10 GHz ($\lambda = 30$ mm). The gain of a circular aperture is $G = \eta(\pi D/\lambda)^2$. With an aperture efficiency $\eta = 0.6$,

$$G = 0.6\left(\frac{\pi \cdot 1}{0.03}\right)^2 = 0.6\,(104.7)^2 \approx 6580,$$

i.e. 38.2 dBi. The half-power beamwidth is HPBW $\approx 70\lambda/D = 70 \cdot 0.03/1 = 2.1°$. Gain and beamwidth are set almost entirely by the size of the aperture in wavelengths.

KEY TAKEAWAYS

- Aperture gain $G = \eta\, 4\pi A/\lambda^2$; for a circular dish $G = \eta(\pi D/\lambda)^2$.
- HPBW $\approx 70\lambda/D$ degrees — gain and beamwidth trade directly against size in wavelengths.
- Horn feeds reach roughly 50% aperture efficiency; illumination taper trades efficiency for lower sidelobes.

Exercises

1. A 0.6 m dish at 12 GHz with 55% efficiency — find the gain in dBi.
2. Estimate the half-power beamwidth of a 2 m dish at 6 GHz.
3. Name three reasons aperture efficiency is typically below 100%.

Further reading: Balanis [14]; Kraus [15].

7.6 Antenna Types

LEARNING OBJECTIVES

- Match antenna types (patch, horn, reflector, log-periodic, PIFA) to requirements.
- Estimate a microstrip patch's resonant length.
- Explain the size–bandwidth–gain trade among antenna families.

Beyond the dipole and helix, the antenna zoo offers a form for every constraint — low profile, high gain, or wide band.

7.7 Wire and Loop

Dipoles, monopoles (a dipole over a ground plane), and loops are the simplest resonant radiators, cheap and omnidirectional-ish, used from HF to UHF.

7.8 Microstrip Patch

A patch is a resonant metal rectangle over a ground plane, roughly a half wavelength long in the substrate,

$$L \approx \frac{c}{2f\sqrt{\varepsilon_{\text{eff}}}}.$$

It is flat, cheap to print, and easily arrayed, but inherently narrowband. Feed by edge line, probe, or aperture coupling.

7.9 Aperture Antennas

Horns and parabolic reflectors turn a large physical aperture into high gain ($G = \eta_a 4\pi A/\lambda^2$), dominating radar and satellite links where directivity is paramount.

7.10 Broadband and Mobile

Log-periodic and spiral antennas trade size for octaves of bandwidth by scaling their geometry. For handsets, the PIFA (planar inverted-F), monopole, and chip antennas fold a resonant structure into a tiny volume against the challenge of a small, detuning ground plane.

WORKED EXAMPLE: PATCH RESONANT LENGTH

A patch on FR-4 ($\varepsilon_r = 4.4$) at $f = 2.4$ GHz resonates near a half wavelength in the substrate. Using $\varepsilon_{\text{eff}} \approx (\varepsilon_r + 1)/2 = 2.7$,

$$L \approx \frac{c}{2f\sqrt{\varepsilon_{\text{eff}}}} = \frac{3 \times 10^8}{2(2.4 \times 10^9)(1.64)} = 38 \text{ mm},$$

trimmed slightly for fringing fields. Patches are low-profile and easily arrayed but narrowband.

KEY TAKEAWAYS

- Patch $L \approx c/(2f\sqrt{\varepsilon_{\text{eff}}})$: flat, cheap, arrayable, narrowband.
- Aperture antennas (horn, reflector) give high gain $G = \eta_a 4\pi A/\lambda^2$.
- Log-periodic/spiral trade size for bandwidth; PIFA/monopole suit handsets.

Exercises

1. Patch length on a substrate with $\varepsilon_{\text{eff}} = 6.25$ at 1 GHz?
2. Which antenna family gives the widest bandwidth for its size?
3. Which gives the highest gain: a patch or a parabolic reflector?

Further reading: Balanis [14] (ch. 14); Pozar [1] (ch. 14).

7.11 Phased Arrays and Beamforming

LEARNING OBJECTIVES

- Write and interpret the array factor of a uniform linear array.
- Compute the progressive phase needed to steer the beam.
- Explain beamwidth, grating lobes, and tapering.

Combining many antenna elements lets a system synthesize and steer a beam electronically — the basis of modern radar and 5G.

7.12 The Array Factor

For N identical elements spaced d along a line, the far-field pattern is the element pattern times the array factor

$$\mathrm{AF}(\theta) = \sum_{n=0}^{N-1} e^{jn(kd\cos\theta+\beta)},$$

where β is the progressive phase between elements. The array factor, not the element, gives the array its directivity.

7.13 Steering

Choosing $\beta = -kd\sin\theta_0$ (for a broadside reference) points the main beam to angle θ_0 — with no moving parts, just phase. The beam can be repointed in microseconds, which is why phased arrays replaced mechanically scanned dishes in radar.

7.14 Beamwidth, Grating Lobes, Tapering

More elements and wider aperture narrow the beam (beamwidth $\propto \lambda/(Nd)$). If $d > \lambda$, extra *grating lobes* appear (usually $d \le \lambda/2$ avoids them). Amplitude tapering across the array trades a little beamwidth for much lower sidelobes.

7.15 Digital Beamforming and MIMO

With a receiver per element, beams are formed in DSP — many simultaneous beams, adaptive nulling of interferers, and, in communications, spatial multiplexing (MIMO/massive MIMO) that sends independent streams on the same frequency.

WORKED EXAMPLE: STEERING PHASE AND BEAMWIDTH

A uniform linear array with $d = \lambda/2$ is steered to $\theta_0 = 30°$. The required progressive phase per element is

$$\beta = -kd\sin\theta_0 = -\frac{2\pi}{\lambda}\cdot\frac{\lambda}{2}\cdot\sin 30° = -\frac{\pi}{2} = -90°.$$

For $N = 8$ elements at $d = \lambda/2$, the broadside half-power beamwidth is about $0.886\,\lambda/(Nd) = 0.886/4 = 0.22$ rad $= 12.7°$.

KEY TAKEAWAYS

- $\mathrm{AF}(\theta) = \sum_n e^{jn(kd\cos\theta+\beta)}$; the array factor sets directivity.
- Steer with $\beta = -kd\sin\theta_0$ — electronic, instantaneous.
- Beamwidth $\propto \lambda/(Nd)$; keep $d \le \lambda/2$ to avoid grating lobes; taper for low sidelobes.

Exercises

1. Steering phase for $d = \lambda/2$ to $\theta_0 = 90°$ (endfire)?
2. Why keep element spacing $d \le \lambda/2$?
3. What does amplitude tapering buy, and at what cost?

Further reading: Mailloux [32]; Balanis [14] (ch. 6).

7.16 Radio Propagation and Fading

LEARNING OBJECTIVES

- Contrast free-space, two-ray, and empirical path-loss models.
- Compute a Fresnel-zone radius and explain clearance.
- Distinguish Rayleigh and Rician fading and relate delay spread to coherence bandwidth.

Between transmitter and receiver the channel adds far more than free-space loss: reflection, diffraction, and multipath shape every real link.

7.17 Beyond Free Space

Near the ground, a direct ray and a ground-reflected ray interfere; beyond a break point the received power falls as $1/d^4$ rather than $1/d^2$. Empirical models (Okumura–Hata, COST-231) fit measured path loss in urban and suburban environments where a full calculation is hopeless.

7.18 Diffraction and Fresnel Zones

Waves bend around obstacles (diffraction), so a link can close even without line of sight. The first Fresnel zone — an ellipsoid around the direct path with radius

$$r_1 = \sqrt{\frac{\lambda\, d_1 d_2}{d_1 + d_2}}$$

at the point between distances d_1 and d_2 — should be kept mostly clear of obstructions to avoid diffraction loss.

7.19 Multipath and Fading

Many reflected paths arrive with different delays and phases, so the received amplitude *fades*. With no dominant path the envelope is Rayleigh distributed; with a line-of-sight component it is Rician. The spread of path delays (delay spread) sets the *coherence bandwidth* beyond which the channel is frequency-selective, and motion causes a Doppler spread that sets how fast the fading changes.

WORKED EXAMPLE: FIRST FRESNEL-ZONE RADIUS

A 2.4 GHz link ($\lambda = 0.125$ m) spans 1 km with the obstacle at midpoint ($d_1 = d_2 = 500$ m). The first Fresnel-zone radius there is

$$r_1 = \sqrt{\frac{\lambda d_1 d_2}{d_1 + d_2}} = \sqrt{\frac{0.125 \cdot 500 \cdot 500}{1000}} = \sqrt{31.25} = 5.6 \text{ m}.$$

Keeping about 60% of this radius clear avoids significant diffraction loss.

KEY TAKEAWAYS

- Two-ray ground reflection gives $1/d^4$ far-field loss; Okumura–Hata/COST-231 fit real environments.
- First Fresnel radius $r_1 = \sqrt{\lambda d_1 d_2/(d_1 + d_2)}$; keep it mostly clear.
- Multipath fades: Rayleigh (no LOS) or Rician (LOS); delay spread sets coherence bandwidth, Doppler sets fade rate.

Exercises

1. First Fresnel radius at the midpoint of a 2 km, 900 MHz link?
2. What fading distribution applies with a strong line-of-sight component?
3. What channel parameter sets the coherence bandwidth?

Further reading: Rappaport [31]; Goldsmith [30].

7.20 Antenna Measurement

LEARNING OBJECTIVES

- Compute the far-field distance for a measurement.
- Describe gain-comparison and three-antenna gain measurement.
- Explain near-field scanning and pattern/polarization measurement.

An antenna's real performance is measured, not just simulated, and RF metrology has its own methods and pitfalls.

7.21 The Far-Field Condition

Pattern and gain are defined in the far field, beyond

$$R \geq \frac{2D^2}{\lambda},$$

where D is the largest aperture dimension. Measurements are made on outdoor ranges or in anechoic chambers whose absorber suppresses reflections.

7.22 Gain Measurement

Two standard methods: the *gain-comparison* method references the antenna to a calibrated gain standard, and the *three-antenna* method solves for three unknown gains from three pairwise transmission measurements — needing no prior standard.

7.23 Pattern, Polarization, and Near-Field

Rotating the antenna traces its radiation pattern and, with a polarized source, its polarization and axial ratio. When a far-field range is impractical, a probe scans the near field over a plane, cylinder, or sphere and a mathematical transform computes the far-field pattern. Impedance and return loss are measured directly on a VNA, and radiation efficiency by methods such as the Wheeler cap.

WORKED EXAMPLE: FAR-FIELD DISTANCE

A reflector of largest dimension $D = 0.3$ m is measured at $f = 10$ GHz ($\lambda = 0.03$ m). The far field begins at

$$R \geq \frac{2D^2}{\lambda} = \frac{2(0.3)^2}{0.03} = 6 \text{ m}.$$

Measuring closer than this corrupts the pattern; if 6 m of range is impractical, a near-field scan with a far-field transform is used instead.

KEY TAKEAWAYS

- Far field begins at $R \geq 2D^2/\lambda$; measure in an anechoic chamber or on a range.
- Gain by comparison (to a standard) or the three-antenna method (no standard needed).
- Near-field scanning + transform yields the far-field pattern when a far-field range is impractical.

Exercises

1. Far-field distance for $D = 1$ m at 3 GHz?
2. Which gain method needs no pre-calibrated reference antenna?
3. How is antenna impedance/return loss measured?

Further reading: Balanis [14] (ch. 17); Pozar [1] (ch. 4).

Chapter 8

RF Systems, Receivers and Communications

8.1 RF Frequency Bands

LEARNING OBJECTIVES

- Recall the IEEE microwave band letters.
- Relate frequency to wavelength and antenna size.
- Match a band to typical applications.

The electromagnetic spectrum from 3 Hz to 300 GHz is divided into named bands by the ITU. Each band has characteristic propagation behaviour, regulatory allocations, and typical applications. This page is a compact reference for the most important RF and microwave bands.

8.1.1 ITU Frequency Band Designations

Band	Name	Frequency	Wavelength	Propagation
ELF	Extremely Low Frequency	3–30 Hz	10,000–100,000 km	Penetrates seawater; submarine communications
SLF	Super Low Frequency	30–300 Hz	1,000–10,000 km	Power grid frequency; submarine comms
LF	Low Frequency	30–300 kHz	1–10 km	Ground wave; LORAN, DECCA navigation, AM longwave
MF	Medium Frequency	300 kHz–3 MHz	100 m–1 km	Ground wave + night-time skywave; AM broadcast
HF	High Frequency	3–30 MHz	10–100 m	Ionospheric skywave; intercontinental amateur, shortwave broadcast, OTHR

VHF	Very High Frequency	30–300 MHz	1–10 m	Line of sight + troposcatter; FM radio, TV, aviation, land mobile
UHF	Ultra High Frequency	300 MHz–3 GHz	10 cm–1 m	Line of sight; cellular, Wi-Fi, GPS, DVB-T, radar
SHF	Super High Frequency	3–30 GHz	1–10 cm	Line of sight (rain scatter); satellite, microwave links, 5G, radar
EHF	Extremely High Frequency	30–300 GHz	1–10 mm	Line of sight (rain/O_2 absorption); mmWave 5G, automotive radar, imaging

8.1.2 IEEE Radar / Microwave Band Letters

Band	Frequency	Applications
L	1–2 GHz	GPS (1.575 GHz), cellular 4G/5G, ATC radar, satellite
S	2–4 GHz	Wi-Fi 2.4 GHz, Bluetooth, weather radar, cellular
C	4–8 GHz	Satellite communications, C-band radar, Wi-Fi 5 GHz
X	8–12 GHz	Airborne/marine radar, SAR, satellite TV, PCB measurement
Ku	12–18 GHz	Satellite broadcast (DBS), VSAT, Ku-band radar
K	18–27 GHz	Automotive radar (24 GHz)
Ka	27–40 GHz	Satellite broadband, 5G mmWave, imaging radar
V	40–75 GHz	60 GHz WiGig, automotive radar (77 GHz)
W	75–110 GHz	Automotive radar (79 GHz), imaging, atmospheric
mm	110–300 GHz	THz imaging, spectroscopy, 5G future bands

8.1.3 Key Frequency Allocations

Frequency	Allocation / Service
433.92 MHz	ISM band (EU): wireless sensors, key fobs, LoRa
868 MHz	ISM band (EU): LoRa, Zigbee, smart meters
915 MHz	ISM band (US/AU): LoRa, RFID
1.575 GHz	GPS L1 (BPSK, C/A code)
1.227 GHz	GPS L2 (military P-code)
2.400–2.4835 GHz	ISM: Wi-Fi (802.11b/g/n), Bluetooth, ZigBee, microwave ovens
5.150–5.850 GHz	U-NII: Wi-Fi (802.11a/n/ac/ax)
5.8 GHz	ISM band: cordless phones, some Wi-Fi

24 GHz	ISM band: short-range radar, motion sensors
60 GHz	ISM band: 802.11ad/ay WiGig, high-speed short-range links
77 GHz	Automotive radar (long-range)

8.1.4 Antenna Size Reference

Frequency	$\lambda/2$ dipole	$\lambda/4$ monopole	Patch (in FR4)
433 MHz	346 mm	173 mm	~82 mm
868 MHz	173 mm	86 mm	~41 mm
915 MHz	164 mm	82 mm	~39 mm
2.45 GHz	61 mm	31 mm	~14.5 mm
5.8 GHz	26 mm	13 mm	~6.1 mm
24 GHz	6.2 mm	3.1 mm	~1.5 mm
77 GHz	2.0 mm	1.0 mm	~0.47 mm

WORKED EXAMPLE: BAND AND WAVELENGTH

The IEEE letter bands split the microwave spectrum: L (1–2 GHz), S (2–4), C (4–8), X (8–12), Ku (12–18), K (18–27), Ka (27–40). A half-wave dipole scales as $\ell = \lambda/2 = c/(2f)$: at S-band 2.45 GHz, $\ell = (3 \times 10^8)/(2 \cdot 2.45 \times 10^9) = 61$ mm; at Ka-band 30 GHz it is only 5 mm.

KEY TAKEAWAYS

- IEEE bands (L, S, C, X, Ku, K, Ka) label microwave ranges.
- Antenna size scales with $\lambda = c/f$; higher bands mean smaller antennas.
- Band choice trades bandwidth/size against rain and atmospheric loss.

Exercises

1. Which IEEE band contains 10 GHz?
2. Half-wave dipole length at 1 GHz?
3. Do higher bands give larger or smaller antennas at the same electrical size?

Further reading: Pozar [1]; IEEE Std 521 [21].

8.2 The Superheterodyne Receiver

LEARNING OBJECTIVES

- Trace the superheterodyne signal flow.
- Explain image rejection and IF choice.
- Understand dynamic-range placement.

The superheterodyne receiver, invented by Edwin Armstrong in 1918, translates the received RF signal to an intermediate frequency (IF) where filtering and amplification are more practical. It remains the dominant architecture in virtually every radio, cellular phone, and test instrument made today.

8.2.1 Block Diagram Signal Flow

The signal path from antenna to baseband:

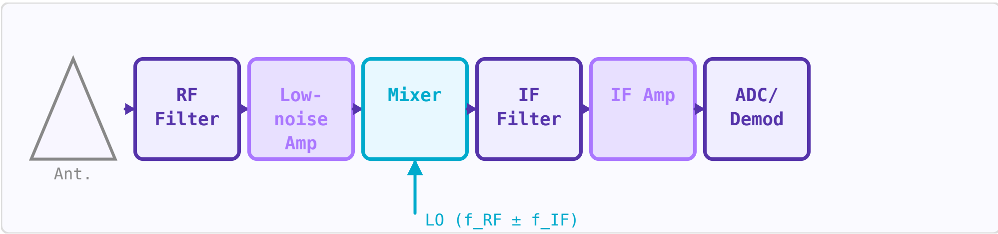


Figure 8.1: Superheterodyne receiver signal flow. The RF filter sets image rejection; the IF filter provides channel selectivity.

1. **Antenna** → receives signal + noise + interference across all frequencies.
2. **RF bandpass filter (BPF)** → passes the desired RF band, rejects out-of-band signals. Critical for preventing image interference and protecting the LNA.
3. **Low-noise amplifier (LNA)** → first amplification stage. Dominates system noise figure (Friis: first stage matters most). Typically 15–25 dB gain, 1–3 dB NF.
4. **Image rejection filter** → rejects the image frequency f_image = f_LO − f_IF (for low-side injection). Must be placed before the mixer.
5. **Mixer** → multiplies RF by local oscillator (LO), producing sum and difference frequencies. Desired output is |f_RF − f_LO| = f_IF.
6. **IF bandpass filter** → selects the IF and provides adjacent-channel rejection. This is where most selectivity comes from — fixed-frequency filters are practical at IF.
7. **IF amplifier** → provides most of the system gain at IF (easier to build stable high-gain amplifiers at lower frequencies).
8. **Demodulator / ADC** → extracts information from the IF signal.

8.2.2 The Image Frequency Problem

The mixer responds to two input frequencies — the desired signal at f_RF and an "image" at f_image = f_LO ± f_IF. Noise or interference at the image frequency is also downconverted to the IF and corrupts the desired signal. Image rejection is the ratio (in dB) of how much the receiver attenuates the image relative to the desired frequency:

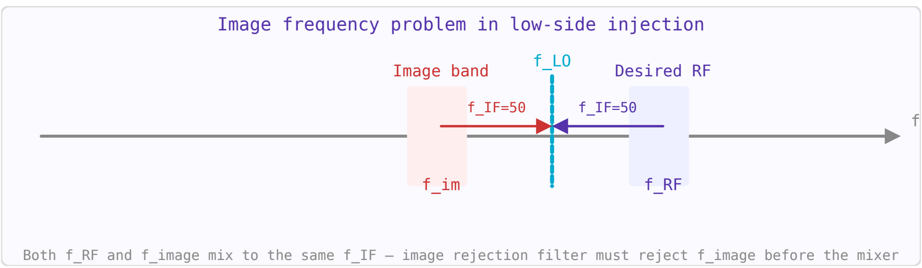


Figure 8.2: The image frequency f_image = f_LO − f_IF (low-side injection) produces the same IF as the desired signal.

$$f_{image} = f_{RF} \pm 2f_{IF}$$

Higher IF → image is further from f_RF → easier to filter. This motivates the choice of IF. A high first IF (e.g. 70 MHz or 1.4 GHz) gives good image rejection; a second lower IF can then improve channel selectivity.

8.2.3 Noise Figure Through the Chain

Using the Friis formula, the cascaded noise figure:

$$F_{sys} = F_{filter} + \frac{F_{LNA} - 1}{G_{filter}} + \frac{F_{mixer} - 1}{G_{filter}G_{LNA}} + \cdots$$

A passive RF filter before the LNA (e.g. with 2 dB insertion loss) adds directly to the noise figure: F_filter = 1 + loss(linear) = 1 + (L−1)/L. This is why a low-noise LNA must come as early in the chain as possible, and why surface-mount filters with high insertion loss degrade sensitivity.

8.2.4 Dynamic Range

The receiver must handle signals from the noise floor (-100 to -120 dBm) to strong in-band signals that could be 80 dB stronger. Two limits define the dynamic range:

- **Minimum detectable signal (MDS)**: noise floor + minimum required SNR. Determined by noise figure and bandwidth.
- **Blocking / desensitisation**: strong off-channel signals that compress the LNA or generate intermodulation products that mask the desired signal. Determined by IIP3 and P_1dB.

$$\text{Dynamic Range} = P_{1dB,input} - \text{MDS} \quad [\text{dB}]$$

8.2.5 Superheterodyne vs Direct Conversion (Zero-IF)

Feature	Superheterodyne	Direct conversion (zero-IF)
IF frequency	Fixed non-zero value	0 Hz (baseband I/Q output)
Image rejection	Achieved by RF filter	No image problem (I/Q cancellation)
Channel filtering	Fixed IF filter (ceramic, SAW)	Baseband low-pass filter (active)
Integration	Harder — off-chip IF filters needed	Easier — fully on-chip in CMOS
Key problems	Image, LO harmonics, IF filter	DC offset, IQ imbalance, 1/f noise
Used in	Test instruments, broadcast receivers	Smartphones, SDR, Wi-Fi, Bluetooth

WORKED EXAMPLE: IF AND IMAGE PLACEMENT

A superhet mixes RF down to a fixed IF for selectivity and gain. With $f_{RF} = 100$ MHz and high-side LO $f_{LO} = 110$ MHz, the IF is 10 MHz and the image sits at $f_{LO}+f_{IF} = 120$ MHz — $2f_{IF} = 20$ MHz from the wanted signal. A higher IF pushes the image farther out (easier to filter) but complicates the IF stage — the classic IF trade-off.

KEY TAKEAWAYS

- The superhet converts RF to a fixed IF where filtering and gain are easier.
- The image lies $2f_{IF}$ from the RF; a higher IF eases image rejection.
- Front-end selectivity and gain distribution set sensitivity and dynamic range.

Exercises

1. $f_{RF} = 1000$ MHz, $f_{LO} = 1090$ MHz. Find the IF and image.
2. Does a higher IF make image rejection easier or harder?
3. How far in frequency is the image from the desired signal?

Further reading: Razavi [8] (ch. 4); Pozar [1].

8.3 Receiver Architectures

LEARNING OBJECTIVES

- Compare superheterodyne, direct-conversion, and low-IF receivers.
- List the impairments unique to zero-IF receivers.
- Explain how an image-reject mixer suppresses the image.

The receiver architecture decides how the RF is brought down to where it can be digitized, and each choice trades image rejection, integration, and impairments.

8.4 Superheterodyne

The classic superhet converts the RF to a fixed intermediate frequency where high-Q filtering and gain are cheap. Its weakness is the image, $2f_{IF}$ from the wanted signal, which a front-end filter must reject before the mixer.

8.5 Direct Conversion (Zero-IF)

Setting $f_{LO} = f_{RF}$ folds the signal straight to baseband ($f_{IF} = 0$). There is no image, and the whole radio integrates on one chip — but new problems appear: DC offset (from LO self-mixing), flicker ($1/f$) noise at baseband, LO leakage, and I/Q imbalance. Modern CMOS transceivers manage these with calibration.

8.6 Low-IF and Image-Reject Mixers

A low-IF receiver uses a small non-zero IF to escape the DC problems while keeping integration; the image, now close by, is removed by *complex* (polyphase) filtering or by an image-reject mixer. The Hartley and Weaver architectures cancel the image by combining two quadrature mixing paths, so the achievable rejection is set by the I/Q amplitude and phase balance.

WORKED EXAMPLE: IMAGE REJECTION FROM I/Q IMBALANCE

An image-reject mixer's rejection is limited by gain error g (fractional) and phase error φ (radians) between its I and Q paths:

$$\text{IRR} \approx 10\log_{10}\frac{4}{g^2+\varphi^2}.$$

For a 1% gain error ($g = 0.01$) and 1° phase error ($\varphi = 0.0175$ rad),

$$\text{IRR} \approx 10\log_{10}\frac{4}{(0.01)^2+(0.0175)^2} = 10\log_{10}(9850) = 39.9\ \text{dB}.$$

Practical analog balance thus caps image rejection near 40 dB, which is why calibration or an external filter is often still needed.

KEY TAKEAWAYS

- Superhet: fixed IF, easy filtering, but an image $2f_{IF}$ away to reject.
- Zero-IF: no image, fully integrated, but DC offset, $1/f$ noise, LO leakage, I/Q imbalance.
- Image-reject (Hartley/Weaver) cancellation is limited by I/Q balance: $\text{IRR} \approx 4/(g^2+\varphi^2)$.

Exercises

1. How far from the wanted signal is a superhet's image?
2. Name two impairments specific to a direct-conversion receiver.
3. With $g = 0.02$ and $\varphi = 0.01$ rad, estimate the image-reject ratio.

Further reading: Razavi [8] (ch. 4); Lee [9].

8.7 Receiver Cascade Analysis

LEARNING OBJECTIVES

- Combine stage gains, noise figures, and IP3.
- Apply Friis for cascaded NF.
- Identify the limiting stage.

A receiver chain is a sequence of amplifiers, filters, and frequency converters from antenna to digital output. The system noise figure determines sensitivity, the gain distribution determines dynamic range, and the IP3 cascade determines linearity. These three analyses — Friis NF, gain chain, and cascaded IP3 — must be done together to optimise the receiver architecture.

8.7.1 The Three-Parameter Cascade

For each stage, you need three numbers: gain G (dB), noise figure NF (dB), and input-referred IP3 (IIP3, dBm). From these, the cascaded quantities are computed.

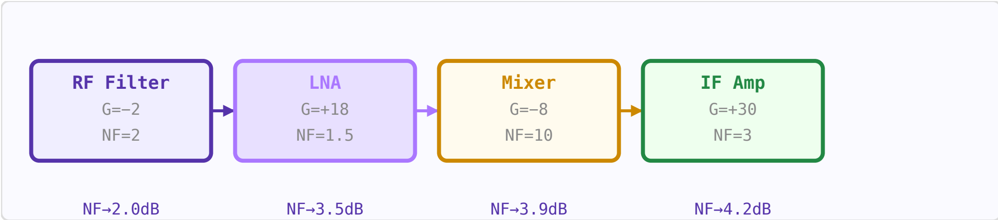


Figure 8.3: Cumulative noise figure stabilises after the LNA. First stage and its gain dominate the system NF.

8.7.2 Friis Formula for Cascaded Noise Figure

$$F_{sys} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots$$

where F = 10^(NF/10) (linear). The first stage dominates if G_1 is large. A low-NF, high-gain LNA first is critical.

8.7.3 Cascaded IIP3

$$\frac{1}{IIP3_{sys}} = \frac{1}{IIP3_1} + \frac{G_1}{IIP3_2} + \frac{G_1 G_2}{IIP3_3} + \cdots$$

(all in linear watts, not dBm). The last high-gain stage dominates linearity — opposite to noise figure.

8.7.4 Worked Example: 2.4 GHz Receiver

Stage	G (dB)	NF (dB)	IIP3 (dBm)	Cumul. NF	Cumul. IIP3
RF filter (insertion loss)	−2	2	∞	2.0 dB	—
LNA	+18	1.5	+10	3.5 dB	+8.5 dBm
Image rejection filter	−2	2	∞	3.5 dB	+8.5 dBm
Mixer	−8	10	+20	4.3 dB	+4.5 dBm
IF amplifier	+30	3	+25	4.3 dB	−6.3 dBm

The cumulative noise figure stabilises at 4.3 dB after the LNA — the LNA gain means all downstream stages add negligible noise. The IIP3 degrades to −6.3 dBm due to the high IF amplifier gain — if this is insufficient, the IF gain must be spread across more stages or reduced.

8.7.5 Minimum Detectable Signal

$$MDS = -174 + NF_{sys} + 10\log_{10}(BW) + SNR_{min} \quad \text{[dBm]}$$

For the example above with 5 MHz bandwidth and 10 dB minimum SNR: MDS = −174 + 4.3 + 67 + 10 = −92.7 dBm.

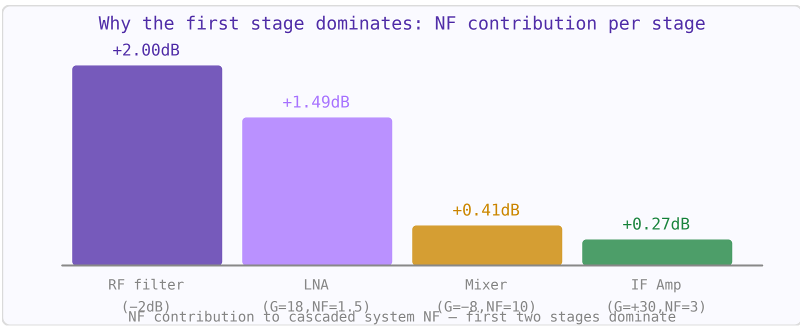


Figure 8.4: Each stage's marginal contribution to cascaded NF (linear units). Later stages contribute nearly nothing after a high-gain LNA.

8.7.6 Spurious-Free Dynamic Range

$$SFDR = \frac{2}{3}(IIP3_{sys} - MDS)$$

With IIP3 = −6.3 dBm and MDS = −92.7 dBm: SFDR = 2/3 × 86.4 = 57.6 dB. This means the receiver can handle a 57.6 dB range of input signal levels without the IM3 products exceeding the noise floor.

8.7.7 Gain Distribution Rules of Thumb

- Place the LNA as early as possible (after only necessary filtering) to establish noise figure.
- Total front-end gain before the ADC should be enough that ADC noise is irrelevant: typically 60–90 dB total gain for a −100 dBm sensitivity target into an ADC with −70 dBm full scale.
- Distribute gain across multiple stages to keep no single stage near compression. Each high-gain stage needs correspondingly high IIP3.
- Use variable gain amplifiers (VGAs) to handle the signal range: the AGC loop keeps the ADC input level constant regardless of received signal power.
- Don't forget the ADC itself: its ENOB and input full-scale power set the final SNR.

8.7.8 AGC and Automatic Gain Control

An AGC loop monitors the signal level (usually after the IF filter) and adjusts attenuators or VGAs to keep the ADC input at a constant level. Good AGC design requires: fast attack time (handles sudden strong signals before saturation), slow decay time (prevents noise pumping), and sufficient control range (> dynamic range of expected input signals).

WORKED EXAMPLE: CASCADED NOISE FIGURE

For stages in series, $F_{tot} = F_1 + (F_2 - 1)/G_1 + (F_3 - 1)/(G_1 G_2) + \ldots$ Take an LNA ($\text{NF}_1 = 2$ dB, $G_1 = 15$ dB) then a mixer ($\text{NF}_2 = 10$ dB). In linear terms $F_1 = 1.585$, $G_1 = 31.6$, $F_2 = 10$:

$$F_{tot} = 1.585 + \frac{10 - 1}{31.6} = 1.87,$$

i.e. $\text{NF}_{tot} = 2.72$ dB — only 0.7 dB above the LNA alone, because its gain suppresses the mixer's noise.

KEY TAKEAWAYS

- Cascaded NF (Friis): later stages' noise is divided by all preceding gain.
- A high-gain, low-NF first stage sets the system NF.
- Cascaded IP3 is instead dominated by the last high-level stages.

Exercises

1. Why does a high-gain LNA first improve system NF?
2. LNA NF = 1 dB, $G = 20$ dB; second stage NF = 10 dB. Approximate system NF?
3. Does the first or last stage tend to dominate cascade IP3?

Further reading: Pozar [1] (ch. 10); Razavi [8] (ch. 2).

8.8 Gain Planning and Dynamic Range

LEARNING OBJECTIVES

- Explain the trade-off in distributing gain along a chain.
- Compute the minimum detectable signal (sensitivity).
- Compute spurious-free dynamic range.

Gain planning distributes gain, noise, and linearity along the receive chain so that the system meets both its sensitivity and its dynamic-range goals.

8.9 The Balancing Act

Put too little gain early and later stages' noise dominates (poor sensitivity); put too much and a strong signal compresses the front end (poor large-signal handling). The plan sets each stage's gain, NF, and IP3 so the cascaded noise figure and cascaded IP3 both land where they must.

8.10 Sensitivity

The minimum detectable signal is the noise floor plus the required SNR:

$$\mathrm{MDS} = -174\ \mathrm{dBm/Hz} + 10\log_{10} B + \mathrm{NF} + \mathrm{SNR}_{min}.$$

Lowering NF (a good LNA first) or narrowing B improves sensitivity.

8.11 Spurious-Free Dynamic Range

The top of the usable range is set by third-order intermodulation. The spurious-free dynamic range — the span from the noise floor to where IM3 products rise out of it — is

$$\mathrm{SFDR} = \tfrac{2}{3}\,(\mathrm{IIP3} - \mathrm{MDS}).$$

Automatic gain control extends the *instantaneous* range by backing off gain on strong signals, but SFDR is the fundamental figure.

WORKED EXAMPLE: SENSITIVITY AND SFDR

A receiver has bandwidth $B = 1$ MHz and noise figure $\mathrm{NF} = 5$ dB. Its noise floor (with $\mathrm{SNR}_{min} = 0$) is

$$\mathrm{MDS} = -174 + 10\log_{10}(10^6) + 5 = -174 + 60 + 5 = -109\ \mathrm{dBm}.$$

If its input third-order intercept is $\mathrm{IIP3} = -10$ dBm, then

$$\mathrm{SFDR} = \tfrac{2}{3}(\mathrm{IIP3} - \mathrm{MDS}) = \tfrac{2}{3}(-10 - (-109)) = 66\ \mathrm{dB}.$$

KEY TAKEAWAYS

- $\mathrm{MDS} = -174 + 10\log B + \mathrm{NF} + \mathrm{SNR}_{min}$.
- $\mathrm{SFDR} = \tfrac{2}{3}(\mathrm{IIP3} - \mathrm{MDS})$.
- Too little early gain hurts NF; too much hurts large-signal handling; AGC extends the instantaneous range.

Exercises

1. Noise floor for $B = 200$ kHz, NF $= 8$ dB ($\text{SNR}_{min} = 0$)?

2. With IIP3 $= 0$ dBm and MDS $= -110$ dBm, find the SFDR.

3. What does AGC improve, and what does it not change?

Further reading: Razavi [8] (ch. 2); Pozar [1] (ch. 10).

8.12 I/Q Modulation and Impairments

LEARNING OBJECTIVES

- Explain quadrature (I/Q) up- and down-conversion.
- Identify how gain/phase imbalance, DC offset, and LO leakage corrupt the signal.
- Relate EVM to SNR.

Almost every modern radio represents its signal as a complex baseband envelope carried on quadrature (I and Q) mixers. Understanding the ideal operation and its impairments is central to receiver and transmitter design.

8.13 Quadrature Up/Down Conversion

An I/Q modulator forms $s(t) = I(t)\cos\omega_c t - Q(t)\sin\omega_c t$, placing independent information on the in-phase and quadrature carriers; the demodulator recovers I and Q by multiplying with cos and sin and low-pass filtering. This is what lets a single RF channel carry a two-dimensional constellation.

8.14 Impairments

Real quadrature paths are imperfect. *Gain and phase imbalance* between I and Q create a spectral image of the signal; *DC offset* adds a tone at the carrier; *LO leakage* radiates carrier feedthrough. Each corrupts the constellation in a characteristic way and must be calibrated out.

8.15 Error Vector Magnitude

EVM is the RMS distance between received and ideal constellation points, expressed as a fraction (or dB) of the signal. For a link limited only by additive noise it ties directly to SNR:

$$\text{EVM}_{\text{rms}} \approx 10^{-\text{SNR}/20}.$$

EVM is the single most useful bench metric of modulation quality, capturing noise, imbalance, distortion, and phase noise together.

WORKED EXAMPLE: EVM FROM SNR

For a link degraded only by additive noise, the error-vector magnitude is $\mathrm{EVM_{rms}} \approx 10^{-\mathrm{SNR}/20}$. At SNR = 25 dB,

$$\mathrm{EVM_{rms}} \approx 10^{-25/20} = 0.056 = 5.6\%.$$

A 64-QAM link typically needs EVM below about 2% (SNR > 34 dB), so this 25 dB link would support only a lower-order constellation.

KEY TAKEAWAYS

- I/Q carries two independent streams: $s = I \cos \omega_c t - Q \sin \omega_c t$.
- Gain/phase imbalance → image; DC offset → carrier tone; LO leakage → carrier feedthrough.
- EVM ties quality to SNR: $\mathrm{EVM} \approx 10^{-\mathrm{SNR}/20}$; it captures all impairments together.

Exercises

1. What EVM corresponds to an SNR of 30 dB?
2. What impairment produces a spectral image of the wanted signal?
3. Why is EVM a more complete quality metric than SNR alone?

Further reading: Razavi [8] (ch. 4); Steer [5].

8.16 Sampling and Data Conversion

LEARNING OBJECTIVES

- Explain aliasing, Nyquist zones, and anti-alias filtering.
- Describe bandpass (undersampling) sampling.
- Compute ADC SNR from resolution and from clock jitter.

Between the analog RF world and the digital signal processor sit the data converters, and their sampling theory sets what the radio can do.

8.17 Nyquist, Aliasing, and Zones

Sampling at f_s replicates the spectrum every f_s; any energy above $f_s/2$ folds (aliases) into the first Nyquist zone. An anti-alias filter must remove out-of-band energy before the ADC.

8.18 Bandpass (Undersampling) Sampling

A band of width B centered at a high frequency need not be sampled at twice its top frequency — only at $f_s \geq 2B$, chosen so the band folds cleanly into a Nyquist zone without overlapping itself. This lets an ADC digitize an IF (or even RF) band directly, the enabling trick of software-defined radio.

8.19 Converter Metrics

An ideal N-bit ADC has quantization-limited SNR

$$\mathrm{SNR} = 6.02\,N + 1.76\ \mathrm{dB}.$$

Real converters fall short (ENOB), limited by distortion (SFDR) and especially by aperture jitter: a clock jitter σ_t caps the SNR at $-20\log_{10}(2\pi f_{in}\sigma_t)$, which dominates at high input frequencies. The DAC on the transmit side produces spectral images that a reconstruction filter must suppress.

WORKED EXAMPLE: ADC SNR AND THE JITTER LIMIT

A 12-bit ADC has an ideal, quantization-limited SNR of

$$\mathrm{SNR} = 6.02(12) + 1.76 = 74\ \mathrm{dB}.$$

But sampling a $f_{in} = 100$ MHz input with $\sigma_t = 1$ ps of clock jitter caps the SNR at

$$-20\log_{10}(2\pi f_{in}\sigma_t) = -20\log_{10}(2\pi\cdot 10^8\cdot 10^{-12}) = 64\ \mathrm{dB}.$$

Here jitter, not quantization, limits performance — a common situation for high-IF sampling.

KEY TAKEAWAYS

- Energy above $f_s/2$ aliases into the first Nyquist zone; filter before the ADC.
- Bandpass sampling needs only $f_s \geq 2B$, folding a high band into a Nyquist zone.
- ADC SNR $= 6.02N + 1.76$ dB; jitter caps it at $-20\log_{10}(2\pi f_{in}\sigma_t)$.

Exercises

1. Ideal SNR of a 10-bit ADC?
2. Jitter-limited SNR for $f_{in} = 1$ GHz, $\sigma_t = 0.5$ ps?
3. Minimum sample rate to bandpass-sample a 5 MHz-wide band?

Further reading: Razavi [8] (ch. 4); Steer [5].

8.20 Software-Defined Radio

LEARNING OBJECTIVES

- Explain the SDR concept and why it is flexible.
- Compare direct-RF-sampling and direct-conversion SDR front ends.
- Describe digital downconversion and decimation.

Software-defined radio pushes the analog-to-digital boundary as close to the antenna as possible and does the rest — filtering, mixing, demodulation — in software.

8.21 The Idea

Instead of a fixed chain of analog filters and mixers, an SDR digitizes a wide band and reconfigures its behavior in DSP. The same hardware can be an FM receiver, a spectrum monitor, or a cellular base station, changed by loading new code.

8.22 Architectures

Two dominate: *direct RF sampling*, where a fast ADC digitizes the RF band outright (using bandpass sampling); and *direct-conversion*, where an analog I/Q front end brings the band to baseband for a slower ADC. A digital downconverter (a numerically controlled oscillator, digital mixer, and decimating CIC/FIR filters) then selects and narrows the channel.

8.23 Processing and Practicalities

The wideband data rate off the ADC is large, so decimation reduces it to the channel bandwidth before heavy processing on an FPGA or general-purpose processor. SDR also enables digital predistortion, digital I/Q-imbalance and DC-offset correction, and rapid retuning — at the cost of ADC/DAC dynamic range and raw data throughput.

WORKED EXAMPLE: DATA RATE AND DECIMATION

An SDR digitizes at $f_s = 100$ Msps with 16-bit samples. Complex (I and Q) sampling produces a raw rate of

$$100 \times 10^6 \times 16 \times 2 = 3.2 \text{ Gbit/s},$$

far too much to process directly. A digital downconverter selects the wanted channel and decimates by 50, leaving 2 Msps (a 2 MHz processing bandwidth) that a modest processor can demodulate in real time.

KEY TAKEAWAYS

- SDR digitizes wide and does filtering/mixing/demod in software — one radio, many waveforms.
- Front ends: direct RF sampling (fast ADC, bandpass sampling) or direct conversion (I/Q to baseband).
- A digital downconverter (NCO + mixer + decimating filter) selects and narrows the channel.

Exercises

1. Raw data rate of a 61.44 Msps, 12-bit, complex ADC stream?
2. What decimation factor turns 100 Msps into a 1 MHz channel?
3. Name one capability SDR gains from doing conversion digitally.

Further reading: Steer [5]; Razavi [8] (ch. 4).

8.24 Link Budgets

LEARNING OBJECTIVES

- Compute free-space path loss.
- Assemble a link budget for received power.
- Evaluate link margin.

A link budget is a systematic accounting of all gains and losses in a communication system from transmitter to receiver. It determines whether a link will achieve adequate signal-to-noise ratio under specified conditions, and by how much margin.

8.24.1 Link Budget Equation

$P_r = P_t + G_t - L_{tx} - FSPL - L_{misc} + G_r - L_{rx}$ *(dBm)*

Then compare received power to the minimum detectable signal (MDS):

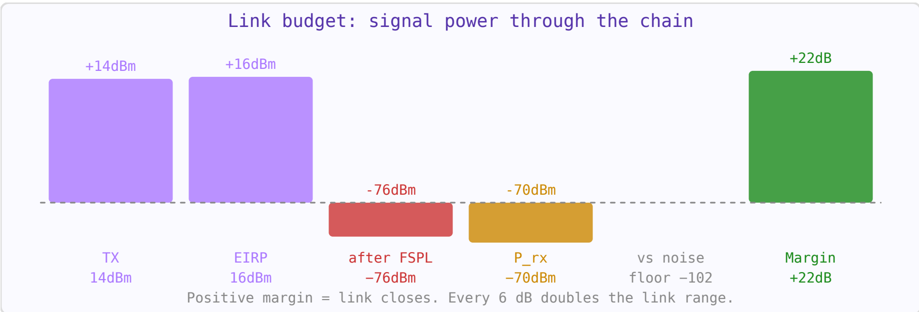


Figure 8.5: Link budget bar chart: TX power flows through gains and losses. Green margin bar = working link.

$MDS = kTB + NF + SNR_{min}$ *(dBm)*

Link margin $= P_r - MDS$. A positive margin means the link works; typical design margins are 3–20 dB depending on the fade margin required.

8.24.2 Free-Space Path Loss

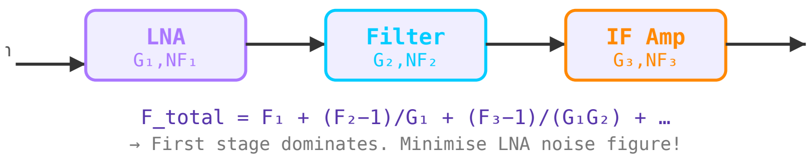


Figure 8.6: Link budget chain: each block contributes gains (antennas) or losses (cables, path loss). The sum determines received power vs noise floor.

$$FSPL = 20\log_{10}(d) + 20\log_{10}(f) + 20\log_{10}\left(\frac{4\pi}{c}\right)$$

At 2.4 GHz, FSPL over 100 m ≈ 80 dB. FSPL grows with distance^2 and frequency^2 — doubling distance adds 6 dB; doubling frequency adds 6 dB.

8.24.3 Additional Loss Mechanisms

- **Multipath fading** — reflections cause constructive/destructive interference. Fast fading margin: 20–30 dB for 99.9% link reliability.
- **Rain attenuation** — significant above 10 GHz; up to several dB/km in heavy rain.
- **Atmospheric absorption** — oxygen absorption peak at 60 GHz (15 dB/km) limits this band to short-range applications.
- **Polarisation mismatch** — misaligned transmit/receive polarisations cause 0–∞ dB loss.

WORKED EXAMPLE: PATH LOSS AND MARGIN

Free-space path loss is FSPL $= 20\log_{10} d + 20\log_{10} f + 32.44$ (d in km, f in MHz). At $d = 10$ km, $f = 2400$ MHz: FSPL $= 20 + 67.6 + 32.44 = 120$ dB. With EIRP $= +30$ dBm and receive gain $+10$ dBi, the received power is $30 + 10 - 120 = -80$ dBm; against a -95 dBm sensitivity the margin is 15 dB.

KEY TAKEAWAYS

- FSPL $= 20\log d + 20\log f + k$; loss grows 20 dB per decade of d or f.
- $P_{rx} = \text{EIRP} + G_{rx} - \text{FSPL} - \text{losses}$.
- Margin $= P_{rx} -$ sensitivity; positive margin closes the link.

Exercises

1. By how many dB does FSPL grow when distance is $\times 10$?
2. EIRP +20 dBm, path loss 100 dB, RX gain +3 dBi. Received power?
3. If sensitivity is -90 dBm, what is the margin?

Further reading: Pozar [1]; Skolnik [17].

8.25 Radar Fundamentals

LEARNING OBJECTIVES

- Apply the radar range equation.
- Relate range to round-trip time.
- Compute Doppler shift.

Radar (Radio Detection And Ranging) uses reflected electromagnetic waves to detect objects and measure their range, velocity, and angular position. The radar equation relates transmitted power to received echo power.

8.25.1 Radar Equation

$$P_r = \frac{P_t G_t G_r \lambda^2 \sigma}{(4\pi)^3 R^4}$$

where σ is the target radar cross section (RCS) in m^2, and R is range. Note the R^4 dependence — radar range halves when range doubles require 16× more transmit power.

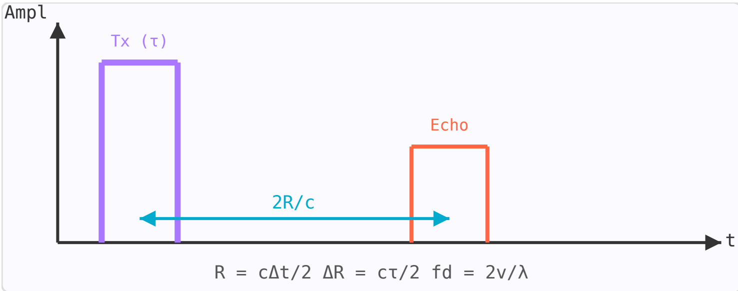


Figure 8.7: Radar timing diagram: transmit pulse (purple, width τ) and echo (orange). Range $\mathrm{R} = \mathrm{c}\Delta\mathrm{t}/2$; resolution $\Delta\mathrm{R} = \mathrm{c}\tau/2$.

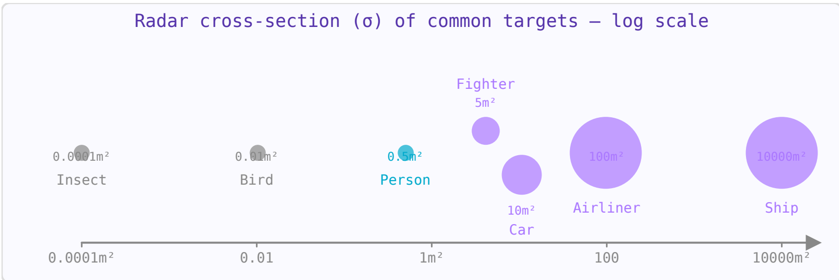


Figure 8.8: Radar cross-section of typical targets. RCS varies enormously — from 10^{-4} m^2 for insects to 10^4 m^2 for ships.

8.25.2 Range Resolution

The ability to distinguish two closely spaced targets in range depends on pulse width τ: $\Delta R = c\tau/2$. Shorter pulses $\to$ better resolution. Pulse compression (chirp or phase coding) allows high range resolution while maintaining high average power.

8.25.3 Doppler Effect

A target moving with radial velocity v_r shifts the echo frequency by the Doppler shift: $f_d = 2v_r/\lambda$. At 10 GHz, a 100 km/h target produces $f_d \approx 1.85$ kHz. Moving target indicator (MTI) and pulse-Doppler processing use this to separate moving targets from stationary clutter.

8.25.4 Radar Cross Section (RCS)

RCS (σ) characterises how effectively a target reflects radar energy back toward the receiver. It depends on target size, shape, material, and radar frequency. Stealth aircraft are designed to minimise RCS through shape (angled surfaces scatter energy away from radar) and radar-absorbing materials.

WORKED EXAMPLE: RANGE FROM ECHO DELAY

Radar range follows from the round-trip time: $R = c\,\Delta t/2$. An echo returning $\Delta t = 20\ \mu$s after transmission puts the target at

$$R = \frac{(3 \times 10^8)(20 \times 10^{-6})}{2} = 3000 \text{ m}.$$

Received power falls as $1/R^4$ (two-way spreading times cross-section), so doubling range cuts echo power by 12 dB.

KEY TAKEAWAYS

- $R = c\Delta t/2$ from the round-trip delay.
- The radar equation gives echo power $\propto 1/R^4$; doubling range loses 12 dB.
- Doppler shift $f_d = 2v/\lambda$ gives radial velocity.

Exercises

1. Echo delay 60 μs: what is the range?
2. Doubling target range changes echo power by how many dB?
3. Doppler shift for $v = 30$ m/s at $\lambda = 3$ cm?

Further reading: Skolnik [17] (ch. 1–2).

8.26 FMCW and Pulsed Radar

LEARNING OBJECTIVES

- Explain the FMCW beat frequency.
- Relate chirp bandwidth to range resolution.
- Understand the range-Doppler picture.

The radar equation gives the received signal power, but the waveform determines how accurately range and velocity can be measured. The three main radar waveform families — pulsed, CW, and FMCW — each make a different tradeoff between range resolution, velocity measurement capability, and hardware complexity.

8.26.1 Pulsed Radar

A pulsed radar transmits short bursts of RF energy and measures the round-trip travel time to determine range. The transmitter is silent while the receiver listens for echoes:

$$R = \frac{c \cdot \tau}{2} \qquad \Delta R = \frac{c}{2B}$$

where τ is the pulse round-trip time, and ΔR is the range resolution set by the signal bandwidth B (not the pulse duration, for pulse-compressed systems). A 1 μs pulse → 150 m range resolution. A 10 MHz bandwidth (pulse-compressed or stepped frequency) → 15 m resolution.

Parameter	Formula	Notes
Max unambiguous range	R_max = c/(2·PRF)	PRF = pulse repetition frequency
Range resolution	ΔR = c/(2B)	B = bandwidth; 150 m/μs for CW pulse
Duty cycle	DC = τ·PRF	Typically 0.001–0.1 for pulsed systems
Average power	P_avg = P_peak · DC	Sets detection range via SNR

8.26.2 CW Radar

Continuous wave radar transmits a single frequency continuously. It cannot measure range (no time reference) but measures velocity extremely well via the Doppler shift:

$$f_D = \frac{2v_r f_0}{c} = \frac{2v_r}{\lambda}$$

where v_r is the radial velocity (positive = approaching). A 10 GHz radar and 100 km/h target: f_D = 2 × 27.8 / 0.03 = 1.85 kHz. Police speed guns and simple motion detectors use CW radar.

8.26.3 FMCW Radar

Frequency-modulated CW (FMCW) combines range and velocity measurement by sweeping the transmit frequency linearly. The echo from a target at range R arrives with a time delay τ = 2R/c, creating a constant "beat frequency" f_b between the current transmit frequency and the received echo:

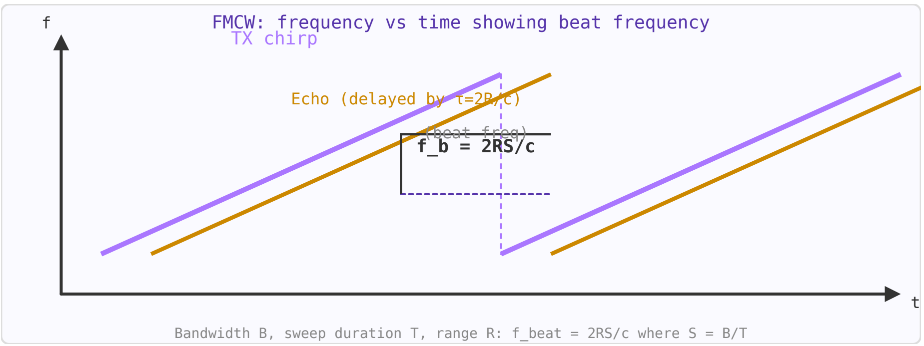


Figure 8.9: FMCW: TX sweeps from f_low to f_high; echo arrives delayed by τ=2R/c; beat frequency f_b $\propto$ range.

$$f_b = \frac{2R \cdot S}{c} \qquad S = \frac{B}{T_{sweep}}$$

where S is the frequency sweep rate (Hz/s), B the sweep bandwidth, and T_sweep the sweep duration. By measuring f_b with an FFT, range is determined directly. Modern automotive radar (77 GHz, 4 GHz bandwidth) achieves ΔR = c/(2×4GHz) = 3.75 cm range resolution.

8.26.4 Range-Doppler Matrix

A moving target creates both a range displacement and a Doppler frequency. In FMCW, velocity causes a phase rotation of the beat signal between sweeps. By taking a 2D FFT (range FFT per sweep, then Doppler FFT across sweeps), the range-Doppler matrix simultaneously shows all targets' range and velocity:

$$\Delta v = \frac{\lambda}{2NT_{sweep}} \qquad v_{max} = \frac{\lambda}{4T_{sweep}}$$

where N is the number of sweeps in the coherent processing interval (CPI).

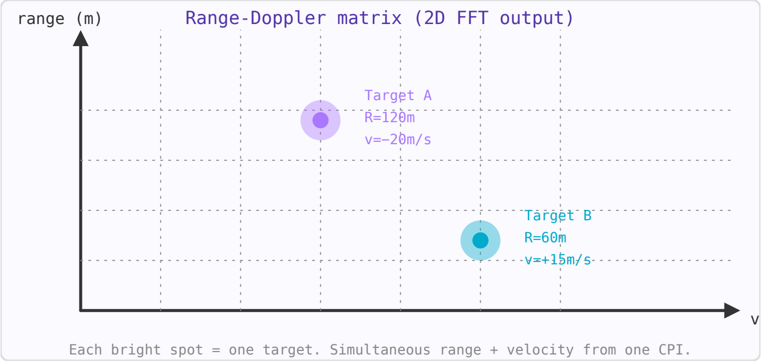


Figure 8.10: Range-Doppler matrix from 2D FFT. Range (vertical axis) and velocity (horizontal) resolved simultaneously.

8.26.5 Ambiguity Function

The radar ambiguity function $\chi(\tau$, f_D) describes the range-Doppler response of a waveform — how much a target at (range τ_0, Doppler f_D$_0$) leaks into adjacent range-Doppler cells. A perfect ambiguity function has a single narrow spike at the origin and zero everywhere else (impossible). The uncertainty principle of radar states: ΔR × Δv ≥ λc/4. Short pulses give good range resolution but poor velocity resolution; long CPI gives good velocity but requires coherence across many sweeps.

8.26.6 FMCW Applications

- **Automotive radar (77 GHz)**: 4 GHz sweep, 3.75 cm range resolution, ±200 km/h velocity, 200 m range. Used for adaptive cruise control, AEB, blind-spot detection.
- **Level measurement**: Industrial FMCW sensors measure tank fill level to mm accuracy over tens of metres.
- **Ground-penetrating radar (GPR)**: 100 MHz–3 GHz sweep through soil to detect buried utilities, unexploded ordnance, or archaeological features.
- **Indoor positioning**: 60 GHz UWB FMCW can locate people to $<$ 10 cm accuracy.

WORKED EXAMPLE: BEAT FREQUENCY AND RESOLUTION

A linear chirp of slope $S = B/T$ produces a beat frequency proportional to range, $f_b = 2RS/c = 2RB/(cT)$. Range resolution depends only on sweep bandwidth: $\Delta R = c/(2B)$. For $B = 150$ MHz,

$$\Delta R = \frac{3 \times 10^8}{2 \times 150 \times 10^6} = 1 \text{ m}.$$

Wider sweeps give finer range resolution; the beat frequency is read off an FFT of the mixed signal.

KEY TAKEAWAYS

- FMCW beat frequency $f_b = 2RS/c$ (slope $S = B/T$) encodes range.
- Range resolution $\Delta R = c/(2B)$ depends only on sweep bandwidth.
- A 2-D FFT over chirps yields the range-Doppler map (range and velocity).

Exercises

1. Range resolution for $B = 1$ GHz?
2. To halve ΔR, how should B change?
3. Does the beat frequency increase or decrease with range?

Further reading: Skolnik [17]; Pozar [1].

8.27 Digital Communication Fundamentals

LEARNING OBJECTIVES

- Relate E_b/N_0, SNR, data rate, and bandwidth.
- State the Shannon capacity limit.
- Explain pulse shaping and spectral efficiency.

Modern RF exists to move bits, so the RF engineer needs the information-theory vocabulary that sets how many bits a channel can carry and how reliably.

8.28 Energy per Bit and SNR

Digital links are compared not by raw SNR but by the energy-per-bit to noise-density ratio E_b/N_0, which normalizes out data rate and bandwidth:

$$\mathrm{SNR} = \frac{E_b}{N_0} \cdot \frac{R_b}{B}.$$

A scheme's bit-error rate is a function of E_b/N_0 alone; for BPSK/QPSK, $\mathrm{BER} = Q(\sqrt{2E_b/N_0})$.

8.29 The Shannon Limit

Shannon's capacity gives the ultimate error-free rate of a bandlimited channel:

$$C = B\ \log_2(1 + \mathrm{SNR}).$$

No modulation or code can exceed it; real systems operate some margin below it and spend coding and bandwidth to approach it.

8.30 Bandwidth, Pulse Shaping, and Efficiency

Sharp symbols would need infinite bandwidth, so transmitters shape pulses (e.g. raised-cosine with roll-off β) to fit a spectral mask while limiting inter-symbol interference. *Spectral efficiency* (bit/s/Hz) rises with constellation order but demands more SNR — the same trade seen throughout digital RF.

WORKED EXAMPLE: SHANNON CAPACITY

A channel has bandwidth $B = 20$ MHz and $\mathrm{SNR} = 100$ (20 dB). Its capacity is

$$C = B \log_2(1 + \mathrm{SNR}) = 20 \times 10^6\ \log_2(101) = 20 \times 10^6 \times 6.66 = 133\ \mathrm{Mbit/s}.$$

A practical link runs a few dB below this, choosing a modulation and code whose required E_b/N_0 fits the available SNR.

KEY TAKEAWAYS

- SNR $= (E_b/N_0)(R_b/B)$; BER depends on E_b/N_0, e.g. QPSK $= Q(\sqrt{2E_b/N_0})$.
- Shannon capacity $C = B\log_2(1 + \mathrm{SNR})$ is the unbeatable ceiling.
- Pulse shaping fits the spectral mask; higher-order modulation trades SNR for spectral efficiency.

Exercises

1. Shannon capacity of a 5 MHz channel at SNR $= 31$ (15 dB)?
2. Does BER depend on absolute SNR or on E_b/N_0?
3. Why can't symbols be made arbitrarily sharp in time?

Further reading: Proakis [28]; Sklar [29].

8.31 Coding, Spread Spectrum and OFDM

LEARNING OBJECTIVES

- Explain forward error correction and coding gain.
- Compute spread-spectrum processing gain.
- Describe OFDM and its cyclic prefix and PAPR trade-offs.

Getting close to the Shannon limit takes more than a good constellation: it takes coding, and often spread-spectrum or multicarrier techniques matched to the channel.

8.32 Forward Error Correction

FEC adds structured redundancy so the receiver can correct errors without retransmission. Block, convolutional, turbo, and LDPC codes each buy a *coding gain* — the reduction in required E_b/N_0 for a target BER — at the cost of overhead and decoder complexity. Modern LDPC/turbo codes approach the Shannon limit within a fraction of a dB.

8.33 Spread Spectrum

Direct-sequence spread spectrum multiplies the data by a much faster pseudo-random chip sequence, spreading the signal over a wide band. The receiver despreads it, gaining a *processing gain*

$$G_p = \frac{R_{chip}}{R_{data}},$$

which suppresses narrowband interference and enables code-division multiple access. Frequency hopping spreads by rapidly changing carrier instead.

8.34 OFDM

Orthogonal frequency-division multiplexing splits a high-rate stream across many narrow, orthogonal subcarriers, each seeing a nearly flat channel. A cyclic prefix longer than the channel delay spread eliminates inter-symbol interference, making equalization trivial — the reason

OFDM underlies Wi-Fi, LTE, and 5G. Its cost is a high peak-to-average power ratio (PAPR) that stresses the power amplifier.

WORKED EXAMPLE: SPREADING PROCESSING GAIN

A direct-sequence system spreads a $R_{data} = 10$ kbit/s stream with a $R_{chip} = 10$ Mchip/s code. The processing gain is

$$G_p = \frac{R_{chip}}{R_{data}} = \frac{10 \times 10^6}{10 \times 10^3} = 1000 = 30 \text{ dB}.$$

Despreading concentrates the wanted signal while spreading interference, so a narrowband jammer is suppressed by ~ 30 dB relative to the signal.

KEY TAKEAWAYS

- FEC (block, convolutional, turbo, LDPC) buys coding gain; LDPC/turbo near the Shannon limit.
- DSSS processing gain $G_p = R_{chip}/R_{data}$ suppresses narrowband interference and enables CDMA.
- OFDM: orthogonal subcarriers + cyclic prefix beat delay spread, at the cost of high PAPR.

Exercises

1. Processing gain for $R_{chip} = 1$ Mchip/s, $R_{data} = 2$ kbit/s?
2. What must the OFDM cyclic prefix be longer than?
3. Name the main power-amplifier drawback of OFDM.

Further reading: Sklar [29]; Goldsmith [30].

Chapter 9

Measurement, EMC and RF Safety

9.1 RF Measurement and Instrumentation

LEARNING OBJECTIVES

- Explain RBW/VBW and the spectrum-analyzer noise floor.
- Measure noise figure by the Y-factor method.
- Describe VNA calibration and time-domain reflectometry.

RF engineering is empirical: designs are proven on the bench, and the instruments have characteristic strengths and traps.

9.2 Spectrum Analyzer

A spectrum analyzer shows power versus frequency. Its resolution bandwidth (RBW) sets both the frequency resolution and the displayed noise floor (narrower RBW lowers the floor); the video bandwidth (VBW) smooths the trace. Swept analyzers trade speed for span; FFT/real-time analyzers capture transients.

9.3 Sources, Power, and Noise

Signal and vector-signal generators provide clean or modulated stimuli; power meters (diode or thermal) measure absolute level. Noise figure is measured by the *Y-factor* method: an excess-noise-ratio (ENR) source is switched on and off, and the hot/cold power ratio Y gives

$$F = \frac{\mathrm{ENR}}{Y - 1}.$$

9.4 Network Analysis and Time Domain

The VNA measures S-parameters after a SOLT or TRL calibration that moves the reference plane to the device and removes systematic error. Time-domain reflectometry (TDR) locates impedance discontinuities along a line by their echo delay, and oscilloscopes/logic analyzers handle the digital side of an SDR.

WORKED EXAMPLE: Y-FACTOR NOISE FIGURE

A noise source with ENR = 15 dB (= 31.6 linear) is switched on and off at a device's input, giving a hot/cold power ratio $Y = 8$ dB (= 6.31). The noise factor is

$$F = \frac{\text{ENR}}{Y-1} = \frac{31.6}{6.31-1} = 5.95,$$

i.e. NF = 7.75 dB. A larger Y (from a lower-noise device) gives a lower, more accurate NF reading.

KEY TAKEAWAYS

- Narrower RBW lowers the analyzer noise floor and sharpens resolution but slows the sweep.
- Y-factor: $F = \text{ENR}/(Y-1)$ from the hot/cold power ratio.
- VNA needs SOLT/TRL calibration; TDR locates discontinuities by echo delay.

Exercises

1. With ENR = 15 dB and $Y = 10$ dB, find the noise figure.
2. What happens to a spectrum analyzer's noise floor when you narrow the RBW?
3. What does a TDR measurement locate?

Further reading: Pozar [1] (ch. 4); Steer [5].

9.5 RF Simulation and Design Flow

LEARNING OBJECTIVES

- Distinguish linear, harmonic-balance, transient, and SPICE circuit simulation.
- Choose among MoM, FEM, and FDTD electromagnetic solvers.
- Outline the schematic-to-measurement design flow and meshing rule.

No modern RF design reaches hardware without simulation. Knowing which tool models what — and its limits — is part of the craft.

9.6 Circuit Simulation

Linear (S-parameter) simulation gives frequency response instantly for matching and filters. Nonlinear behavior needs *harmonic balance* (steady-state response to periodic drive — ideal for amplifiers, mixers, and oscillators) or *transient/envelope* methods; SPICE handles time-domain and start-up. Each answers a different question about the same circuit.

9.7 Electromagnetic Simulation

When layout parasitics or radiation matter, EM solvers compute the fields directly: method-of-moments (planar structures, e.g. microstrip), finite-element (arbitrary 3-D, e.g. connectors and packages), and FDTD (broadband, transient). They are accurate but slow, so they verify critical structures rather than whole boards.

9.8 The Design Flow

A typical flow: write the spec, design and simulate the schematic, lay it out, EM- verify the critical nets, fabricate, then measure and tune. Models are de-embedded against measurements, and Monte-Carlo/corner analysis checks yield against component and process tolerances (design for manufacture). A mesh cell of roughly $\lambda/10$ to $\lambda/20$ is the rule of thumb for EM accuracy.

WORKED EXAMPLE: EM MESH CELL SIZE

An EM solver models a microstrip structure at $f = 10$ GHz ($\lambda_0 = 30$ mm in air). A common accuracy rule is a mesh cell no larger than about $\lambda/10$:

$$\Delta \leq \frac{\lambda_0}{10} = 3 \text{ mm},$$

and finer ($\lambda/20 = 1.5$ mm) where fields vary rapidly (edges, gaps). Too coarse a mesh mis-predicts resonances; too fine wastes hours of compute.

KEY TAKEAWAYS

- Linear S-param for response; harmonic balance for nonlinear steady state; SPICE/-transient for time domain.
- EM solvers: MoM (planar), FEM (3-D), FDTD (broadband); slow, so they verify critical structures.
- Flow: spec $\to$ schematic sim $\to$ layout $\to$ EM verify $\to$ measure/tune; mesh $\sim \lambda/10$–$\lambda/20$.

Exercises

1. Which simulation type best analyzes a mixer's conversion loss and spurs?
2. Recommended EM mesh cell at 5 GHz in air (using $\lambda/10$)?
3. Why EM-simulate only critical nets rather than the whole board?

Further reading: Steer [5]; Pozar [1].

9.9 EMC Shielding Effectiveness

LEARNING OBJECTIVES

- Decompose shielding effectiveness into reflection and absorption.
- Explain aperture leakage.
- Understand cable-shield transfer impedance.

Shielding effectiveness (SE) quantifies how much an enclosure or barrier reduces the electromagnetic field at a point. It is expressed in dB as the ratio of the field without the shield to the field with it. Good shielding is essential for meeting regulatory limits (FCC Part 15, CE/ETSI), preventing interference between circuits, and protecting sensitive receivers from external interference.

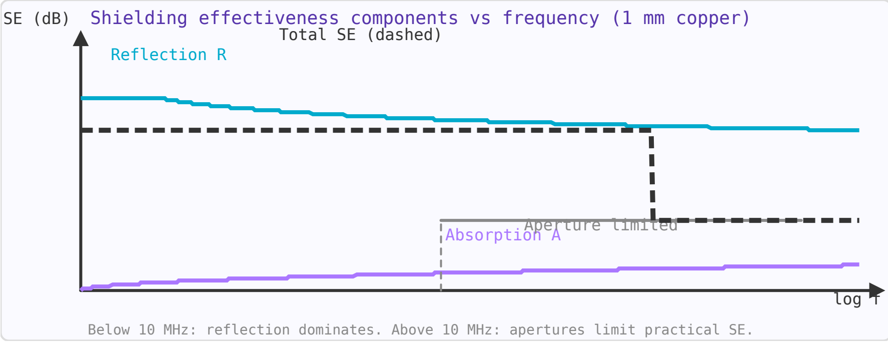


Figure 9.1: Shielding effectiveness components for 1 mm copper: reflection (blue), absorption (purple), and aperture limit (grey).

9.9.1 Shielding Mechanisms

A metallic shield attenuates fields by three mechanisms:

$$SE = R + A + B \quad [\text{dB}]$$

- **Reflection (R)**: At the air-metal interface, impedance mismatch reflects waves. Significant even for thin shields. Dominant at low frequencies for electric fields.
- **Absorption (A)**: Energy is dissipated as the wave propagates through the metal. $A = 20 \cdot \log_{10}(e) \times t/\delta = 8.686 \times t/\delta$ (dB), where t is shield thickness and δ is skin depth. Increases with frequency.
- **Multiple reflections (B)**: For thin shields ($t < \delta$), waves bounce between the two surfaces, reducing the net absorption. B is negative and usually negligible when $A > 15$ dB.

9.9.2 Shielding Effectiveness vs Frequency

Frequency	δ (Cu)	A (1mm Cu)	R (plane wave)	Total SE (1 mm Cu)
1 kHz	2.1 mm	4 dB	132 dB	~136 dB
100 kHz	0.21 mm	40 dB	112 dB	~152 dB
1 MHz	66 μm	131 dB	102 dB	>200 dB (limited by seams)
1 GHz	2.1 μm	>400 dB	62 dB	Limited by apertures

Above ~10 MHz, even a very thin copper sheet is electromagnetically opaque — the limitation is *apertures* (holes, slots, seams), not the metal itself.

9.9.3 Aperture Leakage

Holes and slots in shielding enclosures are the dominant leakage path at high frequencies. A rectangular aperture of length L acts as a half-wave resonant slot antenna at $f = c/(2L)$. Below resonance, SE decreases by 20 dB/decade as frequency increases:

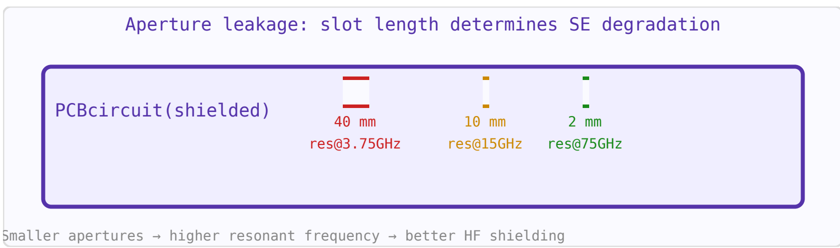


Figure 9.2: Aperture resonant frequency = c/(2L). Smaller holes give better HF shielding at the cost of airflow.

- A 3 mm slot resonates at 50 GHz — fine for cellular but useless at mmWave.
- Replace single large holes with many small holes of the same total area: N holes of diameter d instead of one hole of diameter D=d$\sqrt{}$N gives ~20·$\log_{10}$($\sqrt{}$N) more SE.
- Honeycomb waveguide below cutoff arrays provide ventilation with >100 dB SE.

9.9.4 Seams and Joints

The weakest point of most shields is where panels join. A seam with periodic fasteners (screws every L mm) acts like an array of slots of length L. For good HF shielding, seams must be:

- Soldered or welded for maximum conductivity
- EMI gaskets (beryllium copper finger stock, wire mesh, conductive foam) for removable panels
- Screws spaced to $\lambda/10$ at the highest frequency of concern
- Conductive paint or plating on plastic enclosures (provides 30–50 dB SE)

9.9.5 Cable Shield Transfer Impedance

Cable shields leak energy through their transfer impedance Z_T (Ω/m) — the ratio of voltage induced on the inner conductor to the current flowing on the outer shield:

$$Z_T = \frac{V_{induced}/\text{unit length}}{I_{shield}} \quad [\Omega/\text{m}]$$

Cable type	Z_T @ 10 MHz	SE @ 10 MHz
RG-58 (single braid)	10–50 mΩ/m	50–60 dB
RG-214 (double braid)	1–5 mΩ/m	70–80 dB
Semi-rigid coax	$<$ 0.1 mΩ/m	$>$ 100 dB
Foil + braid	2–10 mΩ/m	60–75 dB

Always connect cable shields at both ends for RF shielding. Single-end grounding (for audio hum prevention) leaves the shield as an antenna above a few kHz.

9.9.6 PCB Shielding Cans

Stamped metal shielding cans solder directly to a PCB ground pour to isolate sensitive circuits (RF front-ends, oscillators). Key design rules: no gaps $> \lambda/20$ at the highest frequency; soldered continuously or with fences of vias spaced $< \lambda/20$; include a test access point (small hole or removable lid) for production testing.

WORKED EXAMPLE: A SLOT AS A LEAKAGE ANTENNA

A shield's effectiveness is set less by wall thickness than by its apertures. A slot radiates efficiently once its length approaches a half wavelength; leakage becomes severe near $f \approx c/(2L)$. A 30 cm seam leaks strongly around

$$f = \frac{3 \times 10^8}{2 \times 0.30} = 500 \text{ MHz}.$$

The rule: keep the longest slot dimension $\ll \lambda/2$ — many small holes beat one long slot of equal area.

KEY TAKEAWAYS

- Shielding effectiveness = reflection + absorption loss; apertures dominate real enclosures.
- A slot leaks strongly as its length nears $\lambda/2$ ($f \approx c/2L$); slot length, not area, governs leakage.
- Cable shields couple via transfer impedance; pigtails ruin high-frequency shielding.

Exercises

1. At what frequency does a 15 cm slot become a strong radiator?
2. For fixed total open area, are many small holes or one long slot better?
3. What ruins a cable shield's HF performance at the connector?

Further reading: Ott [18] (ch. 6); Paul [19].

9.10 RF Safety and Human Exposure

LEARNING OBJECTIVES

- Compute power density versus distance.
- Apply ICNIRP/FCC exposure limits.
- Find a minimum safe distance.

RF electromagnetic fields are non-ionising: their photon energies (eV scale at microwave frequencies) are far too low to break chemical bonds or directly damage DNA. However, they can heat tissue through resistive losses, and at high power densities near high-power transmitters, this heating is the primary safety concern.

9.10.1 Exposure Limits: ICNIRP and FCC

Two sets of exposure guidelines dominate internationally:

- **ICNIRP (International Commission on Non-Ionizing Radiation Protection)**: Used in Europe and most of the world. Sets reference levels in terms of power density (W/m^2) and electric field strength (V/m).
- **FCC OET-65 (US)**: Used in the United States. Defines Maximum Permissible Exposure (MPE) for both occupational/controlled and general public/uncontrolled environments.

Frequency	**ICNIRP general public (W/m^2)**	**FCC uncontrolled (W/m^2)**
100 MHz	2	0.2
400 MHz	2	0.57
1 GHz	5	1.0
2.45 GHz	10	1.0
10 GHz	10 (up to 300 GHz)	5

9.10.2 Specific Absorption Rate (SAR)

SAR measures the rate at which RF energy is absorbed per unit mass of tissue:

$$SAR = \frac{\sigma |E|^2}{\rho} \quad [\mathrm{W/kg}]$$

Whole-body average SAR limits:

- ICNIRP / IEEE: 0.08 W/kg whole-body (general public); 0.4 W/kg (occupational)
- Localised head/torso SAR: 2 W/kg averaged over any 10 g of tissue
- Localised limbs: 4 W/kg averaged over any 10 g of tissue

Mobile phones are tested against a 1.6 W/kg limit (FCC) or 2.0 W/kg limit (ICNIRP) averaged over 10 g of tissue. These limits include a safety factor of ~50× relative to the threshold for measurable biological effects from heating.

9.10.3 Minimum Safe Distance from Transmitters

In the far field, power density decreases as $1/R^2$. The minimum safe distance for an EIRP of P_EIRP watts to achieve power density S (W/m^2) at the limit:

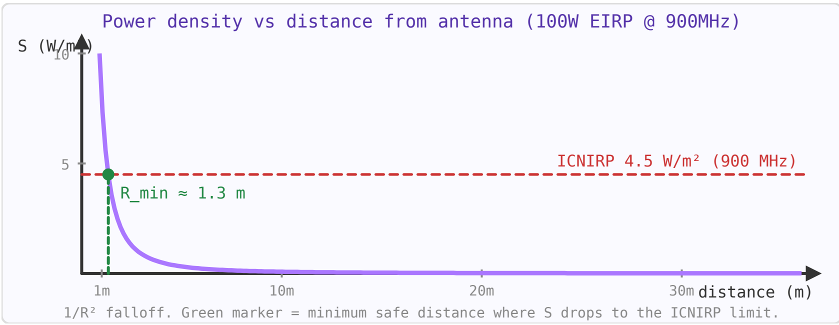


Figure 9.3: Power density falls as $1/R^2$ in the far field. For 100 W EIRP at 900 MHz, the ICNIRP 2 W/m^2 limit is reached at ~2 m.

$$R_{min} = \sqrt{\frac{P_{EIRP}}{4\pi S}}$$

For a 100 W EIRP omnidirectional transmitter at 900 MHz (ICNIRP 2 W/m^2 limit): R_min = $\sqrt{}(100/(4\pi\times2))$ = 2.0 m. For a high-gain antenna system with 1 kW EIRP: R_min = 6.3 m.

9.10.4 Near-Field vs Far-Field Exposure

The transition from near field to far field occurs at the Fraunhofer distance: R_ff = $2D^2/\lambda$ (D = antenna aperture). In the near field, power density does not follow the $1/R^2$ law — it can be higher or lower depending on the antenna. Near-field SAR calculations require full EM simulation (FDTD or FEM), not the simple far-field formula.

Practical near-field exposure situations:

- Holding a mobile phone against the head (3–10 mm from antenna)
- Wearing a smartwatch or fitness tracker
- Working near a base station antenna at maintenance distance
- Industrial RF heating equipment (RF sealers, dielectric heaters)

9.10.5 Practical RF Safety Guidelines

- Never stand in the main beam of a high-gain antenna within the minimum safe distance — be aware that gain increases the EIRP dramatically.
- RF safety limits apply to time-averaged power over 6 minutes (ICNIRP) or 30 minutes (FCC). Pulsed or intermittent transmitters use their duty cycle to calculate the average.
- Reflective surfaces can increase local power density above the direct beam level — consider reflected paths in safety assessments.
- No safety concern at Wi-Fi, Bluetooth, or mobile phone power levels at normal use distances (metres from the source).
- Implanted medical devices (pacemakers, cochlear implants) may be sensitive to strong RF fields at specific frequencies — maintain 20–30 cm distance from transmitters and check device specifications.

Note: This page covers general RF exposure. For MRI-specific SAR limits and the body coil SAR calculation, see the MRI SAR article.

WORKED EXAMPLE: MINIMUM SAFE DISTANCE

In the far field the power density from an EIRP source is $S = \text{EIRP}/(4\pi R^2)$. Setting S to the limit S_{lim} and solving,

$$R_{min} = \sqrt{\frac{\text{EIRP}}{4\pi S_{lim}}}.$$

For EIRP = 100 W and the ICNIRP general-public limit at 900 MHz ($f/200 = 4.5\ \text{W/m}^2$): $R_{min} = \sqrt{100/(4\pi \cdot 4.5)} = 1.33$ m.

KEY TAKEAWAYS

- Far-field power density $S = \text{EIRP}/(4\pi R^2)$ falls as $1/R^2$.
- ICNIRP general-public limit is frequency-dependent (e.g. $f/200\ \text{W/m}^2$, 400 MHz–2 GHz).
- Minimum safe distance $R_{min} = \sqrt{\text{EIRP}/(4\pi S_{lim})}$.

Exercises

1. Power density from 100 W EIRP at 5 m?
2. ICNIRP general-public limit at 1800 MHz?
3. Doubling distance changes power density by how many dB?

Further reading: ICNIRP [20]; Balanis [14].

Chapter 10

MRI Physics and Coil Engineering

10.1 NMR Basics

LEARNING OBJECTIVES

- State the Larmor relationship.
- Relate the gyromagnetic ratio to resonance frequency.
- Understand net magnetization and precession.

Nuclear Magnetic Resonance (NMR) is the physical phenomenon underlying MRI. It exploits the quantum mechanical property of nuclear spin to create detailed maps of atomic nuclei in a sample — in MRI, predominantly the ^{1}H (proton) nuclei in water and fat.

10.1.1 Nuclear Spin

Nuclei with an odd number of protons or neutrons possess a net quantum mechanical spin and an associated magnetic dipole moment $\boldsymbol{\mu}$. In a static magnetic field B_0, these moments precess around the field axis at the Larmor frequency:

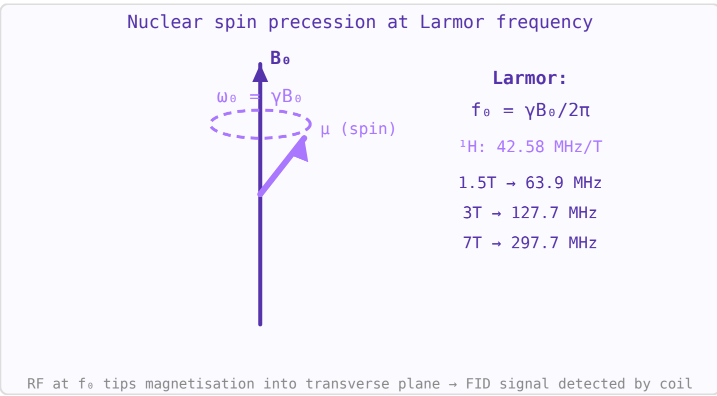


Figure 10.1: Larmor precession: spin μ precesses around B_0 at $\omega_0 = \gamma B_0$. RF excitation at this resonance frequency creates NMR signal.

$$f_0 = \frac{\gamma B_0}{2\pi}$$

where γ is the gyromagnetic ratio (for ^{1}H: $\gamma/2\pi = 42.577$ MHz/T). At 3 T, ^{1}H precesses at ~128 MHz — in the RF range.

10.1.2 Equilibrium Magnetisation

In a static field B_0, spins align preferentially along the field, creating a bulk magnetisation M_0 parallel to B_0 (the z-axis). This magnetisation is the signal source in MRI.

10.1.3 RF Excitation

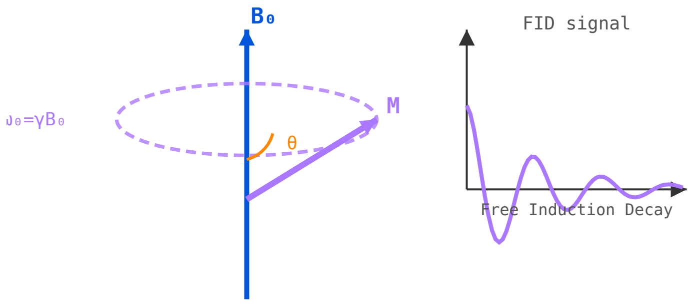


Figure 10.2: Left: M precessing around B_0 at $\omega_0=\gamma B_0$. Right: free induction decay (FID) signal after a 90° pulse.

A radiofrequency pulse at the Larmor frequency (the B1 field) tips M_0 away from z. A 90° pulse rotates M_0 entirely into the transverse (xy) plane where it precesses and induces a signal in the receive coil. A 180° pulse inverts M_0.

10.1.4 Bloch Equations and Relaxation

After excitation, the transverse magnetisation decays (T2 relaxation — spin-spin interactions) while the longitudinal magnetisation recovers (T1 relaxation — spin-lattice interactions). Different tissues have different T1/T2 values, creating the soft-tissue contrast that makes MRI so powerful. Typical values in brain tissue: T1 ≈ 900 ms, T2 ≈ 80 ms at 3 T.

WORKED EXAMPLE: LARMOR FREQUENCY

Nuclear spins precess at $f_0 = \gamma B_0$; for protons $\gamma = 42.58$ MHz/T. At $B_0 = 1.5$ T,

$$f_0 = 42.58 \times 1.5 = 63.9 \text{ MHz};$$

at 3 T it is 127.7 MHz. The RF coil must be tuned to this frequency to excite and receive the spins.

KEY TAKEAWAYS

- Larmor: $f_0 = \gamma B_0$; for protons $\gamma = 42.58$ MHz/T.
- Net magnetization aligns with B_0; an on-resonance B_1 pulse tips it into the transverse plane.
- Higher B_0 raises frequency (and signal), demanding higher-frequency coils.

Exercises

1. Proton Larmor frequency at 3 T?
2. At what field is the proton frequency 300 MHz?

3. Does 7 T need a higher- or lower-frequency coil than 1.5 T?

Further reading: Nishimura [23] (ch. 2); Haacke [22].

10.2 MRI Hardware

LEARNING OBJECTIVES

- Identify the main magnet, gradients, and RF chain.
- Understand field strength versus SNR.
- Recognize shielding and shimming needs.

An MRI scanner is a complex system combining superconducting magnet technology, precision gradient electronics, and RF engineering. Each subsystem must work in concert to produce a diagnostic image.

10.2.1 Main Magnet

Modern clinical MRI scanners use superconducting solenoid magnets operating at liquid helium temperature (4 K) to achieve field strengths of 1.5–3 T (clinical) or 7–14 T (research). The magnet must have exceptional homogeneity — typically <1 ppm over the imaging volume — achieved by shimming with additional coils or ferromagnetic pieces.

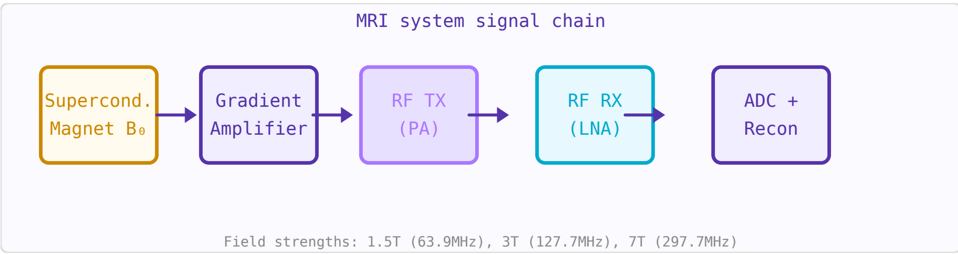


Figure 10.3: MRI system: superconducting magnet, gradient amplifiers, RF transmit power amp, LNA receiver, and digital reconstruction.

10.2.2 Gradient Coils

Three sets of gradient coils (Gx, Gy, Gz) superimpose linear magnetic field gradients on B_0. This makes the Larmor frequency spatially dependent, encoding spatial information. Gradient switching generates the characteristic loud knocking sound during an MRI scan. Gradient amplitude: 20–80 mT/m; slew rate: up to 200 T/m/s.

10.2.3 RF Transmit System

A high-power RF amplifier (typically 1–35 kW) drives the body or volume coil to produce the B1 excitation field at the Larmor frequency. The transmitter must deliver precise pulse shapes and calibrate the flip angle across the imaging volume. Frequency: 64 MHz (1.5 T), 128 MHz (3 T), 300 MHz (7 T).

10.2.4 RF Receive System

Receive coils (surface coils or phased arrays) detect the tiny precessing magnetisation signal. The signal is typically on the order of microvolts. It passes through a low-noise preamplifier

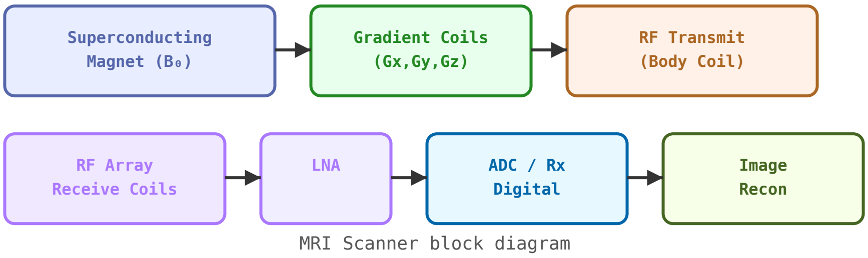


Figure 10.4: MRI scanner block diagram showing magnet, gradient coils, RF transmit/receive chains, and image reconstruction.

(LNA), is downconverted to baseband, digitised, and reconstructed via Fourier transform into an image.

10.2.5 Shielding

The RF system operates inside an RF-shielded room (Faraday cage) to prevent external interference from contaminating the signal. The magnet's fringe field is contained by passive or active shielding to protect nearby equipment and comply with safety zones.

WORKED EXAMPLE: SNR SCALING WITH FIELD

MRI signal-to-noise rises steeply with static field. In the body-noise-dominated regime SNR grows roughly linearly with B_0, so moving from 1.5 T to 3 T roughly doubles SNR. That gain drives high-field imaging — at the cost of higher RF frequency (128 vs. 64 MHz), stronger susceptibility artefacts, and higher SAR (which scales as B_0^2).

KEY TAKEAWAYS

- The magnet sets B_0 (and frequency); gradients encode space; the RF chain excites and receives.
- Higher B_0 improves SNR but raises frequency, SAR ($\propto B_0^2$), and susceptibility artefacts.
- Shimming homogenizes B_0; RF and gradient shielding contain stray fields.

Exercises

1. Roughly how does SNR change from 1.5 T to 3 T?
2. How does whole-body SAR scale with B_0?
3. Which improves with higher field: SNR or susceptibility artefacts?

Further reading: Nishimura [23]; Haacke [22] (ch. 27).

10.3 MRI Gradient Coils

LEARNING OBJECTIVES

- Relate gradient strength to spatial encoding.
- Define slew rate and its PNS limit.
- Understand eddy-current effects.

Gradient coils are the mechanism by which MRI encodes spatial information into the NMR signal. They produce linear spatial variations in the B_0 field (dB_z/dx, dB_z/dy, dB_z/dz) that allow slice selection, frequency encoding, and phase encoding. Their performance — gradient strength and switching speed — directly determines image resolution, scan time, and what pulse sequences are possible.

10.3.1 Why Gradients Are Needed

In a uniform B_0 field, all protons resonate at the same Larmor frequency $f_0 = \gamma B_0$. To determine where a signal comes from, we need to make f_0 vary with position. A gradient G_z adds a field increment G_z·z, so protons at position z resonate at $f = \gamma(B_0 + \text{G\_z} \cdot z)$. By sweeping the gradient and detecting the frequency spectrum, we reconstruct the 1D spin distribution — repeated in three directions gives 3D imaging.

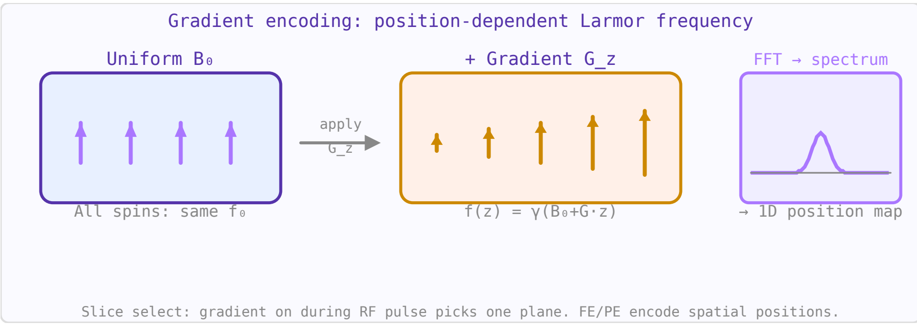


Figure 10.5: Gradient fields make Larmor frequency position-dependent. FFT of the NMR signal then maps frequencies to spatial positions.

10.3.2 Key Performance Metrics

Parameter	Symbol	Typical values	Impact
Gradient strength	G_max	20–80 mT/m	Minimum slice thickness, resolution, diffusion b-value
Slew rate	SR = dG/dt	100–200 T/m/s	Echo time (TE), echo spacing, sequence speed
Rise time	t_rise = G_max/SR	0.3–1 ms	Gradient echo timing
Linearity	% deviation from ideal	< 5% over DSV	Geometric distortion in images
Efficiency	η = G_max/I_max	5–30 mT/m/A	Power amplifier requirements
DSV (diameter spherical volume)		40–50 cm	Field of view with acceptable linearity

10.3.3 Coil Geometries

- **Maxwell pair**: Two circular loops with opposing currents, separated by $\sqrt{3}\cdot r$. Produces a uniform dB_z/dz gradient along the bore axis (z-gradient). Used in early resistive MRI systems.
- **Golay coil (saddle coil)**: Four curved conductor arcs producing transverse gradients (x or y). The Golay geometry minimises higher-order field terms by careful choice of arc angle (120°) and axial spacing.
- **Fingerprint / distributed winding**: Modern gradients are designed numerically (target field method or streamline method) to optimise uniformity, efficiency, and acoustic noise simultaneously.

10.3.4 Eddy Currents

Rapidly switching gradient fields induce eddy currents in the cryostat bore, former, and any nearby conducting structures. These eddy currents create opposing fields that distort the desired gradient waveform — causing B_0 field errors (shift of resonance frequency), gradient pre-emphasis distortion, and image artefacts (ghosting, distortion in EPI). Solutions:

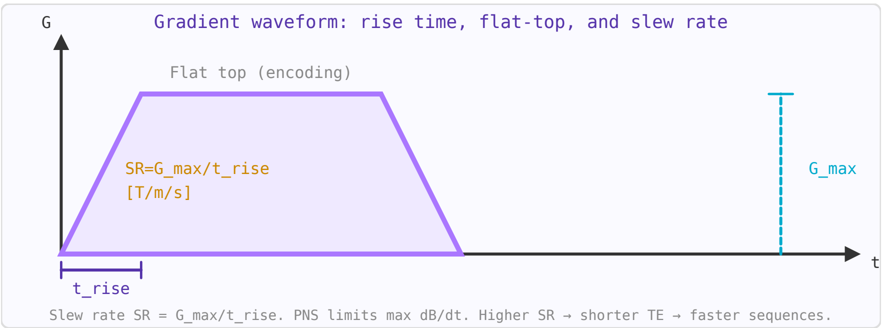


Figure 10.6: Trapezoidal gradient waveform. Rise time t_rise = G_max/SR limits shortest achievable echo time.

- **Actively shielded gradients**: A second "shielding" winding around the primary gradient cancels the field outside the gradient coil set, dramatically reducing eddy currents in the cryostat.
- **Gradient pre-emphasis**: The gradient amplifier applies a pre-distorted drive waveform that compensates for the expected eddy current decay.
- **Slotted/segmented formers**: Break conducting paths to reduce eddy current loops.

10.3.5 Peripheral Nerve Stimulation (PNS)

Very fast gradient switching induces electric fields in the patient's body (by Faraday's law: $E \propto dB/dt$). At high enough dB/dt, these induced fields can directly stimulate peripheral nerves, causing uncomfortable or painful muscular twitching. IEC 60601-2-33 defines PNS limits based on the rheobase (minimum current for stimulation with infinite pulse duration) and the chronaxie (pulse duration for 2× rheobase threshold). Modern scanners monitor dB/dt in real time and limit gradient slew rates to stay within PNS limits, which constrains fast pulse sequences like EPI and diffusion.

10.3.6 Acoustic Noise

Gradient coils experience Lorentz forces (F = IL × B_0) when current flows through them in the main field. Rapidly switching these forces at audio frequencies (typically 1–4 kHz for EPI) causes the coil former to vibrate, generating the characteristic loud banging noise of MRI. Modern scanners use acoustic dampening foam, vacuum-bonded coil formers, and specific pulse sequence designs to reduce peak sound pressure levels below the IEC limit of 140 dB(A) at the patient's ears.

WORKED EXAMPLE: GRADIENT SLEW RATE

A gradient must ramp quickly to keep echo times short. Slew rate is SR = G_{max}/t_{rise}. Ramping to $G_{max} = 40$ mT/m in $t_{rise} = 200$ μs gives

$$\text{SR} = \frac{0.04}{200 \times 10^{-6}} = 200 \text{ T/m/s}.$$

Faster ramps shorten sequences but are capped by peripheral nerve stimulation (PNS), which responds to dB/dt.

KEY TAKEAWAYS

- Gradients make resonance position-dependent: $f(z) = \gamma(B_0 + Gz)$.
- Slew rate = G_{max}/t_{rise}; higher slew shortens TE and TR.
- PNS (from dB/dt) and eddy currents cap how fast gradients can switch.

Exercises

1. Slew rate to reach 30 mT/m in 150 μs?
2. What physiological effect limits maximum slew rate?
3. Faster gradient switching allows shorter what?

Further reading: Nishimura [23] (ch. 5); Haacke [22].

10.4 MRI Pulse Sequences

LEARNING OBJECTIVES

- Distinguish spin echo, gradient echo, and EPI.
- Relate TR/TE to contrast.
- Understand k-space traversal.

A pulse sequence is the pattern of RF pulses and gradient waveforms used to acquire MRI data. Different sequences exploit T1, T2, and proton density contrast to highlight different tissue properties.

10.4.1 Spin Echo (SE)

A 90° excitation pulse followed by a 180° refocusing pulse after time TE/2 creates an echo at time TE. The 180° pulse reverses static field inhomogeneity dephasing but not true T2 decay, making SE sensitive to tissue T2. Repetition time TR controls T1 weighting. SE is robust to B0 inhomogeneity.

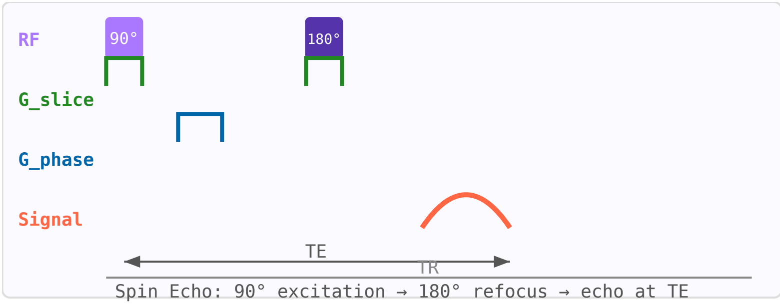


Figure 10.7: Spin echo pulse sequence: 90° excitation, 180° refocus at TE/2, echo at TE. TR sets T1 weighting.

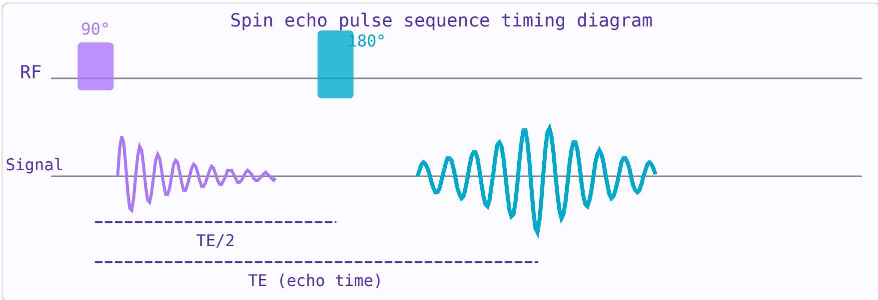


Figure 10.8: Spin echo: 90° excitation creates FID (purple); 180° refocusing pulse at TE/2 forms a spin echo at TE (cyan).

10.4.2 Gradient Echo (GRE)

Uses a flip angle $< 90°$ and a gradient reversal (instead of 180° RF pulse) to form the echo. Faster than SE but sensitive to B0 inhomogeneity (T2* rather than T2). Used for fast imaging, functional MRI (BOLD), and angiography.

10.4.3 Inversion Recovery (IR)

A 180° inversion pulse followed by a delay TI before a spin-echo acquisition. By choosing TI = T1·ln(2) for a tissue, its signal is nulled. FLAIR (Fluid Attenuated Inversion Recovery) nulls CSF to improve lesion conspicuity. STIR nulls fat.

10.4.4 k-Space

MRI data is acquired in the spatial frequency domain (k-space). Each data point corresponds to a different spatial frequency component of the image. The image is reconstructed by a 2D Fourier transform. Low spatial frequencies (near k-space centre) determine image contrast; high frequencies (periphery) determine edge sharpness. Trajectories through k-space define the acquisition strategy: Cartesian, radial, spiral, EPI, etc.

WORKED EXAMPLE: SPIN ECHO VERSUS GRADIENT ECHO

A spin echo uses a 90° excitation and a 180° refocusing pulse that cancels static field inhomogeneity, so its contrast follows true T_2. A gradient echo omits the 180° pulse and refocuses with gradients alone — faster (short TR) but sensitive to T_2^*. Long TR with short TE gives proton-density weighting; short TR emphasizes T_1; long TE emphasizes T_2.

KEY TAKEAWAYS

- Spin echo (with a 180° pulse) images true T_2; gradient echo images T_2^* and is faster.
- Contrast: short TR $\rightarrow T_1$; long TE $\rightarrow T_2$; long TR/short TE $\rightarrow$ proton density.
- Each excitation fills lines of k-space; the trajectory sets speed and artefacts.

Exercises

1. Which sequence refocuses static field inhomogeneity: SE or GRE?
2. Which weighting comes from a short TR?
3. What does the 180° refocusing pulse cancel?

Further reading: Nishimura [23] (ch. 6); Haacke [22].

10.5 MRI Contrast and Relaxation

LEARNING OBJECTIVES

- Define T_1, T_2, T_2^*, and proton density.
- Relate TR/TE to image weighting.
- Understand contrast agents.

MRI contrast arises from differences in tissue relaxation times T1 and T2, proton density (PD), and flow. By choosing appropriate pulse sequence parameters, specific tissue properties are emphasized, allowing discrimination of normal from pathological tissue.

10.5.1 T1 Relaxation (Spin-Lattice)

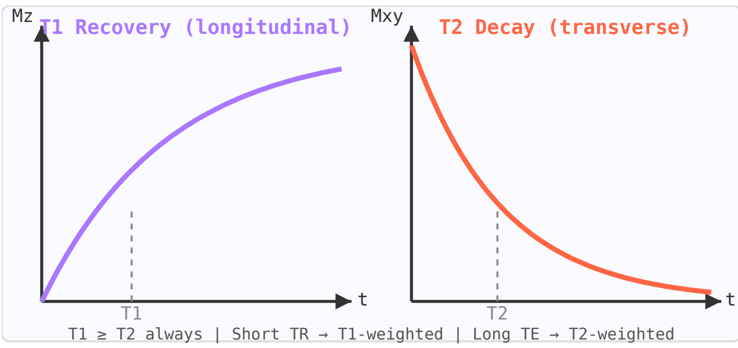


Figure 10.9: T1 recovery (purple) and T2 decay (orange) curves. T1 $\geq$ T2 always. Different tissues have different values.

T1 is the time constant for recovery of longitudinal magnetisation after excitation. Energy is transferred from the spin system to the surrounding lattice (molecular framework). T1 reflects molecular mobility: water in CSF has long T1 (~4 s at 3 T); fat has short T1 (~260 ms) due to efficient energy transfer at the Larmor frequency. Short TR $\rightarrow$ T1-weighted image.

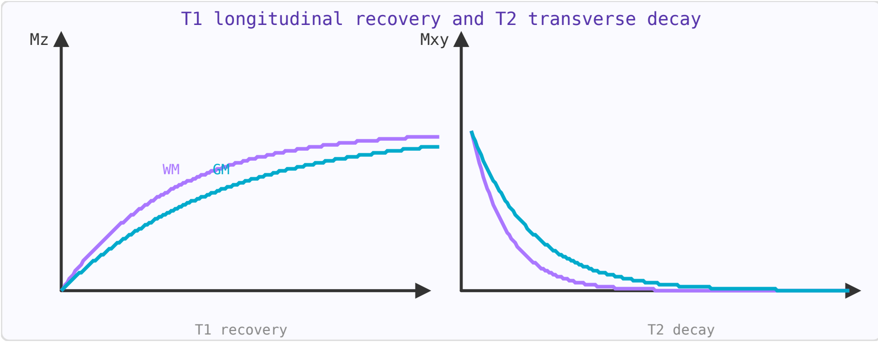


Figure 10.10: T1: longitudinal magnetisation recovers exponentially toward M_0. T2: transverse magnetisation decays. Tissue contrast comes from T1/T2 differences.

10.5.2 T2 Relaxation (Spin-Spin)

T2 is the time constant for decay of transverse magnetisation. Local field inhomogeneities from neighbouring spins cause dephasing. T2 is always $\leq$ T1. Fluids have long T2 (~2 s for CSF) appearing bright on T2-weighted images; solid tissues have short T2. Long TE $\rightarrow$ T2-weighted image.

10.5.3 T2* and Susceptibility

T2* includes additional dephasing from macroscopic B0 field inhomogeneities (susceptibility differences, main magnet imperfections). T2* $<$ T2. Gradient echo sequences are T2*-weighted. Susceptibility-weighted imaging (SWI) exploits T2* to visualise iron deposits, venous blood, and calcifications.

10.5.4 Contrast Agents

Gadolinium-based contrast agents (GBCAs) shorten T1 in their vicinity, making enhancing tissue appear bright on T1-weighted images. Used to detect blood-brain barrier breakdown, tumour vascularity, and inflammation. Iron oxide nanoparticles shorten T2/T2*, creating signal voids.

WORKED EXAMPLE: T1 RECOVERY

Longitudinal magnetization recovers as $M_z(t) = M_0(1 - e^{-t/T_1})$. After a 90° pulse, at $t = T_1$ the recovery is $1 - e^{-1} = 63\%$ of M_0. Short-T_1 tissue (e.g. fat) recovers quickly and looks bright on T_1-weighted images (short TR); long-T_1 tissue (e.g. CSF) recovers slowly and looks dark.

KEY TAKEAWAYS

- T_1 (longitudinal recovery) and T_2 (transverse decay) are tissue-specific; $T_2^* \leq T_2$.
- $M_z(t) = M_0(1 - e^{-t/T_1})$; $M_{xy}(t) = M_0 e^{-t/T_2}$.
- Gadolinium shortens T_1, brightening enhancing tissue.

Exercises

1. Fraction of T_1 recovery at $t = T_1$?
2. Fraction of transverse signal left at $t = T_2$?

3. Does gadolinium predominantly shorten T_1 or T_2 clinically?

Further reading: Nishimura [23]; Haacke [22] (ch. 4).

10.6 MRI Artefacts

LEARNING OBJECTIVES

- Recognize common MRI artefacts.
- Relate chemical shift and susceptibility to field.
- Mitigate motion and aliasing.

MRI artifacts are features in the image that do not correspond to real anatomy. Understanding artifacts is essential for correct image interpretation and for optimising acquisition protocols.

10.6.1 Gibbs Ringing (Truncation Artifact)

Sharp edges in the image (e.g. brain-CSF interface) produce ringing stripes parallel to the edge, caused by truncation of k-space at finite matrix size. Mitigated by acquiring more k-space lines (larger matrix) or applying k-space apodisation filters (at cost of resolution).

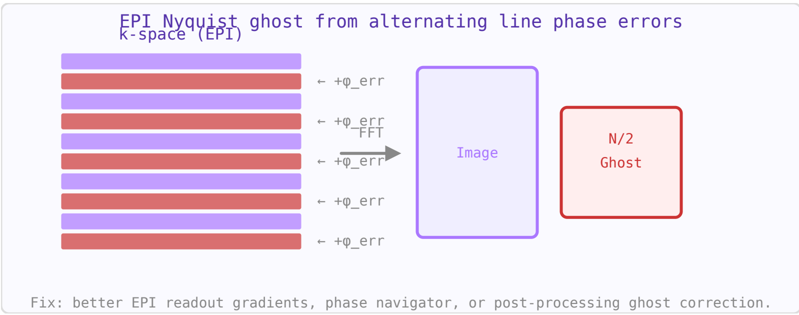


Figure 10.11: EPI Nyquist ghost: alternating k-space lines have opposite phase errors, creating a N/2-shifted ghost image.

10.6.2 Chemical Shift Artifact

Water and fat protons resonate at slightly different frequencies (3.5 ppm $\approx$ 450 Hz at 3 T). In the frequency-encoding direction, fat and water signals are spatially misregistered by this frequency offset, creating a bright/dark band at fat-water interfaces (e.g. kidney-perinephric fat). Mitigated by wider receiver bandwidth or fat suppression.

10.6.3 Motion Artifacts

Patient movement during acquisition causes ghosting (copies of moving structures) in the phase-encoding direction. Respiratory motion causes abdominal blurring. Mitigated by breath-holding, respiratory triggering/gating, navigator echoes, and radial k-space trajectories (more motion-robust).

10.6.4 Susceptibility Artifacts

Interfaces between materials of different magnetic susceptibility (metal implants, air-tissue, bone-tissue) distort the local B0 field, causing signal loss and geometric distortion. Worst near metallic implants. Mitigated by reducing TE, using spin echo (refocuses static inhomogeneity), or MARS (Metal Artifact Reduction Sequences).

10.6.5 RF Inhomogeneity (B1+ Artifact)

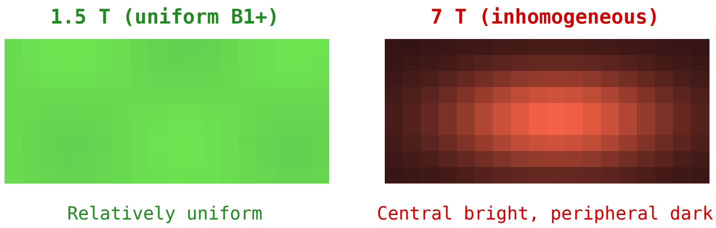


Figure 10.12: B1+ inhomogeneity: uniform at 1.5 T (left) but strong central brightening at 7 T (right) creating signal non-uniformity.

At 3 T and above, wavelength effects cause spatially non-uniform $B1^+$ fields, resulting in flip angle variations across the image. Manifests as signal brightening in the centre and darkening at the periphery of large body parts at 3 T, and complex interference patterns at 7 T. Mitigated by adiabatic pulses, parallel transmit, and B1 mapping-based pulse calibration.

WORKED EXAMPLE: CHEMICAL-SHIFT DISPLACEMENT

Fat and water precess about 3.5 ppm apart. At 1.5 T (64 MHz) that is $3.5 \times 10^{-6} \times 64 \times 10^{6} = 224$ Hz; at 3 T it doubles to 448 Hz. Because the readout maps frequency to position, fat shifts spatially relative to water — worse at high field and with low readout bandwidth.

KEY TAKEAWAYS

- Chemical shift, susceptibility, motion, aliasing, and Gibbs ringing are the common artefacts.
- Chemical-shift and susceptibility effects scale with B_0 (worse at high field).
- Higher readout bandwidth reduces chemical-shift displacement; oversampling cures aliasing.

Exercises

1. Fat–water shift (Hz) at 3 T (3.5 ppm)?
2. Does the susceptibility artefact improve or worsen at 7 T?
3. What reduces chemical-shift displacement in the readout?

Further reading: Haacke [22] (ch. 20); Nishimura [23].

10.7 RF Coil Design for MRI

LEARNING OBJECTIVES

- Tune and match an RF coil to the Larmor frequency.
- Relate coil Q to SNR.
- Apply active/passive detuning.

Designing an MRI RF coil requires careful attention to resonance, matching, decoupling, and coil Q. The goal is to maximise SNR while maintaining safety and compatibility with the scanner's RF system.

10.7.1 Tuning to Resonance

The coil must resonate at the Larmor frequency of the target nucleus. For a single-loop coil of inductance L, a tuning capacitor C_tune = $1/(\omega^2 L)$ brings the coil to resonance. Fixed or variable capacitors (trimmers) are used. The tuning is sensitive to nearby conductive or dielectric objects (loading), so provision for fine adjustment near the sample is necessary.

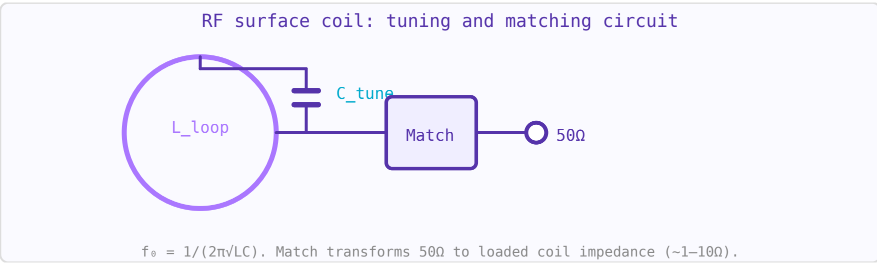


Figure 10.13: RF coil tuning: parallel capacitor C_tune resonates loop inductance L. Matching network presents 50 Ω to the coaxial cable.

10.7.2 Matching to 50 Ω

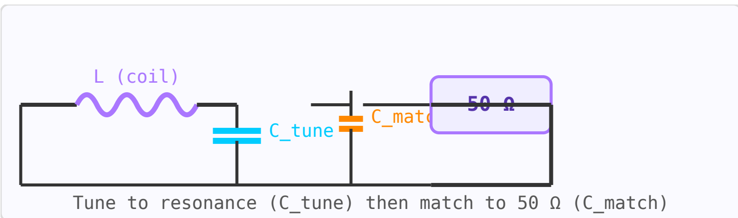


Figure 10.14: Coil tuning and matching circuit: C_tune resonates the coil at f_Larmor; C_match transforms impedance to 50 Ω.

The coil's radiation resistance is much lower than 50 Ω (typically 0.1–10 Ω for small surface coils). A matching network transforms this to 50 Ω to maximise power transfer between the coil and the preamplifier. Common approaches: capacitive tap on the tuning capacitor, additional shunt matching capacitor, or a transformer.

10.7.3 Quality Factor Q

The unloaded Q (Q_u) is determined by the coil's own resistance. The loaded Q (Q_L) includes sample losses. For receive coils, SNR is maximised when Q_L is dominated by sample noise

(Q_L ≈ Q_u/2), not coil resistance. High-conductivity conductor (copper, silver), thick wire, and minimising resistive losses are key. At 3 T, typical unloaded Q for a 10 cm surface coil: 200–500; loaded Q: 30–80.

10.7.4 Decoupling

In phased array coils, adjacent elements must be decoupled to prevent inductive coupling from degrading noise performance. Methods include: overlap decoupling (geometrically overlapping adjacent loops to cancel mutual inductance), capacitive decoupling networks, and preamplifier decoupling (presenting high impedance to the coil current using a low-input-impedance preamp with an additional inductance).

10.7.5 Active Detuning

During RF transmission, receive coils must be detuned (switched off) to prevent them from disturbing the transmit field and from experiencing potentially damaging induced voltages. Active detuning circuits use PIN diodes to insert a large inductance or short circuit that shifts the resonant frequency away from f_0 during the transmit phase.

WORKED EXAMPLE: TUNING A LOOP COIL

A loop of inductance $L = 100$ nH resonates at f_0 when tuned with $C = 1/[(2\pi f_0)^2 L]$. At 1.5 T ($f_0 = 63.9$ MHz),

$$C = \frac{1}{(2\pi \cdot 63.9 \times 10^6)^2(100 \times 10^{-9})} = 62 \text{ pF}.$$

A second (matching) capacitor transforms the coil to 50 Ω. A high unloaded-to-loaded Q ratio means the sample — not the coil — dominates the noise, which is the goal for good SNR.

KEY TAKEAWAYS

- Tune $C = 1/[(2\pi f_0)^2 L]$ to the Larmor frequency; add a matching capacitor for 50 Ω.
- Sample-dominated noise (high $Q_{\text{unloaded}}/Q_{\text{loaded}}$) gives the best SNR.
- Active detuning (PIN diodes) protects the receive coil during transmit.

Exercises

1. Tuning capacitor for $L = 200$ nH at 64 MHz?
2. What does a high unloaded/loaded Q ratio indicate?
3. Why detune a receive coil during transmit?

Further reading: Mispelter [25]; Haacke [22] (ch. 27).

10.8 MRI Coil Types

LEARNING OBJECTIVES

- Distinguish volume, surface, and array coils.
- Trade coverage against SNR.
- Match a coil to the application.

RF coils are the antennas of the MRI scanner. The receive coil must be placed as close as possible to the anatomy of interest to maximise the filling factor (the fraction of coil inductance filled by the sample) and thus the SNR.

10.8.1 Volume Coils

Surround the anatomy completely, providing a uniform B1 field. The birdcage coil is the most common volume coil — a cylindrical structure with rungs and end rings, driven in quadrature to produce circular polarisation (improving SNR by $\sqrt{2}$). Used for head, knee, and wrist imaging.

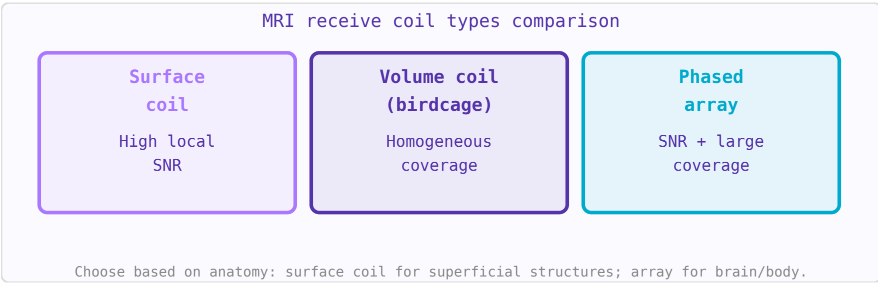


Figure 10.15: MRI coil type comparison. Each represents a different tradeoff between SNR, coverage, and homogeneity.

10.8.2 Surface Coils

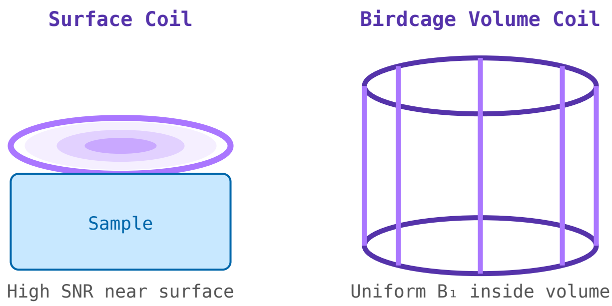


Figure 10.16: Left: surface coil — high local SNR, non-uniform. Right: birdcage volume coil — uniform B_1 throughout volume.

A single loop placed directly on the skin surface. Excellent SNR near the coil but very non-uniform sensitivity — SNR drops rapidly with depth. The depth of sensitivity is roughly equal to the coil radius. Ideal for superficial structures: spine, shoulder, temporomandibular joint.

10.8.3 Phased Array Coils

Multiple receive elements arranged around the anatomy, each with its own preamplifier and receive channel. Combines the high local SNR of surface coils with wider coverage. Also enables parallel imaging (SENSE, GRAPPA) where k-space undersampling is corrected using the spatial sensitivity profiles of the individual elements, reducing scan time.

10.8.4 Birdcage Coil

A cylindrical resonator with multiple rungs connecting two end rings. Driven in two quadrature ports (90° phase offset), it produces a highly uniform rotating $B1^+$ field ideal for transmit. Available in low-pass, high-pass, and band-pass configurations. The high-pass birdcage is preferred at high field (>3 T) as its capacitors are on the end rings rather than the rungs.

10.8.5 Transceiver vs Separate Tx/Rx

Some coils both transmit and receive (transceiver or T/R coil). Others use a separate body coil for transmit and a surface/array coil for receive. The receive-only configuration allows the receive coil to be optimised purely for sensitivity while the transmit coil provides uniform excitation.

WORKED EXAMPLE: SURFACE VERSUS VOLUME COIL

A surface coil sits close to the anatomy, sees little sample noise, and gives high SNR near the surface — but its sensitivity falls off roughly as $1/d^3$ with depth and its B_1 is inhomogeneous. A volume coil (e.g. a birdcage) surrounds the region for uniform B_1 and coverage at lower peak SNR. Phased arrays combine many small elements to get surface-coil SNR over a volume-coil field of view.

KEY TAKEAWAYS

- Surface coils: high local SNR, poor depth penetration and uniformity.
- Volume coils (birdcage): uniform B_1 and coverage, lower peak SNR.
- Phased arrays merge both: many small elements → high SNR over a large FOV.

Exercises

1. Which gives higher SNR near the skin: surface or volume coil?
2. Which gives more uniform B_1?
3. What coil type gets surface-coil SNR over a large FOV?

Further reading: Mispelter [25]; Roemer *et al.* [26].

10.9 The Birdcage Coil

LEARNING OBJECTIVES

- Explain the birdcage resonant modes.
- Relate leg/end-ring reactance to tuning.
- Understand quadrature drive.

The birdcage coil is a cylindrical resonator consisting of a set of parallel conducting rungs connecting two circular end rings. Driven in quadrature (two ports 90° apart in space and in phase), it produces a highly uniform, circularly polarised B1 field ideal for both transmit and receive in MRI. It was proposed by Hayes et al. in 1985 and remains the dominant head and extremity coil design.

10.9.1 Resonant Modes

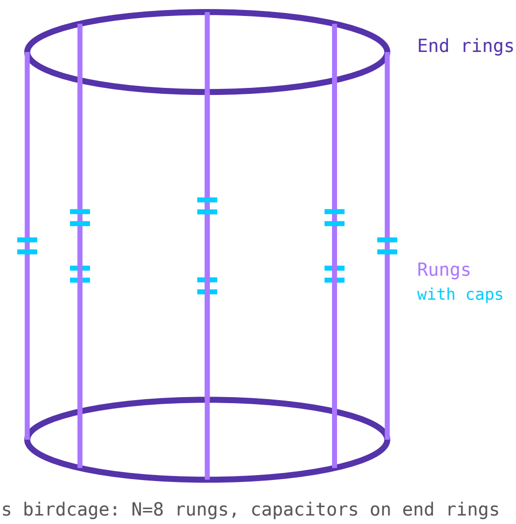


Figure 10.17: High-pass birdcage coil (N=8 rungs). Capacitors (cyan) on end rings. Rungs (purple) carry sinusoidal current distribution.

A birdcage with N rungs supports N resonant modes. The lowest frequency mode (mode 1) produces the sinusoidal current distribution required for field homogeneity. For an N-rung birdcage, the useful mode resonates at:

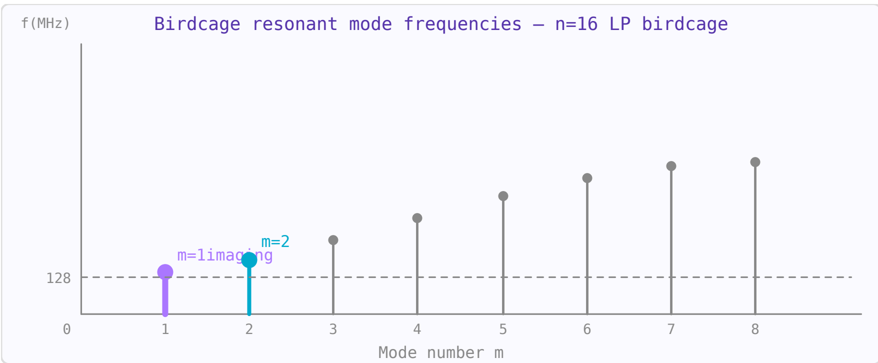


Figure 10.18: Birdcage resonant mode frequencies vs mode number m. Mode m=1 is the imaging mode (uniform field). Higher modes are spurious.

$$f_0 \approx \frac{1}{2\pi\sqrt{L_{rung} \cdot C_{eff}}}$$

where L_{rung} is the rung inductance and C_{eff} is determined by the capacitor placement.

10.9.2 Low-Pass vs High-Pass Configurations

- **Low-pass**: capacitors on the rungs, end rings are continuous. Rungs look like capacitors at low frequency → resonance at low end of mode spectrum.
- **High-pass**: capacitors on the end rings, rungs are continuous inductors. Preferred at high field (3 T and above) because the desired mode is the highest frequency mode, further from spurious modes.
- **Band-pass**: capacitors on both rungs and end rings. Wider bandwidth and used in volume coils for body imaging.

10.9.3 Quadrature Drive

Driving two ports 90° apart in space and phase creates circular polarisation, producing $B1^+$ field in the rotating frame that efficiently tips magnetisation. This gives $\sqrt{2}$ (3 dB) SNR improvement over linear excitation for the same deposited power, or equivalently halves the required transmit power for the same flip angle.

WORKED EXAMPLE: THE HOMOGENEOUS MODE

A birdcage coil is a ladder of legs and end-ring segments forming a resonant network with several modes. The useful *homogeneous* mode produces a uniform transverse B_1 across the bore. Driving two ports 90° apart in space and phase (quadrature) rotates the B_1 field, improving transmit efficiency by $\sqrt{2}$ and receive SNR by up to $\sqrt{2}$ versus a single linear drive.

KEY TAKEAWAYS

- A birdcage supports several modes; the homogeneous ($k = 1$) mode gives uniform B_1.
- Leg and end-ring capacitances set the mode frequencies.
- Quadrature drive (two ports at 90°) rotates B_1, gaining $\sqrt{2}$ in efficiency/SNR.

Exercises

1. Which birdcage mode gives uniform B_1?
2. Quadrature versus linear drive improves SNR by roughly what factor?
3. What components set the mode frequencies?

Further reading: Mispelter [25]; Haacke [22] (ch. 27).

10.10 SNR in MRI Receive Coils

LEARNING OBJECTIVES

- Identify coil and sample noise contributions.
- Relate loaded Q to SNR.
- Diagnose the noise-dominant regime.

Signal-to-noise ratio (SNR) is the fundamental figure of merit of any MRI coil. Understanding the sources of noise and the factors that affect SNR guides coil design decisions.

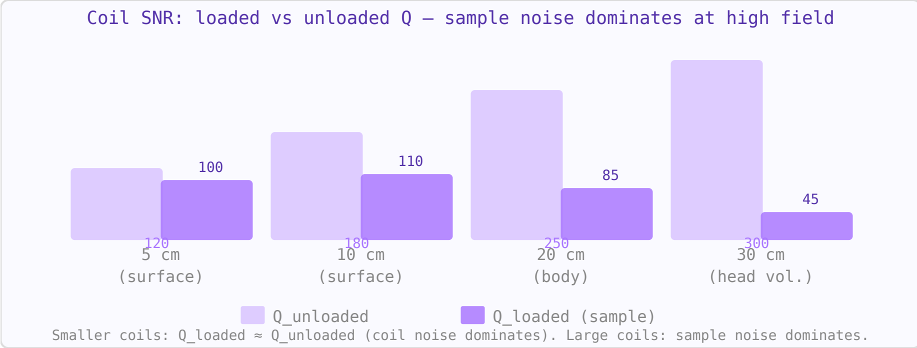


Figure 10.19: Loaded vs unloaded Q for MRI receive coils. When Q_loaded $\ll$ Q_unloaded, sample noise dominates — the coil is in the "sample-noise dominated" regime where SNR $\propto$ ω_0 B_1/coil.

10.10.1 Signal Source

The MRI signal is the EMF induced in the receive coil by the precessing transverse magnetisation in the sample. It scales with: field strength B0 (approximately B0ˆ7/4 for a given experiment time), coil filling factor, and sample magnetisation density.

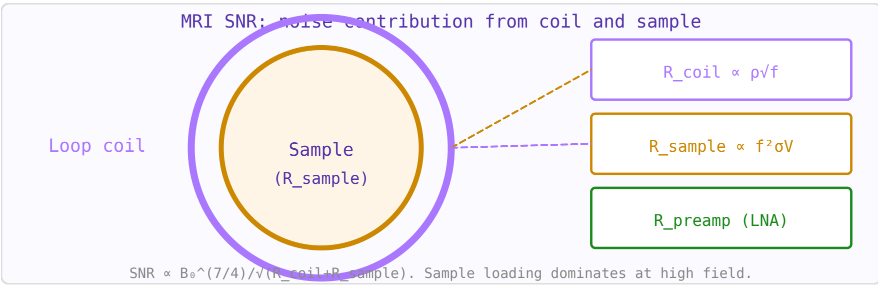


Figure 10.20: MRI coil SNR model: noise sources are coil resistance R_c, sample loading R_s, and preamplifier R_p.

10.10.2 Noise Sources

At MRI frequencies, two main noise sources compete:

- **Coil resistance noise** — Johnson noise from the ohmic resistance of the coil conductor. Dominates for small coils (surface coils <~5 cm at 3 T) or at low field.
- **Sample noise** — noise from induced currents (eddy currents) in the conductive biological tissue. Dominates for large coils, at high field, and near large conductive samples. Unavoidable — it sets the SNR ceiling.

10.10.3 Filling Factor

The filling factor η is the fraction of the coil's inductance that is "filled" by the sample volume of interest. $\eta = \int_{sample} |B_1|^2 dV / \int_{all} |B_1|^2 dV$. Maximising filling factor (placing the coil as close as possible to the anatomy) directly improves SNR.

10.10.4 Optimal Coil Size

A smaller coil placed close to the anatomy intercepts less noise (smaller sensitive volume) but provides high B1 per unit current near the coil. The optimal coil size roughly matches the depth of the structure of interest — a good rule of thumb is: for a structure at depth d below the surface, the optimal surface coil radius is approximately d.

10.10.5 Preamplifier Noise

The preamplifier (preamp) also contributes noise. For a sample-noise-dominated coil, preamp noise figure adds directly to system NF. Modern MRI preamps achieve NF of 0.3–0.7 dB. The preamp should be physically close to the coil (short coax) to minimise cable losses that degrade SNR.

WORKED EXAMPLE: WHEN THE SAMPLE DOMINATES NOISE

A receive coil's noise is the sum of coil (conductor) and sample (body) losses. The ratio of unloaded to loaded quality factor reveals which dominates:

$$\frac{Q_{\text{unloaded}}}{Q_{\text{loaded}}} = 1 + \frac{R_{\text{sample}}}{R_{\text{coil}}}.$$

A ratio of 5 means the sample resistance is 4× the coil resistance — the sample dominates, so polishing the conductor buys little. At low field or for small coils the coil can dominate, and conductor quality matters more.

KEY TAKEAWAYS

- Total noise $\propto R_{\text{coil}} + R_{\text{sample}}$; SNR is best when the sample dominates.
- $Q_{\text{unloaded}}/Q_{\text{loaded}} = 1 + R_{\text{sample}}/R_{\text{coil}}$ diagnoses the regime.
- Sample noise rises with field; small or low-field coils are more coil-noise-limited.

Exercises

1. If $Q_{\text{unloaded}}/Q_{\text{loaded}} = 3$, what is $R_{\text{sample}}/R_{\text{coil}}$?
2. In the sample-dominated regime, does better conductor plating help much?
3. Are small surface coils more coil- or sample-noise-limited?

Further reading: Mispelter [25]; Roemer *et al.* [26].

10.11 Preamplifier Decoupling

LEARNING OBJECTIVES

- Explain preamplifier decoupling in arrays.
- Relate low input impedance to element isolation.
- Understand the $\lambda/4$ transform.

In phased array receive coils, adjacent elements must be isolated from each other to prevent mutual inductive coupling from increasing correlated noise and degrading combined SNR. Preamplifier decoupling is the most effective and widely used technique for achieving this isolation.

10.11.1 The Problem: Mutual Inductance

Two adjacent coil loops have mutual inductance M. If one coil carries current (as part of a resonant circuit), it induces a voltage in its neighbour. This reduces the SNR of both coils and is equivalent to adding correlated noise between array channels.

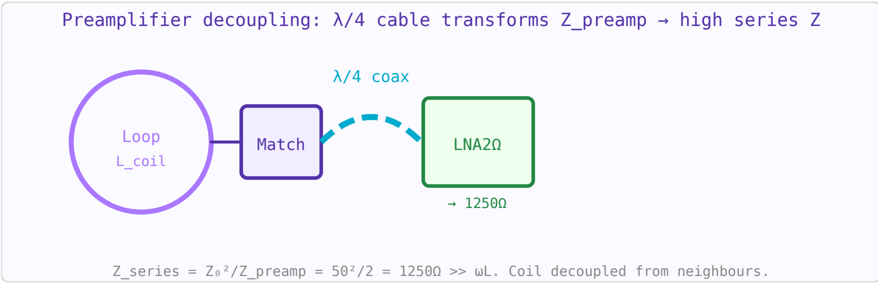


Figure 10.21: Preamplifier decoupling circuit. $\lambda/4$ cable transforms 2 Ω LNA input to 1250 Ω high series impedance in the coil loop.

10.11.2 The Technique

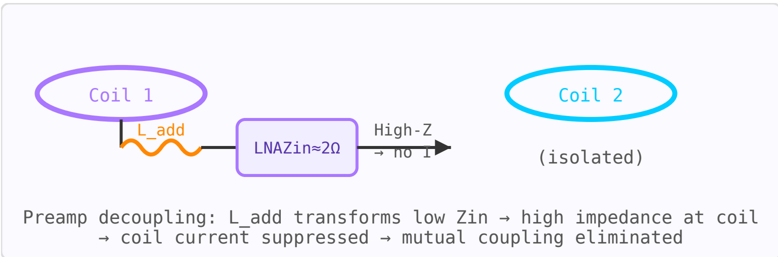


Figure 10.22: Preamplifier decoupling: L_add resonates with C_match, transforming the low preamp Zin to high impedance at the coil loop, suppressing inter-coil coupling.

A low-input-impedance preamplifier, together with a series inductor L_{add} at the coil port, presents a very high impedance to current flowing in the coil loop at the resonant frequency. This "preamplifier decoupling" suppresses the induced current in each coil, effectively eliminating mutual coupling without requiring geometric overlap or additional decoupling networks.

$$Z_{preamplifier\ decoupling} = \frac{(\omega L_{add})^2}{Z_{in,preamp}}$$

If $Z_{in,preamp}$ is very low (~2–5 Ω) and L_{add} is chosen such that $\omega L_{add} = 1/(\omega C_m)$ (resonates with the matching capacitor), the impedance presented to coil current becomes very high (>1000 Ω), suppressing currents and hence coupling.

10.11.3 Geometric Overlap Decoupling

An alternative for nearest-neighbour elements is to physically overlap adjacent loops until their mutual inductance is zero (the induced flux from one loop through the other sums to zero due to partial cancellation). About 15–20% overlap is typical. Used in combination with preamplifier decoupling for arrays with many elements.

WORKED EXAMPLE: DECOUPLING VIA A LOW-Z LNA

In a receive array, current in one element couples to its neighbours. Preamplifier decoupling uses a low-input-impedance LNA: a $\lambda/4$ network turns the LNA's near-short input into a high series impedance in the coil loop, choking the induced current. With line impedance Z_0 and preamp input Z_{in}, the transformed loop impedance is Z_0^2/Z_{in} — large when Z_{in} is small — so the loop current (and inter-element coupling) is suppressed while noise match is preserved.

KEY TAKEAWAYS

- Preamp decoupling suppresses induced loop current with a low-impedance LNA and a $\lambda/4$ transform.
- The $\lambda/4$ line turns the LNA's low input impedance into a high series impedance (Z_0^2/Z_{in}).
- It complements geometric (overlap) decoupling in phased arrays.

Exercises

1. A $\lambda/4$ line transforms a 2 Ω preamp input into what loop impedance in 50 Ω?
2. Does a lower preamp input impedance give more or less decoupling?
3. What other decoupling method does it complement?

Further reading: Roemer *et al.* [26]; Mispelter [25].

10.12 Phased-Array Coils

LEARNING OBJECTIVES

- Explain array coils for parallel coverage.
- Relate element count to SNR and acceleration.
- Understand decoupling requirements.

A phased array is an antenna system consisting of multiple radiating elements whose relative phases (and sometimes amplitudes) are controlled electronically. By varying the phase distribution, the combined beam can be steered in any direction without mechanical movement.

10.12.1 Beam Steering

To steer the main beam to angle θ_0, a progressive phase shift $\psi = kd \sin\theta_0$ is applied across the array, where $k = 2\pi/\lambda$ and d is element spacing. Phase shifters (analog or digital) implement this delay electronically. Beam switching speed: microseconds, compared to seconds for mechanical steering.

10.12.2 Grating Lobes

When element spacing $d > \lambda/2$, additional main beams (grating lobes) appear at angles other than θ_0. Standard design rule: $d \le \lambda/2$ for grating-lobe-free scanning to $\pm 90°$.

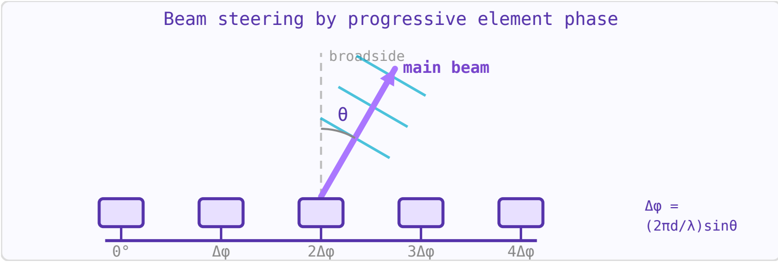


Figure 10.23: Phased array uses progressive phase shifts (implemented by phase shifters) to steer the main beam electronically.

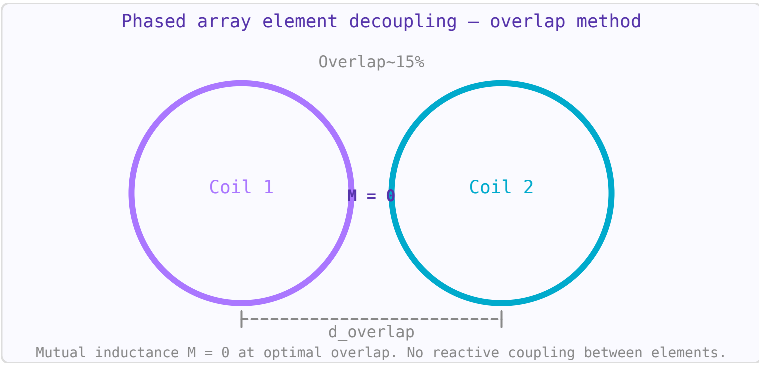


Figure 10.24: Overlapping coil decoupling: optimal overlap distance (≈15% of diameter) sets mutual inductance M = 0.

10.12.3 Beamforming

Digital beamforming (DBF) processes the signals from individual array elements digitally, allowing simultaneous multiple beams, adaptive null steering (pointing nulls toward interference sources), and MIMO operation. 5G base stations use digital beamforming with massive MIMO (64–256 elements) for spatial multiplexing of many users.

10.12.4 Applications

- Military: AESA (Active Electronically Scanned Array) radar — simultaneous multi-function operation
- Satellite communications: electronically steered flat-panel antennas (e.g. Starlink terminals)
- 5G NR mmWave: compact phased array modules at 28/39 GHz for beamforming
- MRI: RF phased array transmitters for parallel transmission (pTx) at 7 T

WORKED EXAMPLE: COVERAGE AND DECOUPLING

An MRI phased array places many small coils to get surface-coil SNR over a large field of view; each element also provides a distinct spatial sensitivity used for parallel imaging. Elements must be decoupled: nearest neighbours by geometric overlap (mutual inductance $M \to 0$) and the rest by preamplifier decoupling. More elements raise SNR and the achievable acceleration, at the cost of more receive channels and reconstruction complexity.

KEY TAKEAWAYS

- Array coils combine small elements: high SNR over a large FOV plus distinct sensitivities for parallel imaging.
- Decoupling is essential: overlap for neighbours, preamp decoupling for the rest.
- More elements → more SNR and acceleration, but more channels and complexity.

Exercises

1. How are nearest-neighbour array elements typically decoupled?
2. What do the distinct element sensitivities enable?
3. More elements generally raise what two things?

Further reading: Roemer *et al.* [26]; Mispelter [25].

10.13 Parallel Imaging

LEARNING OBJECTIVES

- Explain SENSE and GRAPPA.
- Relate acceleration factor to scan time and SNR.
- Understand the g-factor penalty.

Parallel imaging uses the spatial sensitivity profiles of multiple receive array elements to recover images from undersampled k-space data, allowing faster acquisition without image quality penalty (at the cost of some SNR).

10.13.1 The Basic Principle

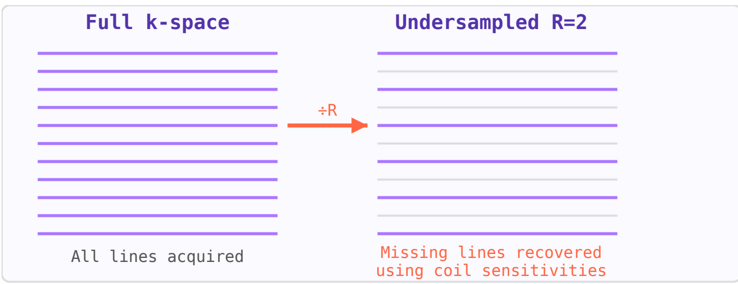


Figure 10.25: k-space: full acquisition (left) vs R=2 undersampled (right). Parallel imaging recovers missing lines from coil sensitivities.

If k-space is undersampled by an acceleration factor R (only every Rth phase-encode line is acquired), the Nyquist criterion is violated and aliasing (fold-over artifacts) occurs in the image. However, if multiple receive channels are available, each with a different spatial sensitivity, the aliased signals from different locations can be separated using the known sensitivity patterns.

10.13.2 SENSE

Sensitivity Encoding (SENSE) operates in image space. The aliased image from each channel is combined using a matrix inversion based on the measured coil sensitivity maps. Requires

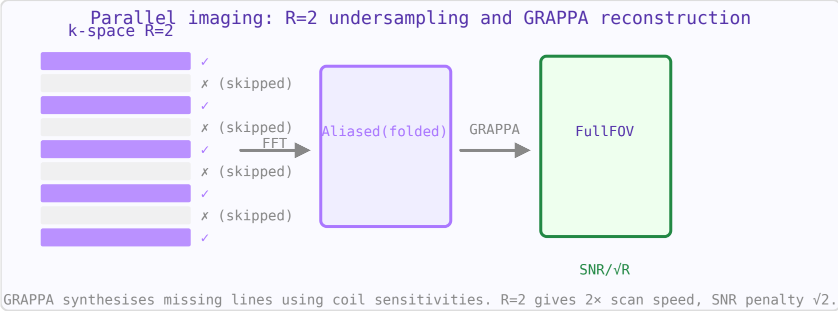


Figure 10.26: Parallel imaging: every 2nd k-space line acquired (R=2), GRAPPA reconstructs missing lines using multi-coil data.

explicit sensitivity calibration and produces a noise amplification characterised by the geometry factor (g-factor).

10.13.3 GRAPPA

GeneRalised Autocalibrating Partial Parallel Acquisition (GRAPPA) operates in k-space. Missing k-space lines are synthesised from acquired lines using calibration coefficients estimated from a central fully-sampled auto-calibration signal (ACS) region. More robust to motion between calibration and imaging than SENSE.

10.13.4 SNR and g-Factor

Parallel imaging reduces scan time by factor R but always reduces SNR by at least $\sqrt{R}$ (less data = more noise). The g-factor (geometry factor) ≥ 1 further penalises SNR due to the conditioning of the unaliasing matrix. $SNR_{PI} = SNR_{full}/(g\sqrt{R})$. High-field MRI (7 T) benefits most from parallel imaging because the SNR advantage at high field partially compensates for the $\sqrt{R}$ SNR loss.

WORKED EXAMPLE: ACCELERATION AND ITS SNR PENALTY

Parallel imaging undersamples k-space by an acceleration factor R and uses the coil sensitivities to unfold the aliasing. Scan time scales as $1/R$, but SNR drops:

$$\mathrm{SNR}_{acc} = \frac{\mathrm{SNR}_{full}}{g\sqrt{R}},$$

where the geometry factor $g \geq 1$ depends on the array. For $R = 2$ and $g = 1.2$, SNR falls to $\mathrm{SNR}_{full}/(1.2\sqrt{2}) = 0.59\,\mathrm{SNR}_{full}$ — a bit worse than the $1/\sqrt{2}$ from undersampling alone.

KEY TAKEAWAYS

- SENSE unfolds in image space; GRAPPA fills missing k-space lines — both use coil sensitivities.
- Acceleration R cuts scan time by $1/R$ and SNR by $g\sqrt{R}$.
- The g-factor (≥ 1) penalizes SNR where element sensitivities overlap.

Exercises

1. Scan-time factor for $R = 3$?

2. Ideal SNR loss ($g = 1$) for $R = 4$?

3. What array property keeps the g-factor near 1?

Further reading: Roemer *et al.* [26]; Haacke [22].

10.14 B1 Field Mapping

LEARNING OBJECTIVES

- Explain why B_1 must be measured.
- Describe the double-angle method.
- Relate B_1 to flip angle.

B1 mapping refers to measuring the spatial distribution of the RF transmit field (B1$^+$) inside the subject. At 3 T and above, wavelength effects cause significant B1$^+$ inhomogeneity, which manifests as flip angle variations across the image. B1 maps are used for quantitative MRI, SAR estimation, and parallel transmit calibration.

10.14.1 Why B1 Mapping Matters at High Field

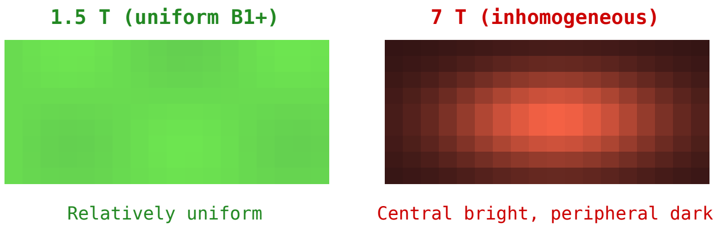


Figure 10.27: Simulated B1+ maps: 1.5 T (left, uniform) vs 7 T (right, central brightening due to constructive EM interference).

At 3 T (128 MHz), the wavelength in tissue (~25 cm) becomes comparable to body dimensions, causing constructive and destructive interference of the RF field. A nominal 90° pulse may deliver 70° in some regions and 110° in others. Quantitative MRI techniques (T1 mapping, T2 mapping, MR spectroscopy) require accurate flip angles for valid results.

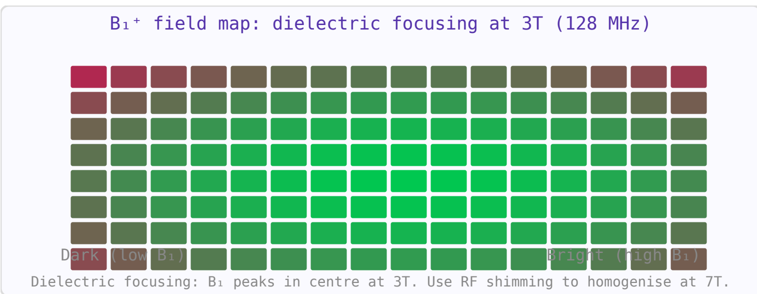


Figure 10.28: B$_1^+$ field map simulation at 3T. Constructive interference of RF waves creates a bright central spot in the phantom.

10.14.2 Double Angle Method (DAM)

Two spin-echo images are acquired with the same TR but with nominal flip angles α and 2α. The signal ratio gives: $|S(2\alpha)/S(\alpha)| = 2\cos(\alpha_{actual})$, from which the actual flip angle can be calculated pixel-by-pixel. Simple but slow (requires long TR for T1 recovery).

10.14.3 Actual Flip Angle Imaging (AFI)

Two interleaved acquisitions with different TRs (TR1 $\ll$ TR2) but the same flip angle. The signal ratio depends only on flip angle and the TR ratio (not T1 for reasonable conditions), allowing fast B1 mapping.

10.14.4 Parallel Transmit (pTx)

At 7 T, multiple transmit channels (typically 8) each drive a separate coil element. By controlling the amplitude and phase of each channel, the B1$^+$ distribution can be shaped (RF shimming) to improve uniformity or focus power. B1 maps for each channel are a prerequisite for pTx calibration.

WORKED EXAMPLE: FLIP ANGLE FROM B1

The flip angle produced by an RF pulse is $\alpha = \gamma \int B_1(t)\, dt$ — proportional to the transmit field. Because B_1 is non-uniform (especially at high field), the actual flip angle varies across the image, altering contrast. B_1 mapping measures this: the double-angle method acquires images at nominal α and 2α and uses the signal ratio $S_{2\alpha}/S_\alpha = 2\cos\alpha$ to solve for the true local flip angle.

KEY TAKEAWAYS

- Flip angle $\alpha = \gamma \int B_1\, dt$; a non-uniform B_1 makes α vary spatially.
- B_1 inhomogeneity worsens at high field (shorter RF wavelength in tissue).
- Double-angle method: $S_{2\alpha}/S_\alpha = 2\cos\alpha$ recovers the true flip angle.

Exercises

1. Flip angle is proportional to which field?
2. What does $S_{2\alpha}/S_\alpha = 2\cos\alpha$ let you solve for?
3. Does B_1 non-uniformity get worse at higher field?

Further reading: Haacke [22]; Nishimura [23].

10.15 SAR and RF Safety in MRI

LEARNING OBJECTIVES

- Define SAR and its field dependence.
- Compute SAR scaling with B_0 and flip angle.
- Apply the IEC operating modes.

The interaction of RF electromagnetic fields with biological tissue at MRI frequencies deposits energy as heat. Safety standards limit this deposition to protect patients from tissue heating. SAR (Specific Absorption Rate) is the primary metric.

10.15.1 Specific Absorption Rate (SAR)

SAR is the RF power absorbed per unit mass of tissue:

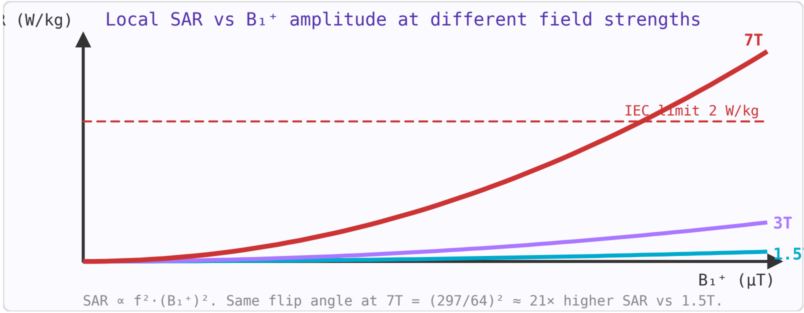


Figure 10.29: Local SAR grows with B_1^+ squared and with frequency squared. Managing SAR is the key challenge at ultra-high field (7T+).

$$SAR = \frac{\sigma|E|^2}{2\rho} \ (W/kg)$$

where σ is tissue conductivity, E the electric field amplitude, and ρ tissue density. IEC and FDA limits for clinical MRI:

- Whole-body average: 2 W/kg (normal), 4 W/kg (first level)
- Head: 3.2 W/kg; local (any 10 g): 10 W/kg

10.15.2 B1+ Field

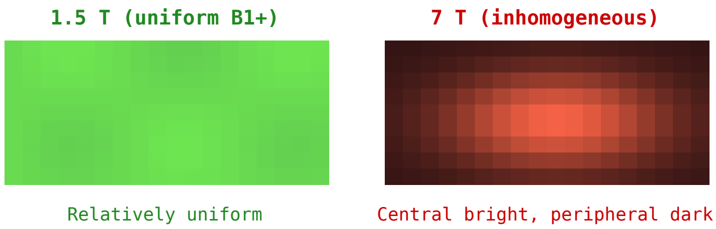


Figure 10.30: B1+ field maps at 1.5 T (uniform, left) and 7 T (inhomogeneous, right), illustrating how wavelength effects create SAR hotspots at high field.

The rotating component of the RF magnetic field that drives spin excitation is $B1^+$. At higher field strengths ($\geq$3 T), wavelength effects cause inhomogeneous $B1^+$ distribution in the body, leading to non-uniform flip angles. At 7 T (300 MHz), the wavelength in tissue (~12 cm) is comparable to body dimensions, causing significant $B1^+$ non-uniformity and hotspots in SAR.

10.15.3 Implant Safety

Conductive implants (pacemaker leads, orthopaedic implants, surgical clips) can act as RF antennas, concentrating the induced electric field and causing local tissue heating far exceeding whole-body SAR limits. MRI-conditional implants are tested and labelled with specific conditions (field strength, coil type, SAR limits) under which they are safe.

10.15.4 SAR Reduction Strategies

- Reducing flip angle (lower SAR $\propto$ flip^2)
- Increasing TR to allow pulse duty cycle reduction
- Parallel transmit (pTx) — using multiple transmit channels to shape B1$^+$ and reduce local SAR hotspots
- Lower field strength systems for implant patients

WORKED EXAMPLE: SAR SCALING WITH FIELD

Specific absorption rate is the RF power deposited per unit mass. Because heating goes as the square of the RF field and frequency,

$$\text{SAR} \propto f^2 B_1^2 \propto B_0^2\,\alpha^2.$$

Moving from 1.5 T to 3 T at the same flip angle raises SAR about fourfold (f doubles). The IEC caps whole-body SAR (normal mode 2 W/kg), so high-field sequences must lengthen TR, lower flip angles, or use low-SAR refocusing to stay within limits.

KEY TAKEAWAYS

- SAR $\propto f^2 B_1^2$, so it scales as B_0^2 for a given flip angle.
- IEC normal-mode whole-body limit is 2 W/kg; first-level mode allows more under supervision.
- High field forces SAR management: longer TR, lower/variable flip angles, low-SAR refocusing.

Exercises

1. How does SAR change from 1.5 T to 3 T at fixed flip angle?
2. Halving the flip angle changes SAR by what factor?
3. What is the IEC normal-mode whole-body SAR limit?

Further reading: ICNIRP [20]; IEC 60601-2-33 [27].

Glossary

Concise definitions of the terms used throughout this book. Symbols and their units are collected separately in the *Notation* section at the front.

ABCD parameters	Chain (cascade) matrix of a two-port; the matrix of cascaded networks is the product of the individual matrices.
Admittance (Y)	Reciprocal of impedance, $Y = 1/Z = G + jB$, with conductance G and susceptance B.
Aperture efficiency	Ratio of an antenna's effective area to its physical area; sets the realised gain of a dish or horn.
Attenuation constant (α)	Real part of the propagation constant; the rate (Np/m or dB/m) at which a wave loses amplitude.
Balun	Balanced-to-unbalanced transformer that couples a balanced load (e.g. a dipole) to an unbalanced line (e.g. coax).
Butterworth filter	Maximally-flat amplitude response; no passband ripple, gentle roll-off.
Chebyshev filter	Equal-ripple passband giving a steeper roll-off than Butterworth for the same order.
Circulator	Non-reciprocal three-port ferrite device that routes power from each port to the next in one direction only.
Conversion loss	Ratio of available RF input power to available IF output power in a mixer (dB).
Coupling factor	Fraction of forward power a directional coupler taps to its coupled port (dB).
Cutoff frequency	Frequency at which a filter response falls to -3 dB, or below which a waveguide mode does not propagate.
dB / dBm / dBc / dBi	Decibel ratio; power referenced to 1 mW; level relative to a carrier; antenna gain relative to an isotropic radiator.
Dielectric constant (ε_r)	Relative permittivity; how much a material slows and stores an electric field versus vacuum.
Directivity	Ratio of an antenna's peak radiation intensity to its average; gain is directivity times efficiency.
Dispersion	Frequency dependence of phase velocity or group delay, which spreads a pulse or symbol in time.
Effective aperture (A_e)	Equivalent capture area of a receiving antenna, $A_e = G\lambda^2/4\pi$.
EIRP	Effective isotropic radiated power: transmit power plus antenna gain (dBm + dBi).
Ferrite	Ceramic ferrimagnetic material whose biased permeability is a tensor, enabling isolators, circulators and phase shifters.
Group delay (τ_g)	Negative slope of phase versus frequency, $\tau_g = -d\phi/d\omega$; constant group delay means distortionless transmission.

Gyromagnetic ratio (γ)	Proportionality between magnetic field and precession frequency; ≈ 2.8 MHz/Oe for electrons, 42.58 MHz/T for protons.
Characteristic impedance (Z_0)	Ratio of voltage to current of a single travelling wave on a line; commonly 50 or 75 Ω.
Insertion loss	Signal power lost by inserting a component into a matched line (dB).
Intermodulation / IP3	Mixing products from non-linearity; the third-order intercept (IP3) is the extrapolated level where third-order products equal the fundamentals.
Isolation	Attenuation between two nominally decoupled ports (e.g. an off switch, or a coupler's isolated port).
Isolator	Non-reciprocal two-port that passes power one way and absorbs the reverse wave; protects sources from reflections.
Kittel resonance	Ferromagnetic (gyromagnetic) resonance frequency of a biased magnetic sample, shifted by its shape (demagnetisation).
Larmor frequency	Precession frequency of magnetic moments in a field, $f_0 = \gamma B_0$; the resonant frequency in NMR/MRI.
Load line	Locus of instantaneous voltage and current at a device's output; sets the achievable power and efficiency of an amplifier.
Loss tangent ($\tan\delta$)	Ratio of a dielectric's loss to its storage; quantifies dielectric heating and attenuation.
Matching network	Lossless network that transforms a load to a source impedance for maximum power transfer / minimum reflection.
Mixer	Non-linear/switching device that multiplies two signals to translate frequency ($f_{RF} \pm f_{LO}$).
Noise figure (NF)	Degradation of signal-to-noise ratio through a component (dB); the linear form is the noise factor F.
Noise temperature (T_e)	Equivalent input noise of a device expressed as a temperature, $T_e = (F-1)T_0$.
Permeability (μ)	Magnetic constitutive parameter relating B and H; a tensor in biased ferrites.
Phase constant (β)	Imaginary part of the propagation constant, $\beta = 2\pi/\lambda$; radians of phase per metre.
Phase noise	Short-term random phase fluctuation of an oscillator, quoted as dBc/Hz at an offset from the carrier.
PIN diode	Current-controlled RF resistor used as a switch or attenuator: low resistance when forward-biased, small capacitance when reverse-biased.
Polarisation	Orientation of an antenna's electric field (linear, circular, elliptical); mismatch between link ends causes loss.
Q **factor**	Quality factor; ratio of energy stored to energy dissipated per cycle, $Q = f_0/\Delta f$ for a resonator.
Quarter-wave transformer	A $\lambda/4$ line of impedance $\sqrt{Z_0 Z_L}$ that matches two real impedances at one frequency.
Radar cross section (RCS)	Effective area (m^2) that characterises how strongly a target scatters radar energy back to the receiver.
Reflection coefficient (Γ)	Ratio of reflected to incident wave, $\Gamma = (Z_L - Z_0)/(Z_L + Z_0)$.

Resonator	Structure storing energy at a resonant frequency (LC tank, cavity, dielectric puck, crystal); characterised by f_0 and Q.
Return loss	Reflected power relative to incident, $-20\log_{10}\lvert\Gamma\rvert$ (dB); larger is better.
S-parameters	Scattering parameters relating incident and reflected wave amplitudes at a network's ports, referenced to Z_0.
SAR	Specific absorption rate (W/kg); RF power deposited per unit mass of tissue, the limit on MRI transmit power.
Schottky diode	Metal–semiconductor diode with fast response and low forward drop, used in detectors and mixers.
Skin depth (δ)	Depth at which current density falls to $1/e$ in a conductor; shrinks as $1/\sqrt{f}$.
Smith chart	Polar chart of the reflection coefficient overlaid with impedance/admittance circles; a graphical matching tool.
Stub	Short length of open- or short-circuited line used as a reactance for matching or filtering.
Superheterodyne	Receiver architecture that mixes the RF signal to a fixed intermediate frequency for selectivity and gain.
TSS	Tangential signal sensitivity; the input level at which a detector's output is just discernible above noise.
VNA	Vector network analyser; instrument that measures magnitude and phase of S-parameters after calibration.
VSWR	Voltage standing-wave ratio, $(1+\lvert\Gamma\rvert)/(1-\lvert\Gamma\rvert)$; the ratio of standing-wave maxima to minima on a mismatched line.
Waveguide	Hollow metal (or dielectric) structure guiding waves above a cutoff frequency with very low loss.
Wavelength (λ)	Spatial period of a wave, $\lambda = v_p/f$; the natural length scale of every RF structure.
YIG	Yttrium iron garnet; a low-loss ferrite used for tunable filters and oscillators ($4\pi M_s \approx 1780$ G).

Companion Calculators

Every formula in this book has a companion interactive calculator at rftoolbox.ca — enter your own numbers and read the answer, with the same diagrams used here. The tools are listed below by category, each cross-referenced to the chapter of this book that develops its theory. All are free and run in the browser.

Fundamentals & EM

See Chapter 1 (Electromagnetic Foundations).

Wavelength	Free-space λ from frequency and dielectric constant.
Freq / λ / Energy	Live converter: frequency, wavelength, period, photon energy, B_0.
Skin Depth	Conductor skin depth vs frequency and conductivity.
dBm $\leftrightarrow$ Watts	Convert between dBm and watts in either direction.
Plane Wave / Power Density	Wave impedance, E/H fields and power density (Poynting) for a plane wave.
Field Strength Converter	V/m $\leftrightarrow$ dBμV/m $\leftrightarrow$ A/m $\leftrightarrow$ W/m^2, or field from EIRP and distance.
Dielectric Boundary (Fresnel)	Reflection/transmission vs angle, Brewster and critical angles at an interface.
Dielectric Loss Tangent	Complex permittivity, dielectric Q, attenuation and wavelength in a material.
Thermal (kTB) Noise	Johnson–Nyquist noise voltage/current and available noise power kTB.
Shielding Effectiveness	Absorption + reflection loss and total SE of a metal barrier (far-field plane wave).
Ferrite Resonance	Gyromagnetic (Kittel) resonance and the Polder permeability tensor μ, κ of a biased ferrite.

Resonance & Passives

See Chapter 4 (Passive Components, Resonators and Filters).

LC Circuit Solver	Solve for L, C, or f_0 — fill any two, get the third.
LC Tuning Capacitor	Capacitance needed to resonate a given inductor at a target frequency.
Coupled Resonators	Mode splitting, bandwidth, and coupling regime for two magnetically coupled LC tanks.
Bandpass Calculator	f_0, Q, and bandwidth from R, L, C for series or parallel RLC.

Solenoid Inductance	Inductance of a single-layer air-core or cored coil (Wheeler's formula) from turns, diameter, and length.
Varactor C–V Tuning	Varactor capacitance vs bias, tuning ratio, and resonant-frequency tuning range.
Parallel-Plate Capacitor	Capacitance from plate area, gap, dielectric and plate count, plus field and stored energy.
Bond Wire Inductance	Parasitic inductance and reactance of a bond wire (Grover formula).
Wire Inductance	Self-inductance of a straight wire, circular loop, or two-wire line.
Toroid Inductor	Turns from the core AL value and target inductance ($L = AL \cdot N^2$), plus a flux-density check.
Capacitor SRF	Self-resonant frequency from C and parasitic ESL, plus Q and reactance at frequency.
Resonator Q	Loaded/unloaded/external Q, 3-dB bandwidth, coupling and insertion loss.
Ferrite Bead / Choke	Approximate bead impedance from a series R–L model or a datasheet point.

Transmission Lines

See Chapters 2–3 (Transmission Lines; Guided-Wave Structures).

Microstrip	Z_0 and εeff from geometry, or trace width from target Z_0.
Coaxial Line	Z_0, velocity factor, and cutoff frequency for coax cable.
Stripline	Fully shielded TEM stripline between two ground planes.
Coplanar Waveguide	CPW and GCPW Z_0 using elliptic integral conformal mapping.
Differential Pair	Z_diff and Z_odd for edge-coupled microstrip pairs.
Stub Calculator	Open/short stub length for a target shunt susceptance.
Quarter-Wave Transformer	$\lambda/4$ impedance transformer Z and physical length.
TL Loss & Delay	Conductor + dielectric insertion loss and propagation delay for microstrip and coax.
TDR Fault Distance	Locate a cable fault from the echo delay and velocity factor, plus the fault impedance.
PCB Via Parasitics	Via inductance, antipad capacitance, and self-resonant frequency.
Coax Power Handling	Peak voltage-breakdown power and Z_0 of a coaxial line from geometry and dielectric.
Transmission-Line Resonator	$\lambda/2$ and $\lambda/4$ open/short line resonators: length, unloaded Q from attenuation, equivalent RLC.

Matching & Dividers

See Chapter 2 (Transmission Lines and Impedance).

Interactive Smith Chart	HiDPI Smith Chart — click to read, pin points, rotate with stubs.
VSWR / Return Loss	Live converter between VSWR, $\lvert\Gamma\rvert$, return loss, and mismatch loss.

L-Network Matcher	L, C values for single-frequency impedance matching.
Pi / T Network	Pi and T matching networks with free choice of loaded Q.
Impedance Matching Network	L-network for complex Z_S Z_L with Smith Chart trajectories.
Capacitor Network	Total capacitance for series and parallel combinations.
LC BALUN	Lattice and Pi BALUN design with impedance transformation.
Wilkinson Divider	Equal and unequal power splits with isolation resistor.
Coupler & Hybrid	Coupled-line, branchline (90°), and rat-race (180°) hybrid design.
Power Combining	N-way combiner output power with efficiency and failure analysis.
Digital Phase Shifter	LSB step, number of states, and RMS quantization error of an N-bit phase shifter.
Mismatch Loss	Mismatch loss, return loss, VSWR, $\lvert\Gamma\rvert$ and reflected/transmitted power.
Single-Stub Matching	Shunt-stub position and open/short length to match a complex load (both solutions).
Double-Stub Tuner	Two-stub matching for a fixed spacing, both solutions, with the forbidden-region check.
Multisection $\lambda/4$ Transformer	Binomial or Chebyshev multisection transformer: section impedances and bandwidth.
Tapered-Line Matching	Exponential, triangular and Klopfenstein taper reflection response and cutoff.
Bode–Fano Limit	Theoretical best matching bandwidth (or minimum reflection) for a reactive load.

Filters & Attenuators

See Chapter 4 (Passive Components, Resonators and Filters).

First-Order Filter	RC & RL low-pass / high-pass: cutoff frequency, τ, and roll-off.
LC Filter Designer	Butterworth/Chebyshev LP/HP up to 9th order with BOM.
Attenuator Designer	Pi, T, L-pad, and Bridged-T resistor values for any attenuation.
Stepped-Z LPF	Microstrip stepped-impedance LPF: Butterworth or Chebyshev, with trace dimensions.
Filter Group Delay	Band-center group delay of a Butterworth or Chebyshev lowpass prototype from its poles.
Filter Prototype g-Values	Butterworth & Chebyshev lowpass g-values of any order, plus denormalised L/C.
Coupled-Line Filter	Parallel coupled-line bandpass: J-inverters and even/odd-mode impedances per section.
Stub Filter (Richards)	$\lambda/8$ short- and open-circuit stub impedances from a lumped lowpass prototype.

Amplifiers & Devices

See Chapters 5–6 (Active Devices; Signal Generation).

Cascaded Noise Figure	System NF and gain for up to 8 cascaded stages (Friis).
Amplifier Stability	Rollett K-factor, μ-factor, MAG/MSG from S-parameters. Unconditional stability check.
IP3 & Intermodulation	OIP3, IM3 product power, and IMD ratio from IIP3 and gain.
Transistor fT / fmax	Transition and maximum oscillation frequency from gm, capacitances and gate resistance.
Unilateral Transducer Gain	Maximum gain from S-parameter magnitudes, with source and load match contributions.
LNA Noise Match	Noise figure for any source reflection from the noise parameters Fmin, Rn and Γopt.
Noise Figure (Y-Factor)	Noise figure and noise temperature from the Y-factor measurement and source ENR.
PLL Loop Filter	Type-2 3rd-order passive loop filter component values from bandwidth and phase margin.
Phase Noise Jitter	Integrate a piecewise phase noise profile to RMS jitter.
DDS Frequency	Output frequency, tuning word and resolution of a direct digital synthesizer.
Noise Temp ↔ NF	Convert noise figure (dB) ↔ noise factor ↔ equivalent noise temperature (K), plus added noise power.
Leeson Phase Noise	Predict oscillator L(fm) from loaded Q, noise figure, power and flicker corner.
PA Load-Line (Cripps)	Optimum load, maximum output power and drain efficiency for an RF power amplifier.
Junction Temperature	Device junction temperature and margin from the θjc/θcs/θsa thermal chain.
Quartz Crystal Resonator	Series/parallel resonance, Q, capacitance ratio and pullability from motional L1/C1/R1/C0.
Amplifier Design Circles	Stability, available-gain and noise-figure circles on the source-plane Smith chart from S-parameters.
Simultaneous Conjugate Match	Source/load reflections and maximum transducer gain for a bilateral amplifier.
Neg-Resistance Oscillator	One-port oscillator load design: Z_L = −Z_in, reflection condition and start-up margin.
PIN Diode Switch	Series/shunt PIN diode SPST insertion loss and isolation from R_f and C_j.

Receivers & DSP

See Chapter 8 (RF Systems, Receivers and Communications).

Receiver Sensitivity / MDS	Noise floor, minimum detectable signal, and spurious-free dynamic range from bandwidth, NF, SNR and IIP3.
Image Frequency & Rejection	Image and LO frequencies plus image-rejection ratio from I/Q gain and phase imbalance.
ADC SNR / ENOB	Quantization SNR, aperture-jitter-limited SNR, combined SNR and effective number of bits.
EVM ↔ SNR Converter	Convert error vector magnitude to/from SNR/MER, with the densest supportable QAM order.

Bandpass Sampling	Nyquist zone, aliased IF, spectral inversion and minimum sample rate for undersampling.
Spectrum Analyzer DANL	Displayed average noise level and sensitivity from RBW, noise figure and attenuation.
RF Cascade Analyzer	Cumulative gain, cascaded noise figure, IIP3/IIP2, noise floor and SFDR for a multi-stage chain.
Mixer Spur Chart	m·LO ± n·RF spurious products landing in the IF band, plus image and half-IF frequencies.
Modulation Bandwidth & Eb/N0	Symbol rate, raised-cosine occupied bandwidth, spectral efficiency and Eb/N0 ↔ SNR.
BER vs Eb/N0	Bit error rate for BPSK/QPSK/M-PSK/M-QAM in AWGN, with the waterfall curve and required Eb/N0.
OFDM Parameters	Symbol time, cyclic-prefix overhead, occupied bandwidth, PAPR bound and data rate.
Mixer Conversion Loss	SSB/DSB mixer noise figure and cascaded receiver noise figure from conversion loss.
Schottky Detector	Detector responsivity, video resistance, NEP and tangential signal sensitivity.

Antennas & Arrays

See Chapter 7 (Antennas and Propagation).

Dipole Antenna	Half-wave dipole arm length and shortening factor.
Patch Antenna	Patch width, length, and effective permittivity for any substrate.
Array Factor Plotter	Polar radiation pattern for N-element linear arrays with steering.
Helical Antenna	Axial-mode helix gain, HPBW, and dimensions via Kraus formulas.
Yagi-Uda Antenna	Gain, element lengths, and boom size for 2–9 element Yagis (NBS/Viezbicke tables).
Aperture Antenna Gain	Dish/horn gain from area and efficiency, plus the far-field (Fraunhofer) distance.
Parabolic Dish	Reflector gain, half-power/first-null beamwidth and effective aperture from diameter.
Horn Antenna	Pyramidal/conical horn gain, effective aperture and beamwidth from the aperture size.
Antenna Gain Converter	dBi ↔ dBd, EIRP, effective aperture ($G\lambda^2/4\pi$) and the far-field distance.
Monopole & Small Loop	$\lambda/4$ monopole length/Rr/gain and small-loop radiation resistance and gain.

Propagation, Radar & Links

See Chapter 8 (RF Systems, Receivers and Communications).

Friis Equation	Received power from EIRP, path loss, and antenna gains.
Link Budget	End-to-end RF link: EIRP, path loss, noise floor, SNR margin.

Fresnel Zone & Clearance	First-zone radius and the 60% clearance needed for a line-of-sight link.
Path Loss Models	Free-space, two-ray, Okumura-Hata and COST-231 propagation loss.
Radar Range Equation	R_max and received power from transmit power, RCS, and gains.
Shannon Capacity & BER	Channel capacity, spectral efficiency, Eb/N0↔SNR and BPSK/QPSK bit error rate.
Processing Gain & Jamming Margin	Spread-spectrum processing gain from chip/data rate, plus jamming margin.
Doppler Shift	Doppler frequency shift for one-way links and two-way radar from speed and angle.
G/T Figure of Merit	Receive-system G/T and system noise temperature from gain, Tant and receiver NF.
FMCW Radar	Range/velocity resolution, beat frequency and chirp slope for a frequency-modulated CW radar.
Radar PRF & Ambiguity	Unambiguous range and velocity, blind speeds, PRI and duty cycle from the PRF.
Radar Cross Section	RCS of a sphere, flat plate, cylinder or trihedral corner in m^2 and dBsm.

Waveguide

See Chapter 3 (Guided-Wave Structures and Interconnect).

Rectangular Waveguide	TE/TM mode cutoffs, guide wavelength, wave impedance, and conductor loss.
Circular Waveguide	TE/TM mode cutoffs using Bessel zeros, guide wavelength, wave impedance, and conductor loss.
Cavity Resonator	Rectangular/circular cavity resonant frequencies and the unloaded Q of the $TE_{10}\ell$ mode.
Dielectric Resonator	Approximate TE01δ resonant frequency of a cylindrical dielectric puck.

MRI Coils & Tools

See Chapter 10 (MRI Physics and Coil Engineering).

Loop Inductance	Inductance of a single circular loop (Rosa formula).
Simple RF Coil Designer	Single-turn loop inductance L and the tuning capacitor C for resonance at f.
Advanced RF Coil Designer	Complete MRI surface receive coil design with matching network.
Birdcage Coil	LP/HP birdcage MRI volume coil: element values and mode frequencies.
SAR Estimator	Local SAR from B_1^+, frequency, tissue type vs IEC limits.
Array Decoupling	Mutual inductance and overlap/capacitive/preamp decoupling.

Converters

See Chapters 2 and 9 (Impedance; Measurement).

S/Z/Y/ABCD Converter	Convert between all four 2-port parameter representations.
S-Parameter Plotter	Plot Touchstone .s1p–.s4p files with Chart.js.
E-Series Finder	Nearest standard R, C, or L value in E6–E192 with % error.
ABCD Matrix Cascade	Multiply two-port ABCD matrices (series Z, shunt Y, line, transformer) and convert to S-parameters.

Simulators

See Chapters 2–4 (Impedance; Components).

RF Circuit Simulator	ABCD-matrix ladder network simulator: S-params, VSWR, group delay.
SPICE Simulator	MNA-based SPICE engine: .AC, .DC, .OP, .TRAN analyses in browser.

Solutions to Exercises

The answers below are keyed to the end-of-section exercises. Try each problem before checking here; many can be verified directly with the companion calculator named in the section.

Section 1.1: Maxwell's Equations

1. $\eta_0 = \sqrt{\mu_0/\varepsilon_0} = \sqrt{1.2566 \times 10^{-6}/8.854 \times 10^{-12}} = 376.7\ \Omega \approx 120\pi\ \Omega$.
2. $v = c/\sqrt{\varepsilon_r} = c/2 = 1.5 \times 10^8$ m/s; $\lambda = v/f = 0.15$ m $= 15$ cm.
3. Faraday's law: the changing magnetic flux set up by the primary induces an EMF in the secondary winding.

Section 1.2: Electromagnetic Waves

1. $E_0 = \sqrt{2\eta_0 S} = \sqrt{2(376.7)(10)} = \sqrt{7534} = 86.8$ V/m.
2. $\lambda = c/f = (3 \times 10^8)/(2.45 \times 10^9) = 0.1224$ m $= 12.2$ cm.
3. $90°$ (a quarter cycle); the sign of the phase sets the handedness (RHCP vs. LHCP).

Section 1.3: Dielectrics and Permittivity

1. $\mathrm{VF} = 1/\sqrt{2.1} = 0.69$ (about 69%).
2. $\lambda = c/(f\sqrt{\varepsilon_r}) = (3 \times 10^8)/[(5 \times 10^9)(1.913)] = 31.4$ mm.
3. PTFE — its loss tangent is roughly $20\times$ lower, so FR-4 is generally avoided above a few GHz.

Section 1.4: Magnetic Materials

1. $\times 4$, since $L \propto N^2$.
2. Eddy-current and hysteresis losses rise steeply with frequency; a high-resistivity ferrite is used instead.
3. Mainly resistive — the bead operates where the ferrite is lossy, so it absorbs (dissipates) the unwanted RF rather than storing it.

Section 1.5: Skin Effect

1. $\delta \propto 1/\sqrt{f}$, so lowering f by 10× raises δ by $\sqrt{10}$: $2.1 \times 3.16 = 6.6$ μm.
2. It falls by $\sqrt{100} = 10\times$.
3. $\delta_{\mathrm{Au}} = \delta_{\mathrm{Cu}}\sqrt{\sigma_{\mathrm{Cu}}/\sigma_{\mathrm{Au}}} = 2.1\sqrt{5.8/4.1} = 2.5$ μm; a plating a few skin depths thick (∼5–8 μm) carries essentially all the current.

Section 2.1: Decibels, dBm and RF Power Units

1. $10^{43/10} = 1.995 \times 10^4$ mW ≈ 20 W.
2. $-90 - (-70) = -20$ dBc.
3. dBW = dBm − 30 = 0 dBW (= 1 W).

Section 2.2: Complex Impedance and Reactance

1. $X_C = 1/(2\pi f C) = 1/[2\pi(2.4 \times 10^9)(2 \times 10^{-12})] = 33.2\ \Omega$.
2. $f = X_L/(2\pi L) = 50/[2\pi(10 \times 10^{-9})] = 796$ MHz.
3. $|Z| = \sqrt{30^2 + 40^2} = 50\ \Omega$; $\angle Z = \arctan(-40/30) = -53.1^\circ$ (capacitive).

Section 2.3: Transmission Line Theory

1. $\Gamma = (75 - 50)/(75 + 50) = 0.20$.
2. $Z_1 = \sqrt{Z_0 Z_L} = \sqrt{50 \times 25} = 35.4\ \Omega$.
3. $Z_{in} = Z_1^2/Z_L = 70.7^2/50 = 100\ \Omega$.

Section 2.4: Transmission Line Loss and Attenuation

1. $0.15 \times 20 = 3$ dB.
2. $10^{-3/10} = 0.50$ (50%; 3 dB is a halving).
3. $t = 5/(0.66 \times 3 \times 10^8) = 25.3$ ns.

Section 2.5: VSWR, Return Loss and Reflection Coefficient

1. $|\Gamma| = (3 - 1)/(3 + 1) = 0.5$; RL $= -20\log_{10} 0.5 = 6.0$ dB; reflected $= |\Gamma|^2 = 25\%$.
2. $|\Gamma| = 10^{-20/20} = 0.1$; VSWR $= 1.1/0.9 = 1.22$.
3. VSWR = 1 and RL $= \infty$ ($\Gamma = 0$).

Section 2.6: The Smith Chart

1. $z = Z_L/Z_0 = 0.5 - j1.0$.
2. Short: $z = 0$ (left-most point); open: $z = \infty$ (right-most point).
3. Its resistive part already equals Z_0; the right series reactance cancels x and lands at the center (matched).

Section 2.7: S-Parameters

1. $\mathrm{RL} = -20\log_{10}(0.316) = 10$ dB.
2. $\mathrm{IL} = -20\log_{10}(0.5) = 6.0$ dB.
3. $|S_{11}|^2 + |S_{21}|^2 = 1$ (a lossless network conserves power).

Section 2.8: Impedance Matching

1. $Q = \sqrt{200/50 - 1} = \sqrt{3} = 1.73$.
2. The multi-section transformer (more degrees of freedom broaden the band).
3. $Z_L = Z_s^* = 50 - j30\ \Omega$.

Section 3.1: Microwave Transmission Lines

1. $\varepsilon_{\text{eff}} = \varepsilon_r = 4$ (all field is in the dielectric).
2. Lower Z_0 (more capacitance per unit length).
3. $\lambda_g = \lambda_0/\sqrt{\varepsilon_{\text{eff}}} = 15/\sqrt{3.34} = 8.2$ cm.

Section 3.2: RF PCB Design

1. $X_L = 2\pi f L = 2\pi(5 \times 10^9)(2 \times 10^{-9}) = 62.8\ \Omega$.
2. It halves (two inductors in parallel).
3. A solid plane gives an uninterrupted return path under the trace, keeping impedance controlled and loop area (radiation, crosstalk) small; a slot forces a detour.

Section 3.3: Coaxial Cable

1. $\mathrm{VF} = 1/\sqrt{2.25} = 0.667$.
2. Raises Z_0 ($Z_0 \propto \log(D/d)$).
3. $0.5 \times 6 = 3$ dB (a halving of power).

Section 3.4: RF Connectors

1. 2.92 mm — SMA is only specified to about 18 GHz.
2. Yes; the 2.92 mm (“K”) interface is mechanically compatible with SMA, though careful use preserves precision.
3. It deforms or wears the mating surfaces, degrading the impedance interface and raising reflections (worse return loss).

Section 3.5: Waveguides

1. $f_c = c/(2a) = (3 \times 10^8)/(2 \times 0.015) = 10$ GHz.
2. No — 5 GHz is below cutoff, so the wave is evanescent (does not propagate).
3. About $2f_c \approx 13$ GHz.

Section 3.6: VNA Measurements and Calibration

1. $|\Gamma| = 10^{-20/20} = 0.1$; VSWR $= 1.1/0.9 = 1.22$.
2. Short, open, and matched load.
3. It places the reference plane at the device, calibrating out the cables and connectors.

Section 4.1: E-Series Standard Component Values

1. $10^{1/24} = 1.10$ (about a 10% step; E24 is the 5% series).
2. E12 neighbours are 3.9 and 4.7 kΩ; 4.7 kΩ is closest.
3. E96 (the 1% series).

Section 4.2: Coupled Resonators

1. BW $\approx kf_0 = 0.05 \times 1$ GHz $= 50$ MHz.
2. Halve it (BW $\propto k$).
3. Critical coupling.

Section 4.3: RF Attenuators

1. $K = 10^{6/20} = 2.0$.
2. $10^{-10/10} = 0.10$ (10%).
3. About 0.99 W — only 1% ($10^{-20/10}$) reaches the load.

Section 4.4: Power Dividers and Combiners

1. $Z_{\text{arm}} = 75\sqrt{2} = 106\ \Omega$.
2. 3 dB (half the power to each of two outputs).
3. $R = 2Z_0 = 100\ \Omega$.

Section 4.5: BALUNs and Hybrids

1. $\sqrt{9} = 3{:}1$.
2. $300/75 = 4{:}1$.
3. A defined 180° phase difference between outputs (sum/difference ports); the Wilkinson gives in-phase outputs.

Section 4.6: Microwave Passive Components

1. $10^{-10/10} = 0.10$ (10%).
2. An isolator (a circulator with one port terminated) absorbs the reflected wave.
3. $30 - 20 = +10$ dBm.

Section 4.7: RF Filters

1. $20 \times 5 = 100$ dB/decade.
2. Three.
3. Chebyshev (steeper per order, at the price of passband ripple).

Section 4.8: Group Delay and Phase Linearity

1. $\tau_g = -\,d\phi/d\omega = \tau$ — constant, equal to the delay τ.
2. Bessel (Thomson).
3. It disperses the symbol energy in time, causing intersymbol interference and raising the bit-error rate.

Section 4.9: Stepped-Impedance Low-Pass Filters

1. A series inductor.
2. A shunt capacitor.
3. Raise Z_0 (use a narrower line).

Section 4.10: Advanced Filters

1. $Q_L = f_0/\text{BW} = 1/0.01 = 100$.
2. Acoustic filters (SAW and especially BAW/FBAR).
3. Their high unloaded Q gives the low insertion loss and steep skirts needed to separate closely spaced TX/RX bands.

Section 4.15: Switches, Phase Shifters and Ferrite Devices

1. $360°/2^4 = 22.5°$.
2. An isolator (a circulator with one port terminated).
3. RF MEMS switches.

Section 5.1: RF Transistors

1. $f_T = 0.08/[2\pi(0.1 \times 10^{-12})] = 127$ GHz.
2. GaN HEMT (high breakdown voltage and power density).
3. f_{max} — it is the frequency of unity *power* gain, the real limit for amplification.

Section 5.6: RF Diodes and Varactors

1. $C = 8/\sqrt{1 + 3/0.7} = 8/\sqrt{5.29} = 3.48$ pF.
2. A PIN diode.
3. Its stored charge in the intrinsic region cannot follow the fast RF, so it presents a fixed resistance set by the DC bias rather than rectifying.

Section 5.11: RF Amplifiers

1. $-30 + 12 = -18$ dBm.
2. Falls — the amplifier compresses.
3. The LNA, so its low noise figure dominates the cascade (Friis) before later stages add noise.

Section 5.12: Amplifier Stability

1. Yes — both $K > 1$ and $|\Delta| < 1$ hold.
2. Potentially unstable: some source/load impedances will oscillate.
3. Add a stabilizing resistor (series or shunt, often at the output) or resistive feedback.

Section 5.13: Small-Signal Amplifier Design

1. $G_{TU,\max} = 9/[(1-0.16)(1-0.09)] = 9/0.764 = 11.8 = 10.7$ dB.
2. It is only defined when the device is unconditionally stable ($K > 1$); otherwise no simultaneous conjugate match exists.
3. The locus of source (or load) reflection coefficients giving a chosen constant gain.

Section 5.18: Low-Noise Amplifier Design

1. $F = F_{min}$ exactly (the mismatch term vanishes).
2. A resistor would add its own thermal noise; the inductor sets a real input impedance without adding noise.
3. The first stage (the LNA), by the Friis cascade formula.

Section 5.23: Noise in RF Systems

1. $-174 + 10\log_{10}(10^7) = -174 + 70 = -104$ dBm.
2. Stage 1 — with $G_1 = 100$ the $(F_2 - 1)/G_1$ term is small, so $F_{\text{tot}} \approx F_1$.
3. The LNA: as the first stage with substantial gain, its NF sets the system NF (Friis).

Section 5.24: Power Amplifier Design

1. 50%.
2. $\pi/4 = 78.5\%$.
3. Improves linearity (lower distortion) at the cost of drain efficiency.

Section 5.25: Thermal Management and Junction Temperature

1. $T_j = 30 + 8 \cdot 6 = 78$ °C.
2. θ_{sa}, the sink-to-ambient resistance.
3. To gain reliability margin — lifetime falls roughly exponentially with T_j, so headroom greatly extends the mean time to failure.

Section 5.26: Intermodulation Distortion and IP3

1. $\text{IMD} = 2(10 - (-20)) = 60$ dBc.
2. IM3 grows with slope 3, so $3 \times 3 = 9$ dB.
3. The late, high-gain stage (its intercept is referred back divided by the preceding gain).

Section 6.1: Oscillators

1. $0°$ modulo $360°$.
2. Gain > 1 lets the oscillation build from noise; nonlinear compression then holds it at unity in steady state.
3. Helps — higher Q lowers phase noise (Leeson).

Section 6.2: Quartz Crystal Resonators

1. Parallel resonance ($f_p > f_s$).
2. Its acoustic (mechanical) resonance is extremely low-loss — the effective motional inductance is very large and R_1 small, so $Q = \omega L_1/R_1$ is huge.
3. It lowers the frequency, pulling it back toward f_s; parallel crystals are specified at a stated load capacitance.

Section 6.3: Oscillator Design

1. $C_s = 2$ pF; $f_0 = 1/(2\pi\sqrt{(5 \times 10^{-9})(2 \times 10^{-12})}) = 1.59$ GHz.
2. So the load cannot pull (shift) the oscillation frequency.
3. Its Q is orders of magnitude higher, and Leeson's model makes phase noise fall as $1/Q^2$.

Section 6.7: Phase Noise and Jitter

1. Falls (it is highest close to the carrier).
2. 20 dB/decade (the $1/f^2$ slope).
3. Less — $\sigma_t = \sigma_\phi/(2\pi f_c)$, so higher f_c reduces jitter for the same phase error.

Section 6.8: Phase-Locked Loops and Frequency Synthesis

1. $f_{out} = N f_{ref} = 100 \times 10$ MHz $= 1$ GHz.
2. $f_{ref} = 1$ MHz.
3. VCO noise — a wider loop tracks the reference over a broader offset, suppressing VCO phase noise inside the loop (at the cost of more reference-spur feedthrough).

Section 6.9: PLL and Synthesizer Design

1. $\Delta f = 50 \times 10^6/2^{24} = 2.98$ Hz.
2. Fractional-N (or DDS).
3. More reference-spur feedthrough (and it must stay well below f_{ref} for stability).

Section 6.13: Mixers and Frequency Conversion

1. IF $= |f_{RF} - f_{LO}| = 100$ MHz.
2. At $f_{LO} - f_{IF} = 800 - 100 = 700$ MHz.
3. An RF band-select / image-reject filter ahead of the mixer (or an image-reject mixer).

Section 6.14: Modulation Schemes

1. $\log_2 4 = 2$ bits/symbol.
2. $R_b = R_s \log_2 M = 2$ Msym/s $\times 6 = 12$ Mbit/s.
3. 256-QAM — denser constellations need higher SNR for the same bit-error rate.

Section 7.1: Antenna Fundamentals

1. $10 \log_{10}(100) = 20$ dBi.
2. $G \propto f^2$, so $\times 4$, i.e. $+6$ dB.
3. $12 + 2.15 = 14.15$ dBi.

Section 7.2: Antenna Polarisation

1. $-10 \log_{10}(\cos^2 30°) = -10 \log_{10}(0.75) = 1.25$ dB.
2. 3 dB (fixed, independent of rotation).
3. 0 dB (axial ratio $= 1$).

Section 7.3: Yagi-Uda Antennas

1. The reflector (about 0.505λ).
2. Shorter — adding more directors raises gain with diminishing returns.
3. About 10 dBi.

Section 7.4: Helical Antennas

1. About one wavelength ($C \approx \lambda$).
2. $G \propto N$, so $\times 2$, i.e. $+3$ dB.
3. Circular polarization.

Section 7.5: Aperture Antennas: Reflectors and Horns

1. $\lambda = 0.025$ m; $G = 0.55(\pi \cdot 0.6/0.025)^2 = 0.55\,(75.4)^2 \approx 3130$, i.e. 35.0 dBi.
2. $\lambda = 0.05$ m; HPBW $\approx 70 \cdot 0.05/2 = 1.75°$.
3. Spillover past the reflector edge, illumination (amplitude) taper, aperture blockage by the feed, and surface (phase) errors.

Section 7.6: Antenna Types

1. $L \approx c/(2f\sqrt{\varepsilon_{\text{eff}}}) = 3 \times 10^8/(2 \times 10^9 \times 2.5) = 60$ mm.
2. Log-periodic or spiral (frequency-independent geometries).
3. The parabolic reflector (a large aperture focuses a much narrower, higher-gain beam).

Section 7.11: Phased Arrays and Beamforming

1. $\beta = -kd \sin 90° = -(2\pi/\lambda)(\lambda/2)(1) = -\pi = -180°$.
2. To avoid grating lobes (additional main-beam-strength lobes) that appear once $d > \lambda$.
3. Lower sidelobes, at the cost of a slightly wider main beam (and lower peak directivity).

Section 7.16: Radio Propagation and Fading

1. $\lambda = 0.333$ m; $r_1 = \sqrt{0.333 \cdot 1000 \cdot 1000/2000} = \sqrt{166.7} = 12.9$ m.
2. Rician.
3. The delay spread (its inverse is roughly the coherence bandwidth).

Section 7.20: Antenna Measurement

1. $\lambda = 0.1$ m; $R = 2(1)^2/0.1 = 20$ m.
2. The three-antenna method.
3. Directly with a vector network analyzer (S_{11}).

Section 8.1: RF Frequency Bands

1. X-band (8–12 GHz).
2. $\ell = c/(2f) = (3 \times 10^8)/(2 \times 10^9) = 15$ cm.
3. Smaller ($\lambda = c/f$ shrinks with frequency).

Section 8.2: The Superheterodyne Receiver

1. IF = 90 MHz; image at $f_{LO} + f_{IF} = 1180$ MHz.
2. Easier — the image moves farther from the passband.
3. $2f_{IF}$.

Section 8.3: Receiver Architectures

1. $2f_{IF}$.
2. Any two of: DC offset, flicker ($1/f$) noise, LO leakage/re-radiation, I/Q imbalance.
3. IRR $= 10\log_{10}[4/(0.0004 + 0.0001)] = 10\log_{10}(8000) = 39.0$ dB.

Section 8.7: Receiver Cascade Analysis

1. Its gain G_1 divides every later stage's excess noise $(F_i - 1)$, so early gain suppresses later noise.
2. $F_1 = 1.26$, $G_1 = 100$, $F_2 = 10$: $F_{tot} = 1.26 + 9/100 = 1.35$, so NF = 1.3 dB.
3. The last (high-level) stage.

Section 8.8: Gain Planning and Dynamic Range

1. $-174 + 10\log_{10}(2 \times 10^5) + 8 = -174 + 53 + 8 = -113$ dBm.
2. SFDR $= \frac{2}{3}(0 - (-110)) = 73.3$ dB.
3. AGC extends the instantaneous dynamic range (handling strong signals); it does not change SFDR, set by IIP3 and the noise floor.

Section 8.12: I/Q Modulation and Impairments

1. EVM $\approx 10^{-30/20} = 3.2\%$.
2. I/Q gain and/or phase imbalance.
3. It lumps together noise, I/Q imbalance, nonlinear distortion, and phase noise into one measured number.

Section 8.16: Sampling and Data Conversion

1. $6.02(10) + 1.76 = 61.96$ dB.
2. $-20\log_{10}(2\pi \cdot 10^9 \cdot 0.5 \times 10^{-12}) = -20\log_{10}(3.14 \times 10^{-3}) = 50$ dB.
3. $f_s \geq 2B = 10$ Msps (placed so the band folds without self-overlap).

Section 8.20: Software-Defined Radio

1. $61.44 \times 10^6 \times 12 \times 2 = 1.47$ Gbit/s.
2. $100/1 = 100\times$ decimation (to $\sim$1–2 Msps for a 1 MHz channel).
3. Any of: reconfigurable waveform, digital predistortion, digital I/Q/DC-offset correction, instant retuning.

Section 8.24: Link Budgets

1. 20 dB (FSPL $\propto d^2$).
2. $P_{rx} = 20 + 3 - 100 = -77$ dBm.
3. Margin $= -77 - (-90) = 13$ dB.

Section 8.25: Radar Fundamentals

1. $R = c\Delta t/2 = (3 \times 10^8)(60 \times 10^{-6})/2 = 9000$ m.
2. -12 dB (power $\propto 1/R^4$).
3. $f_d = 2v/\lambda = 2(30)/0.03 = 2000$ Hz.

Section 8.26: FMCW and Pulsed Radar

1. $\Delta R = c/(2B) = (3 \times 10^8)/(2 \times 10^9) = 0.15$ m.
2. Double the bandwidth ($\Delta R \propto 1/B$).
3. Increases ($f_b \propto R$).

Section 8.27: Digital Communication Fundamentals

1. $C = 5 \times 10^6 \log_2(32) = 5 \times 10^6 \times 5 = 25$ Mbit/s.
2. On E_b/N_0 (which normalizes out rate and bandwidth).
3. Sharp (short) pulses require large bandwidth; pulse shaping limits bandwidth while controlling inter-symbol interference.

Section 8.31: Coding, Spread Spectrum and OFDM

1. $G_p = 10^6/(2 \times 10^3) = 500 = 27$ dB.
2. The channel's delay spread (so inter-symbol interference is absorbed).
3. Its high peak-to-average power ratio (PAPR), which forces PA back-off.

Section 9.1: RF Measurement and Instrumentation

1. $Y = 10$ dB $= 10$; $F = 31.6/(10 - 1) = 3.51$, so NF $= 5.5$ dB.
2. It drops (less noise power in a narrower bandwidth), revealing weaker signals.
3. Impedance discontinuities (and their distance) along a transmission line, from the echo delay.

Section 9.5: RF Simulation and Design Flow

1. Harmonic balance (nonlinear steady-state response to a periodic drive).
2. $\lambda_0 = 60$ mm, so $\Delta \leq 6$ mm.
3. EM simulation is computationally expensive; verifying only critical structures keeps runtimes practical.

Section 9.9: EMC Shielding Effectiveness

1. $f \approx c/(2L) = (3 \times 10^8)/(2 \times 0.15) = 1$ GHz.
2. Many small holes — leakage depends on the longest slot dimension, not total area.
3. A pigtail (long, inductive shield termination); a 360° shield bond is needed instead.

Section 9.10: RF Safety and Human Exposure

1. $S = 100/(4\pi \cdot 25) = 0.318$ W/m^2.
2. $f/200 = 1800/200 = 9$ W/m^2.
3. -6 dB (power density $\propto 1/R^2$).

Section 10.1: NMR Basics

1. $f_0 = 42.58 \times 3 = 127.7$ MHz.
2. $B_0 = f_0/\gamma = 300/42.58 = 7.05$ T.
3. Higher frequency (~ 298 MHz vs. 64 MHz).

Section 10.2: MRI Hardware

1. Roughly doubles (SNR scales about linearly with B_0).
2. As B_0^2 (SAR $\propto f^2 \propto B_0^2$).
3. SNR improves; susceptibility artefacts get worse.

Section 10.3: MRI Gradient Coils

1. $\mathrm{SR} = 0.03/(150 \times 10^{-6}) = 200$ T/m/s.
2. Peripheral nerve stimulation (driven by dB/dt).
3. Echo time and repetition time (faster sequences).

Section 10.4: MRI Pulse Sequences

1. Spin echo (its 180° pulse refocuses static inhomogeneity).
2. T_1 weighting.
3. Phase dispersion from static B_0 inhomogeneity, so the echo decays with true T_2 rather than T_2^*.

Section 10.5: MRI Contrast and Relaxation

1. $1 - e^{-1} = 63\%$.
2. $e^{-1} = 37\%$.
3. Predominantly T_1 — it brightens enhancing tissue on T_1-weighted images.

Section 10.6: MRI Artefacts

1. $3.5 \times 10^{-6} \times 127.7 \times 10^6 = 447$ Hz.
2. Worsens (it scales with B_0).
3. A higher readout (receive) bandwidth (more Hz per pixel).

Section 10.7: RF Coil Design for MRI

1. $C = 1/[(2\pi \cdot 64 \times 10^6)^2(200 \times 10^{-9})] = 31$ pF.
2. That the sample (not the coil) dominates losses — the SNR-optimal regime.
3. To decouple it from the transmit field, preventing coupling, heating, and B_1 distortion.

Section 10.8: MRI Coil Types

1. The surface coil (close coupling, low sample noise).
2. The volume coil (e.g. birdcage).
3. A phased array of small elements.

Section 10.9: The Birdcage Coil

1. The homogeneous ($k = 1$) mode.
2. About $\sqrt{2}$ ($\approx 1.41\times$).
3. The leg and end-ring capacitors together with the conductor inductances.

Section 10.10: SNR in MRI Receive Coils

1. $R_{\text{sample}}/R_{\text{coil}} = 3 - 1 = 2$.
2. Little — the sample already dominates, so lowering R_{coil} barely moves the total.
3. More coil-noise-limited (small coils and low field see relatively less sample loss).

Section 10.11: Preamplifier Decoupling

1. $Z_0^2/Z_{in} = 50^2/2 = 1250\ \Omega$.
2. More — a lower Z_{in} gives a higher transformed series impedance, choking the loop current harder.
3. Nearest-neighbour geometric (overlap) decoupling.

Section 10.12: Phased-Array Coils

1. By geometric overlap that nulls the mutual inductance ($M \to 0$).
2. Parallel imaging (SENSE/GRAPPA) for scan acceleration.
3. SNR and the achievable acceleration factor.

Section 10.13: Parallel Imaging

1. One third the scan time ($1/R$).
2. A factor of $1/\sqrt{4} = 1/2$.
3. Distinct, weakly overlapping element sensitivities (good spatial separation).

Section 10.14: B1 Field Mapping

1. The transmit RF field B_1 ($\alpha = \gamma \int B_1\, dt$).
2. The actual local flip angle α (hence the local B_1).
3. Yes — the RF wavelength in tissue shortens with field, so B_1 becomes less uniform.

Section 10.15: SAR and RF Safety in MRI

1. About $\times 4$ (SAR $\propto f^2$, and f doubles).
2. A factor of $1/4$ (SAR $\propto \alpha^2$).
3. 2 W/kg (whole body, normal operating mode).

References

[1] D. M. Pozar, *Microwave Engineering*, 4th ed. Hoboken, NJ: Wiley, 2012.

[2] R. E. Collin, *Foundations for Microwave Engineering*, 2nd ed. New York: IEEE Press, 2001.

[3] J. D. Jackson, *Classical Electrodynamics*, 3rd ed. New York: Wiley, 1998.

[4] S. Ramo, J. R. Whinnery, and T. Van Duzer, *Fields and Waves in Communication Electronics*, 3rd ed. New York: Wiley, 1994.

[5] M. Steer, *Microwave and RF Design*, 3rd ed. Raleigh, NC: NC State Univ., 2019.

[6] R. Ludwig and G. Bogdanov, *RF Circuit Design: Theory and Applications*, 2nd ed. Upper Saddle River, NJ: Prentice Hall, 2009.

[7] W. Hayward, *Introduction to Radio Frequency Design*. Newington, CT: ARRL, 1994.

[8] B. Razavi, *RF Microelectronics*, 2nd ed. Upper Saddle River, NJ: Prentice Hall, 2011.

[9] T. H. Lee, *The Design of CMOS Radio-Frequency Integrated Circuits*, 2nd ed. Cambridge, U.K.: Cambridge Univ. Press, 2004.

[10] G. Gonzalez, *Microwave Transistor Amplifiers: Analysis and Design*, 2nd ed. Upper Saddle River, NJ: Prentice Hall, 1997.

[11] S. C. Cripps, *RF Power Amplifiers for Wireless Communications*, 2nd ed. Norwood, MA: Artech House, 2006.

[12] S. A. Maas, *Microwave Mixers*, 2nd ed. Norwood, MA: Artech House, 1993.

[13] U. L. Rohde, *Microwave and Wireless Synthesizers: Theory and Design*. New York: Wiley, 1997.

[14] C. A. Balanis, *Antenna Theory: Analysis and Design*, 4th ed. Hoboken, NJ: Wiley, 2016.

[15] J. D. Kraus, *Antennas*, 2nd ed. New York: McGraw-Hill, 1988.

[16] P. P. Viezbicke, "Yagi antenna design," Nat. Bur. Stand., Tech. Note 688, 1976.

[17] M. I. Skolnik, *Introduction to Radar Systems*, 3rd ed. New York: McGraw-Hill, 2001.

[18] H. W. Ott, *Electromagnetic Compatibility Engineering*. Hoboken, NJ: Wiley, 2009.

[19] C. R. Paul, *Introduction to Electromagnetic Compatibility*, 2nd ed. Hoboken, NJ: Wiley, 2006.

[20] International Commission on Non-Ionizing Radiation Protection, "Guidelines for limiting exposure to electromagnetic fields (100 kHz to 300 GHz)," *Health Phys.*, vol. 118, no. 5, pp. 483–524, 2020.

[21] IEEE Standard 521-2019, *Standard Letter Designations for Radar-Frequency Bands*. New York: IEEE, 2019.

[22] E. M. Haacke, R. W. Brown, M. R. Thompson, and R. Venkatesan, *Magnetic Resonance Imaging: Physical Principles and Sequence Design*. New York: Wiley, 1999.

[23] D. G. Nishimura, *Principles of Magnetic Resonance Imaging*. Stanford, CA: Stanford Univ., 2010.

[24] Z.-P. Liang and P. C. Lauterbur, *Principles of Magnetic Resonance Imaging: A Signal Processing Perspective*. New York: IEEE Press, 2000.

[25] J. Mispelter, M. Lupu, and A. Briguet, *NMR Probeheads for Biophysical and Biomedical Experiments*, 2nd ed. London, U.K.: Imperial College Press, 2015.

[26] P. B. Roemer, W. A. Edelstein, C. E. Hayes, S. P. Souza, and O. M. Mueller, "The NMR phased array," *Magn. Reson. Med.*, vol. 16, no. 2, pp. 192–225, 1990.

[27] IEC 60601-2-33, *Medical Electrical Equipment—Part 2-33: Particular Requirements for the Basic Safety and Essential Performance of Magnetic Resonance Equipment for Medical Diagnosis*. Geneva, Switzerland: IEC, 2015.

[28] J. G. Proakis and M. Salehi, *Digital Communications*, 5th ed. New York: McGraw-Hill, 2008.

[29] B. Sklar, *Digital Communications: Fundamentals and Applications*, 2nd ed. Upper Saddle River, NJ: Prentice Hall, 2001.

[30] A. Goldsmith, *Wireless Communications*. Cambridge, U.K.: Cambridge Univ. Press, 2005.

[31] T. S. Rappaport, *Wireless Communications: Principles and Practice*, 2nd ed. Upper Saddle River, NJ: Prentice Hall, 2002.

[32] R. J. Mailloux, *Phased Array Antenna Handbook*, 3rd ed. Norwood, MA: Artech House, 2018.

[33] J.-S. Hong, *Microstrip Filters for RF/Microwave Applications*, 2nd ed. Hoboken, NJ: Wiley, 2011.